

(12) **United States Patent**
Ma et al.

(10) **Patent No.:** **US 12,389,629 B2**
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(54) **SOURCE/DRAIN REGIONS IN INTEGRATED CIRCUIT STRUCTURES**

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(73) Assignee: **Intel Corporation**, Santa Clara, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 1012 days.

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(65) **Prior Publication Data**

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(51) **Int. Cl.**

H01L 29/417 (2006.01)

H10D 30/62 (2025.01)

H10D 30/67 (2025.01)

(52) **U.S. Cl.**

CPC **H10D 30/6219** (2025.01); **H10D 30/62** (2025.01); **H10D 30/6757** (2025.01)

(58) **Field of Classification Search**

CPC H01L 29/41791; H01L 29/785; H01L 29/78696; H01L 29/41733; H01L 29/42392; H01L 21/823418; H01L 21/823412; H01L 21/823437; H01L 21/823462; H01L 29/0673; H01L 29/0847; H01L 29/41725-41791; H01L 29/7839; H01L 29/806; H01L 29/665-66507; H01L 29/7845; H01L 29/41783; H01L 21/823814; H01L 29/7842-7849; H01L 29/1054; H01L

29/66636; H10D 30/6219; H10D 30/62; H10D 30/6757; H10D 30/6729; H10D 30/6735; H10D 62/121; H10D 62/151; H10D 84/0128; H10D 84/038; H10D 84/013; H10D 84/0135; H10D 84/0144

See application file for complete search history.

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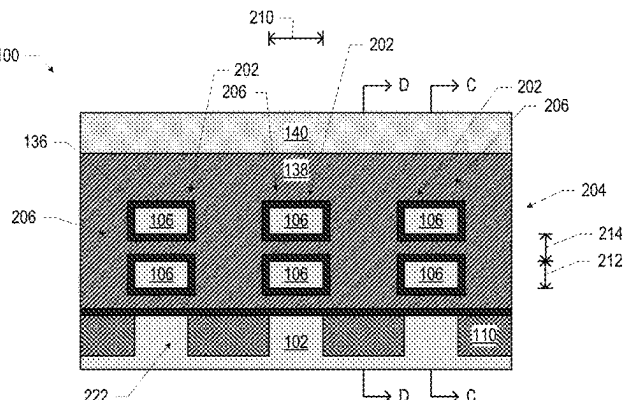
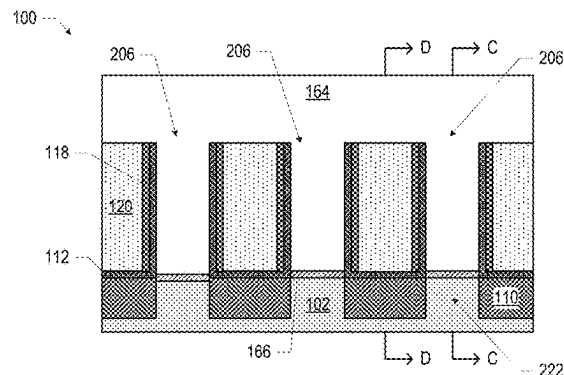
Primary Examiner — Cuong B Nguyen

(74) *Attorney, Agent, or Firm* — Akona IP PC

(57) **ABSTRACT**

Disclosed herein are source/drain regions in integrated circuit (IC) structures, as well as related methods and components. For example, in some embodiments, an IC structure may include: a channel region including a semiconductor material; and a source/drain region at a side face of the channel region, wherein the source/drain region includes a semiconductor portion and a contact metal, and the semiconductor portion is between the contact metal and the semiconductor material.

19 Claims, 95 Drawing Sheets



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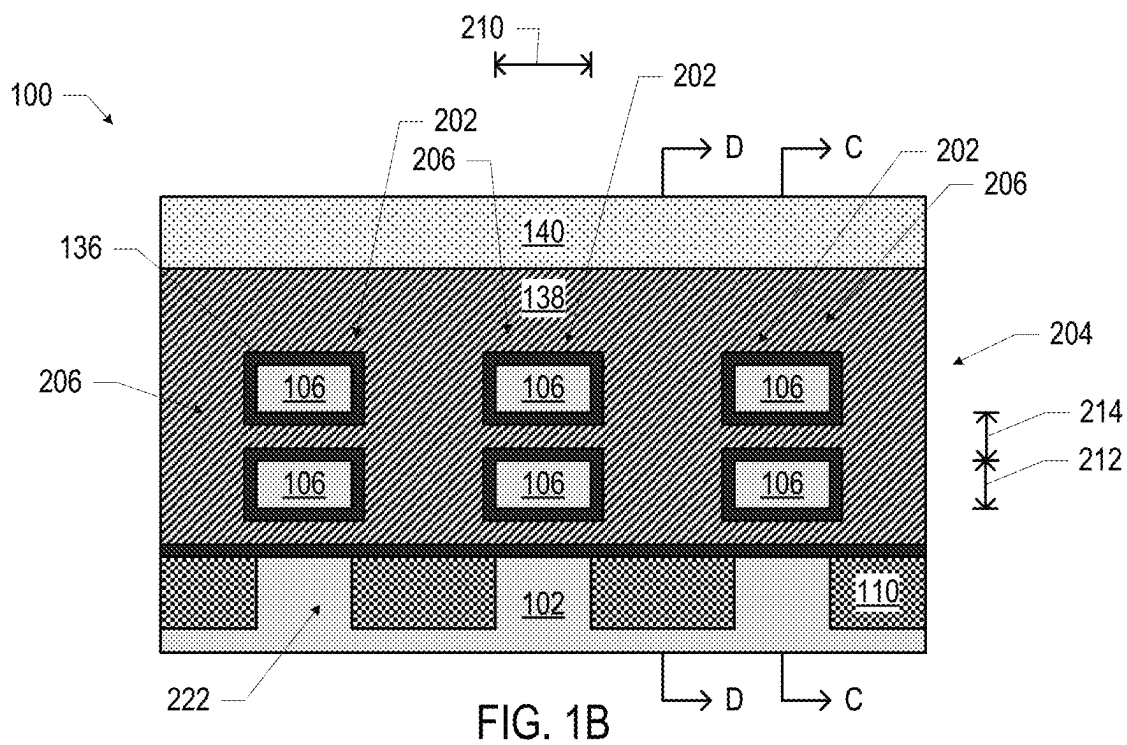
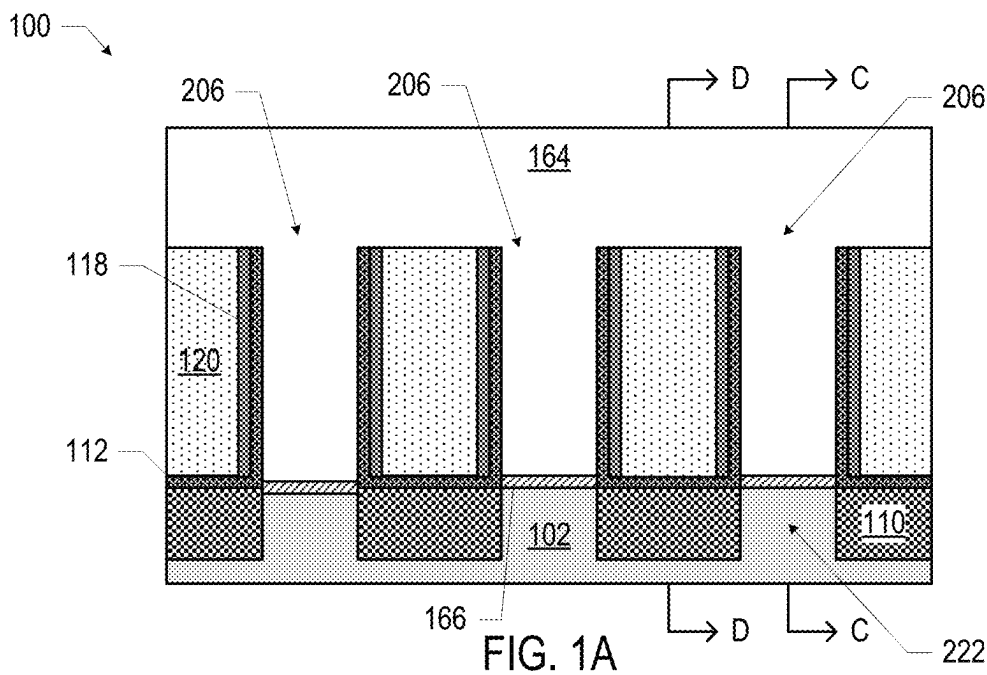
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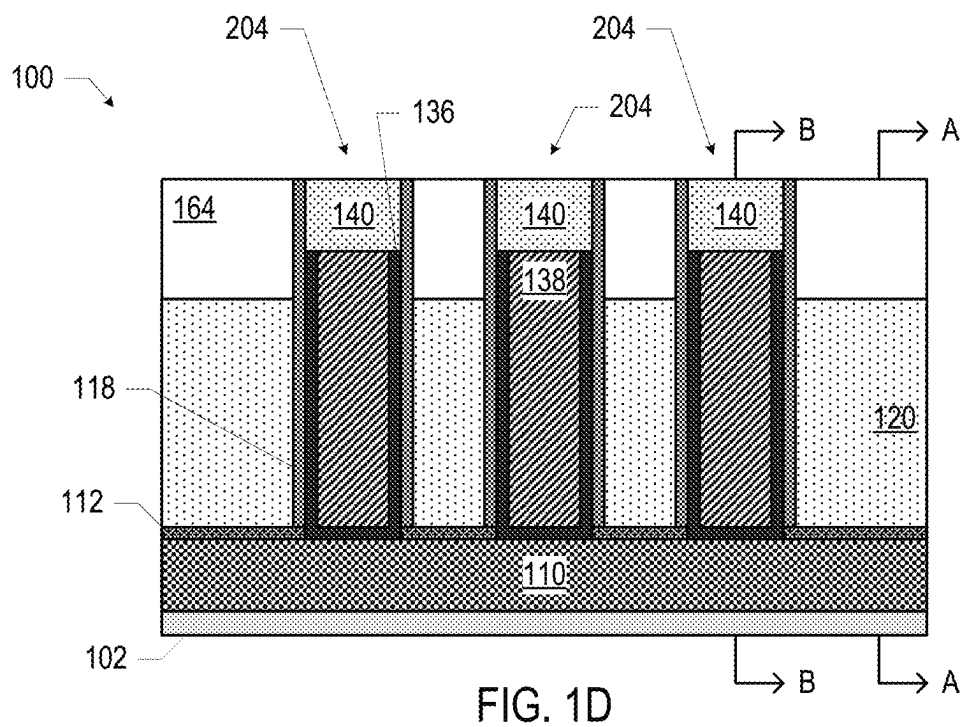
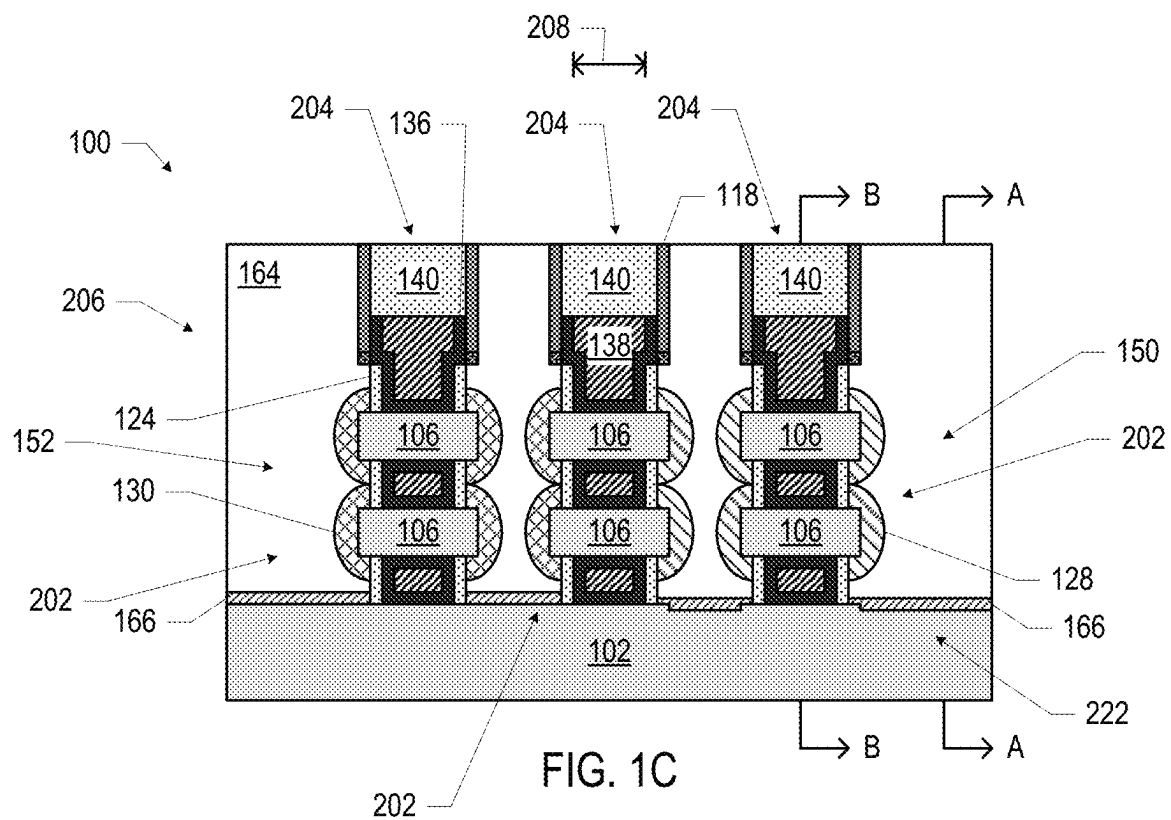
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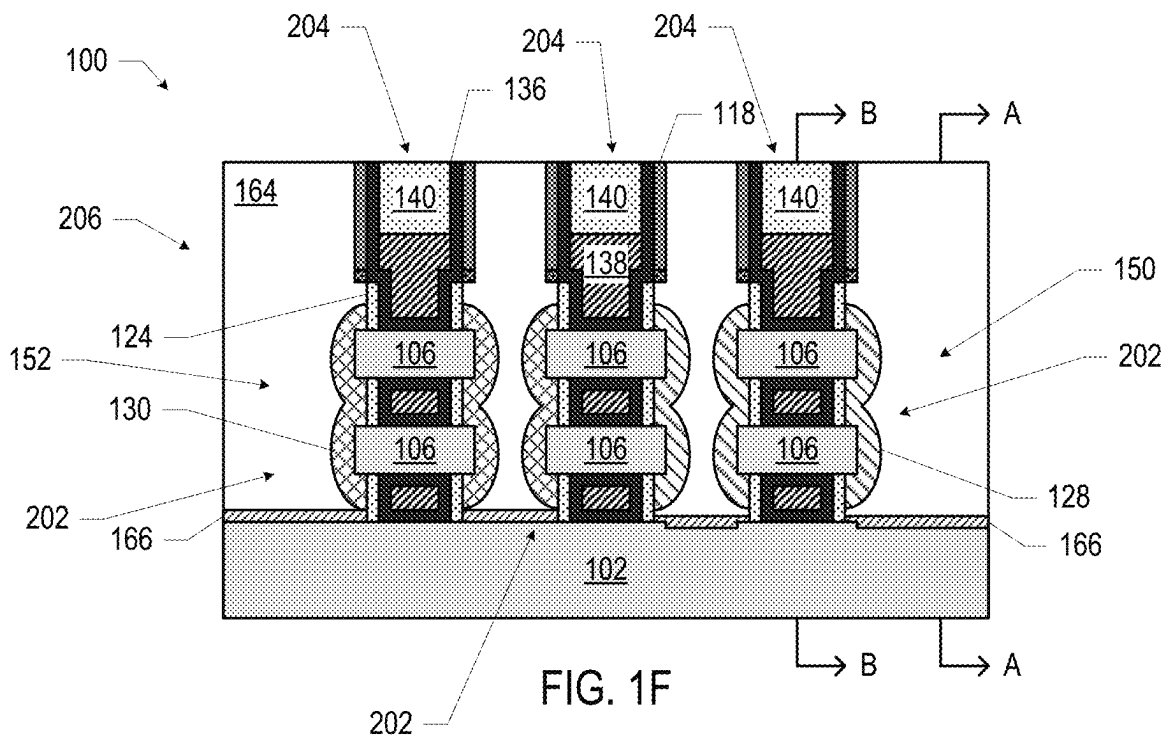
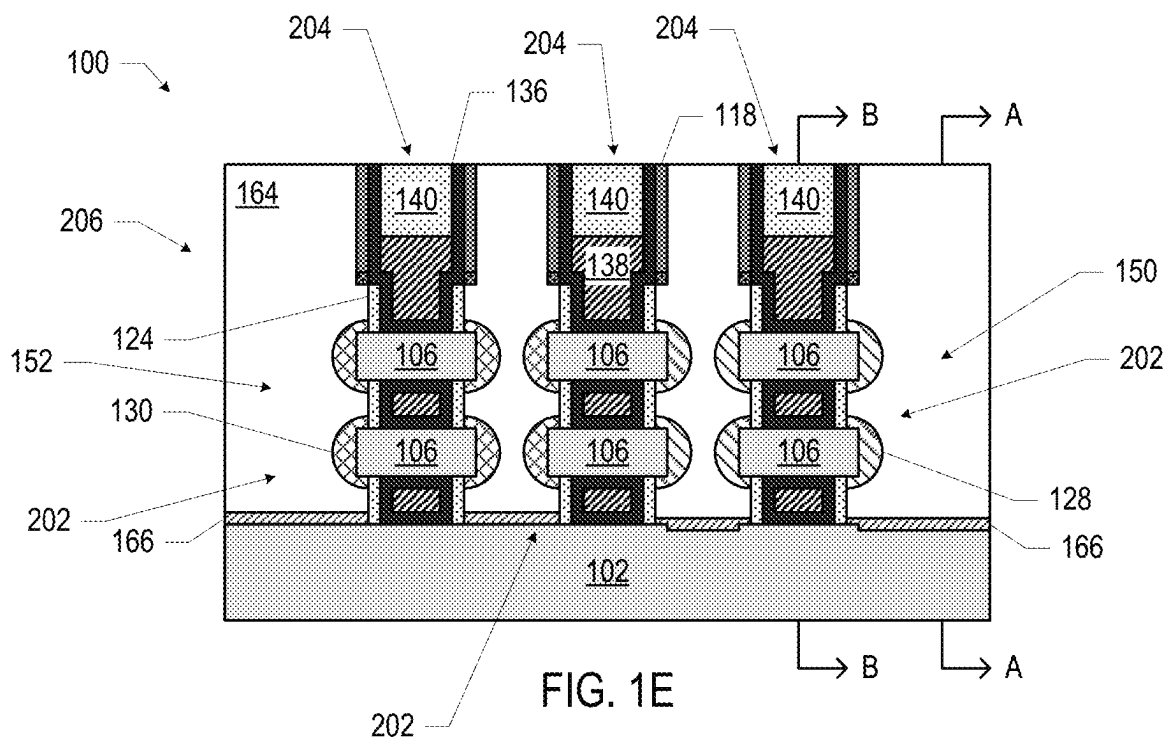
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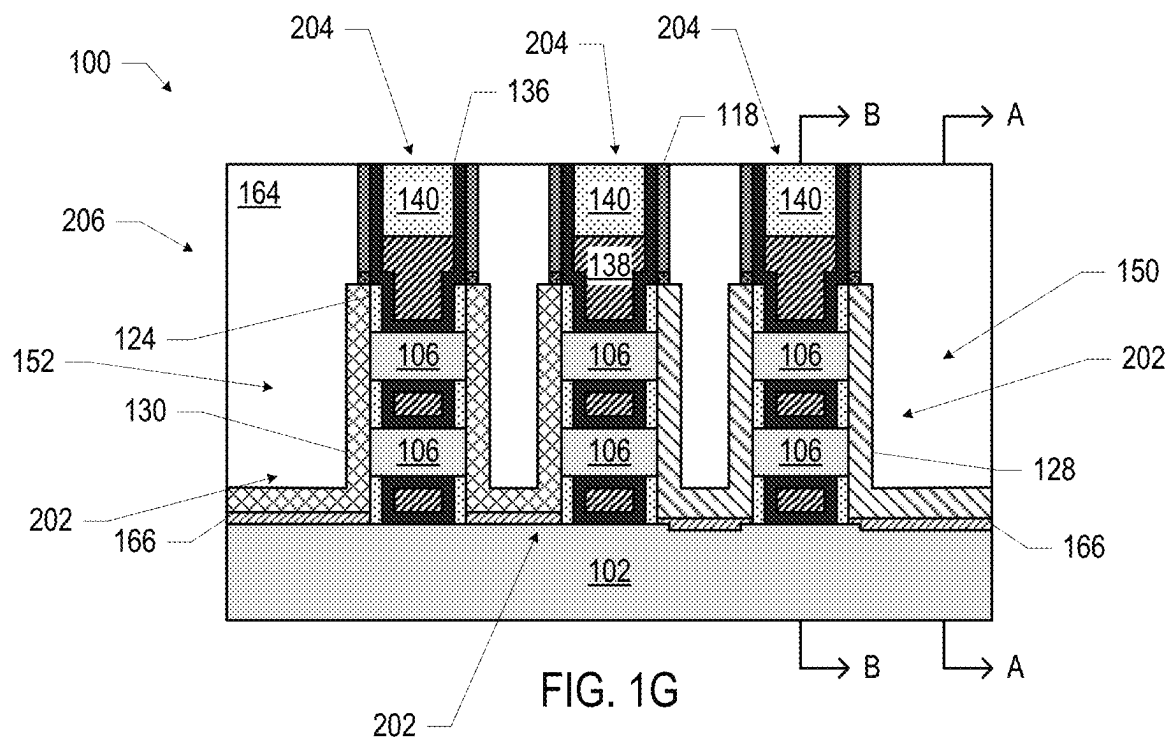
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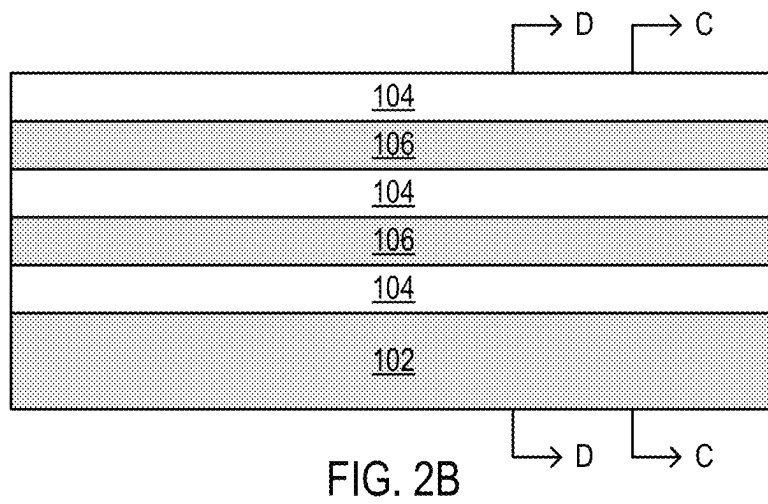
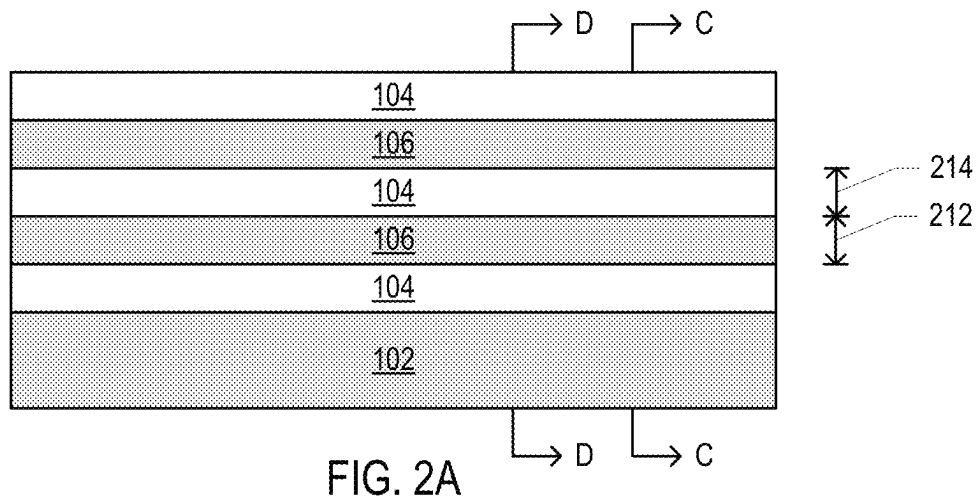
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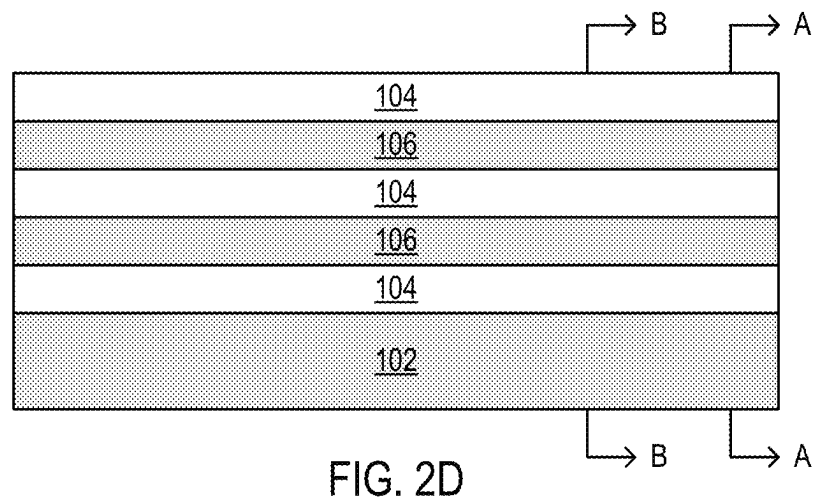
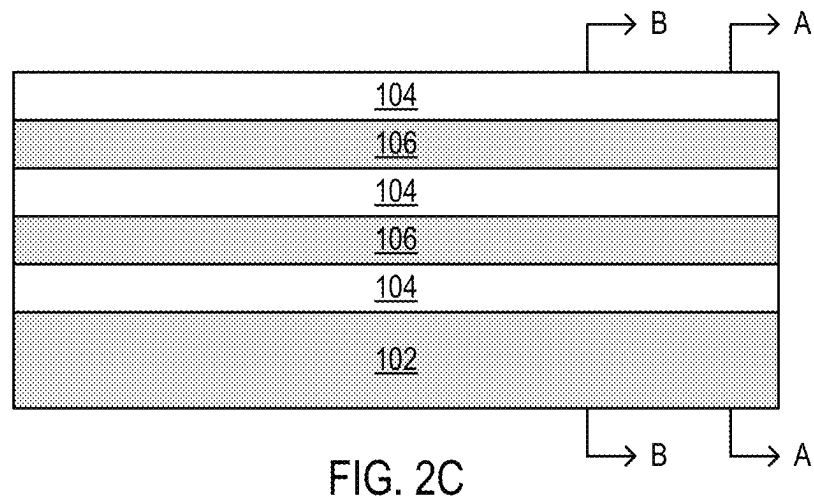












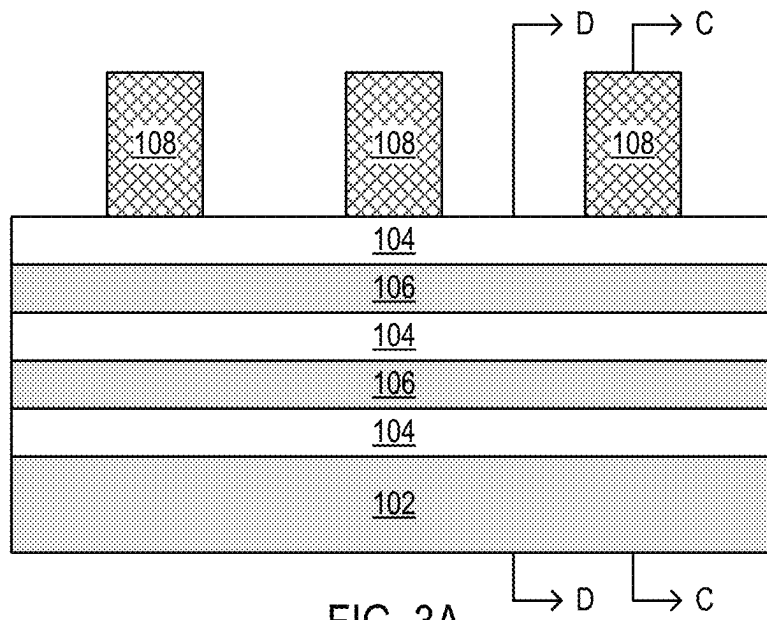


FIG. 3A

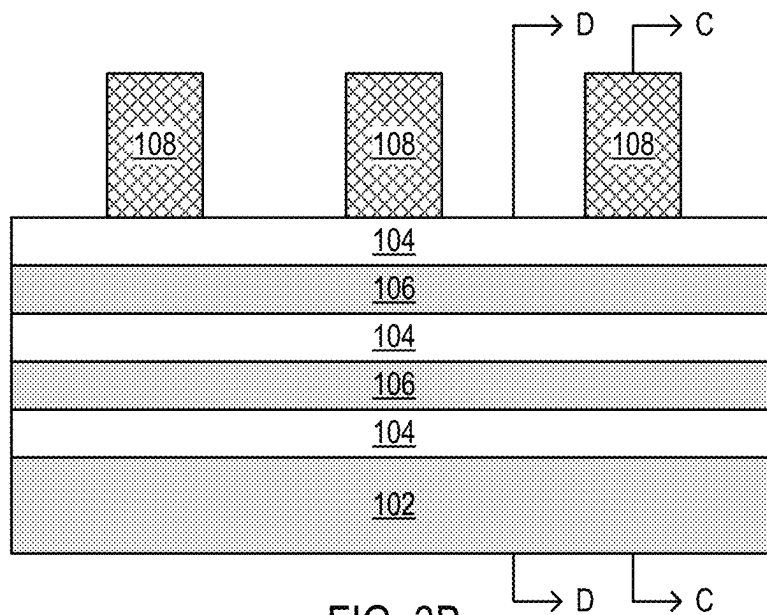


FIG. 3B

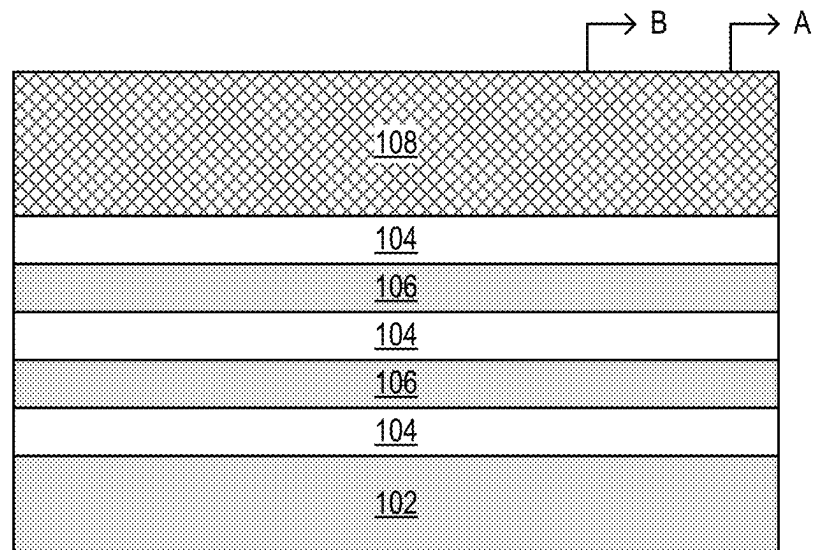


FIG. 3C

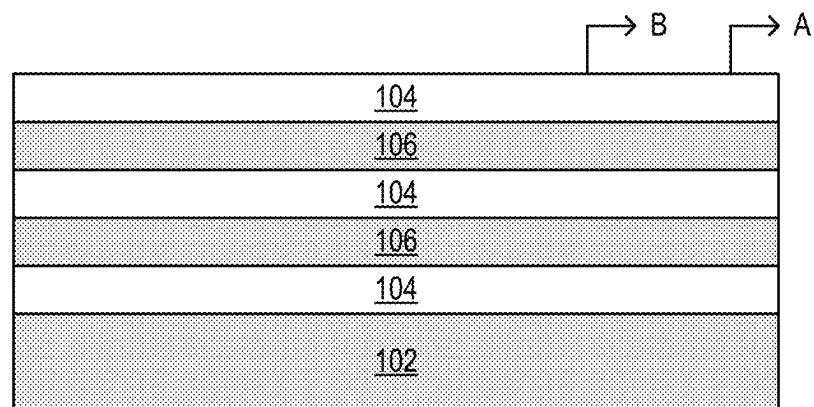
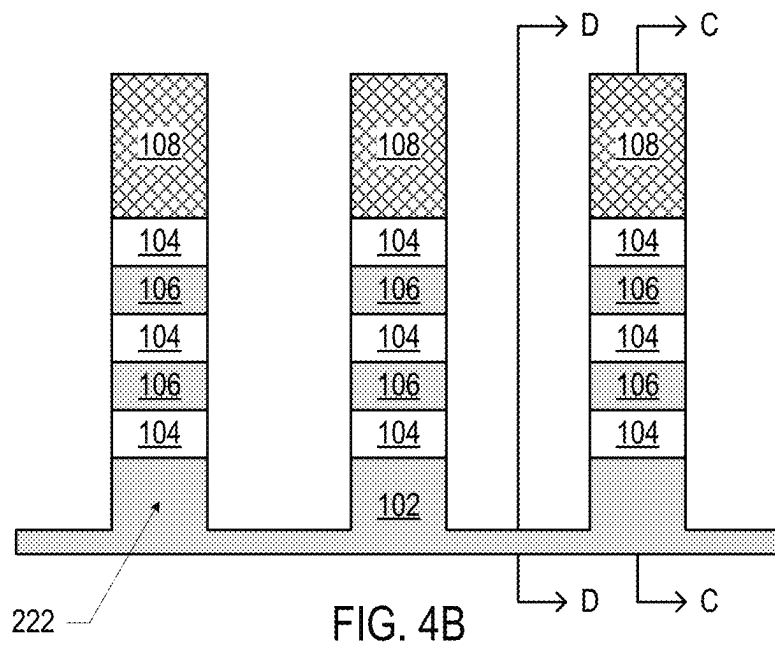
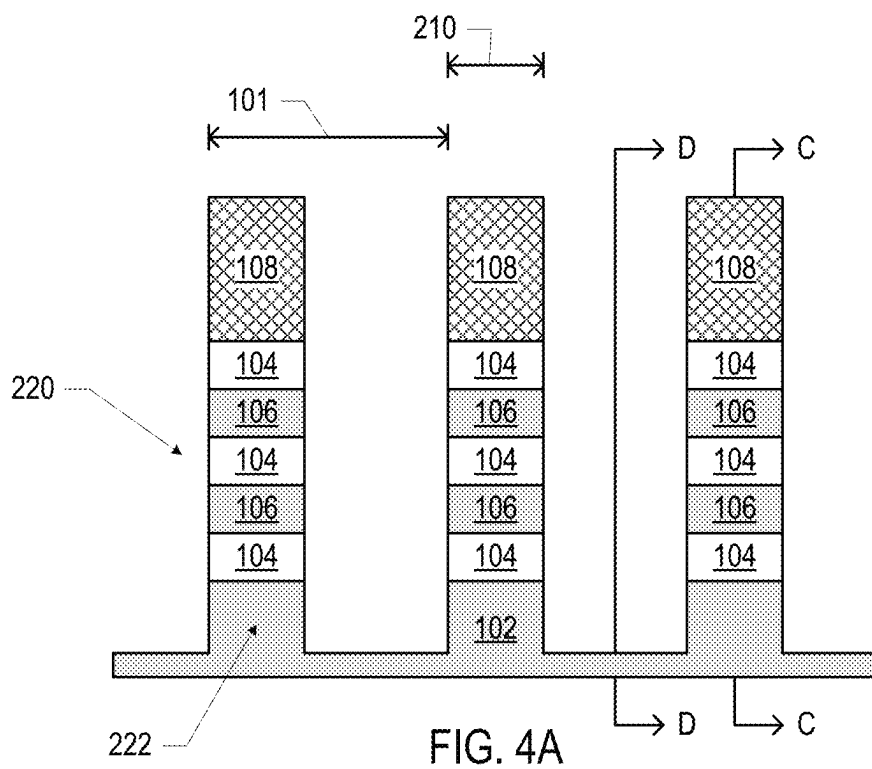
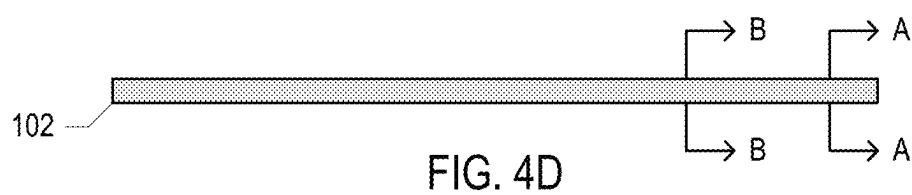
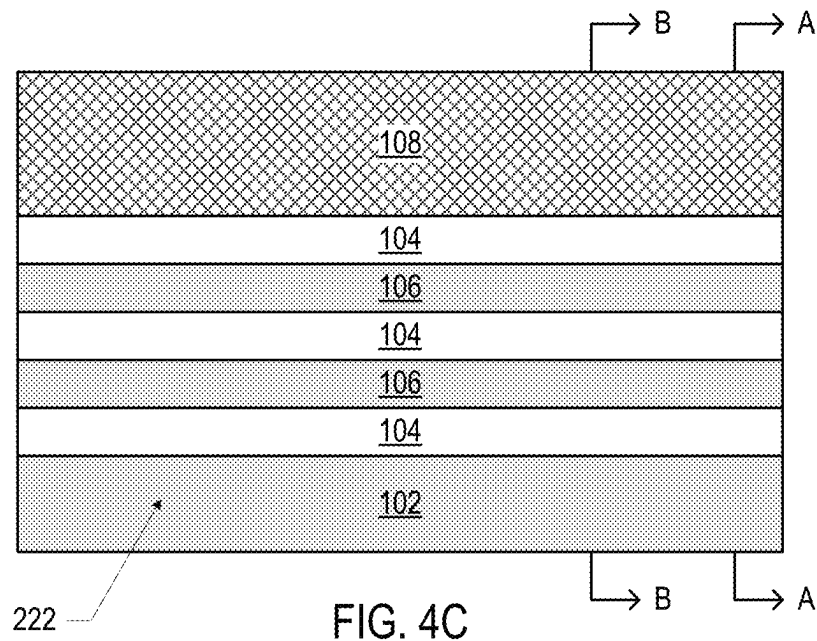
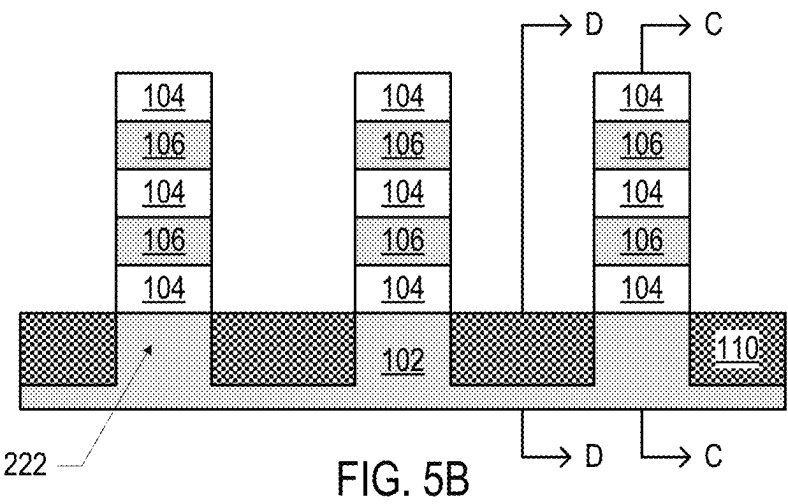
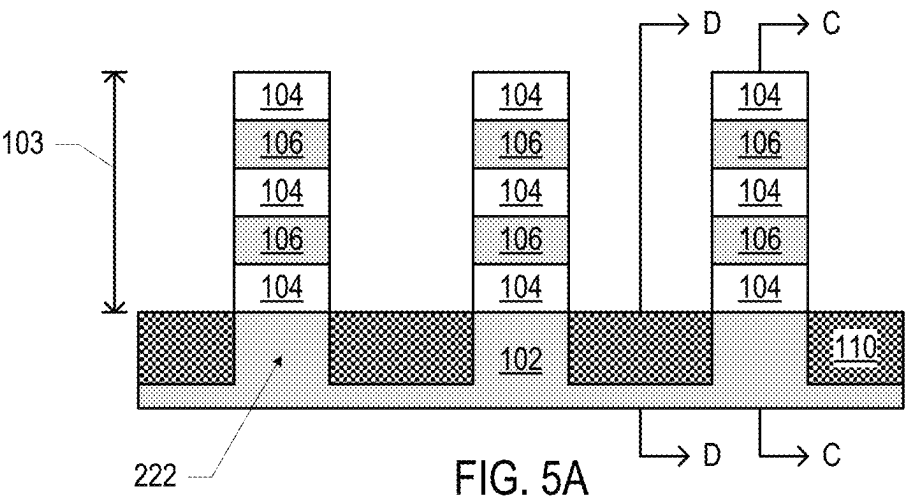
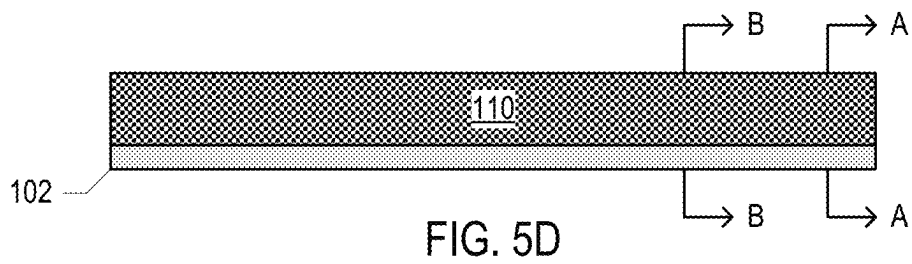
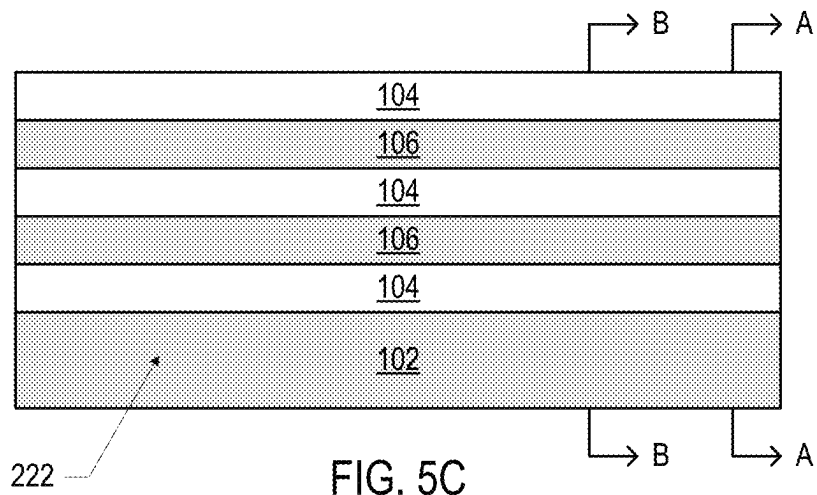


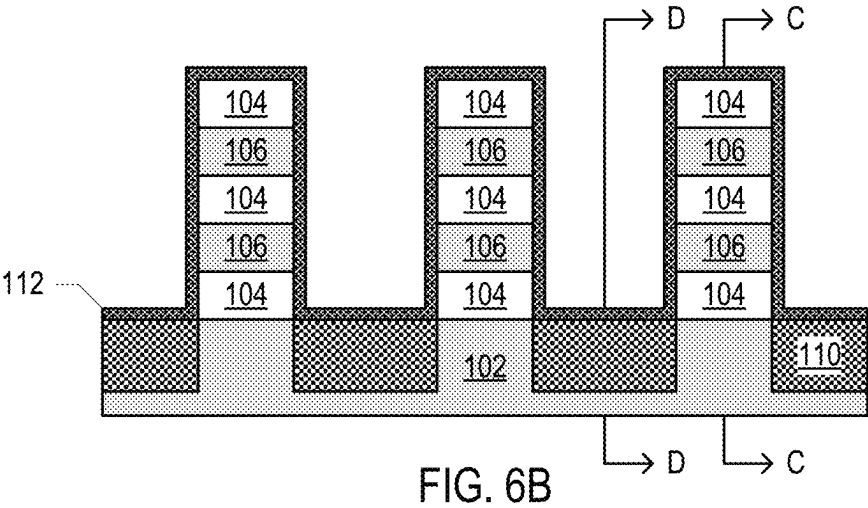
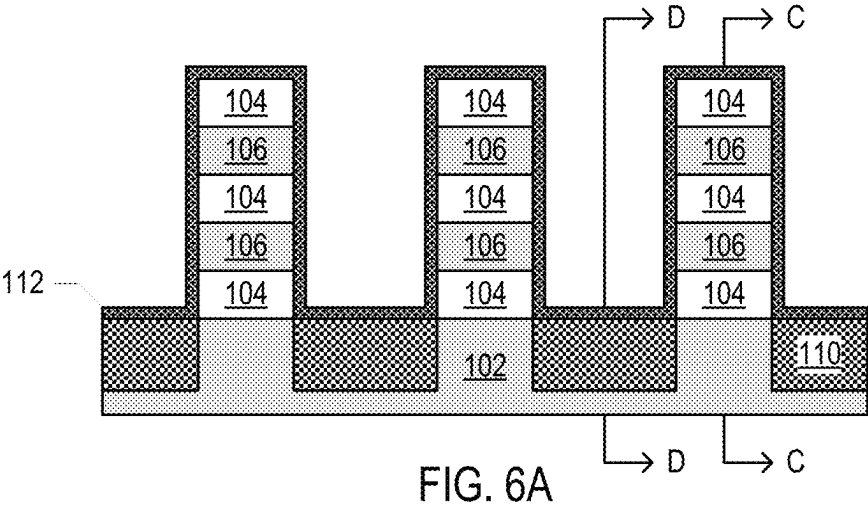
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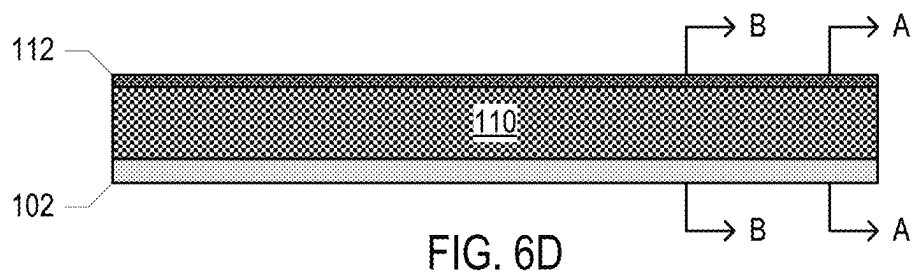
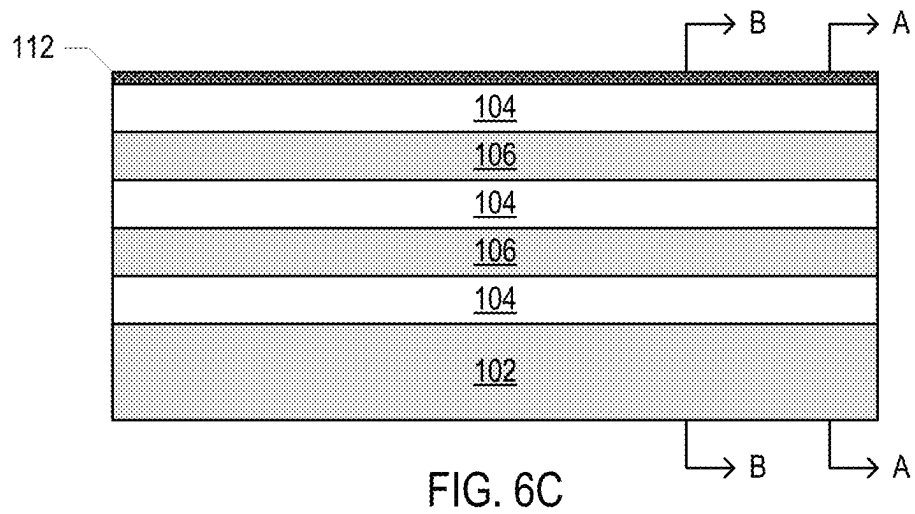


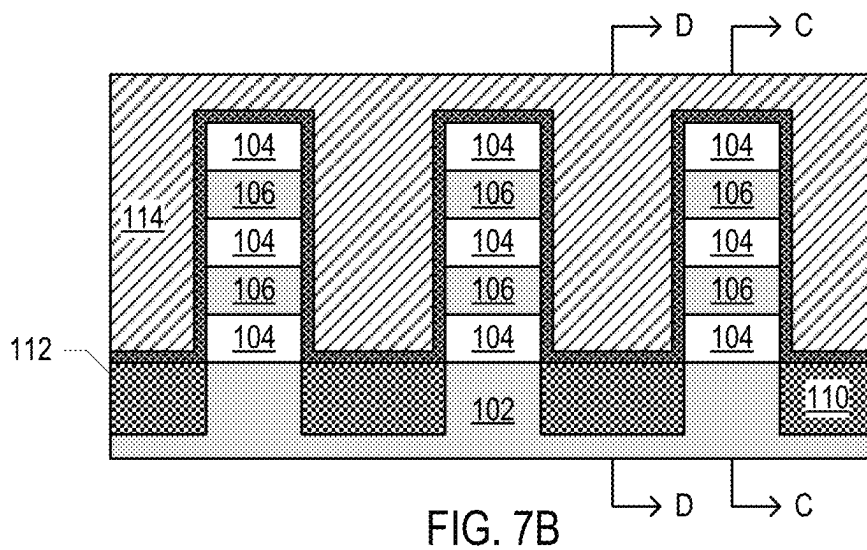
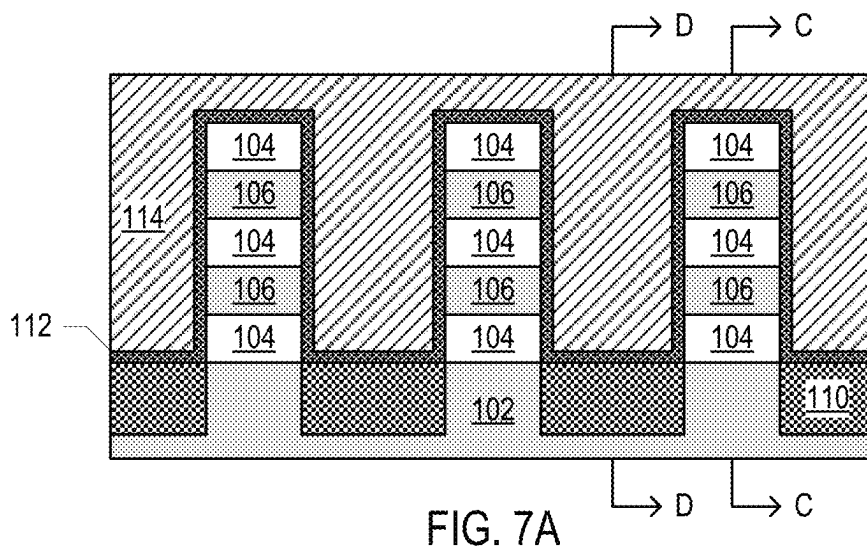


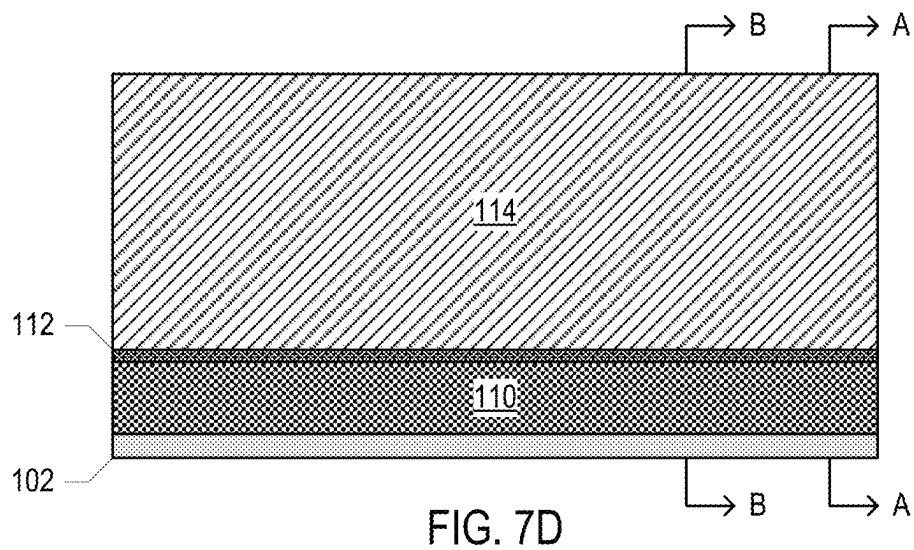
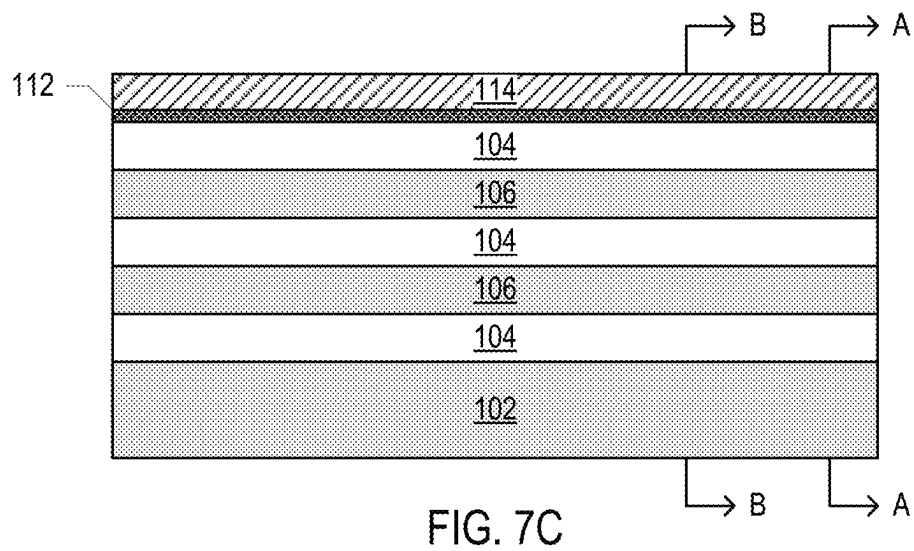


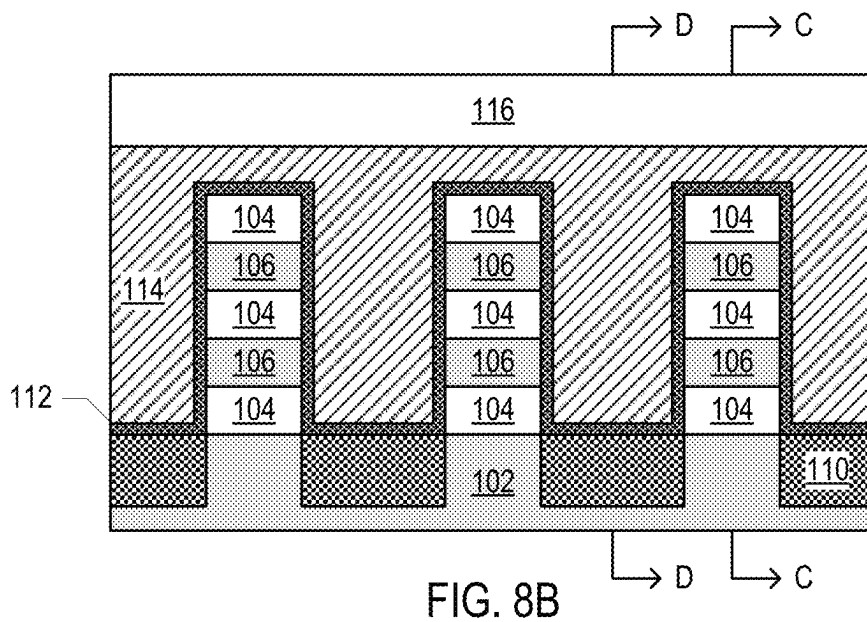
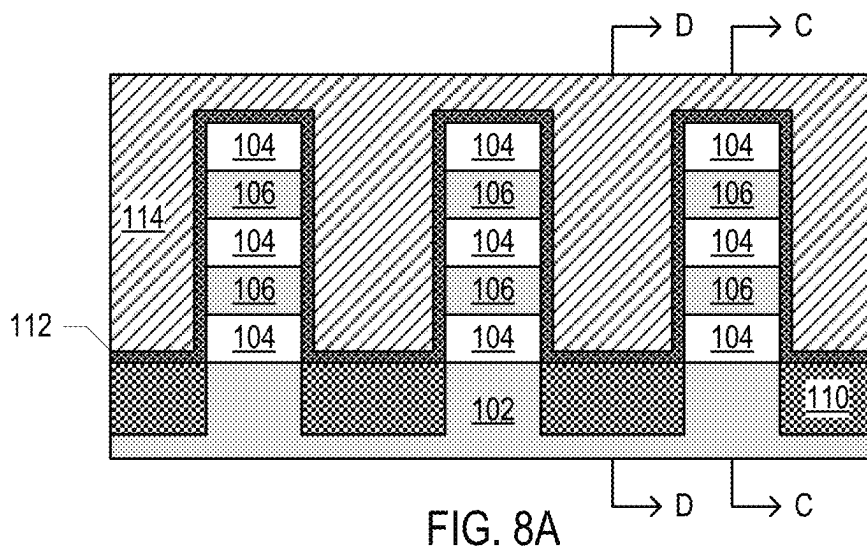












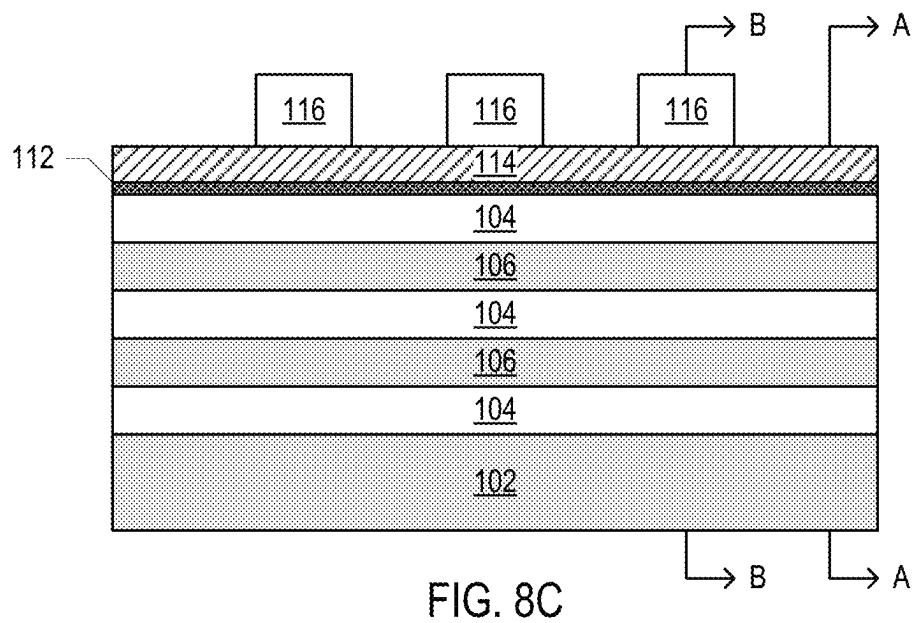


FIG. 8C

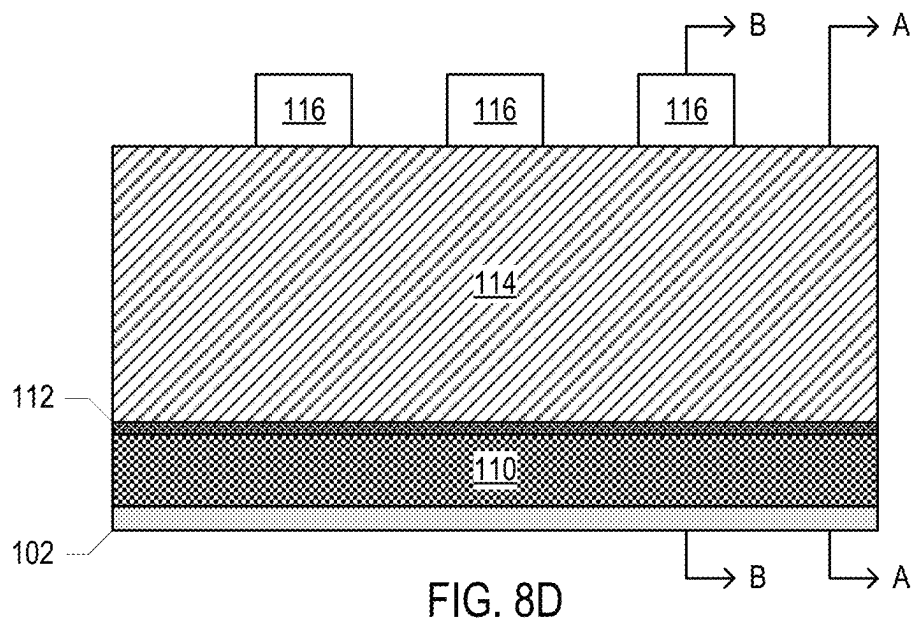
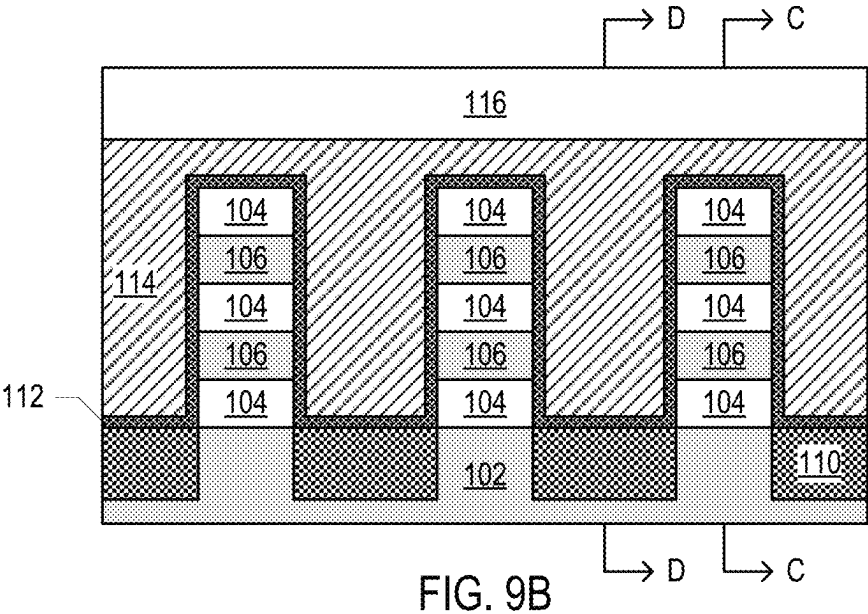
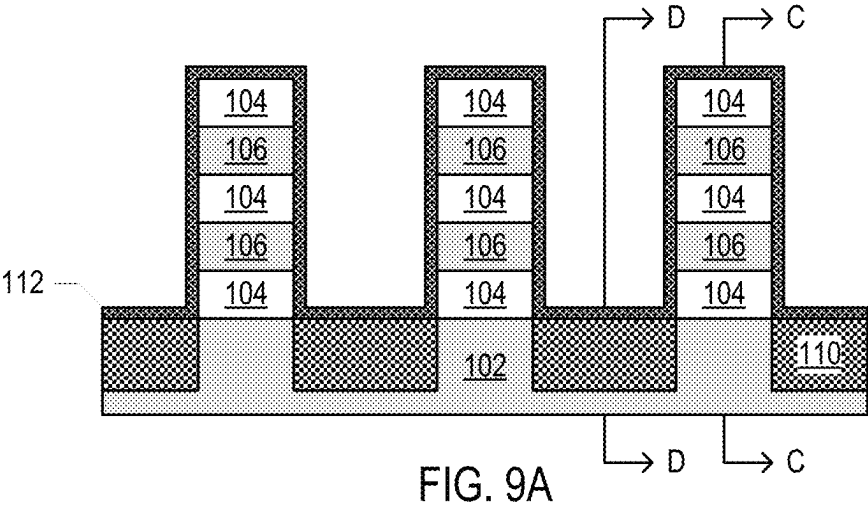
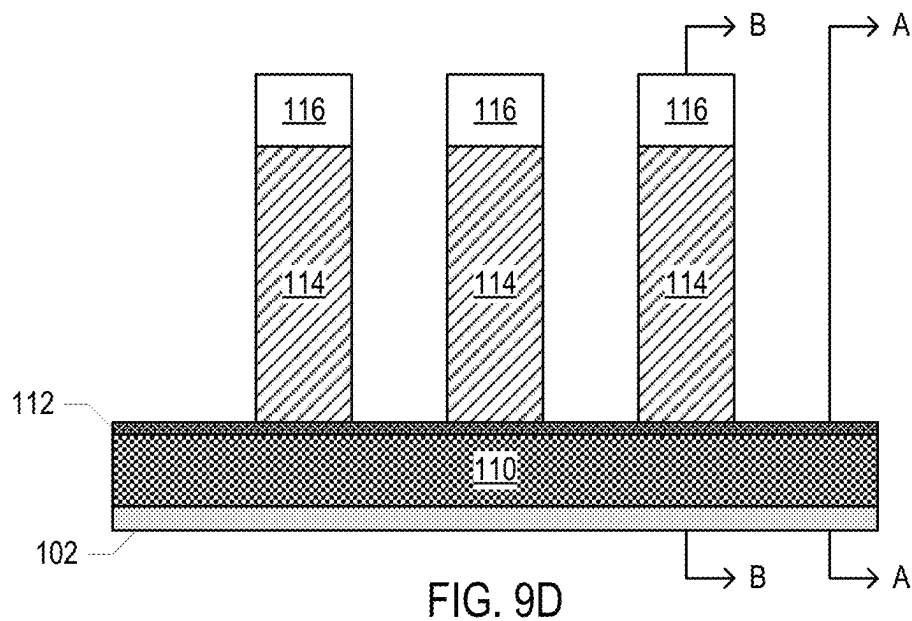
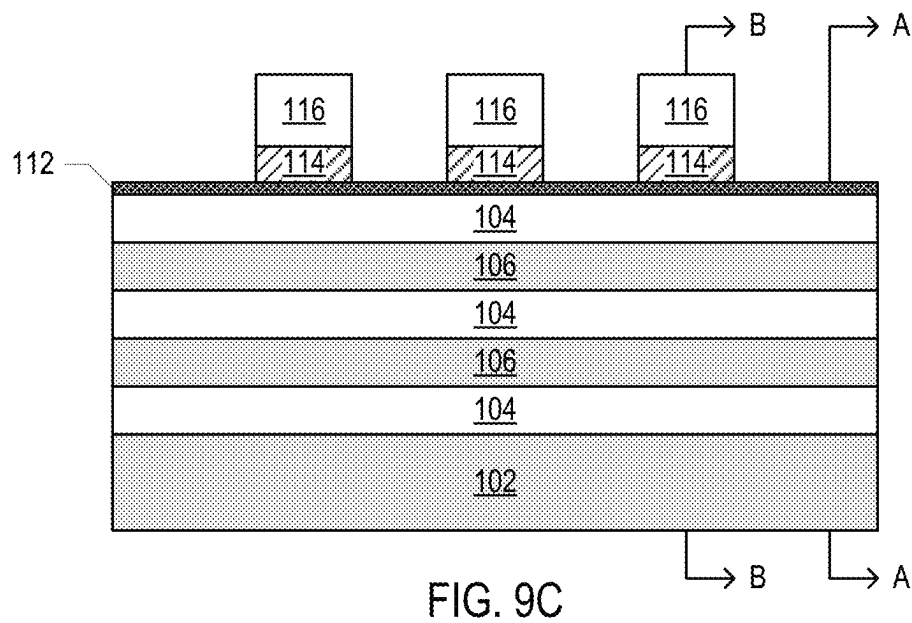
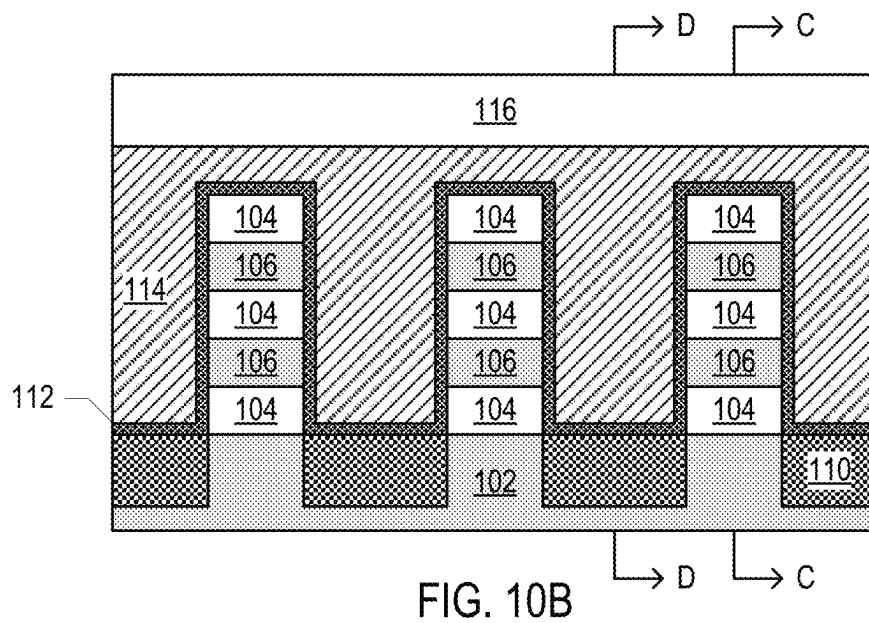
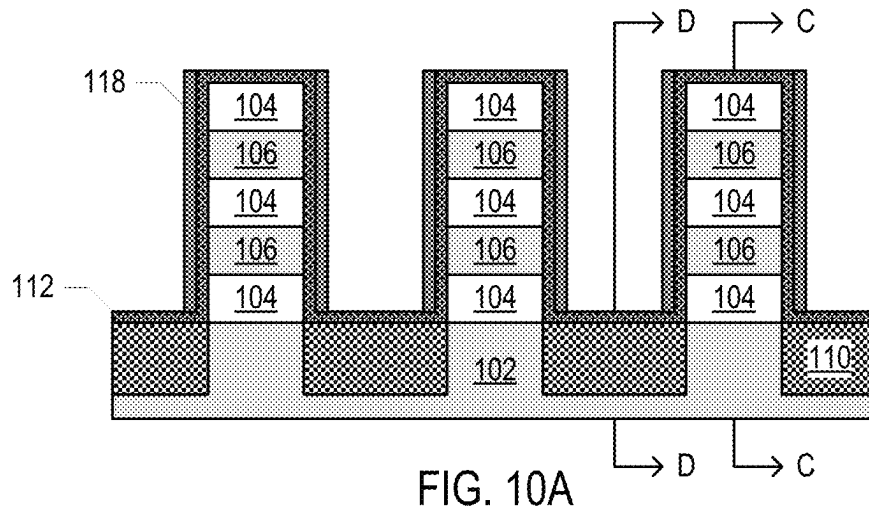
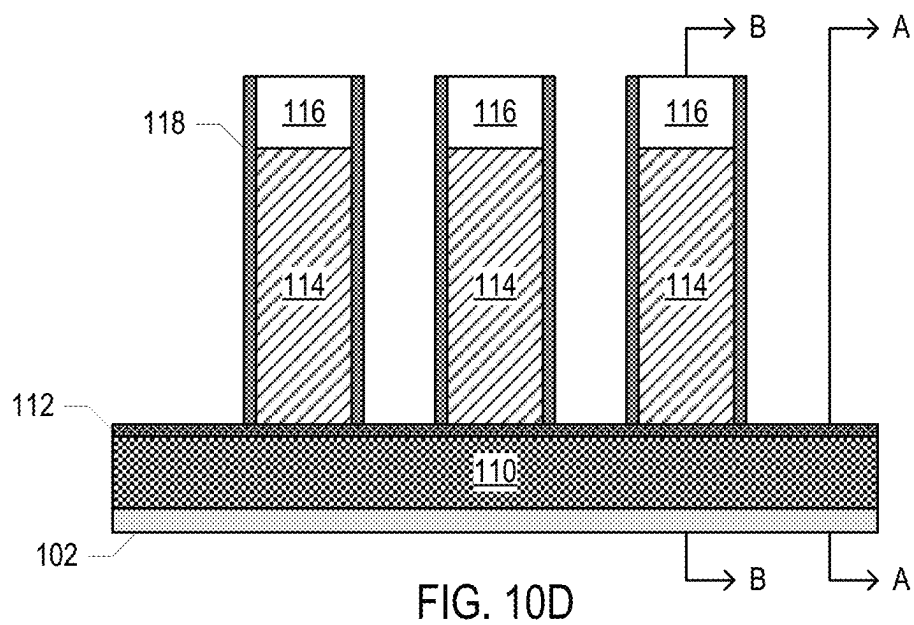
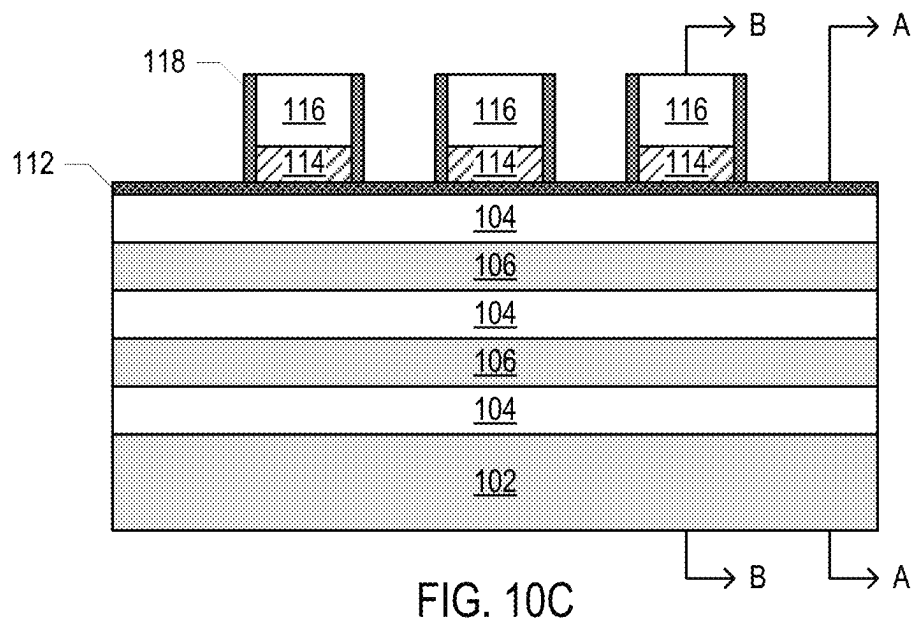


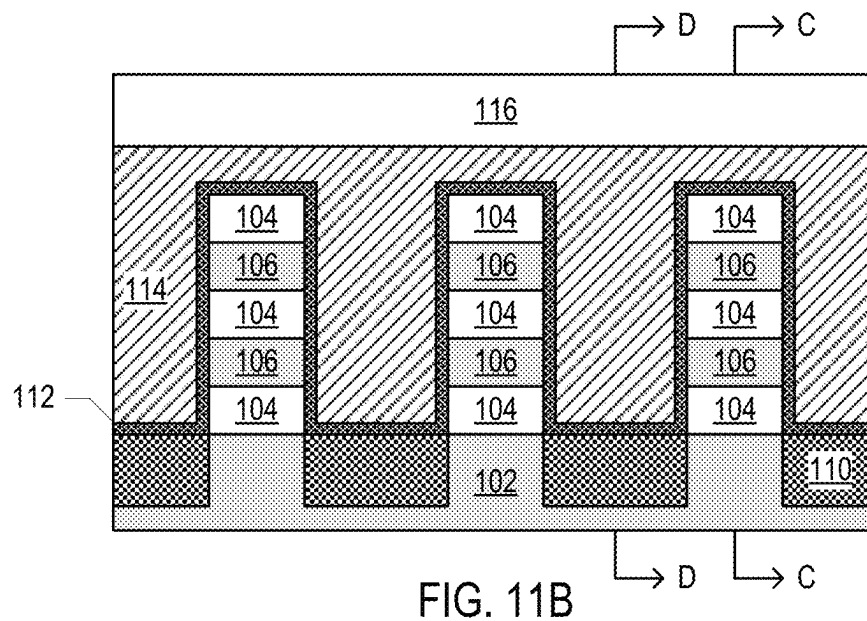
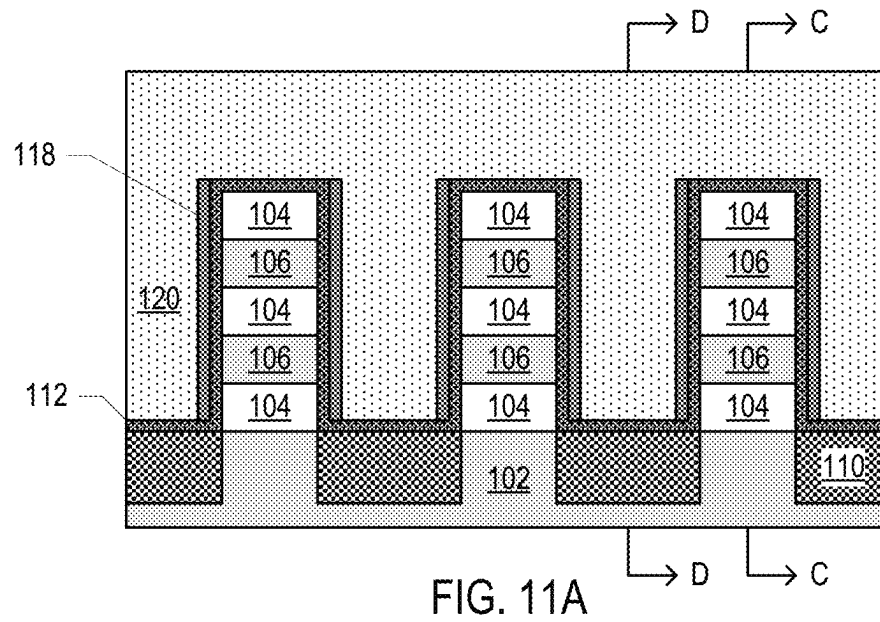
FIG. 8D











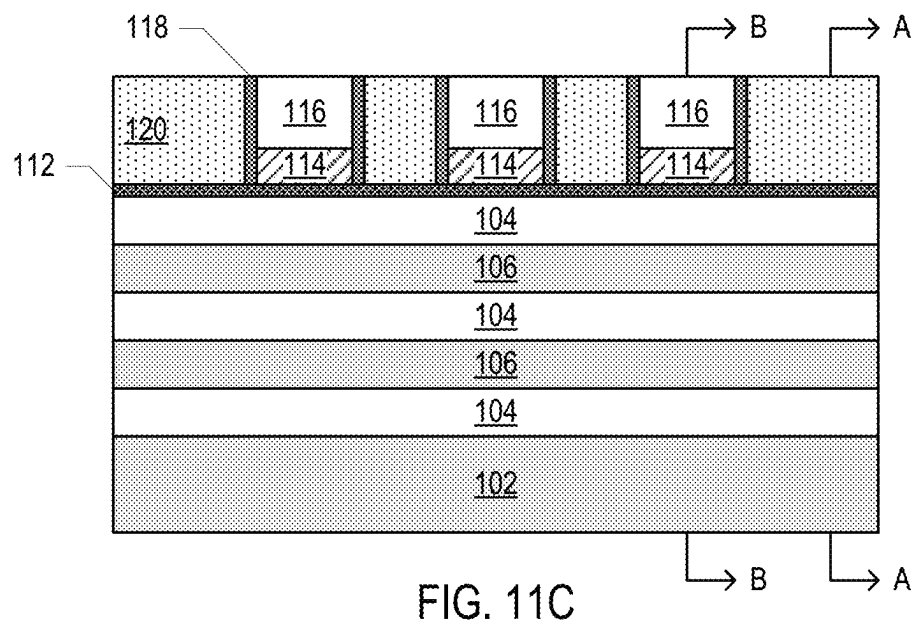


FIG. 11C

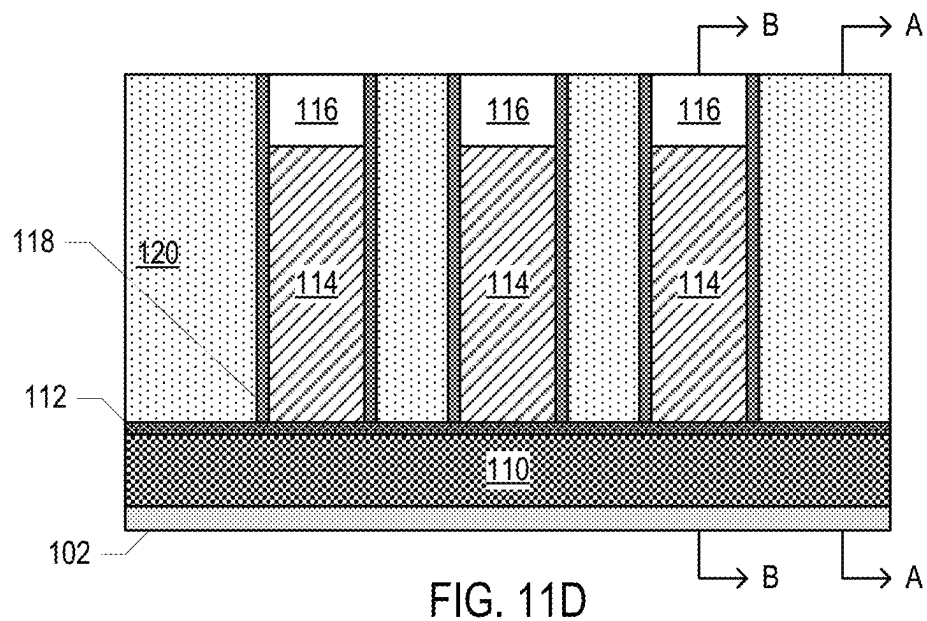
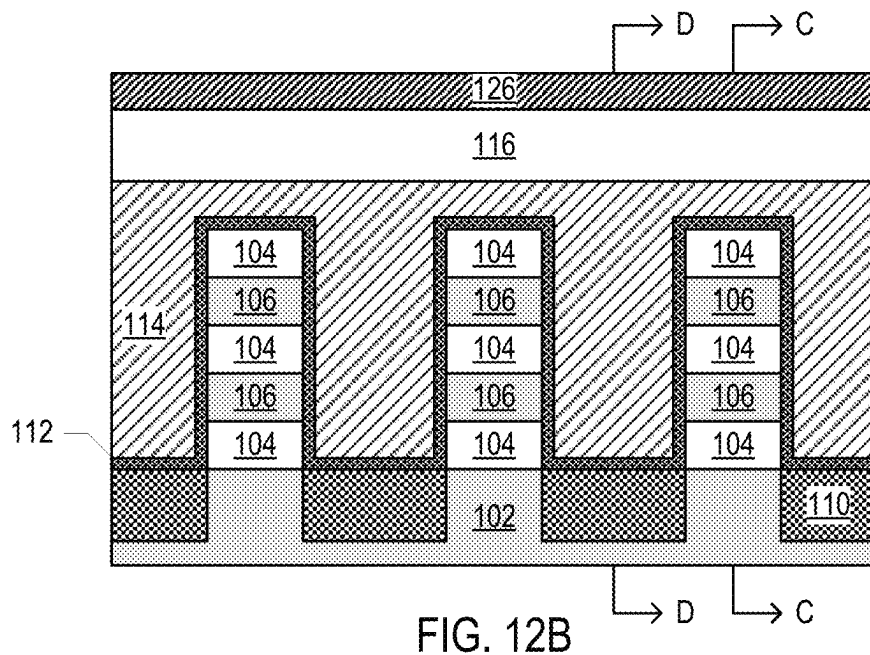
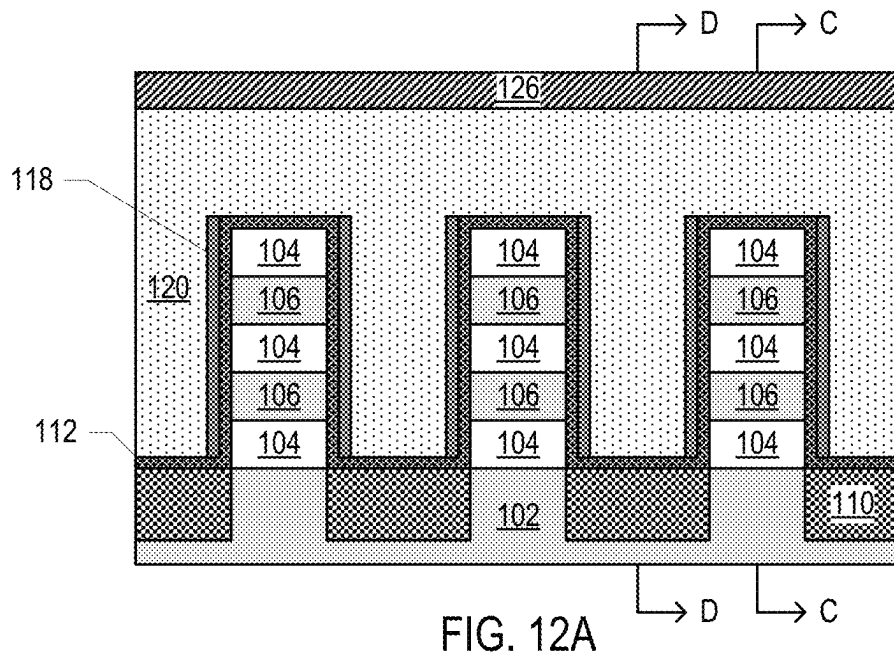


FIG. 11D



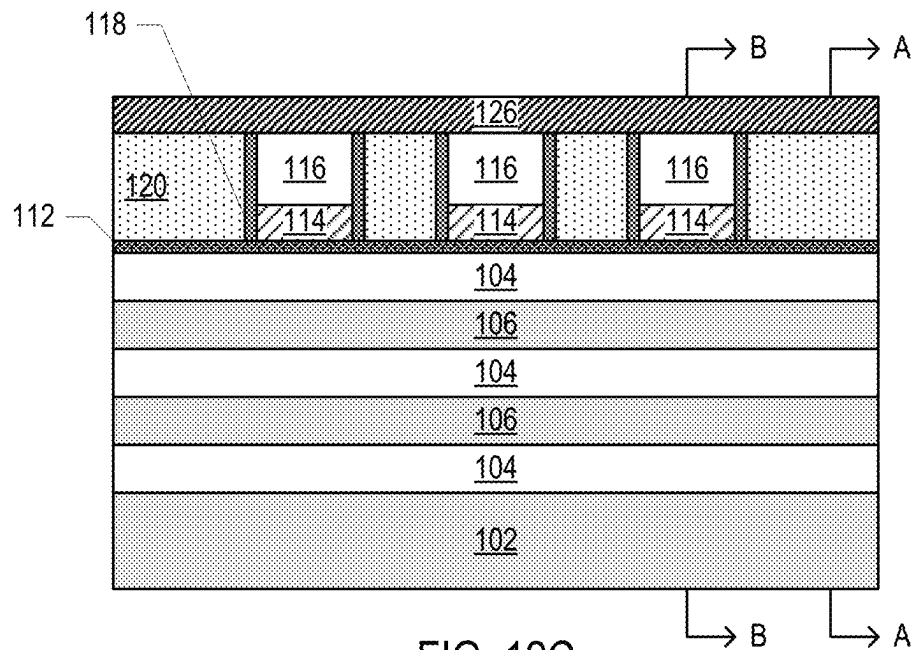


FIG. 12C

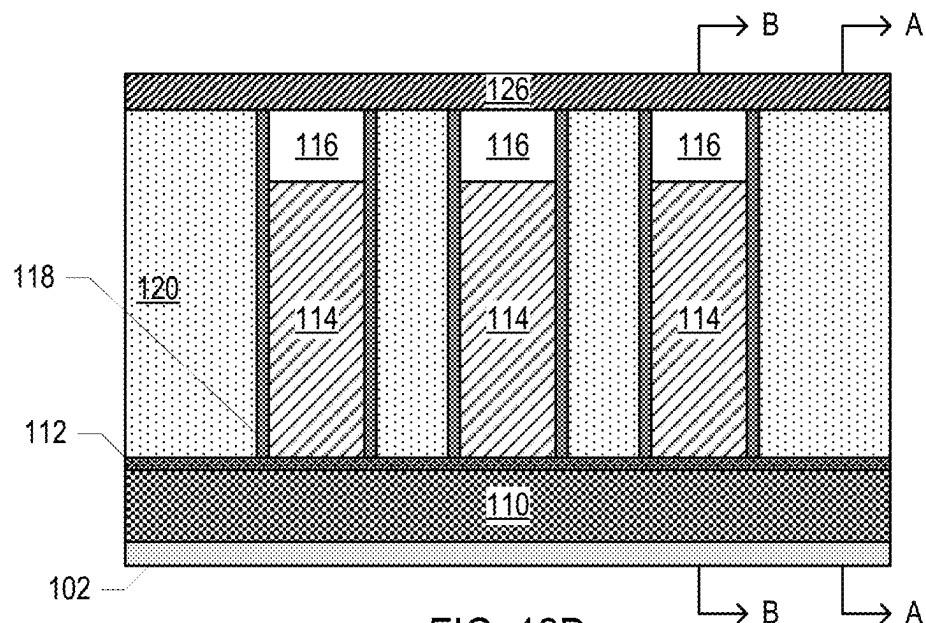


FIG. 12D

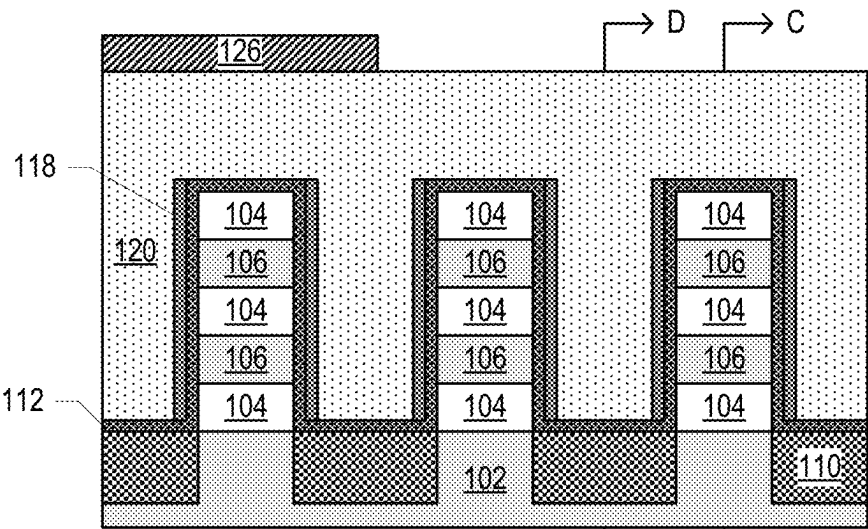


FIG. 13A

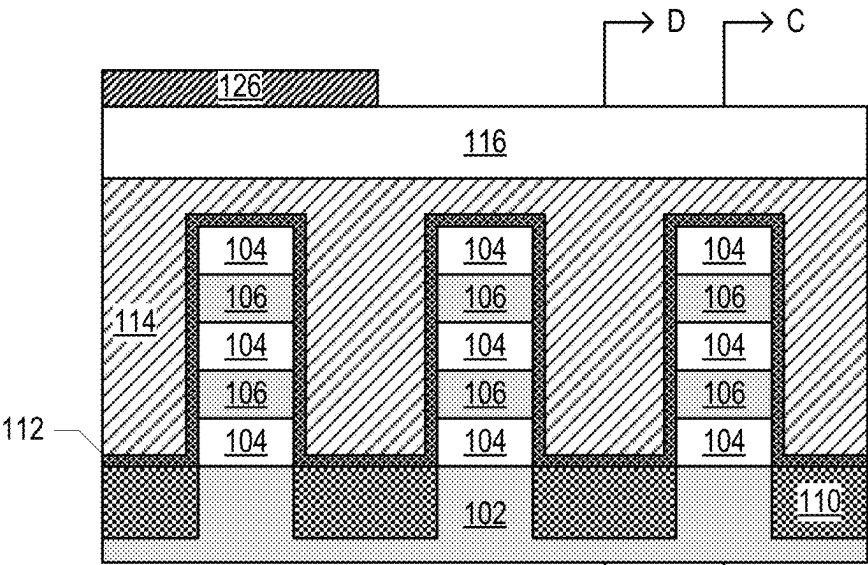
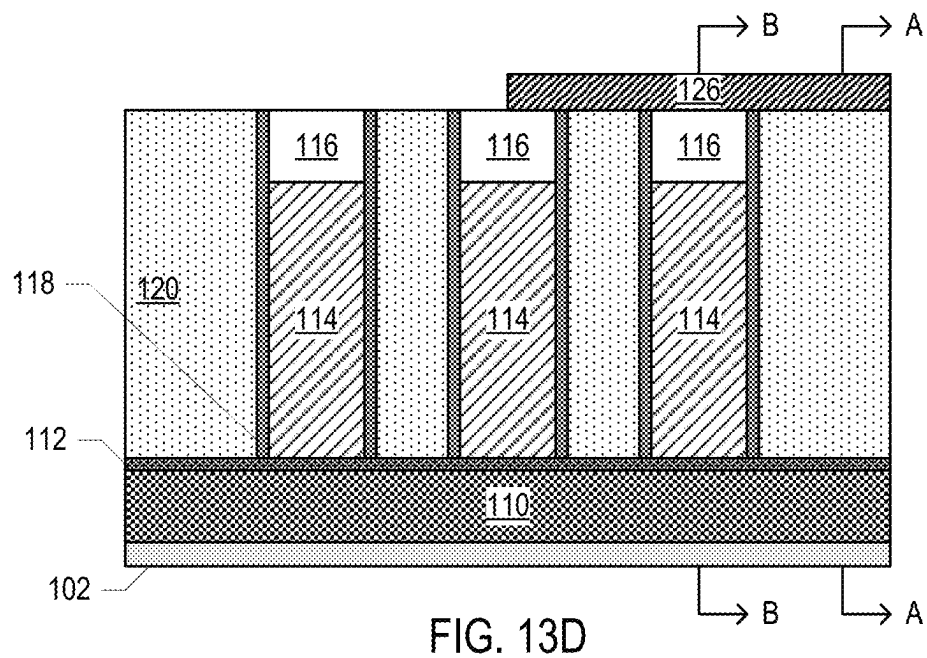
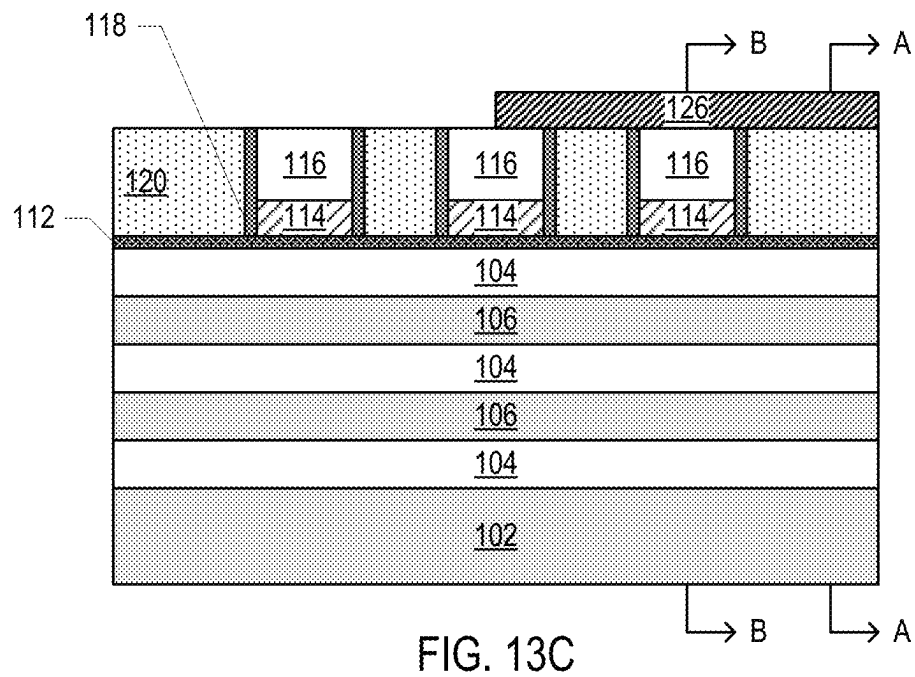
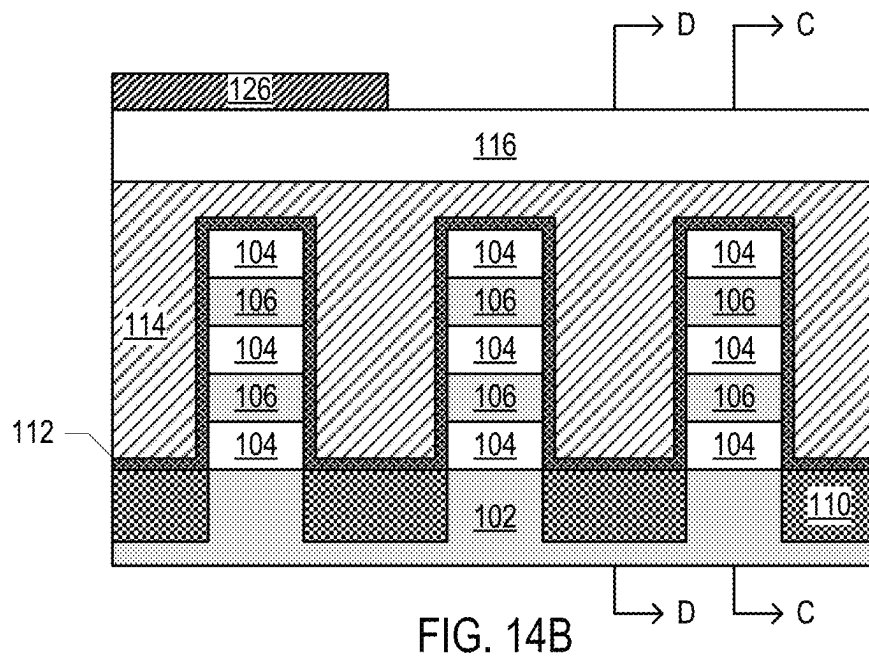
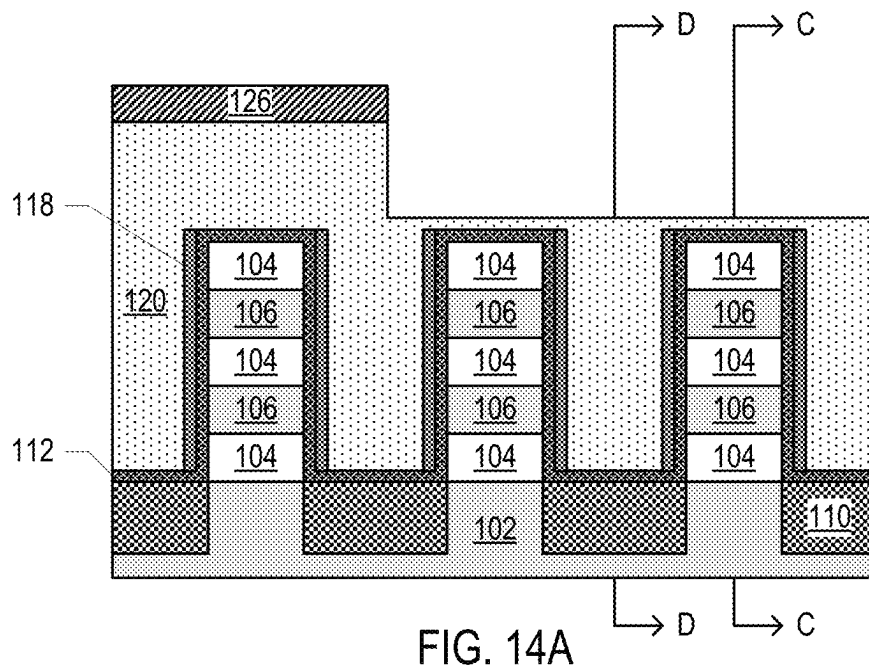
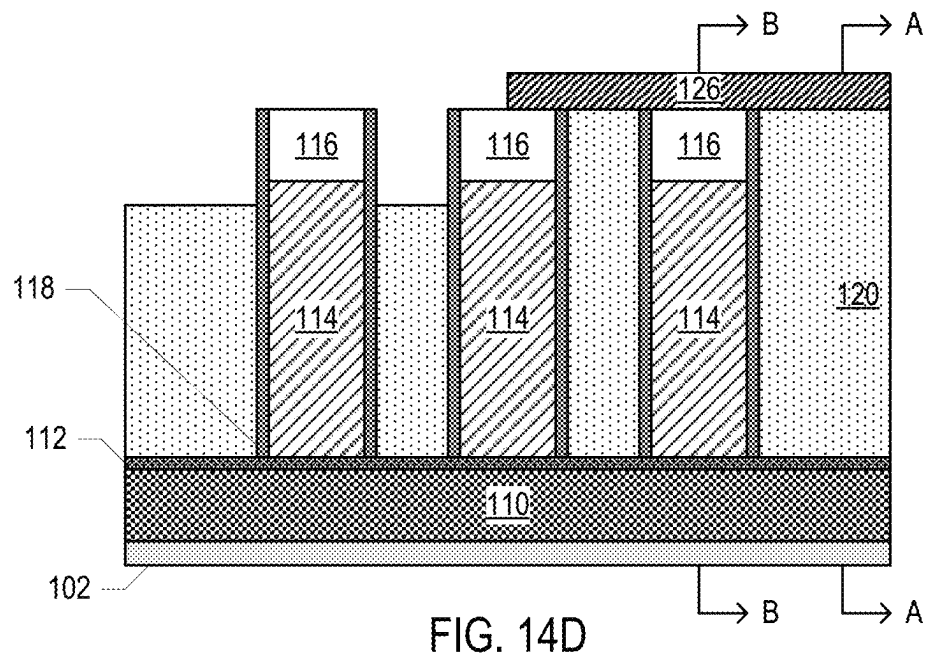
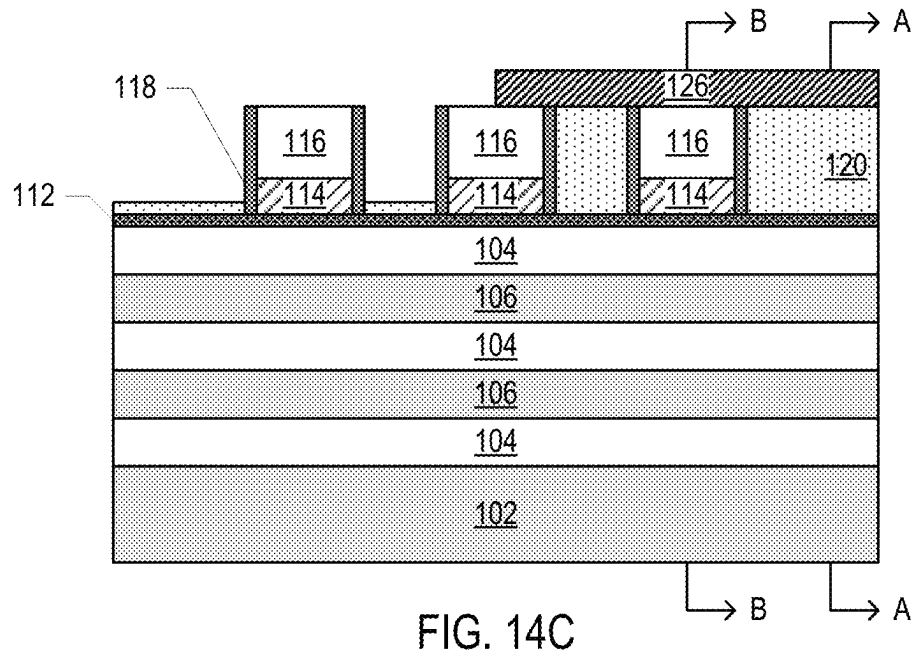
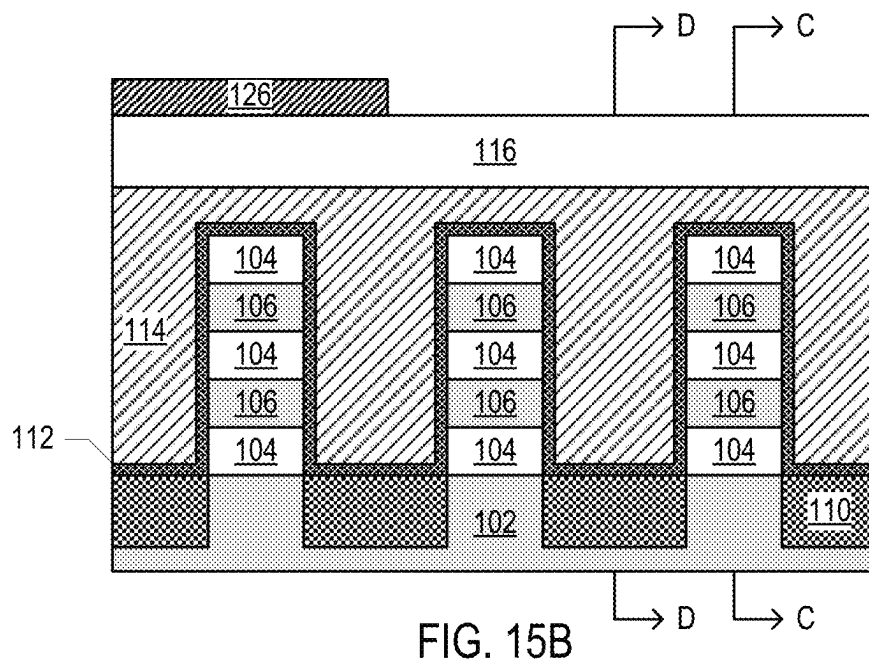
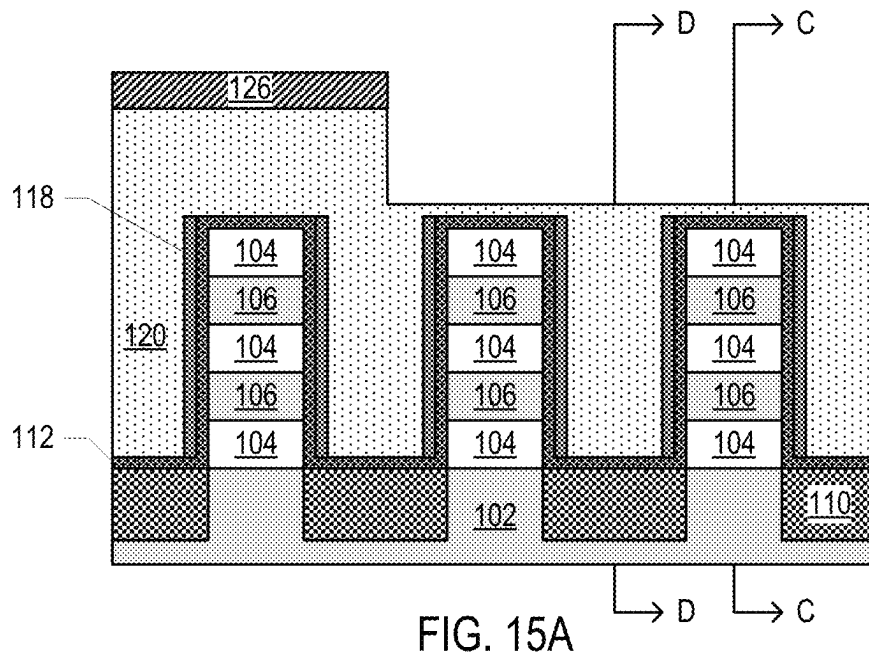


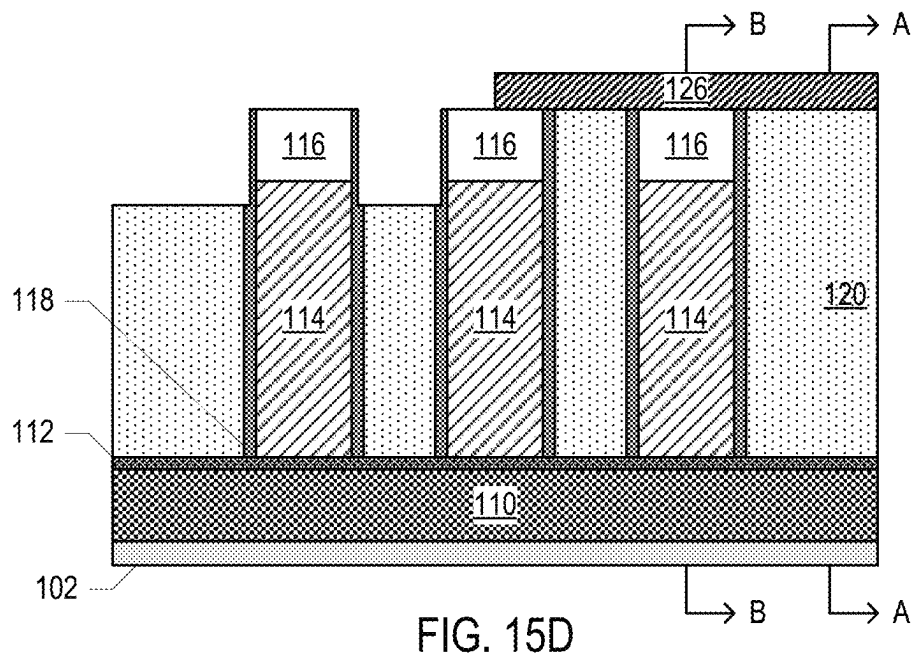
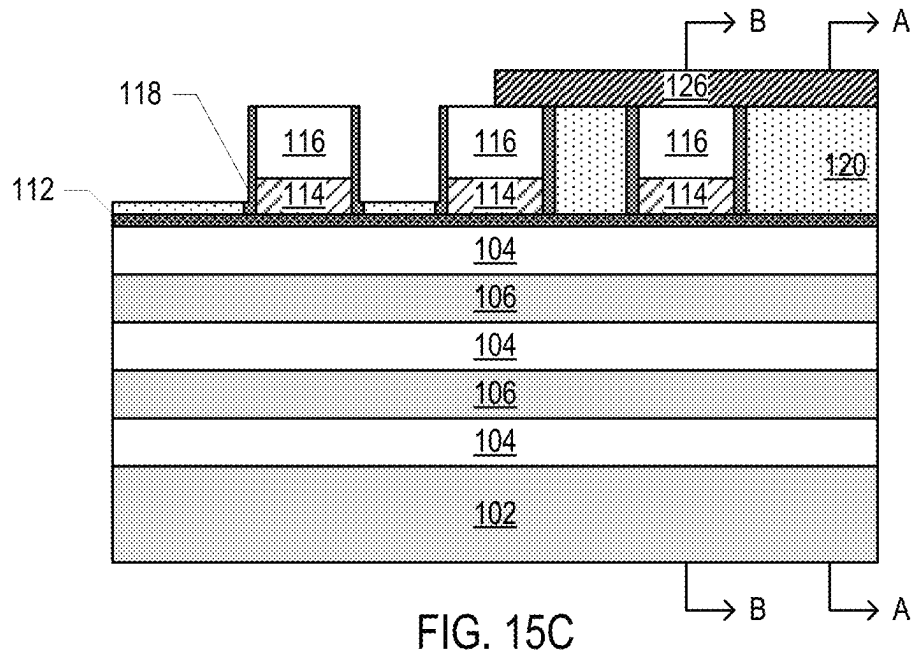
FIG. 13B

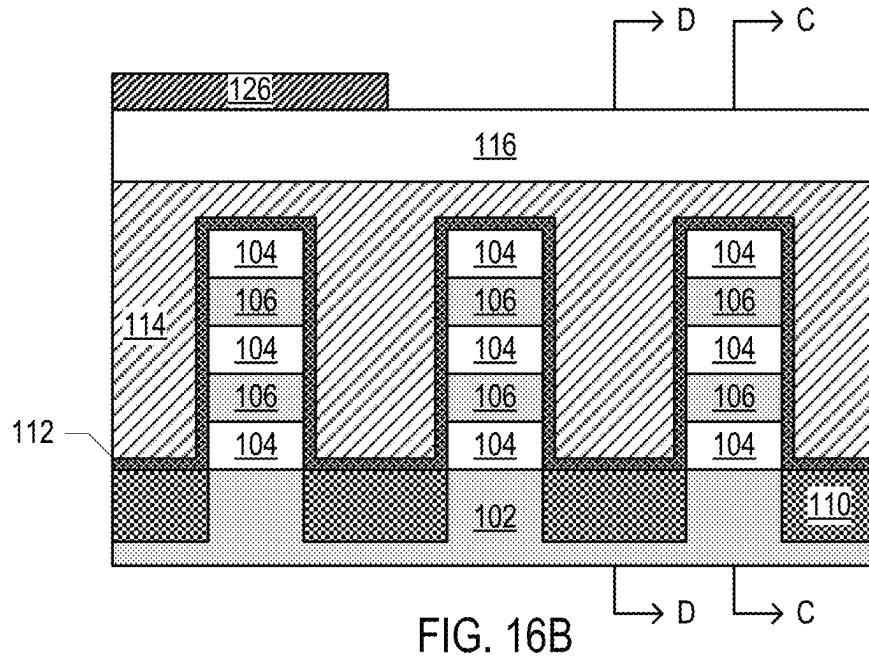
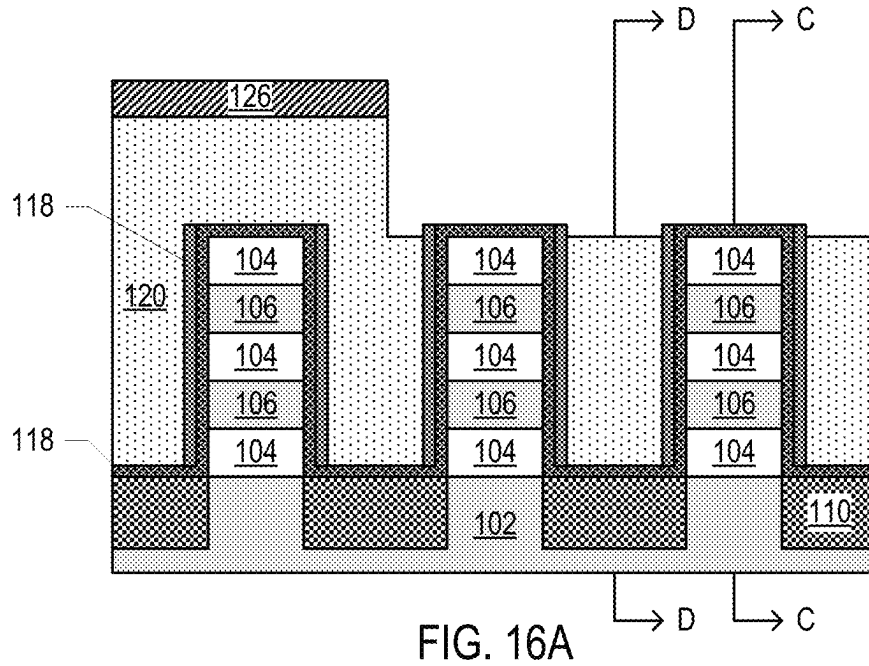


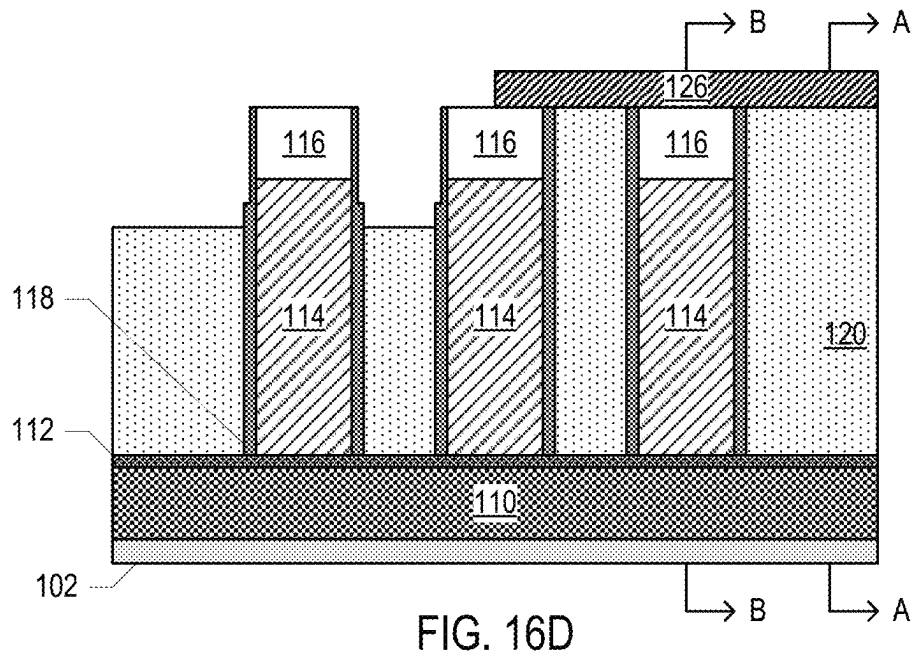
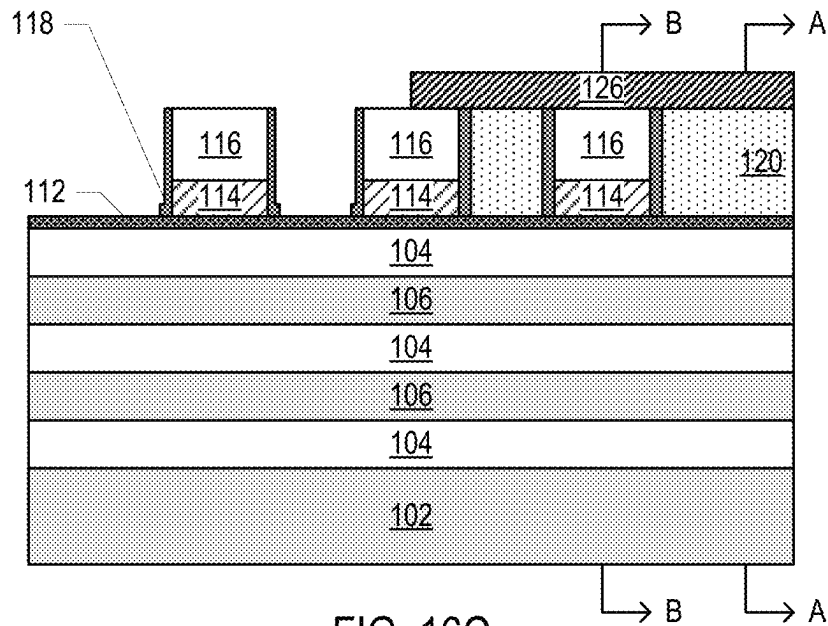












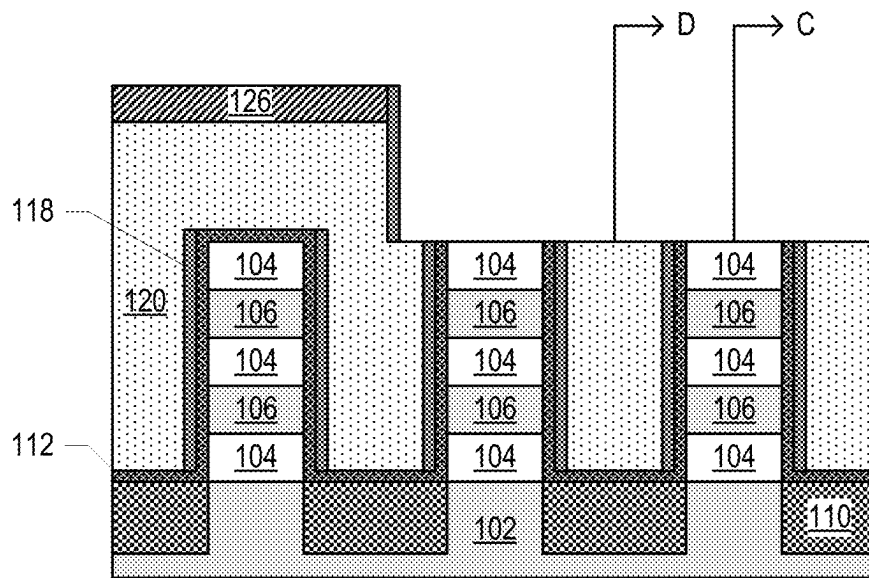


FIG. 17A

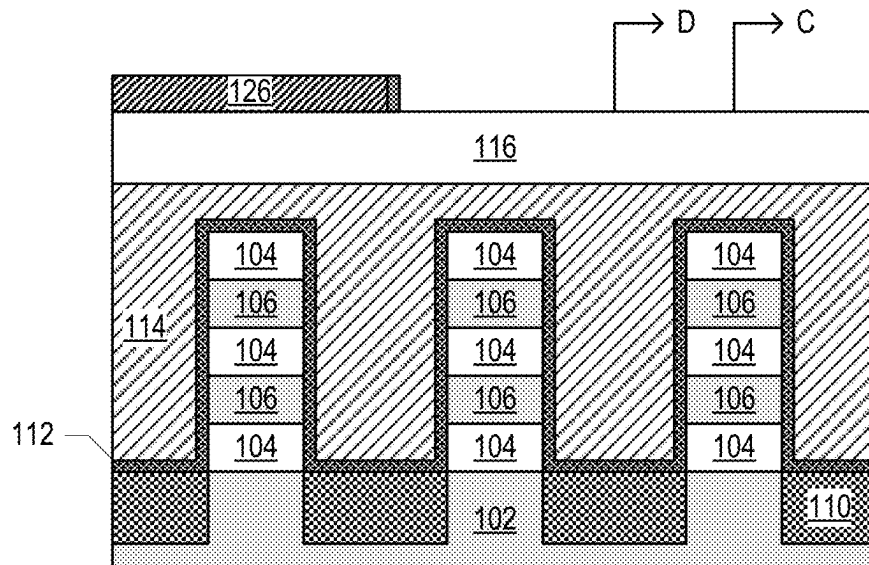
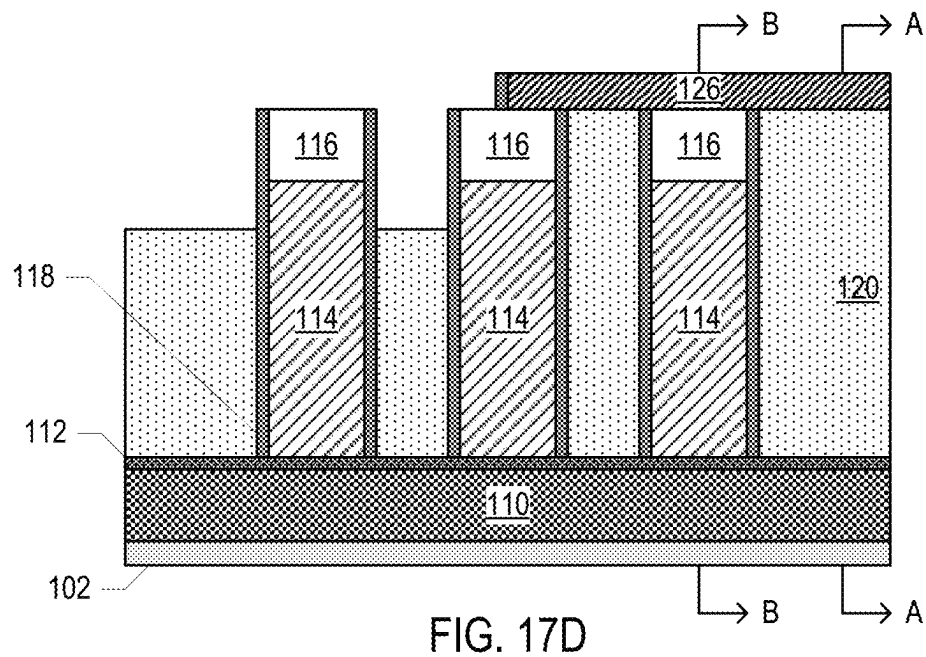
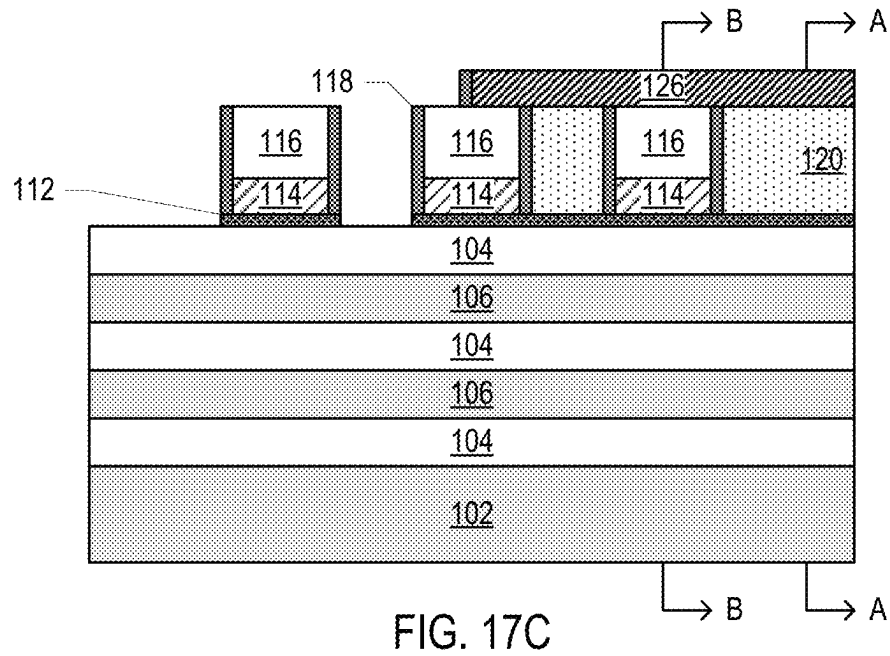
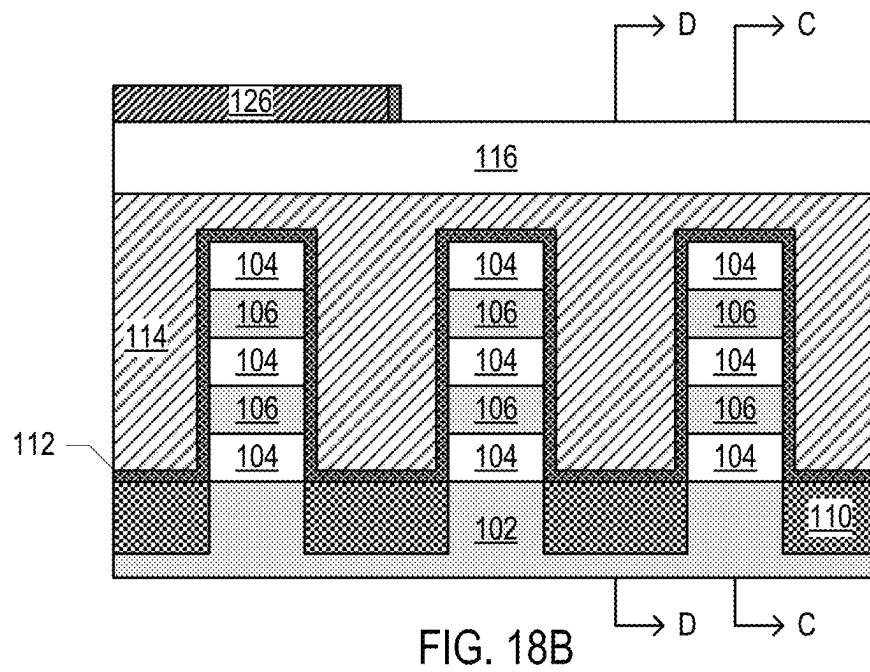
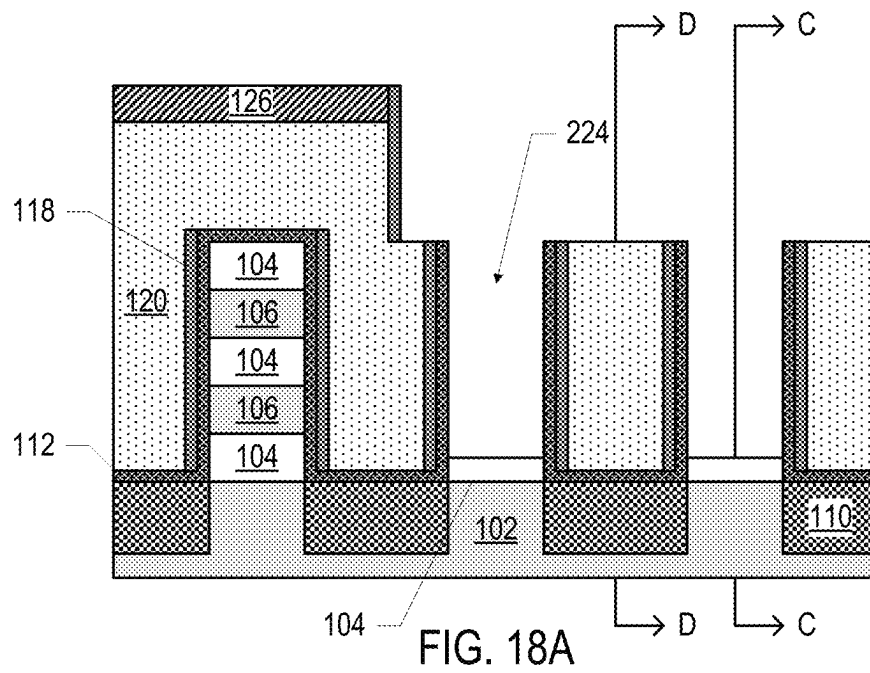
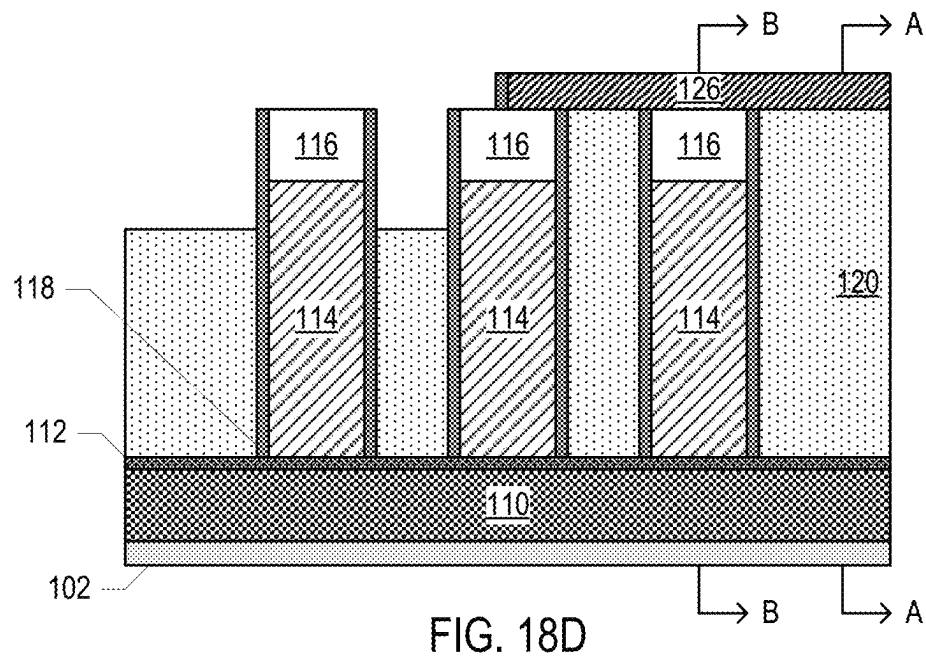
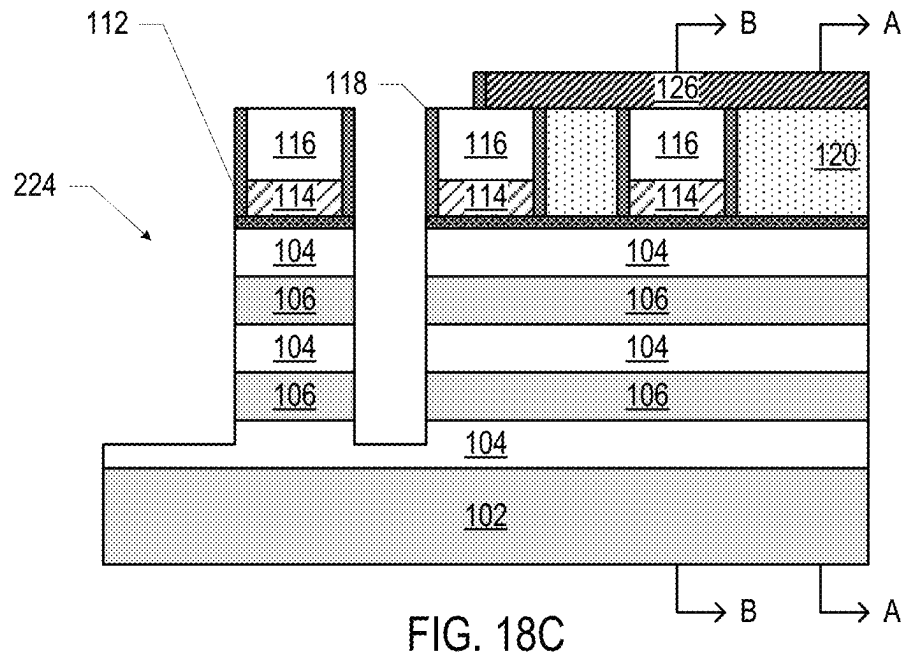
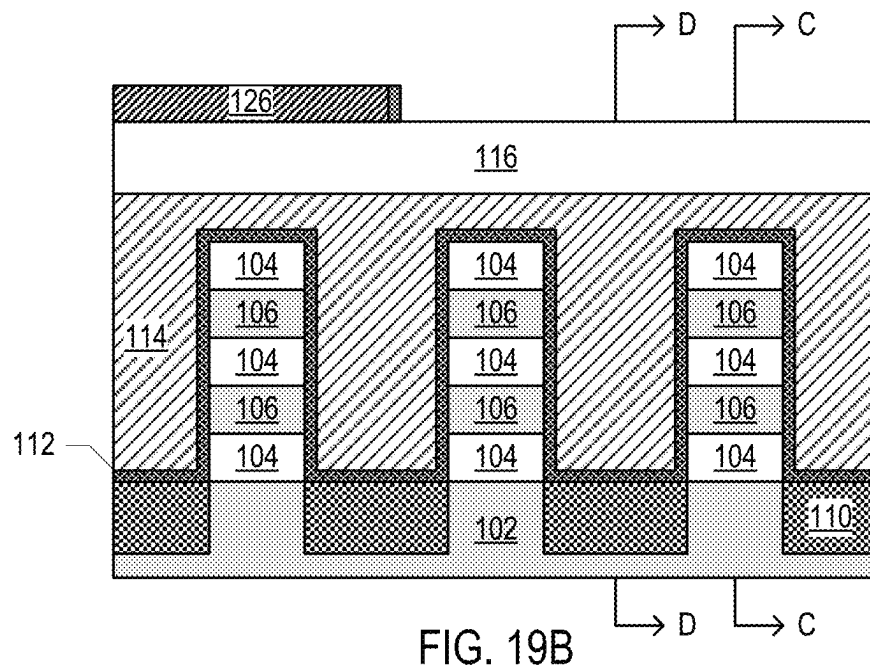
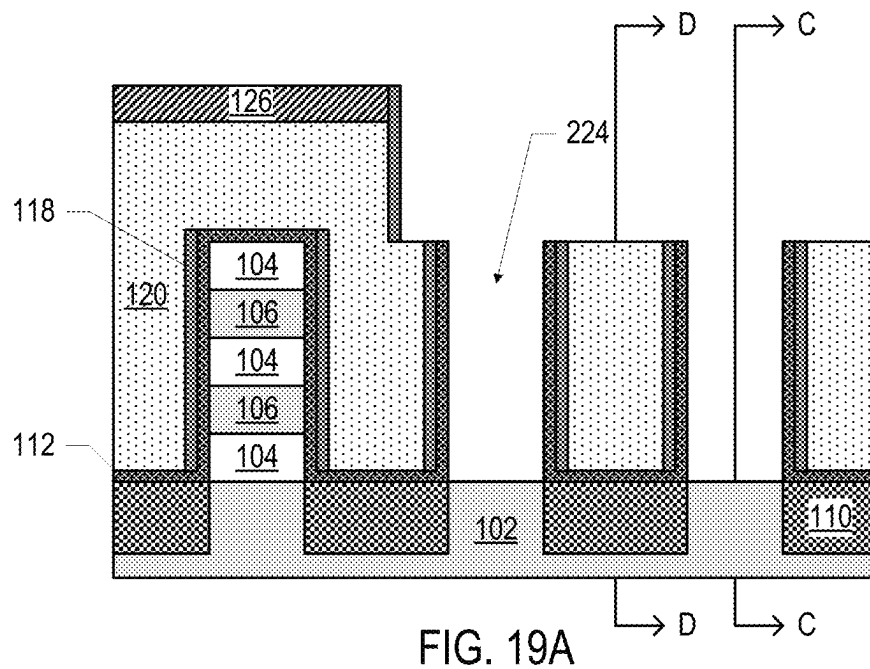


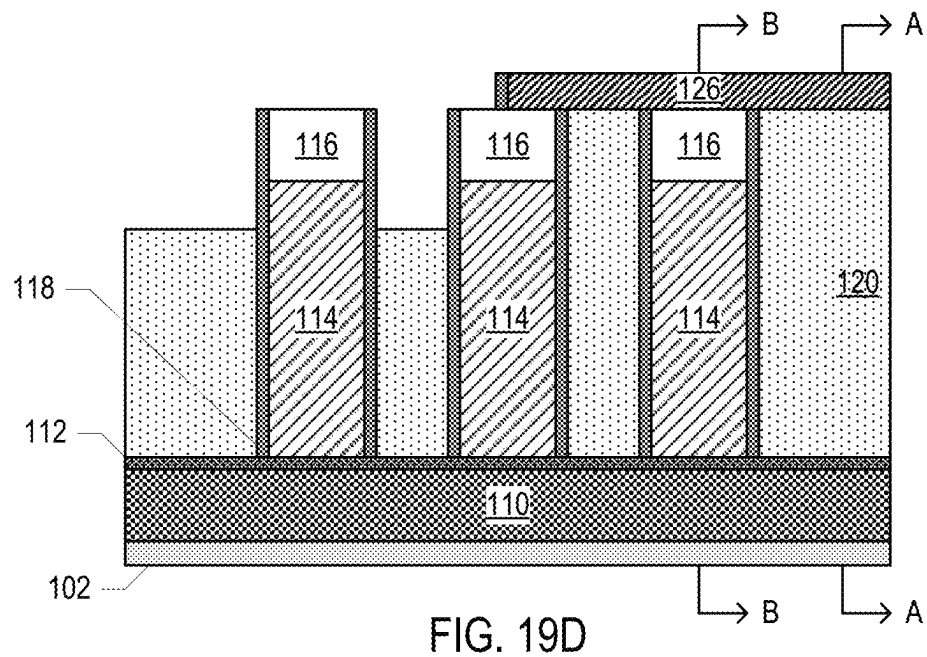
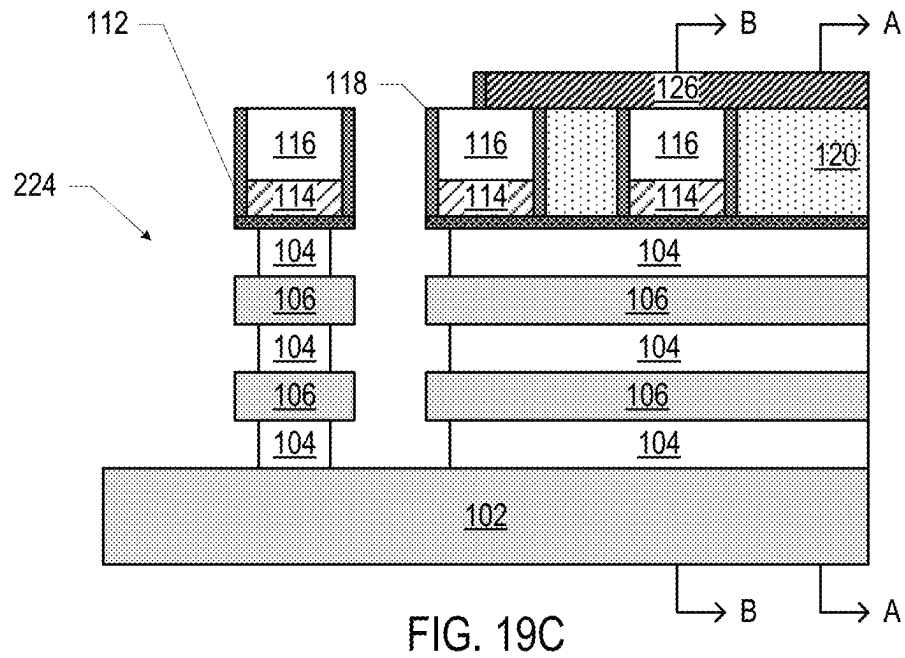
FIG. 17B











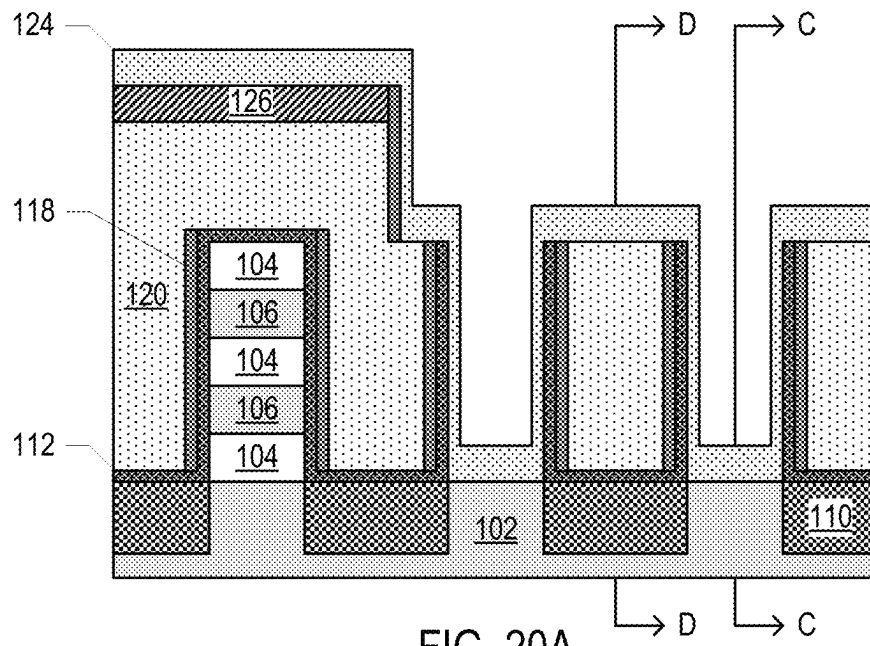


FIG. 20A

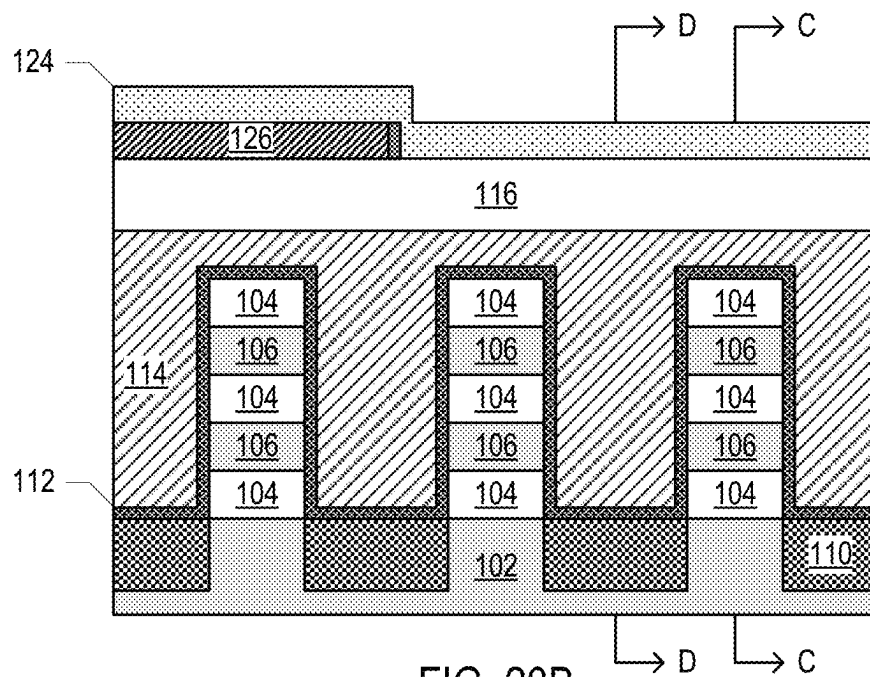
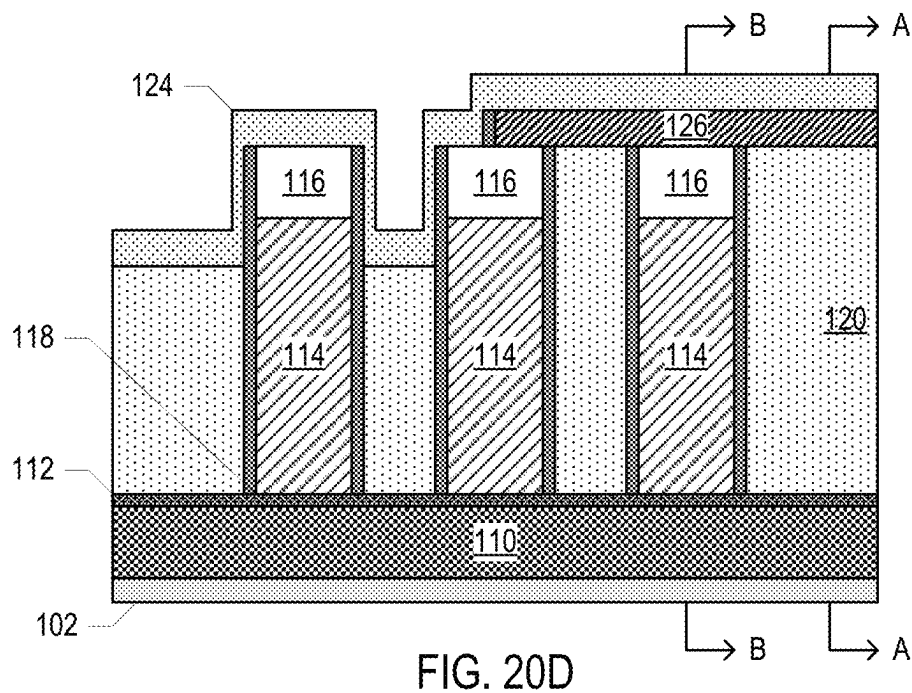
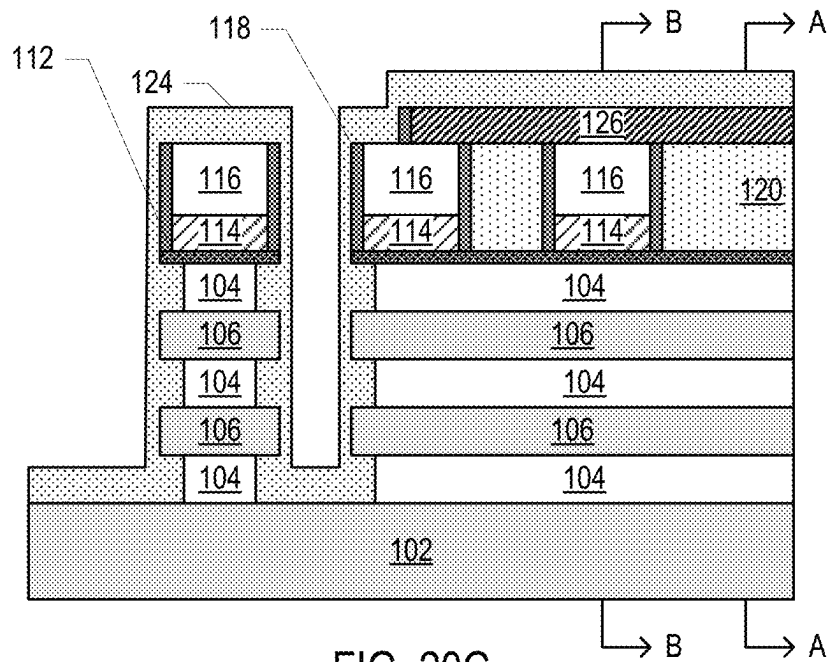
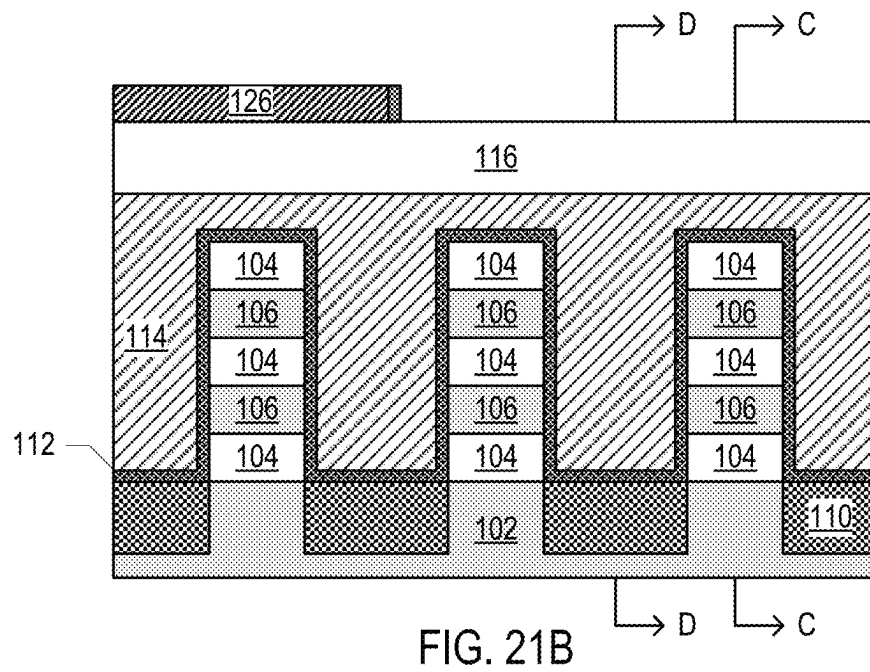
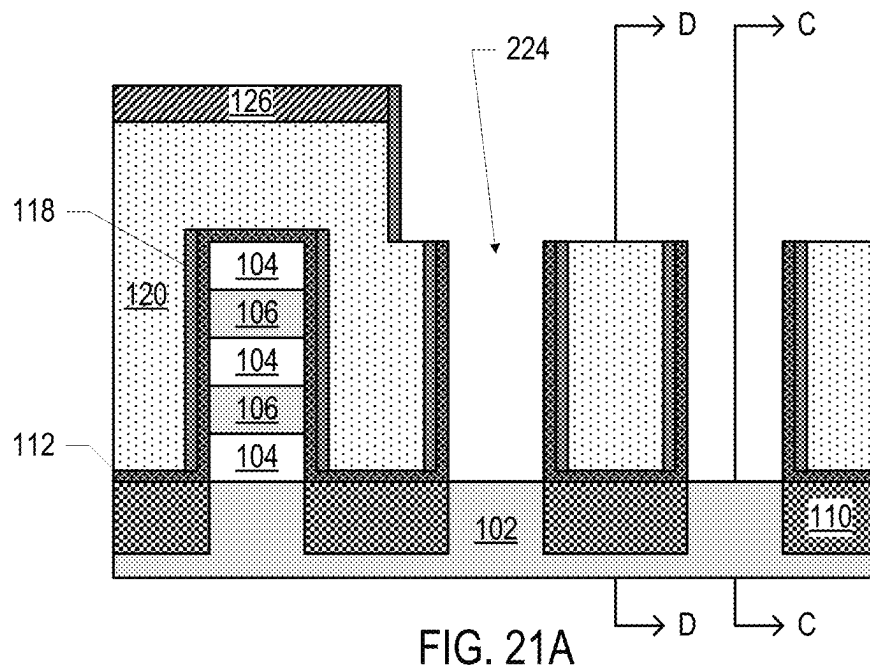
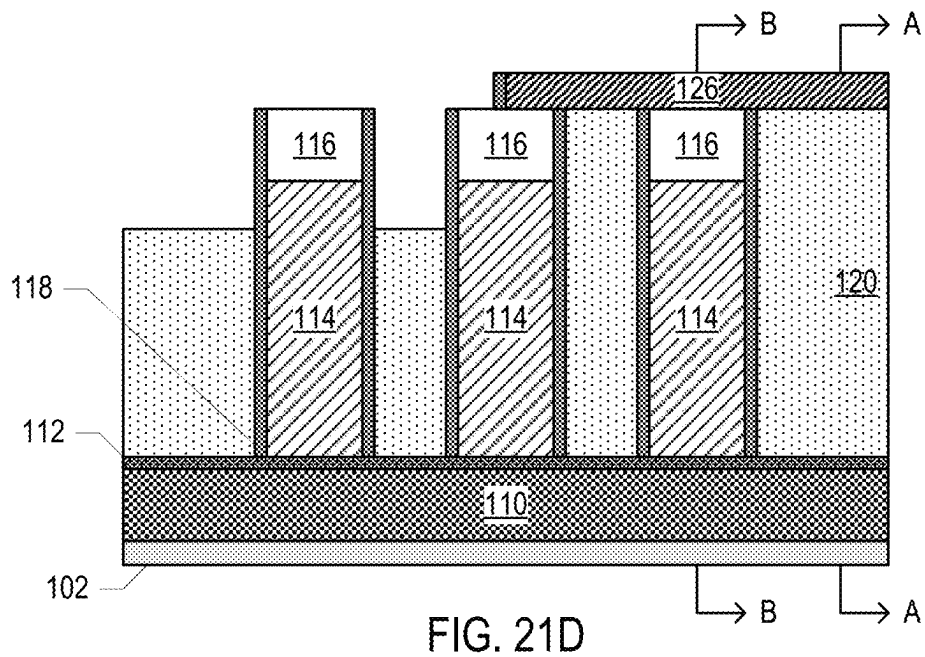
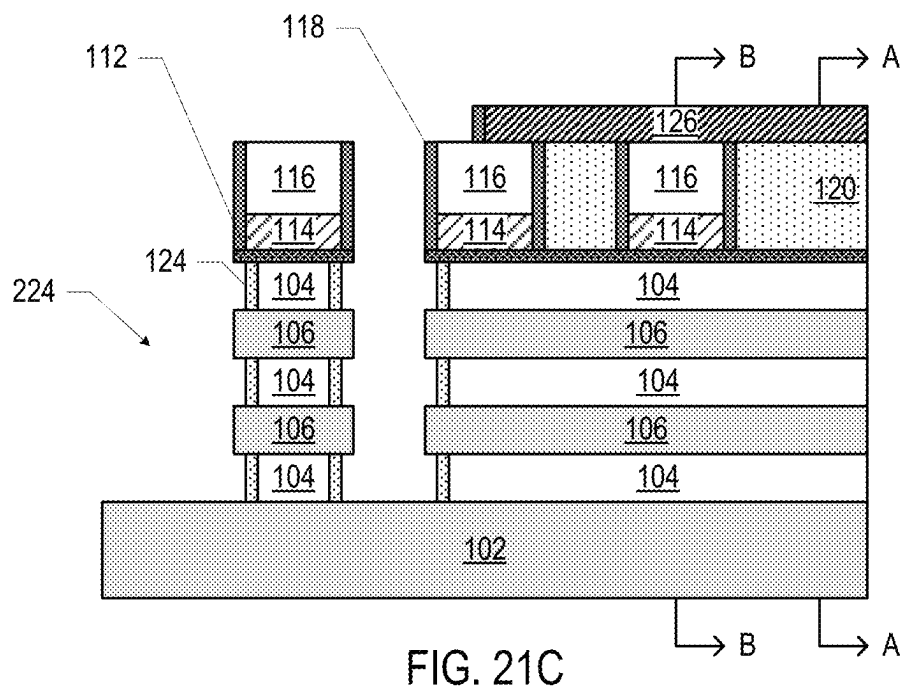
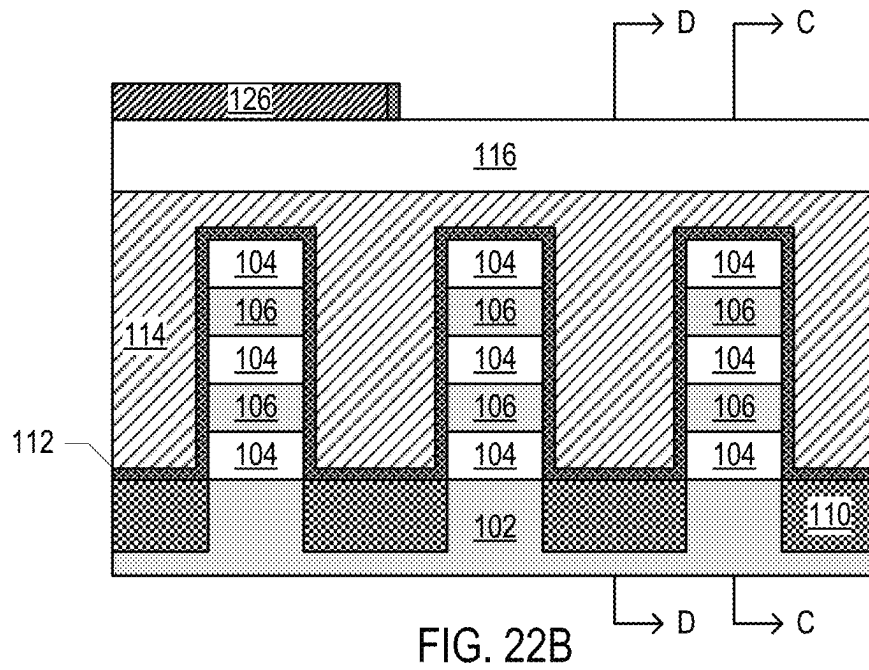
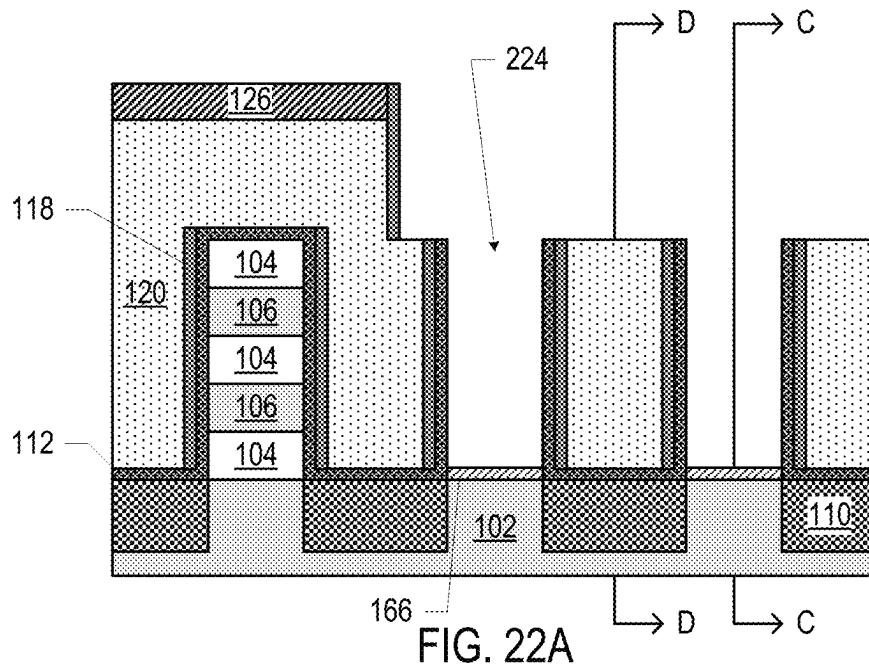


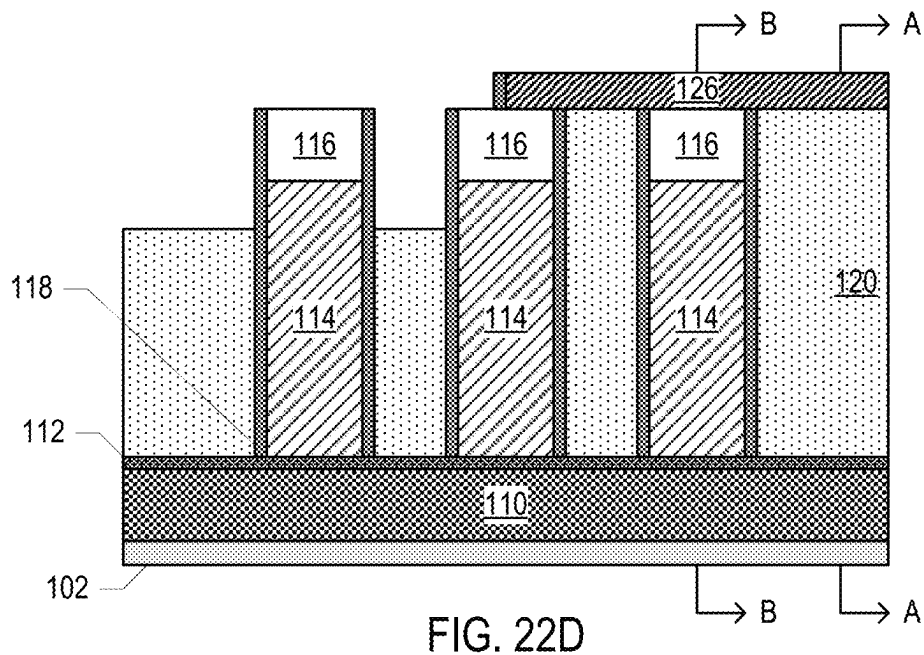
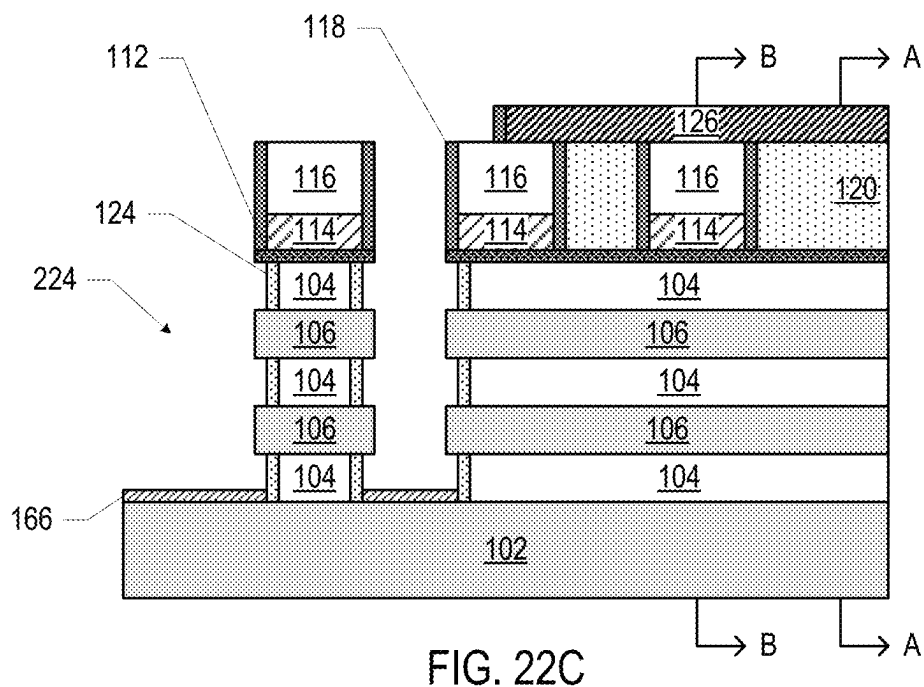
FIG. 20B

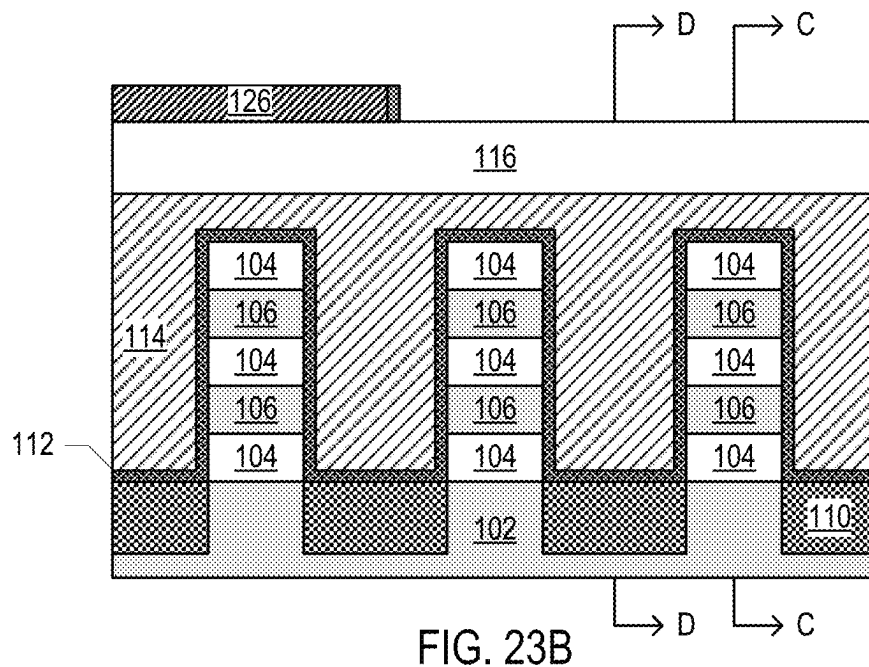
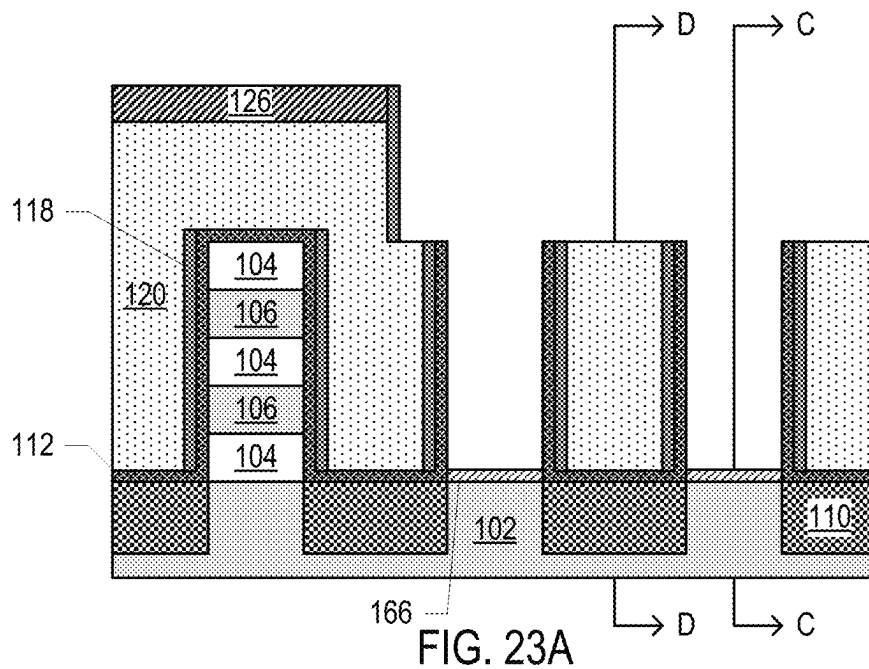


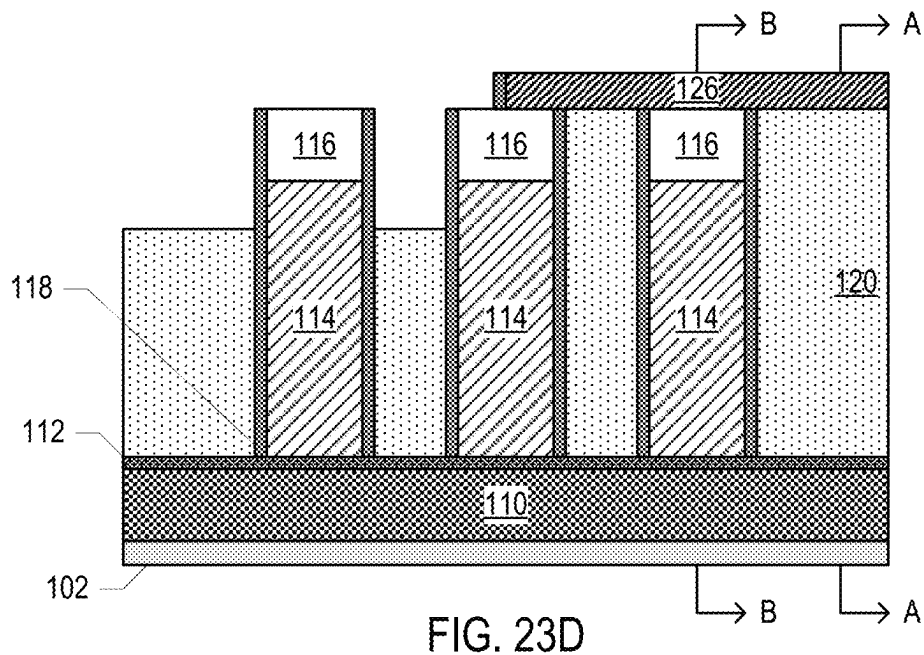
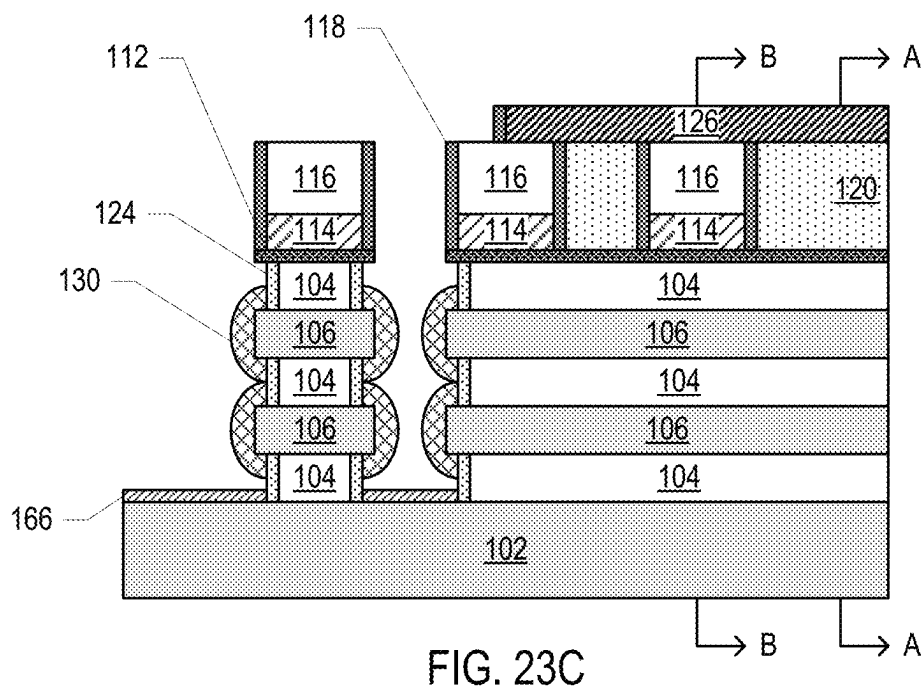


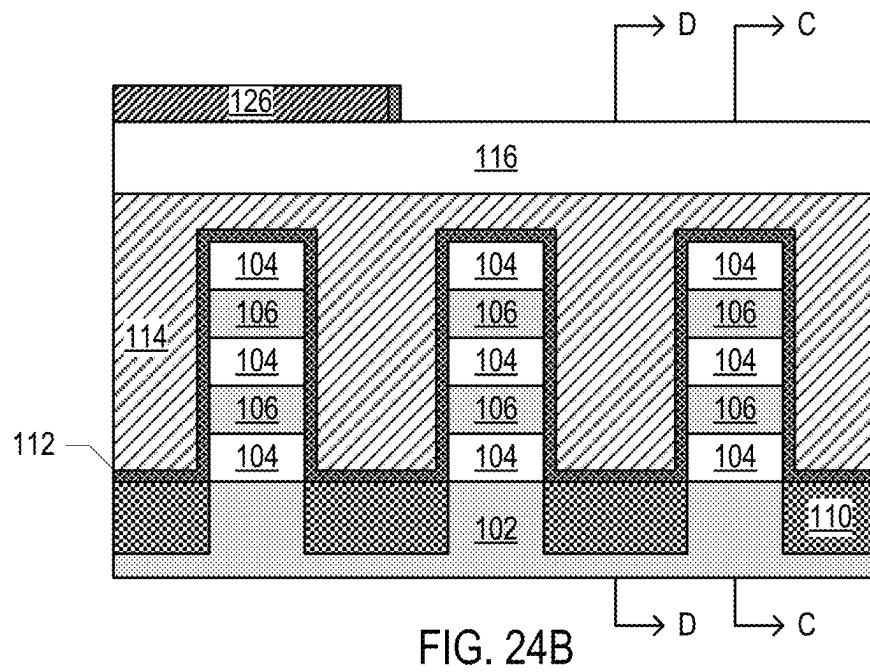
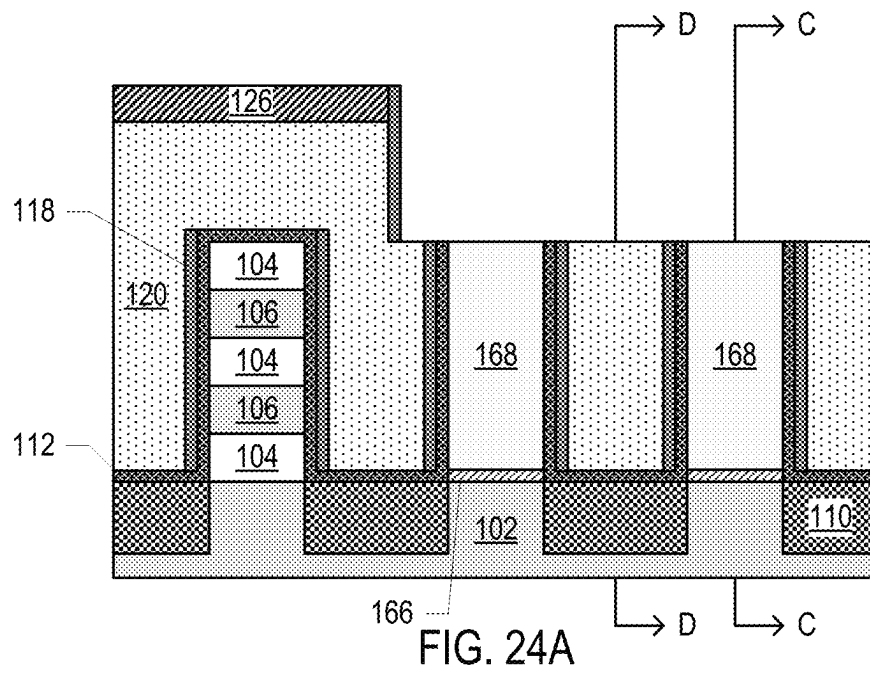


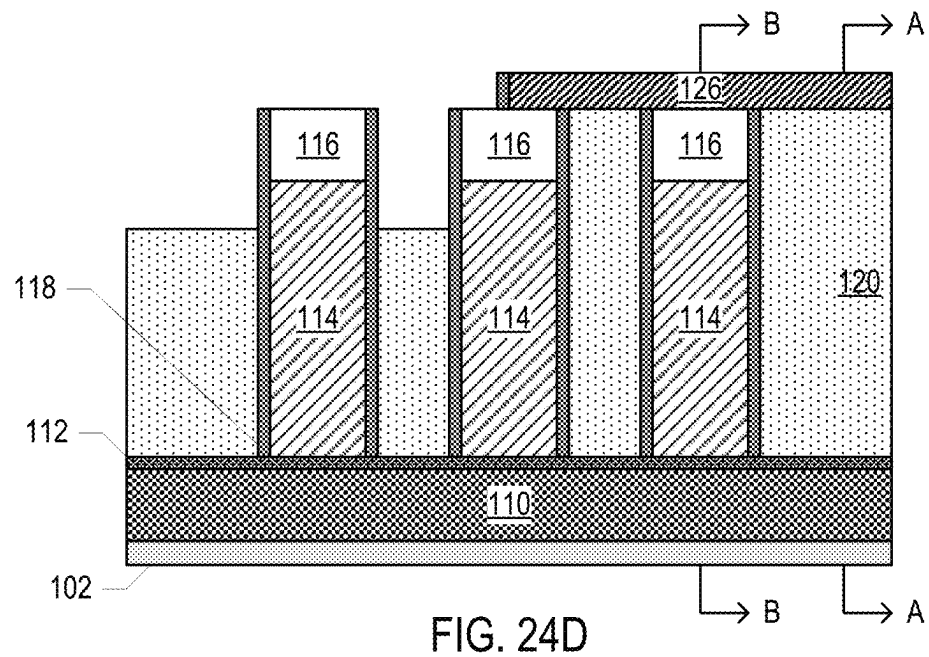
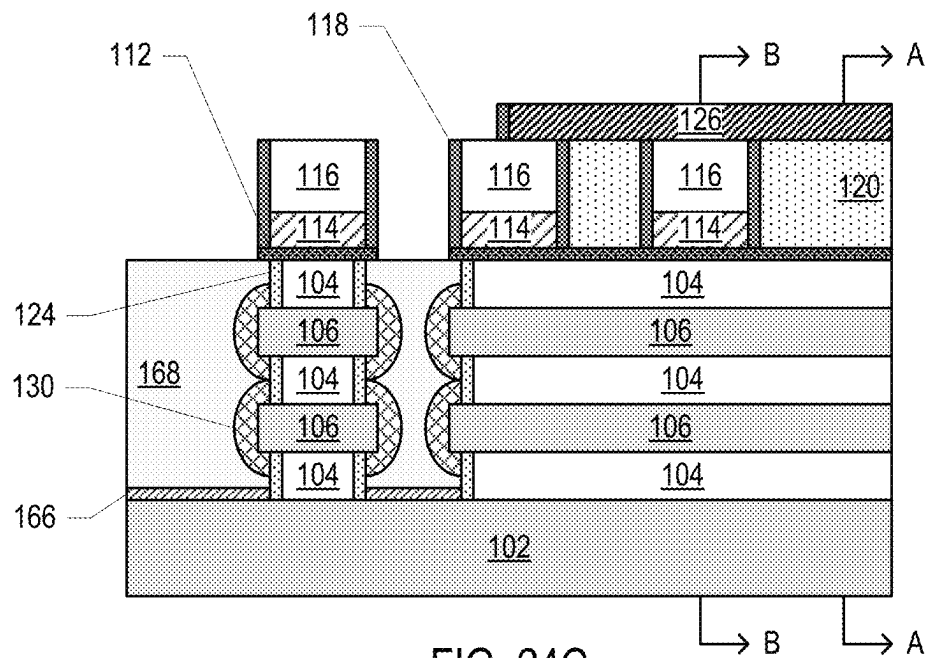


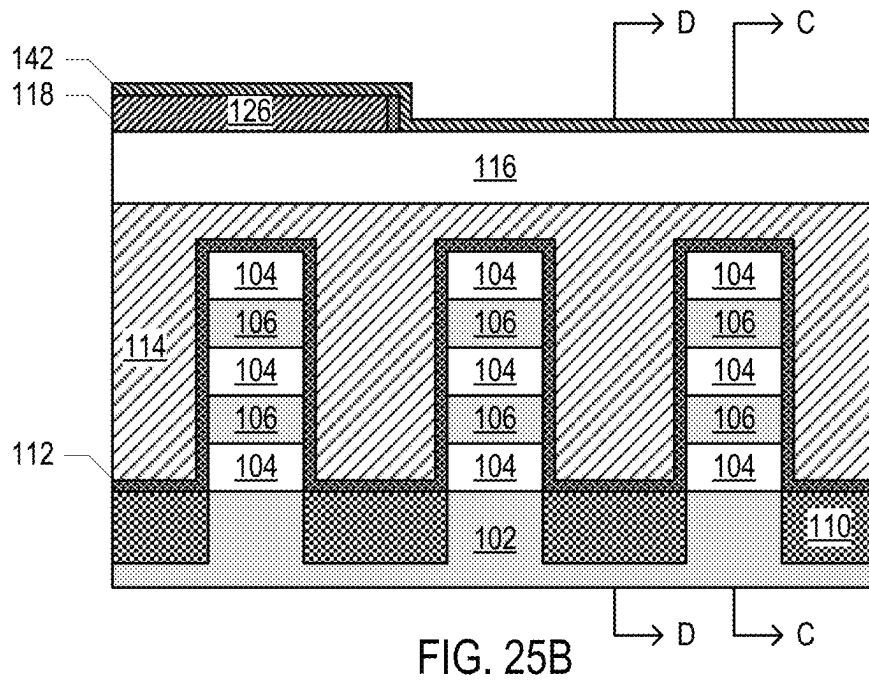
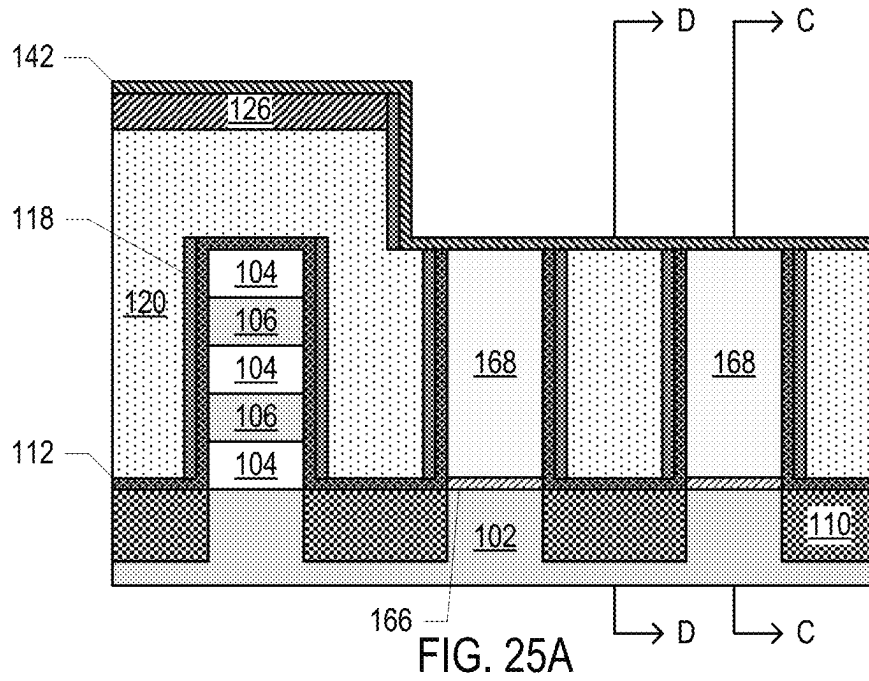


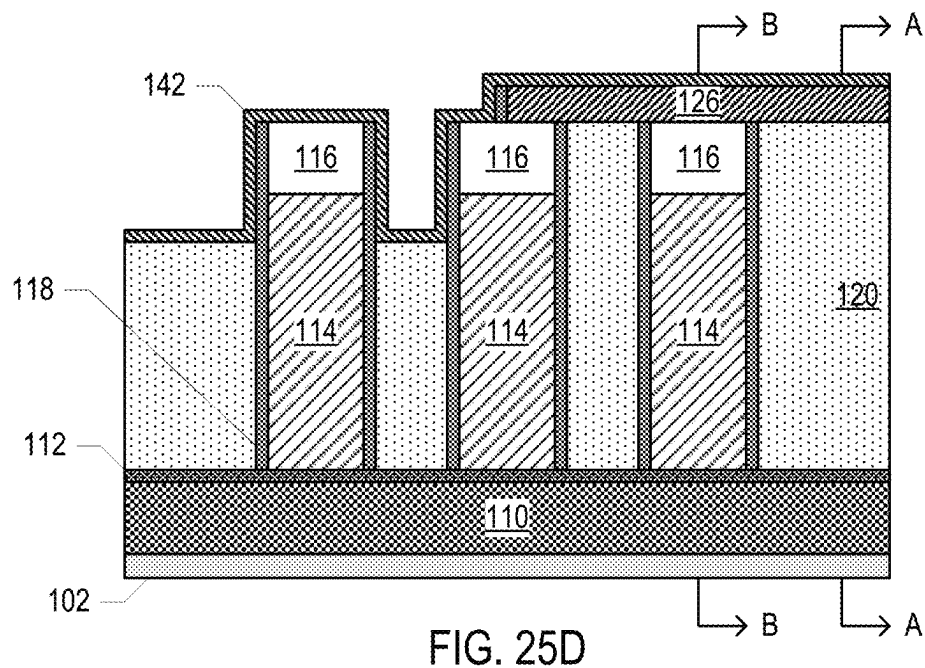
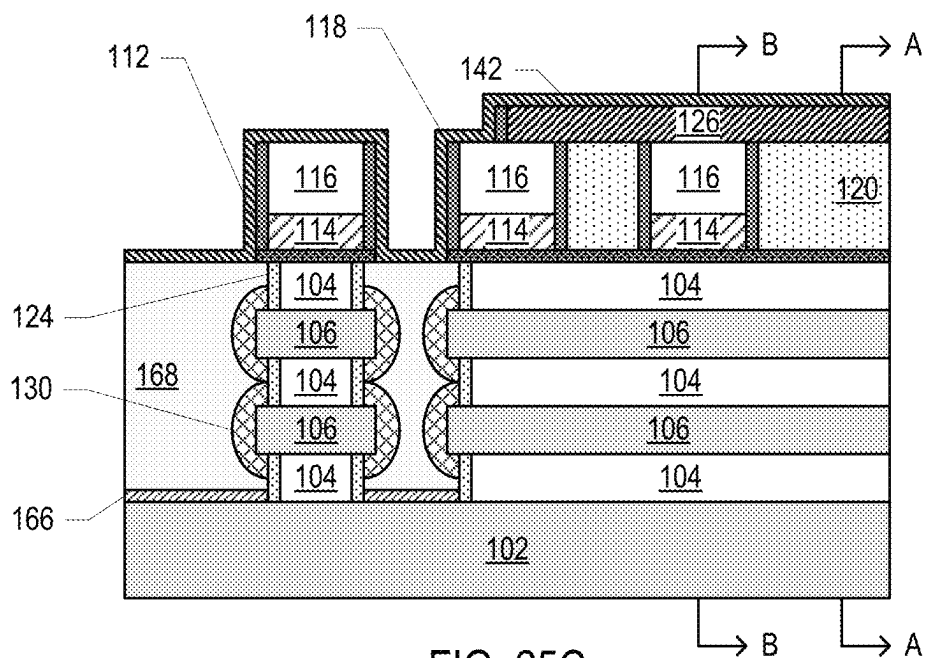


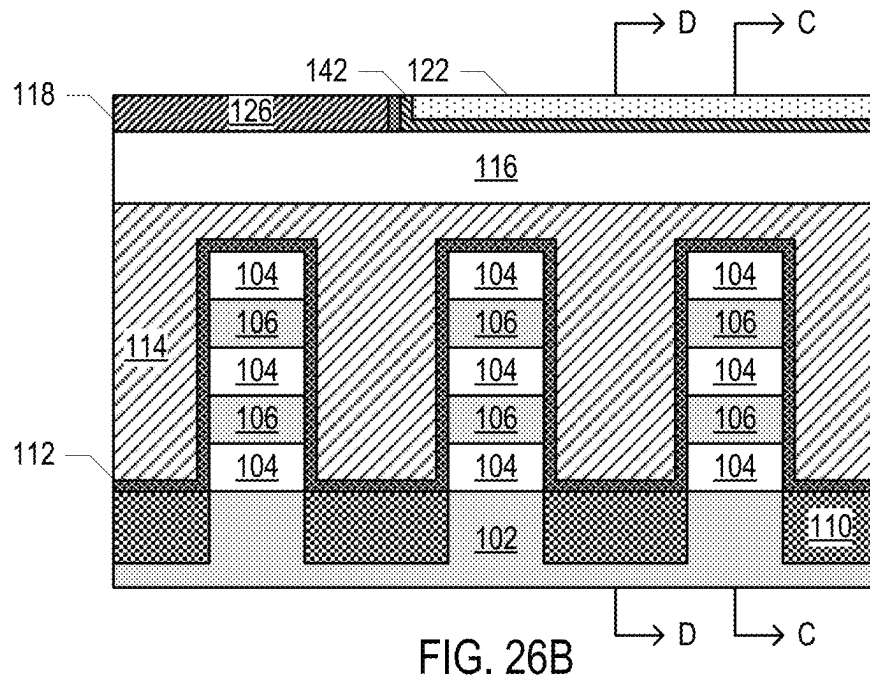


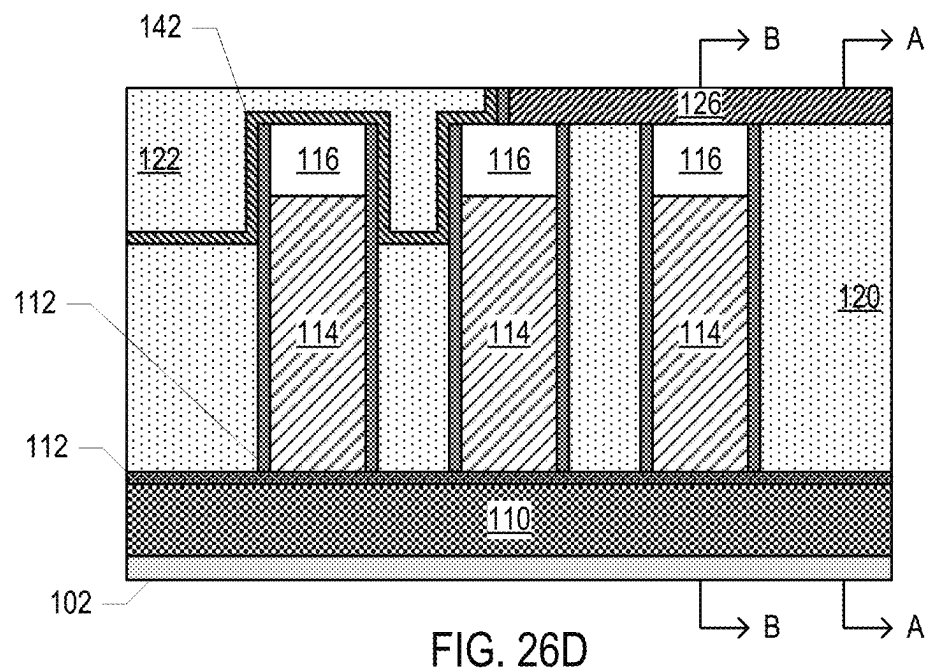
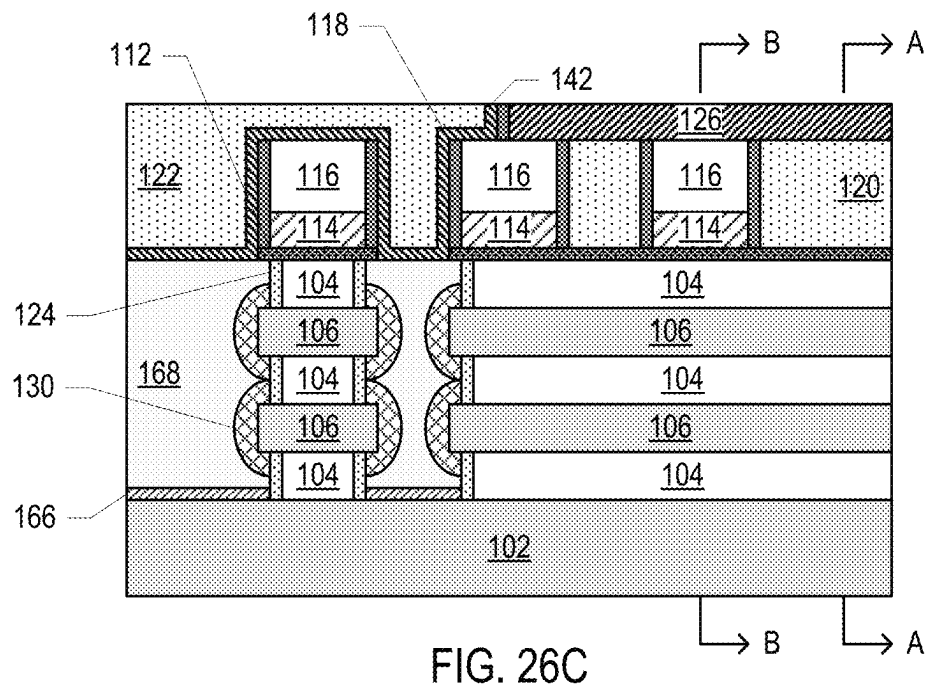


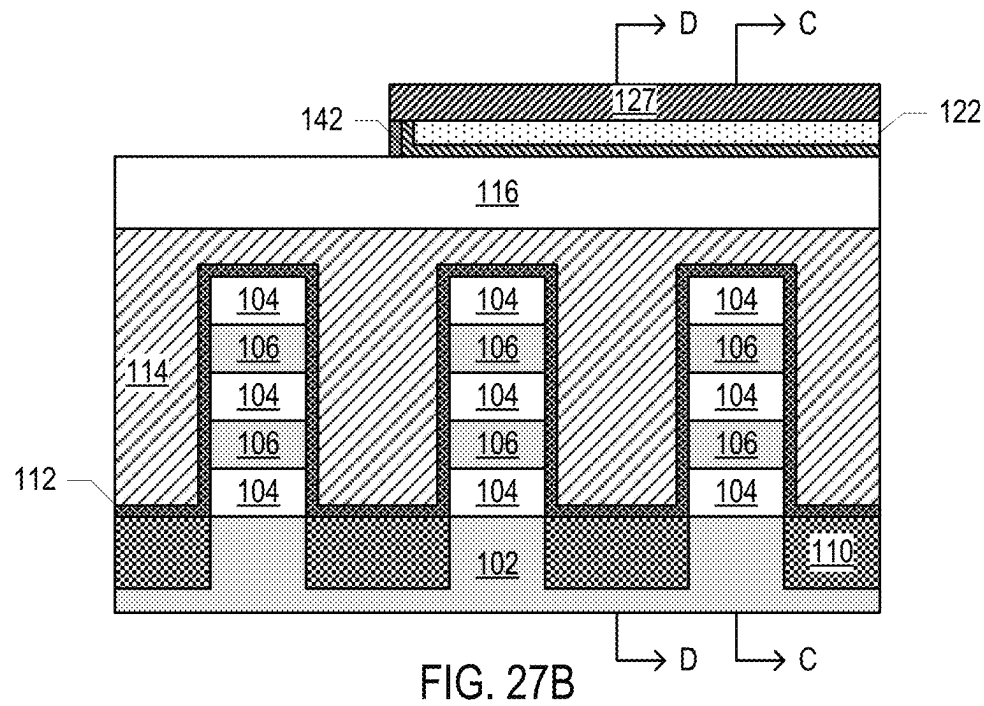
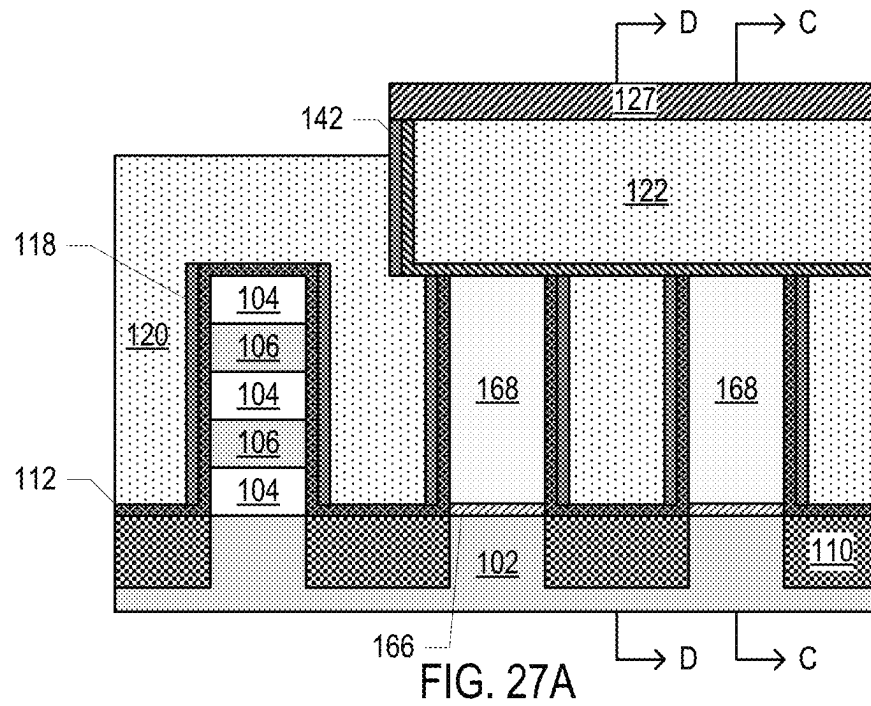


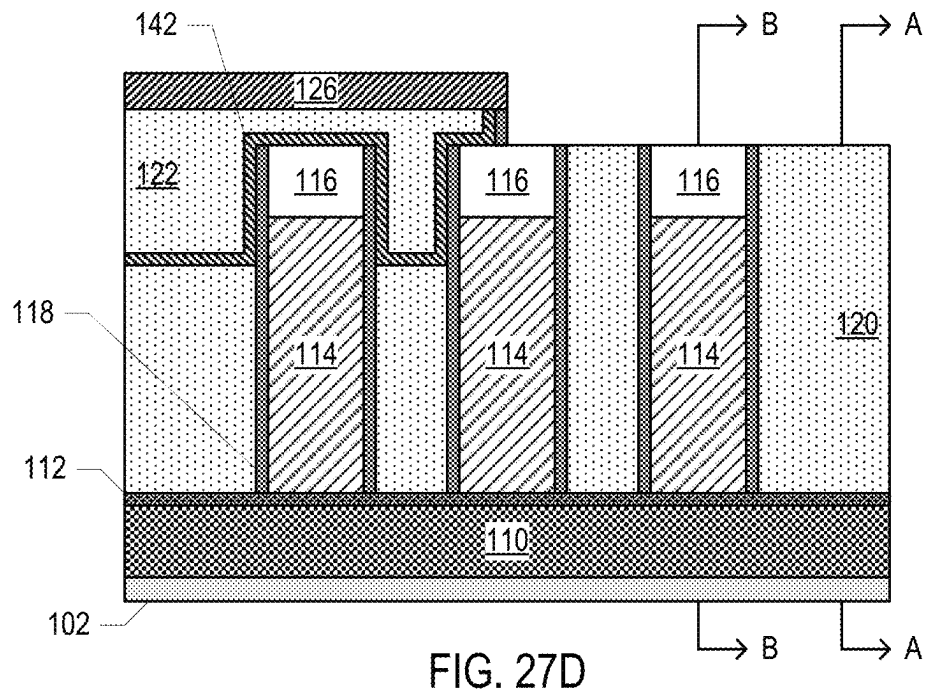
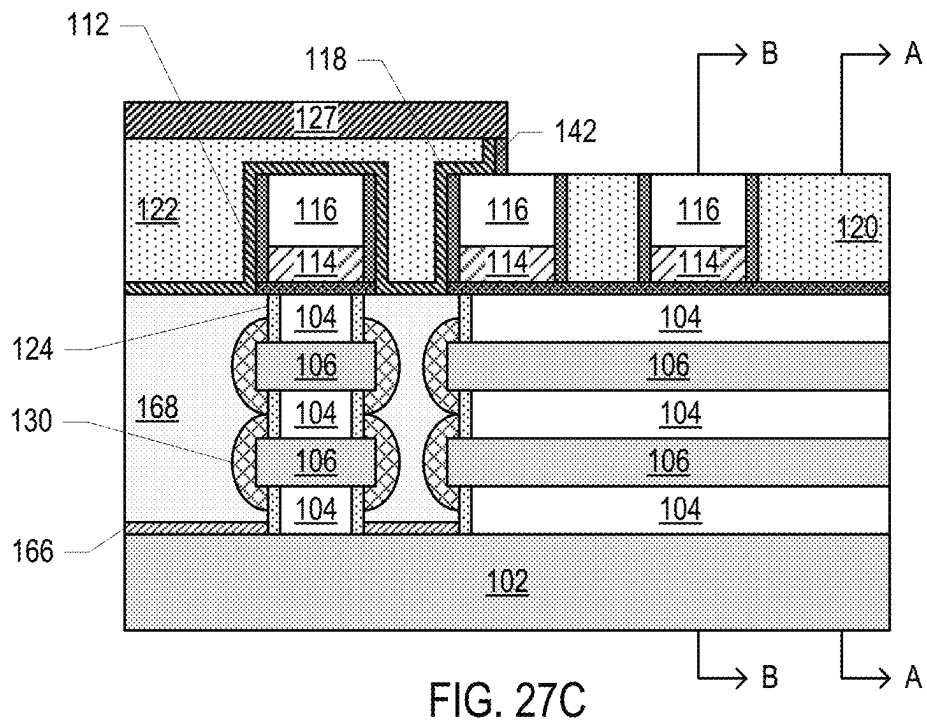


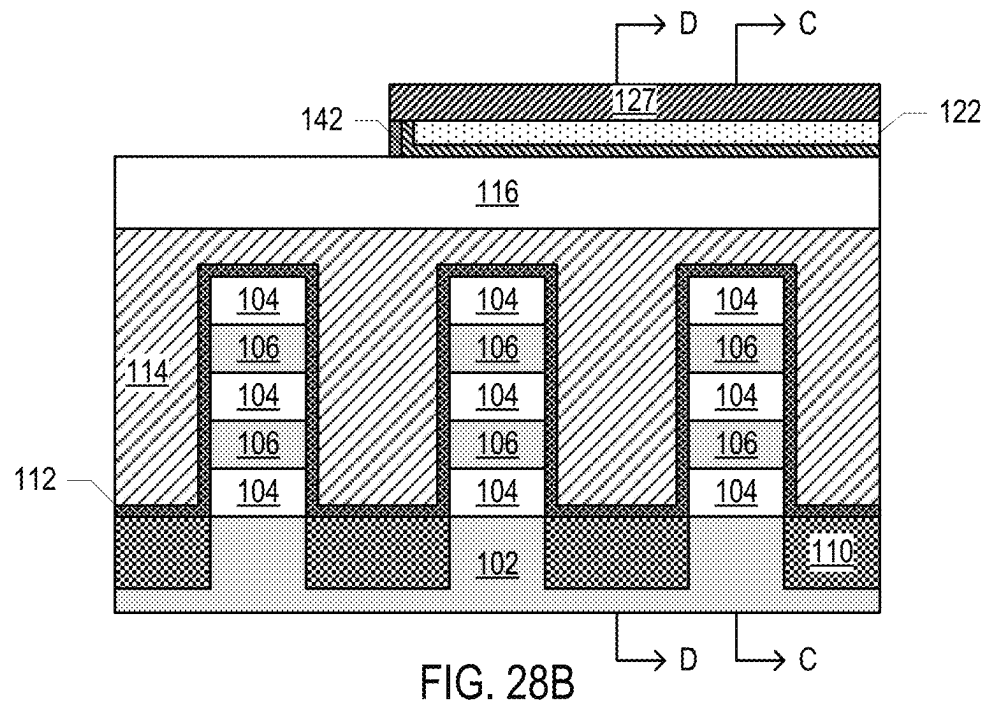
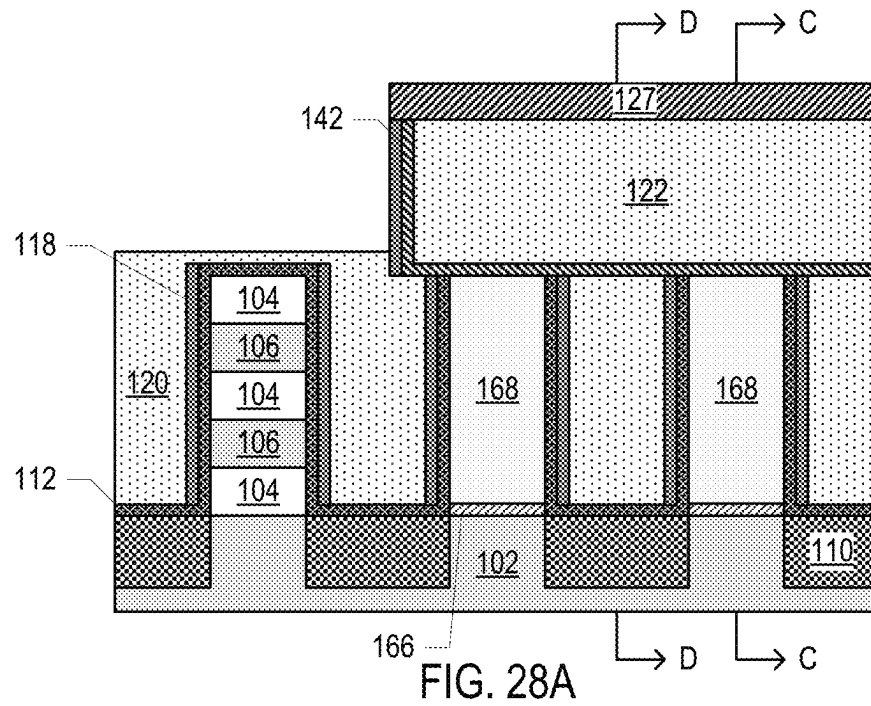












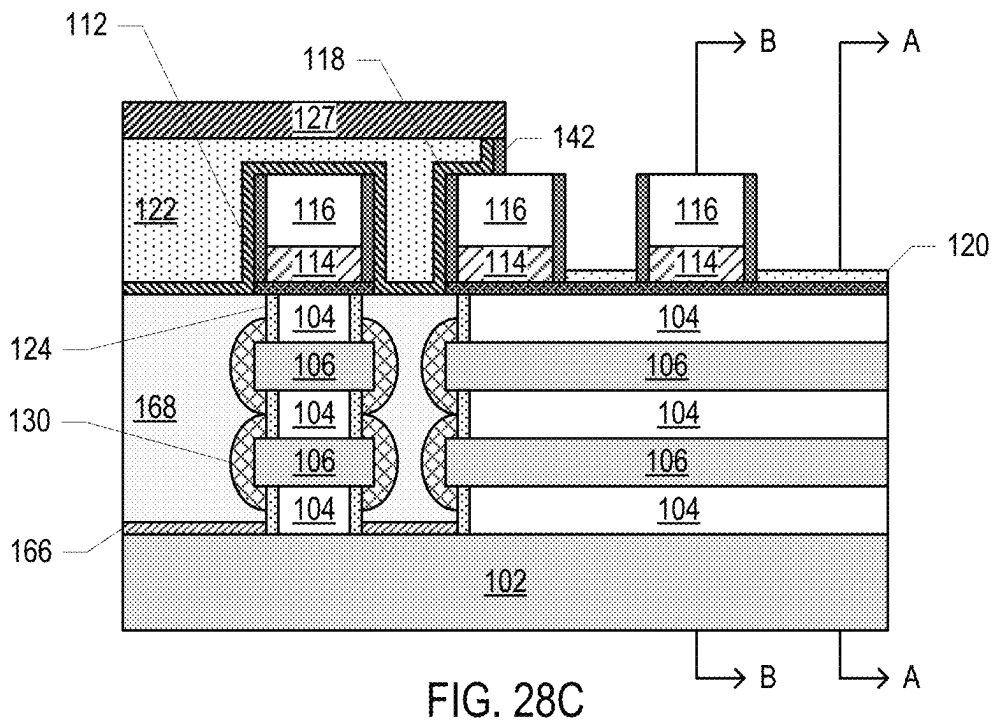


FIG. 28C

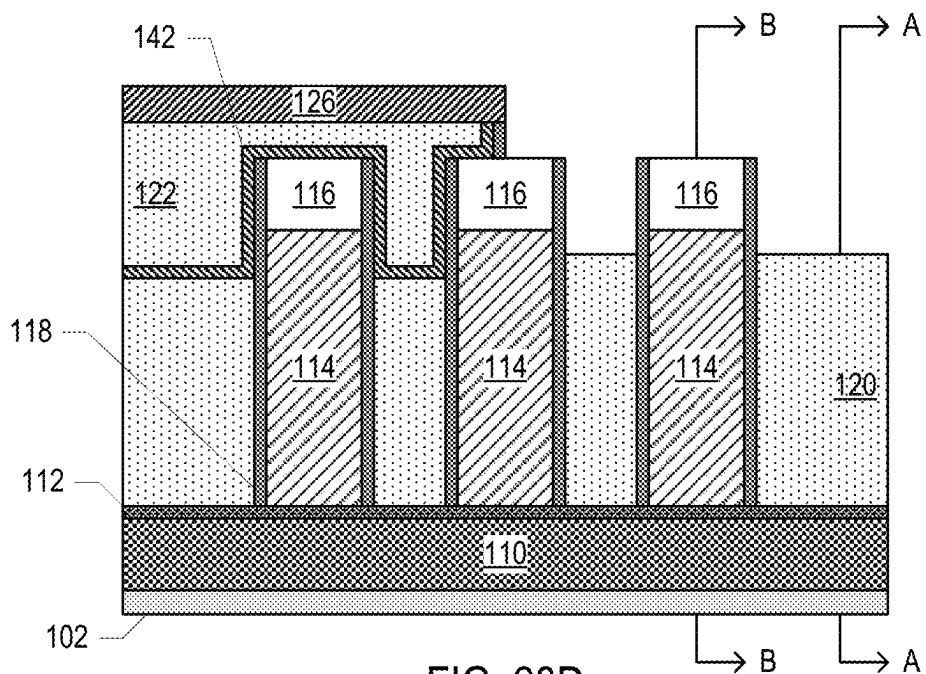
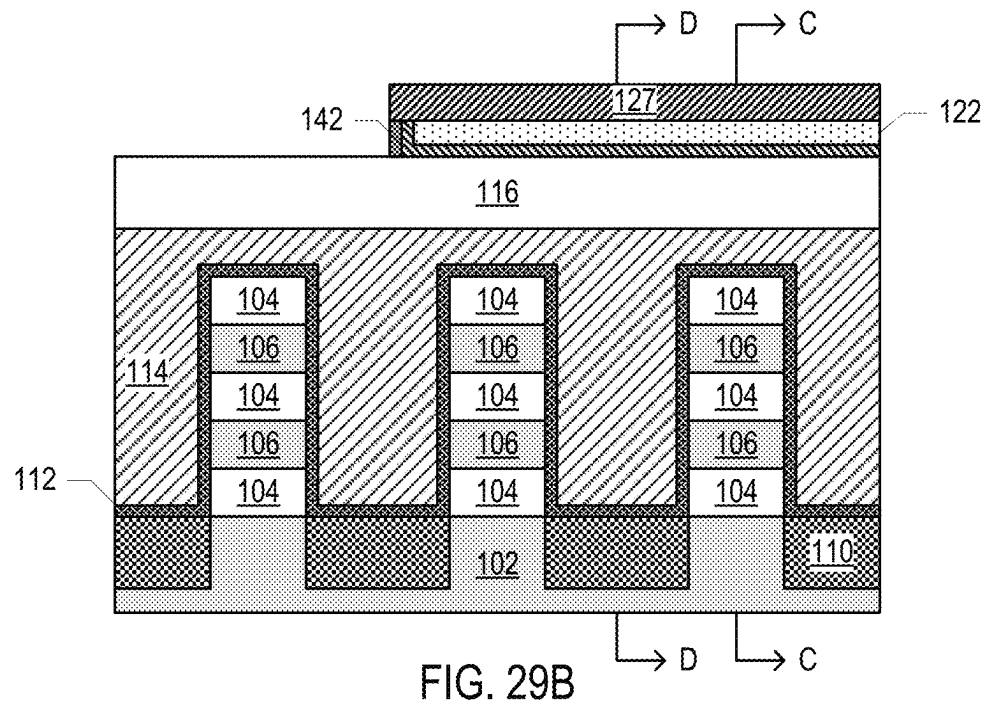
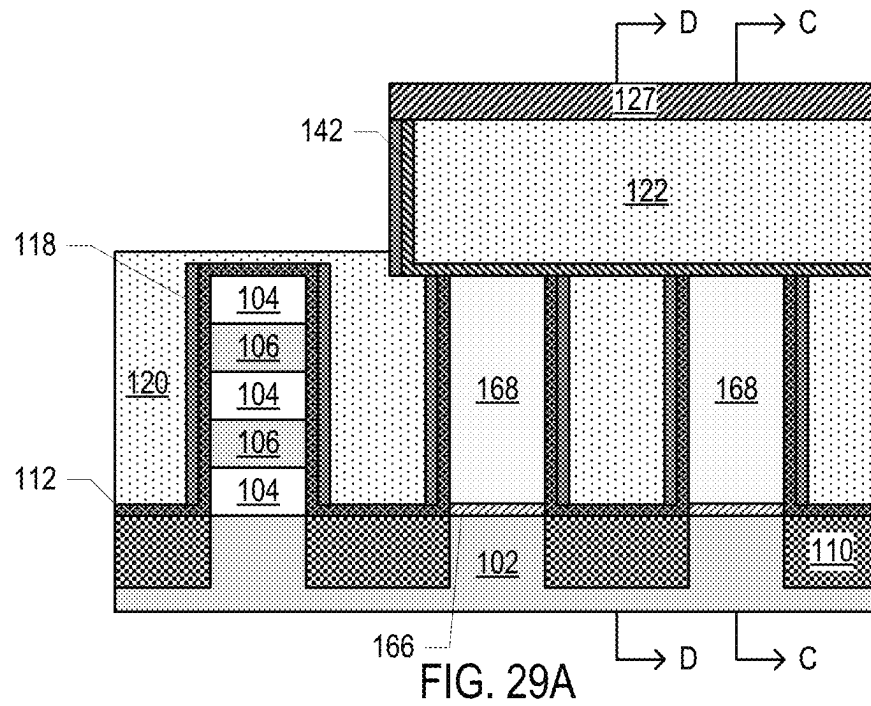
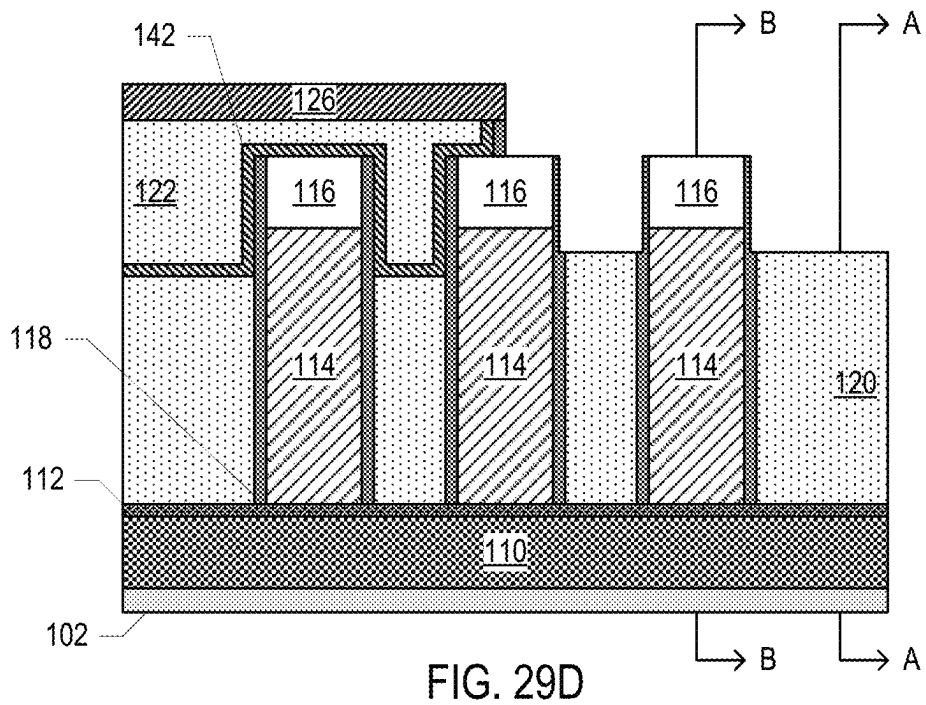
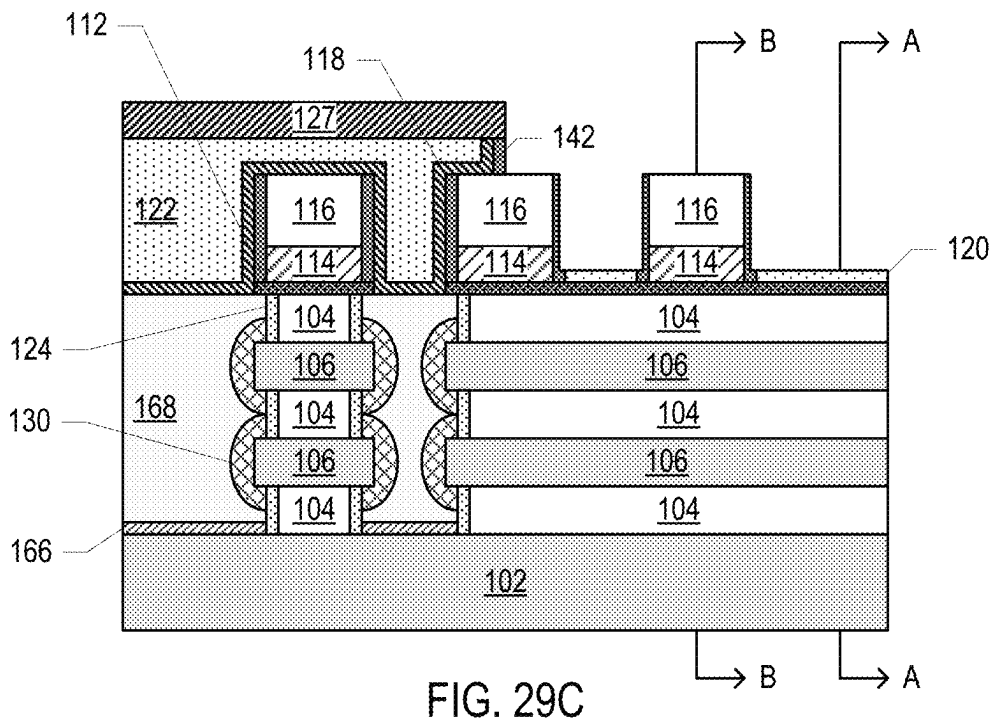
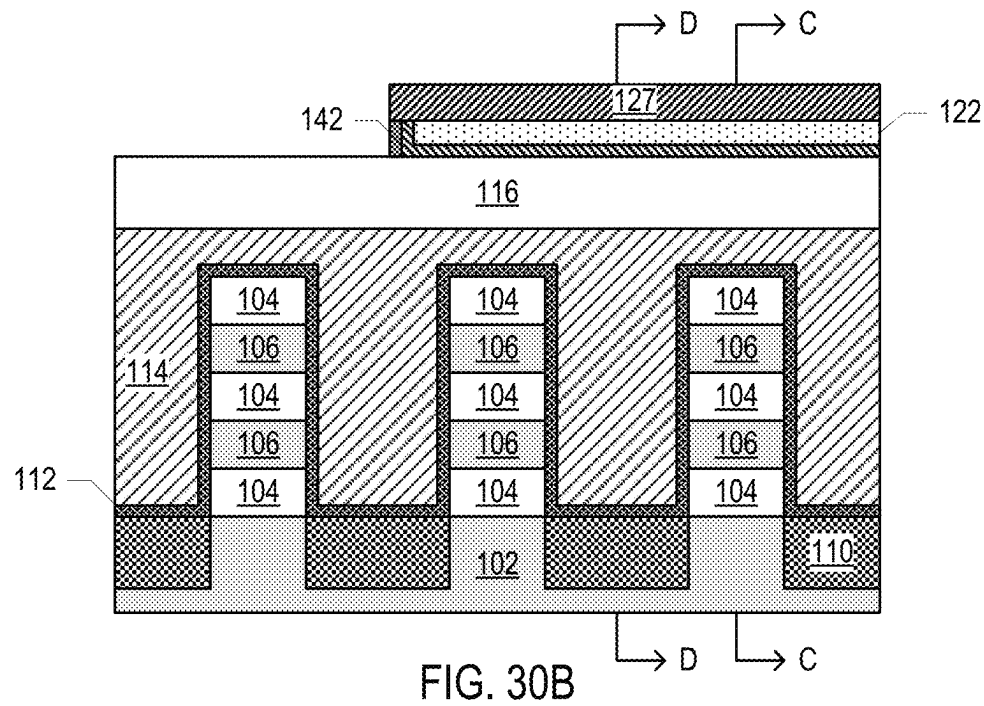
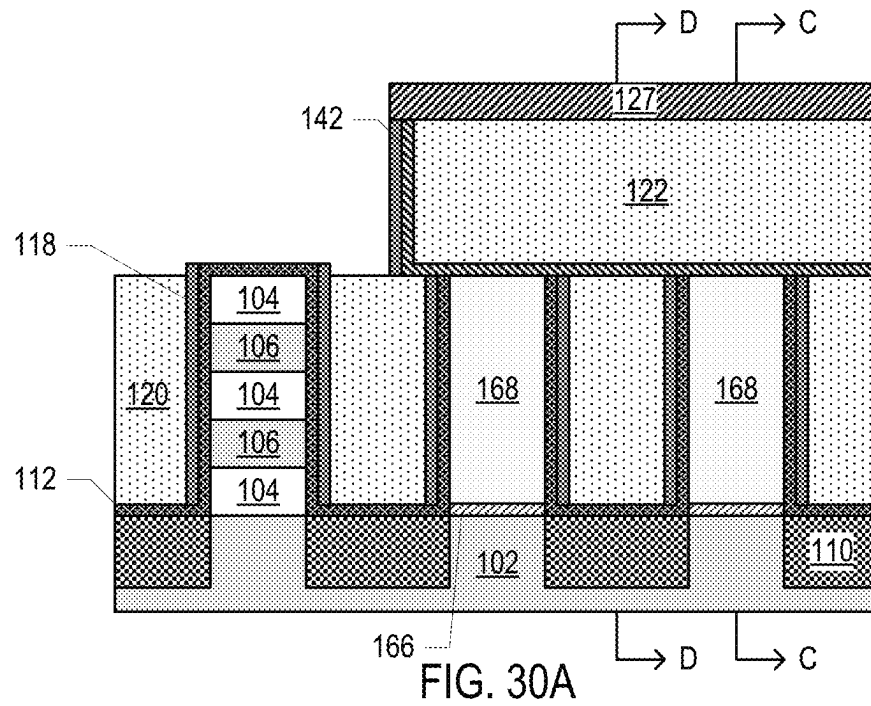
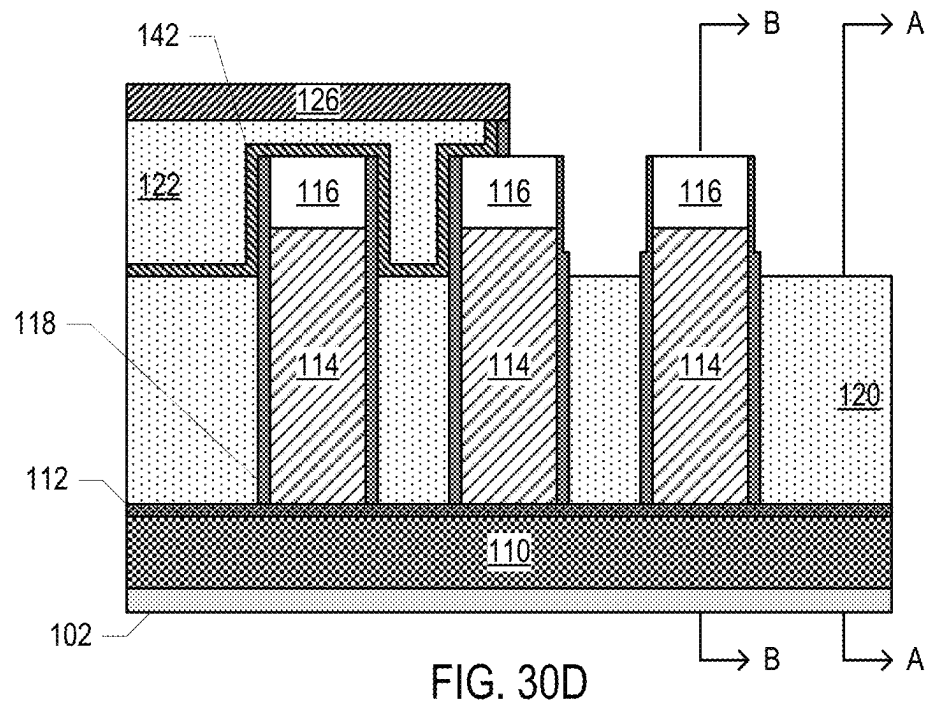
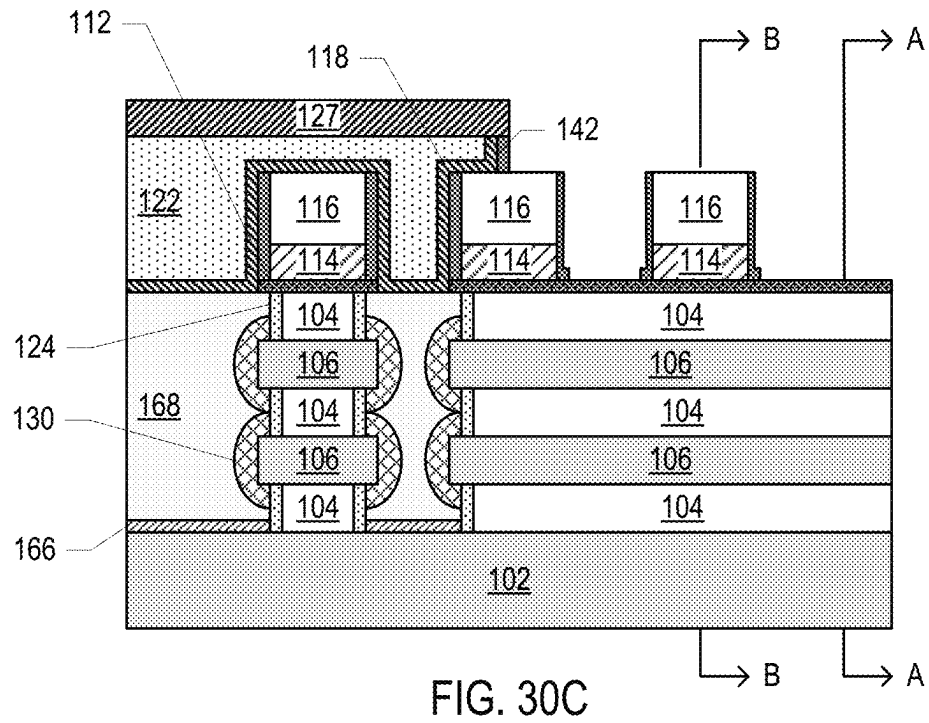


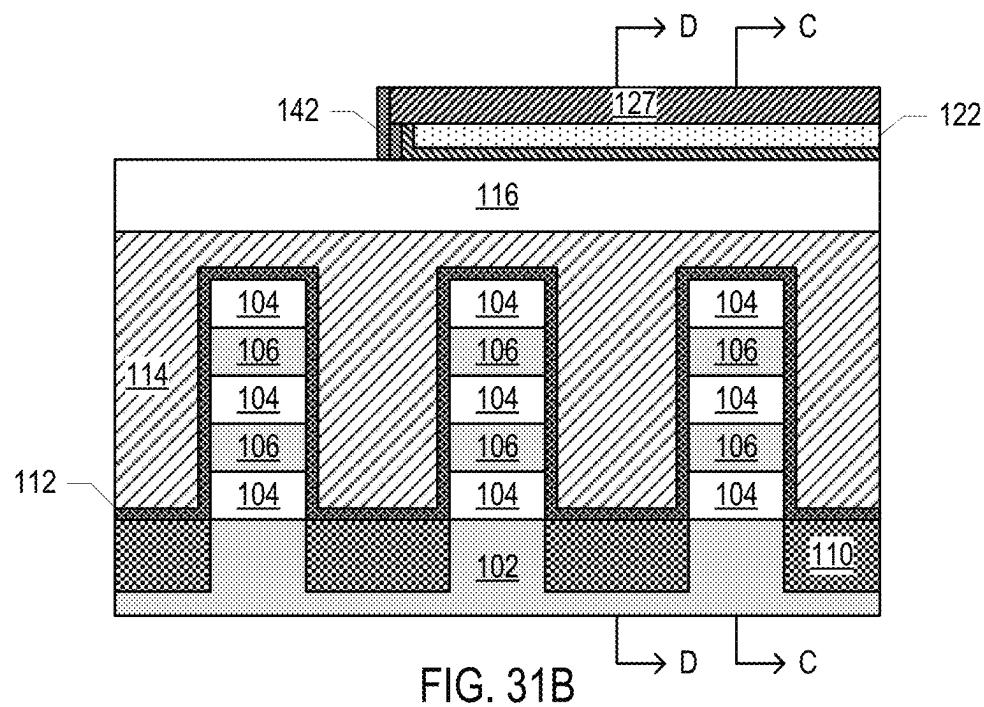
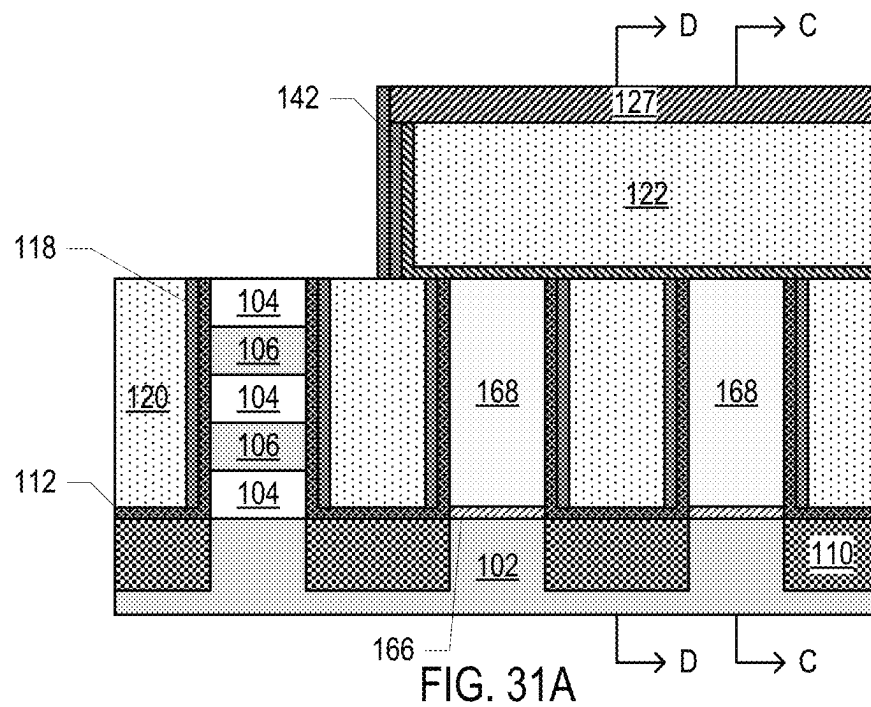
FIG. 28D

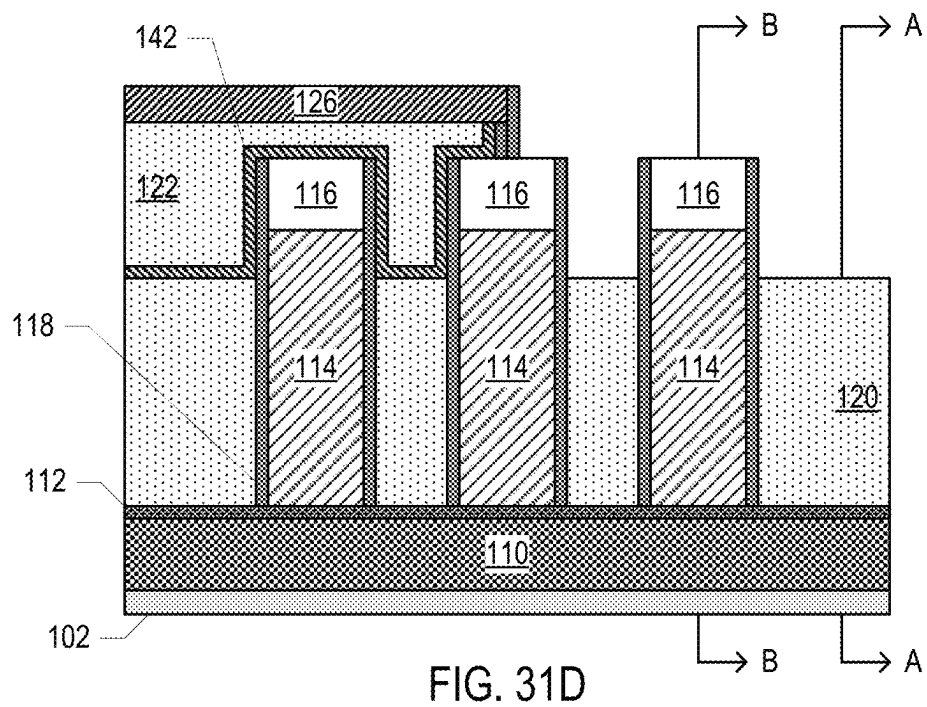
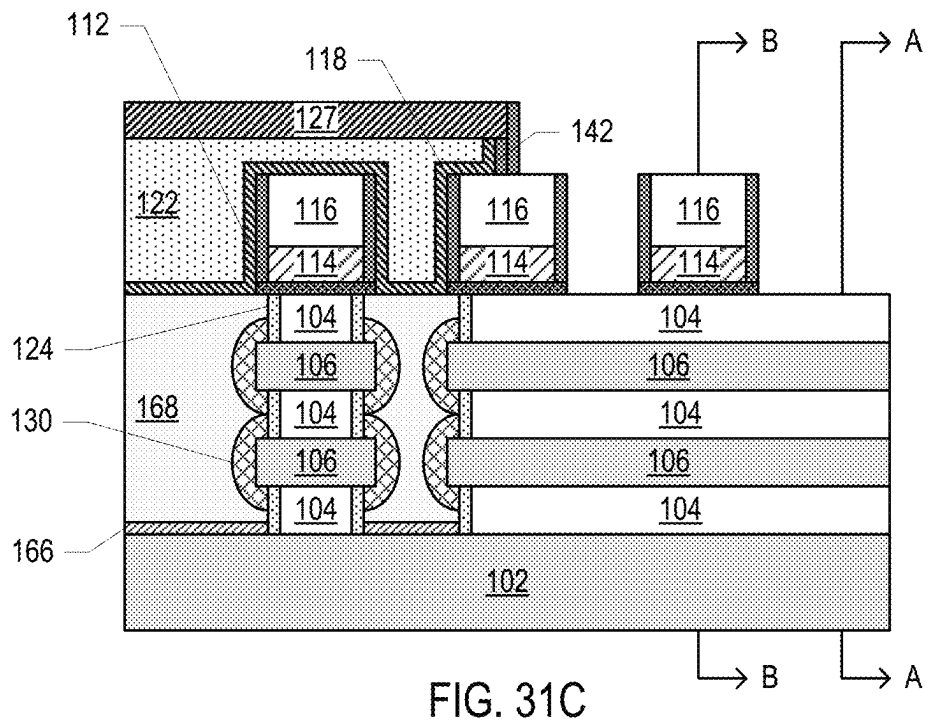


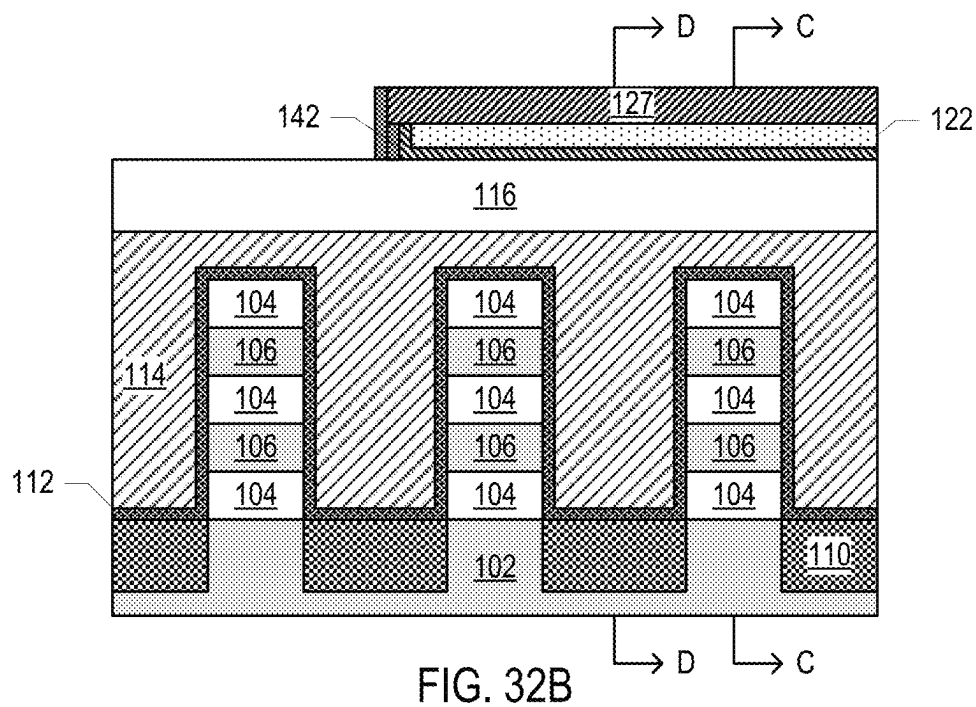
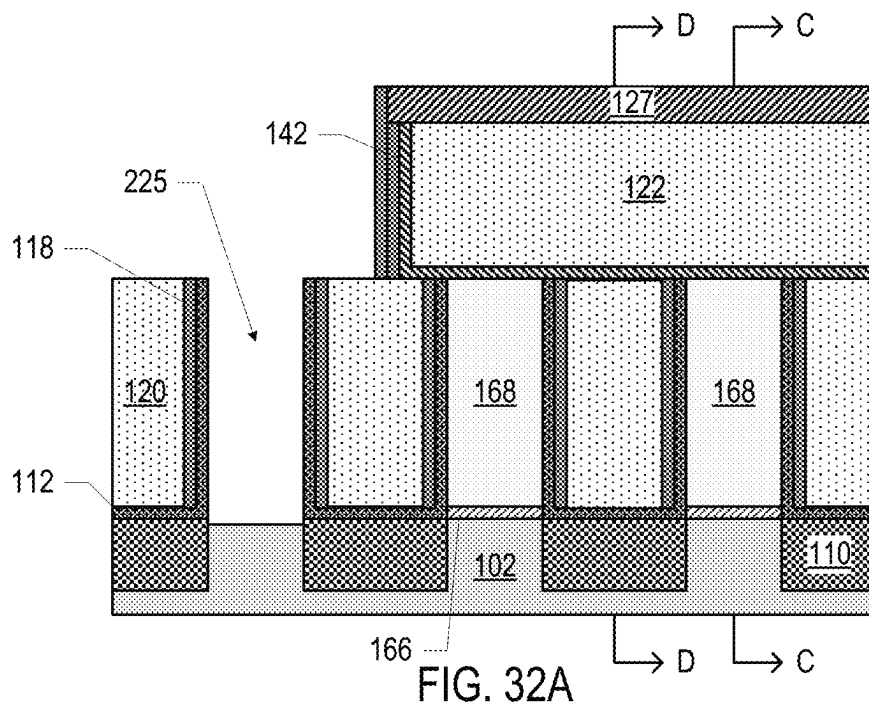


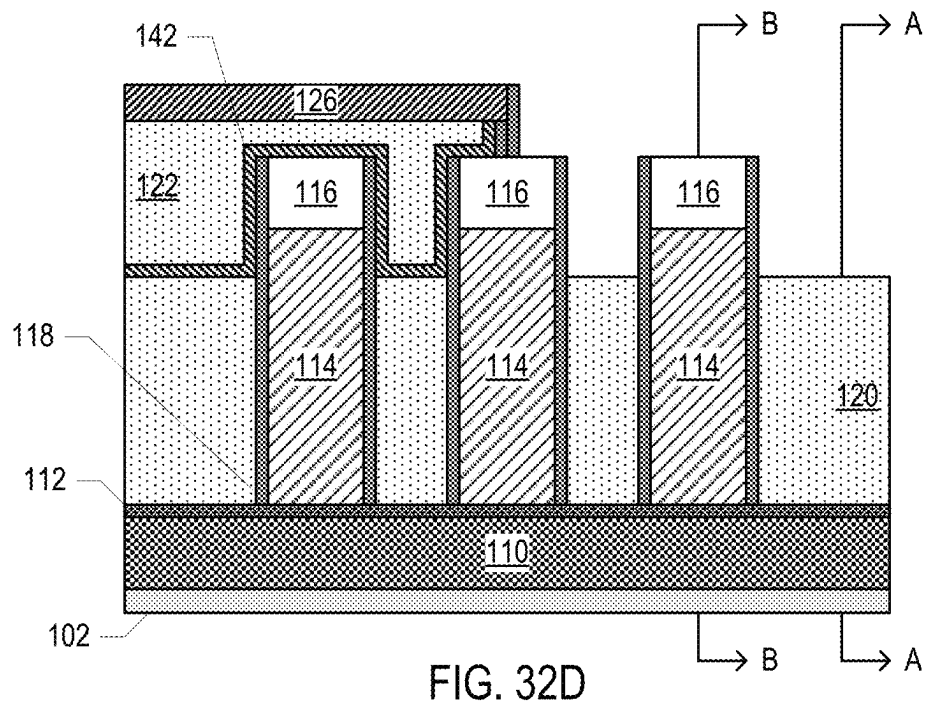
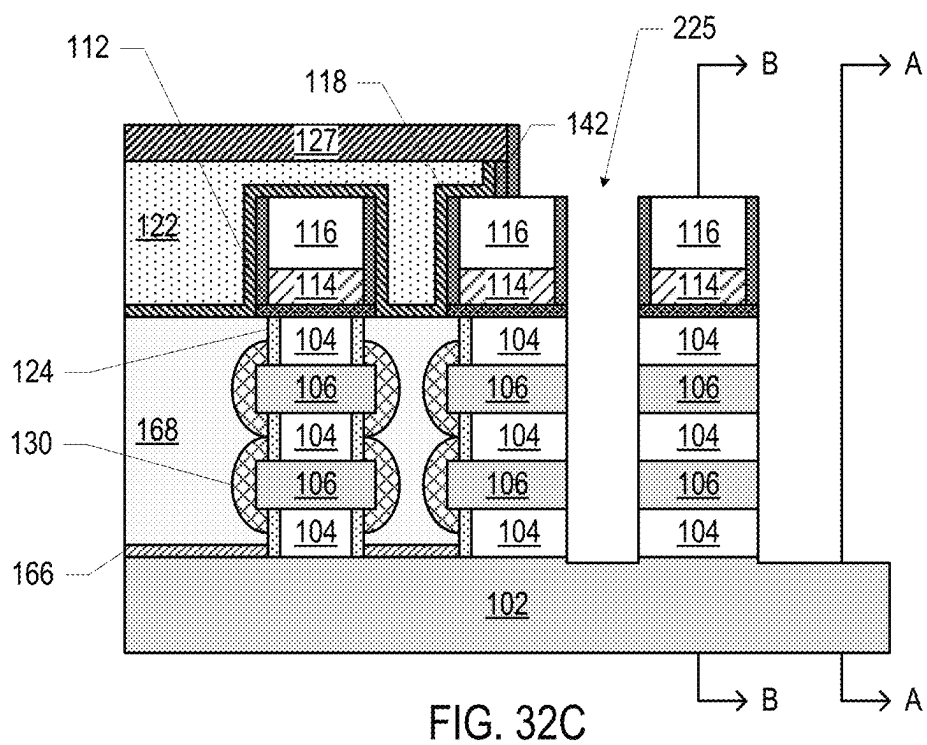


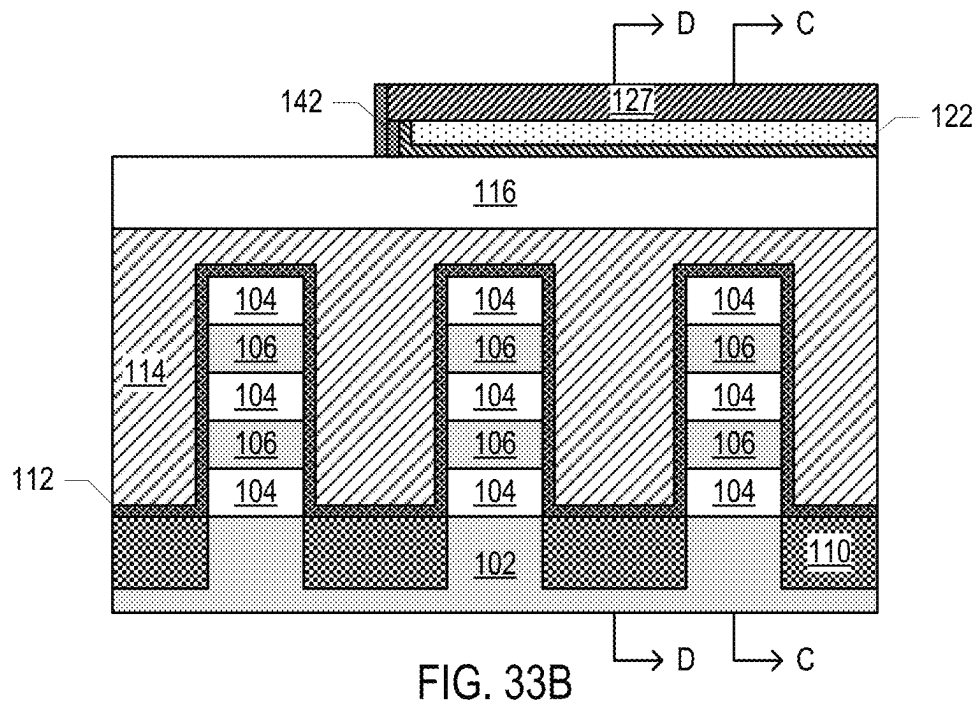
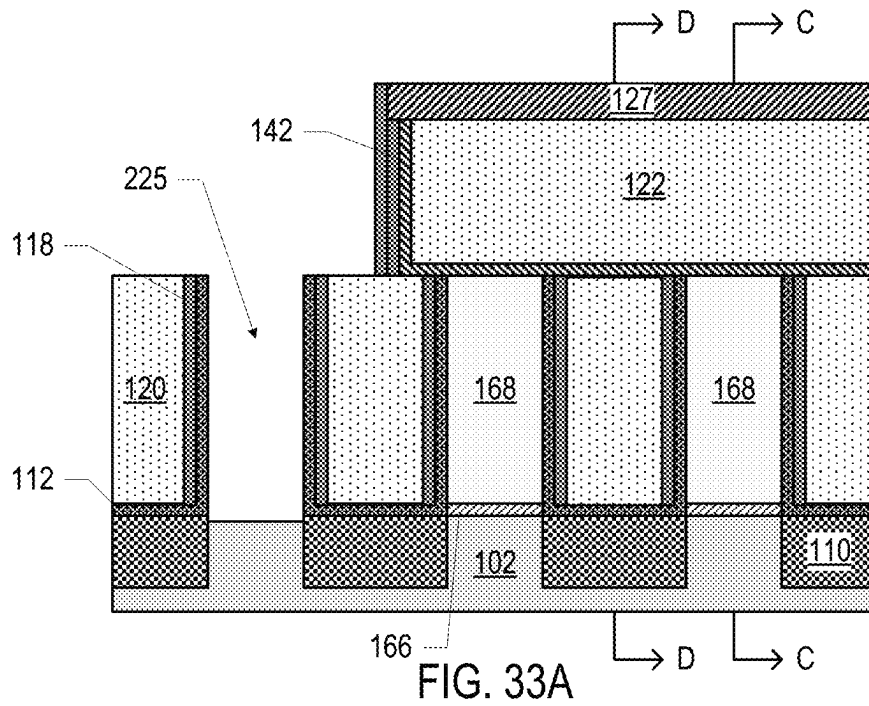


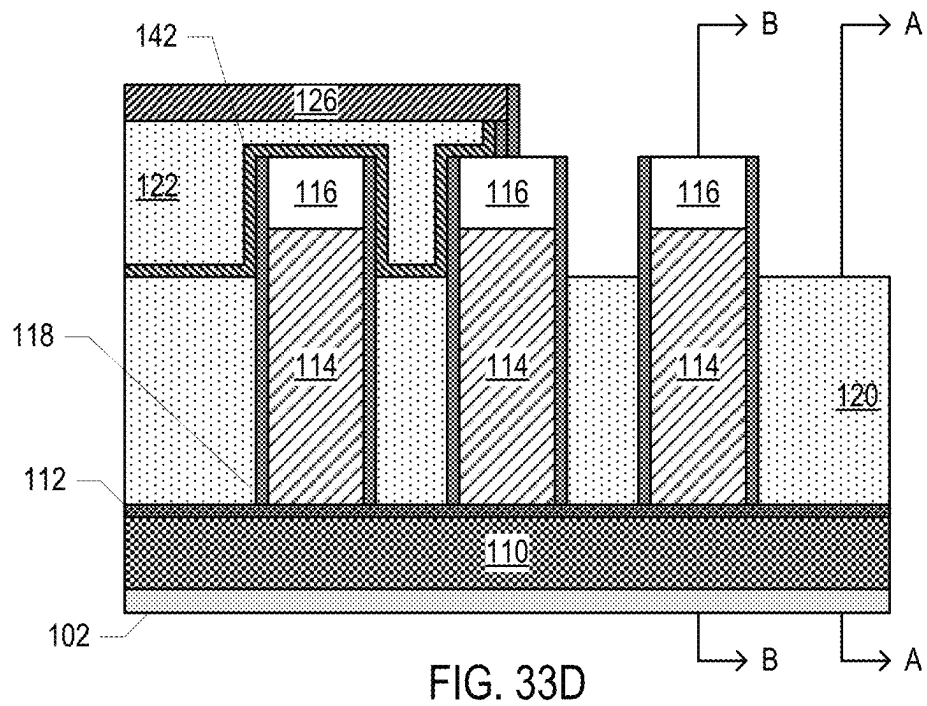
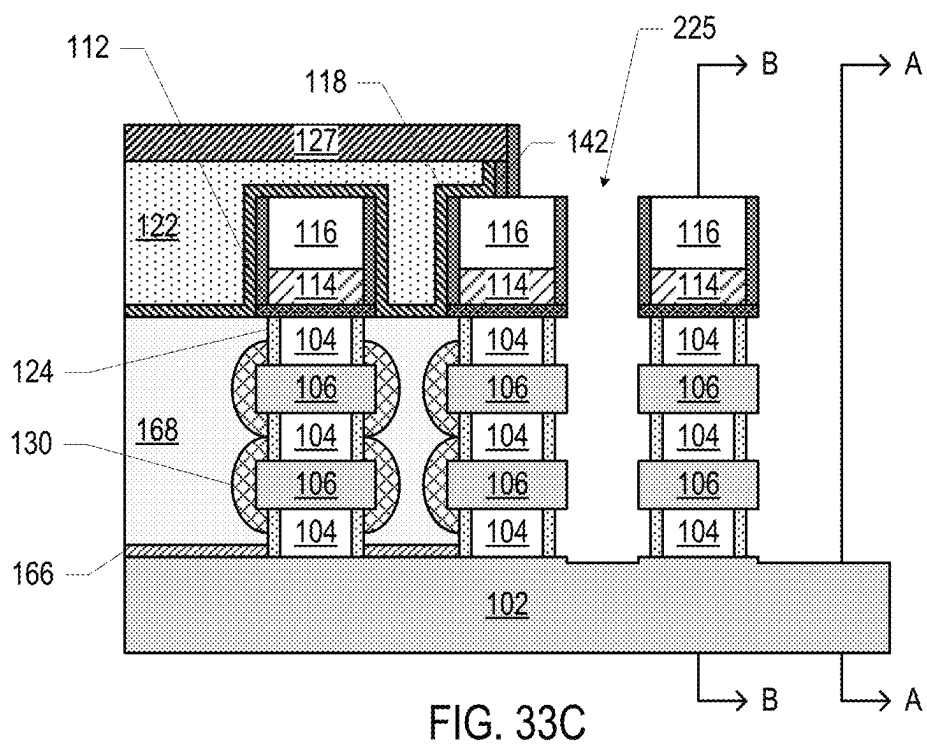


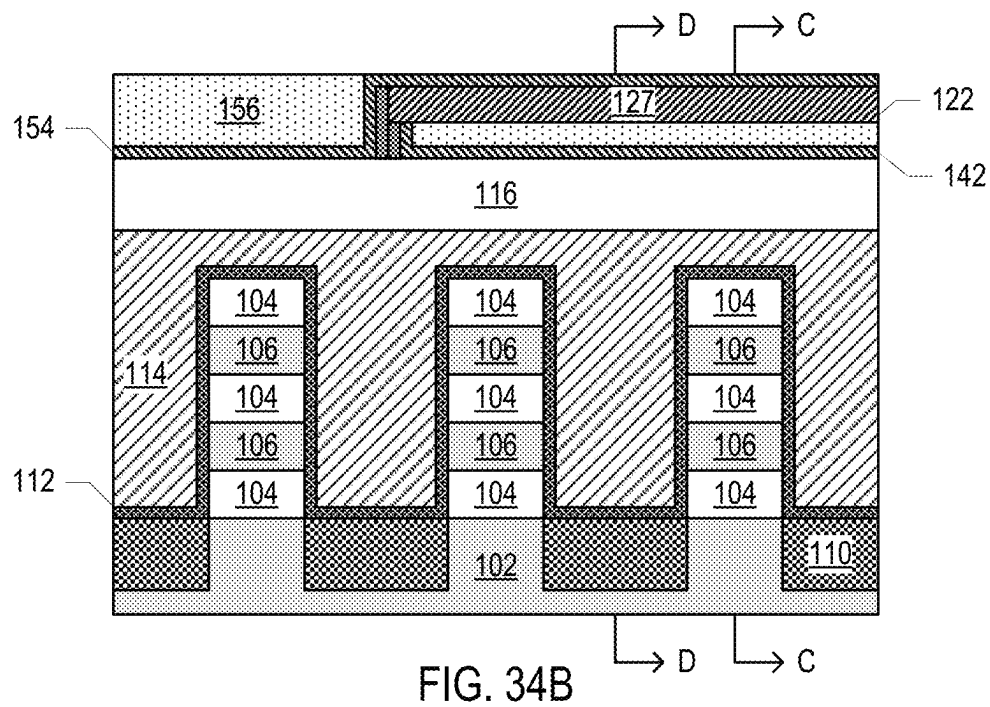
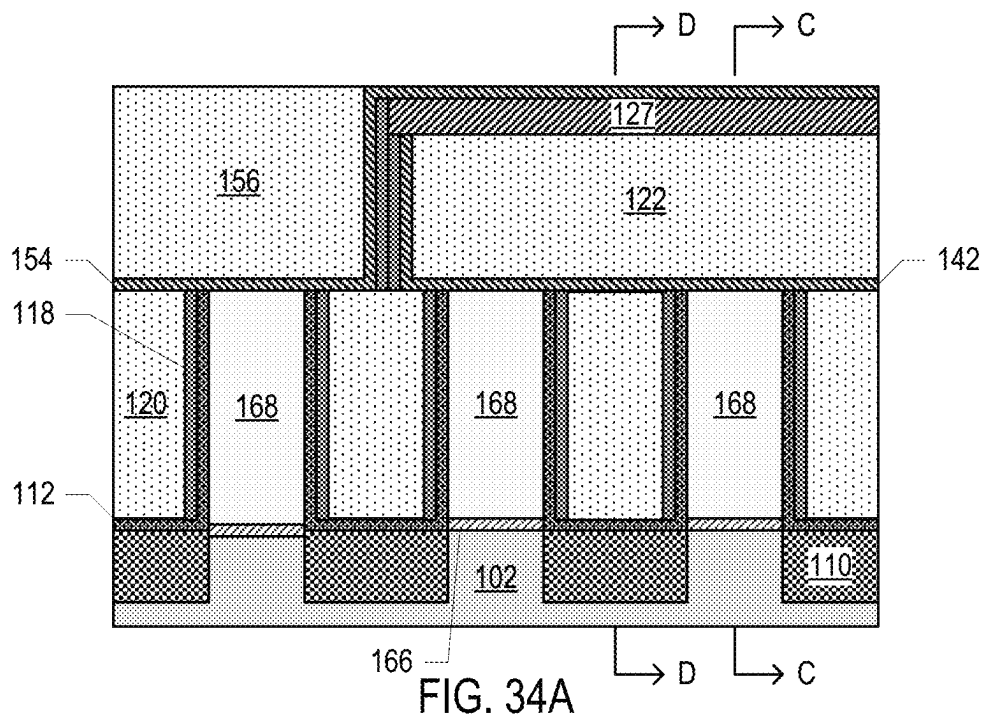


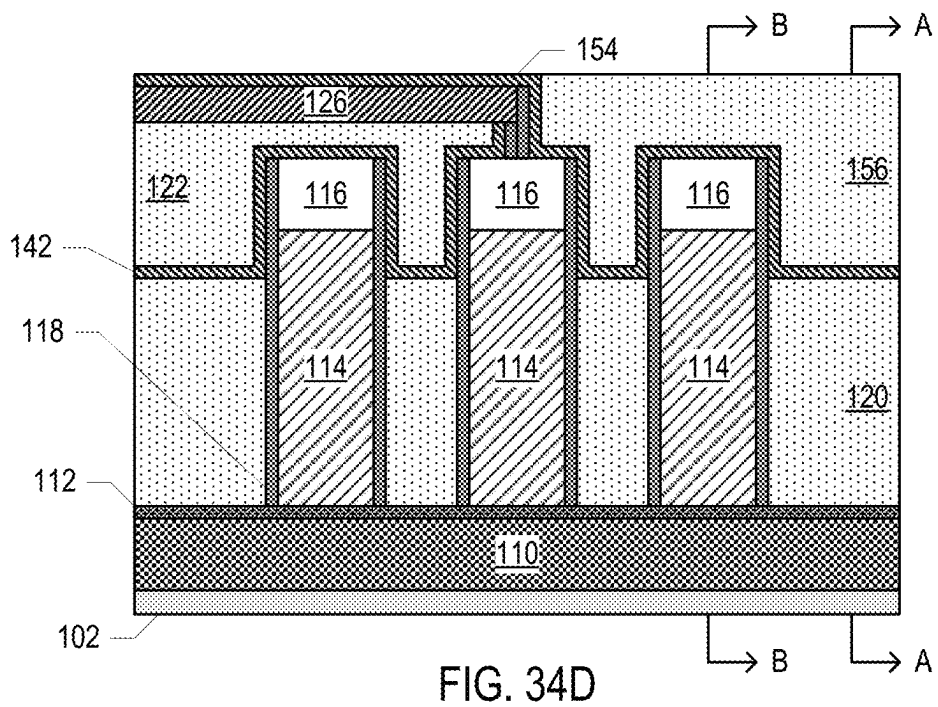
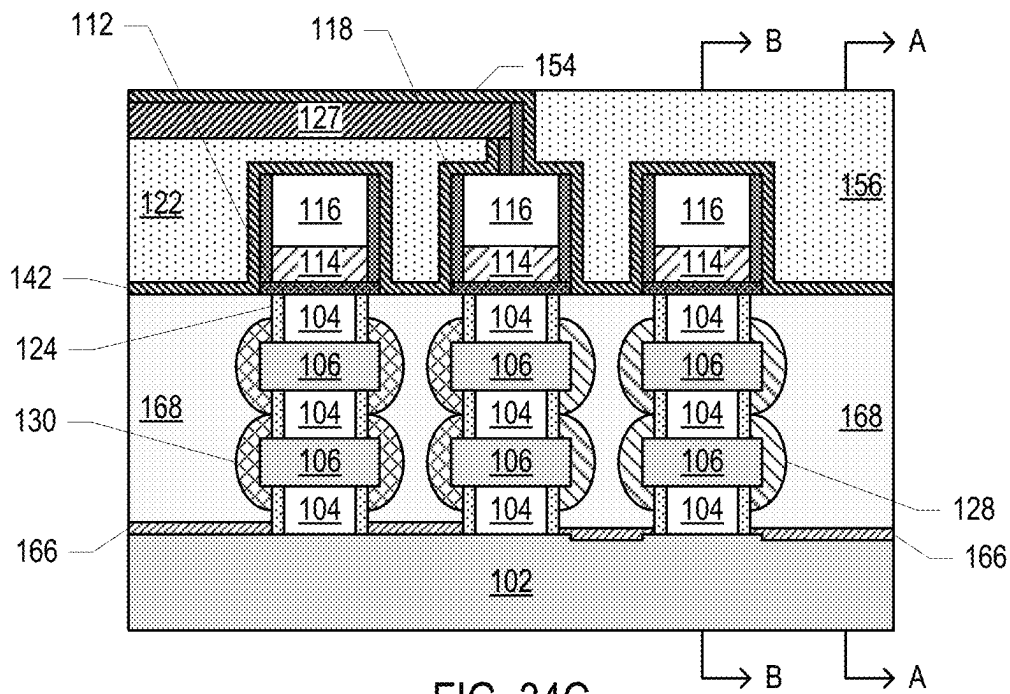












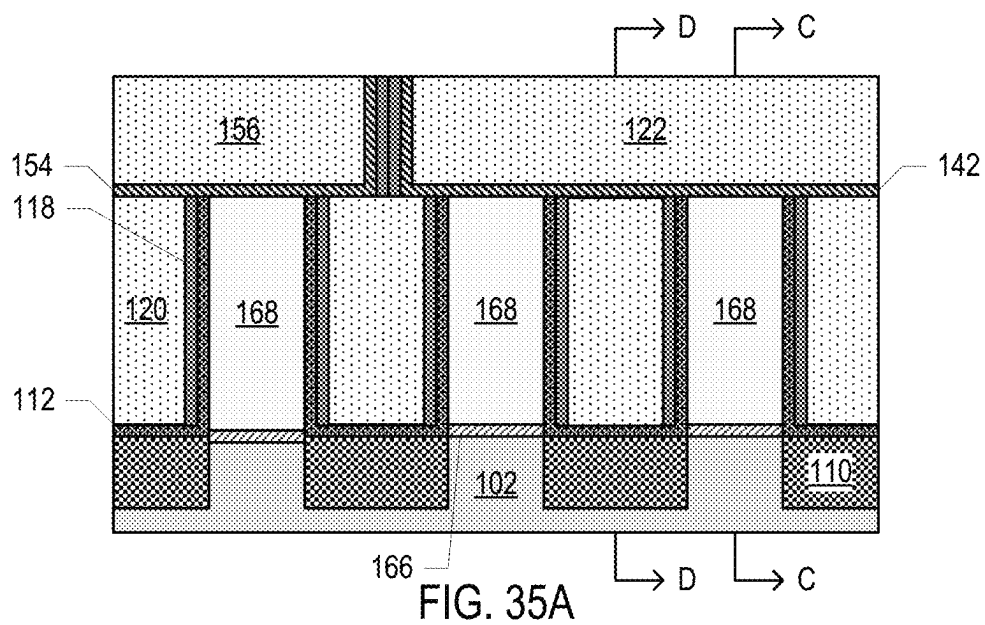


FIG. 35A

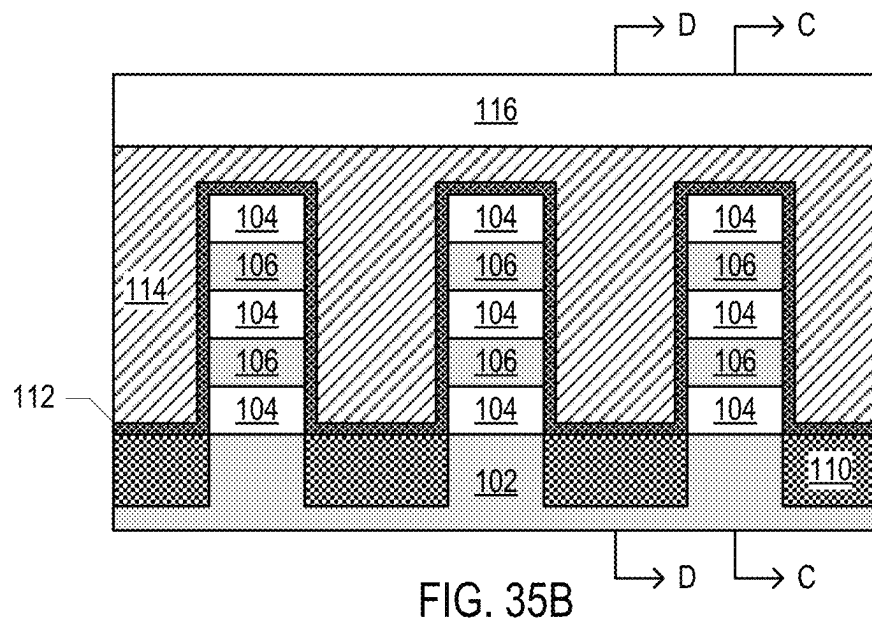
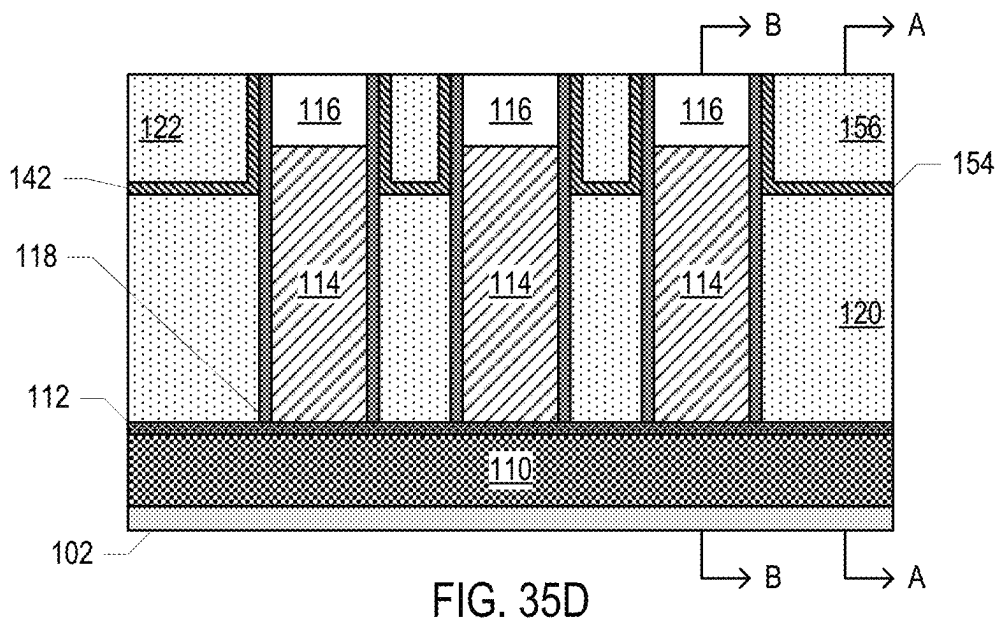
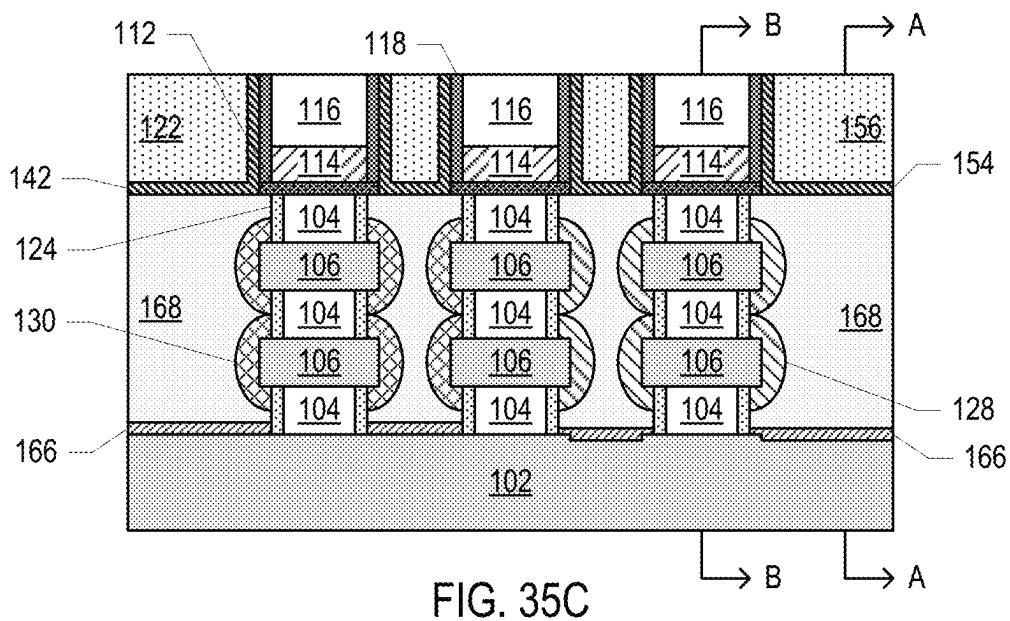
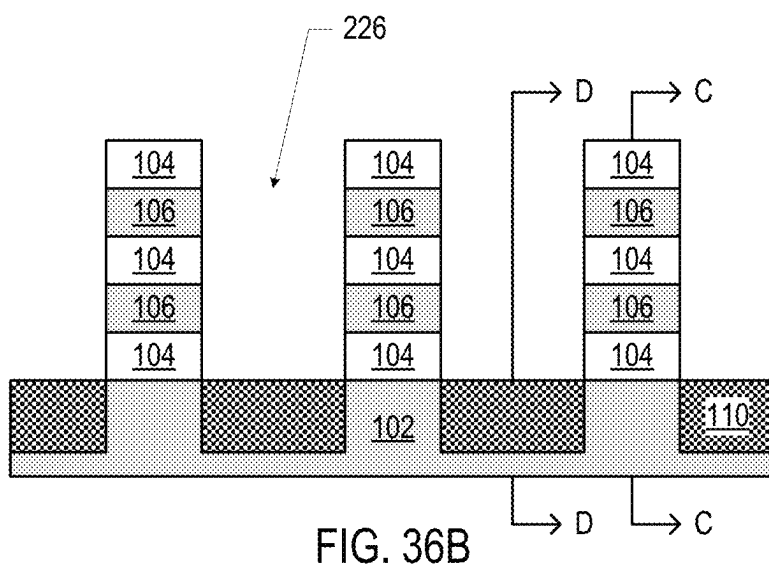
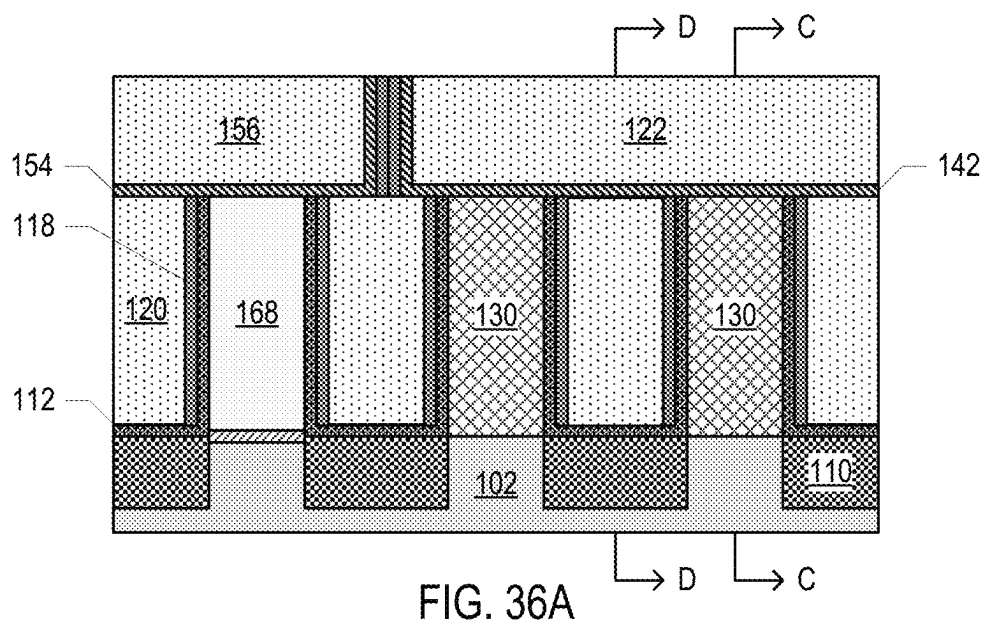
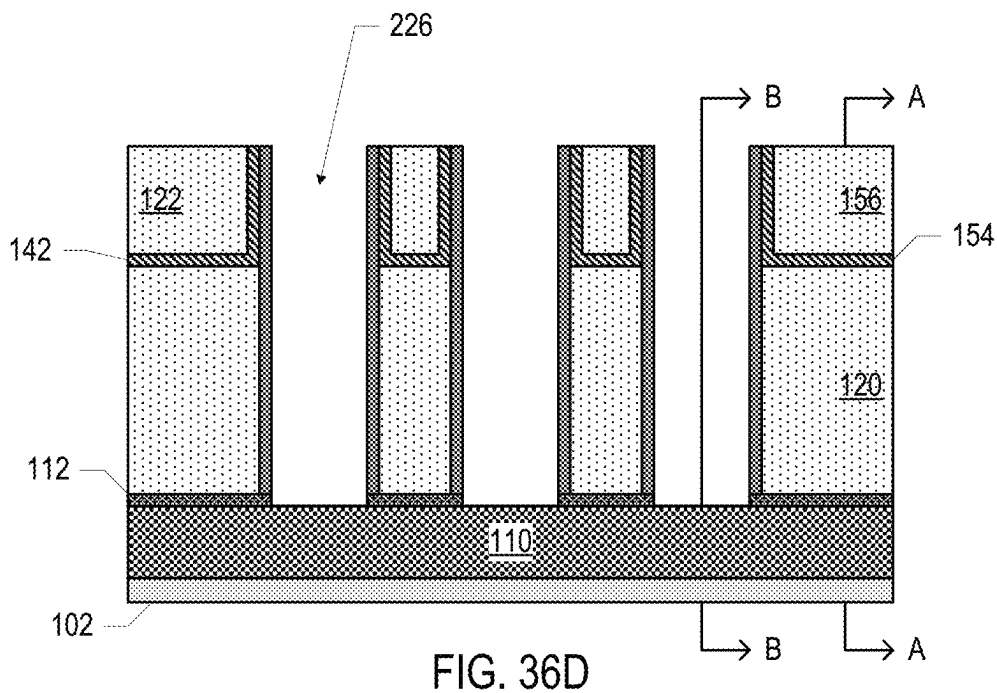
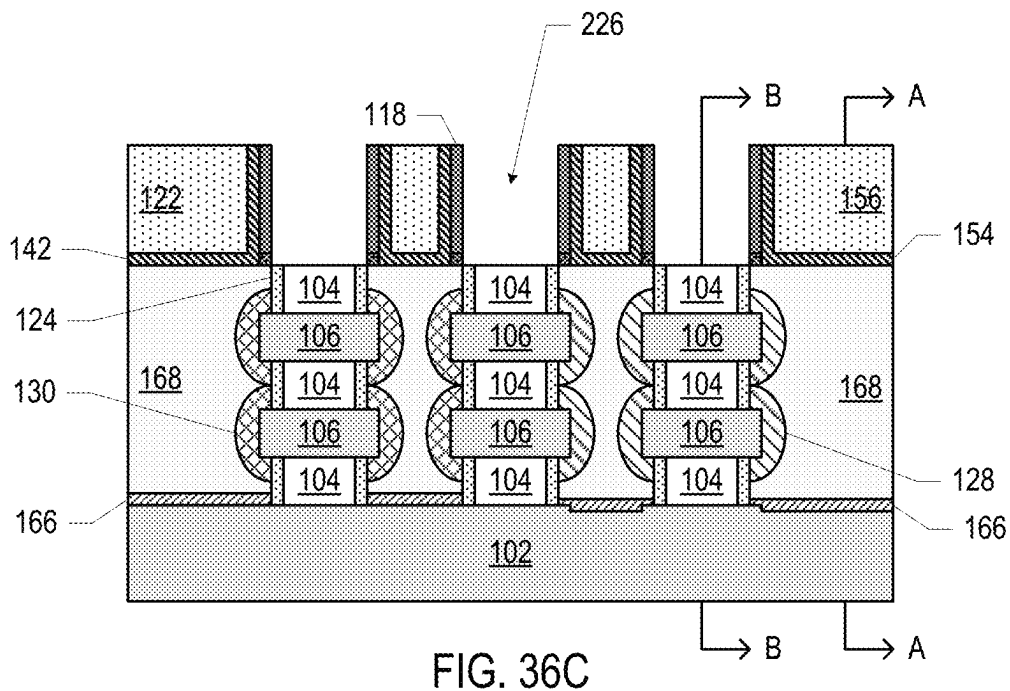
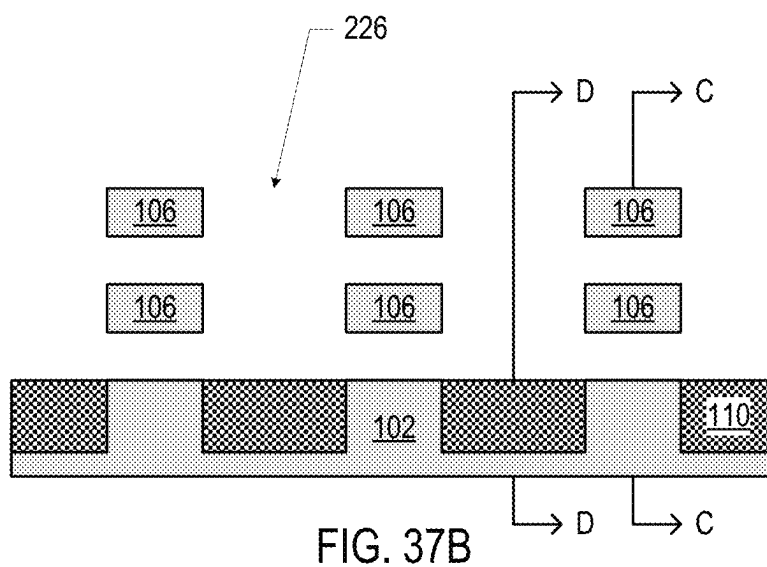
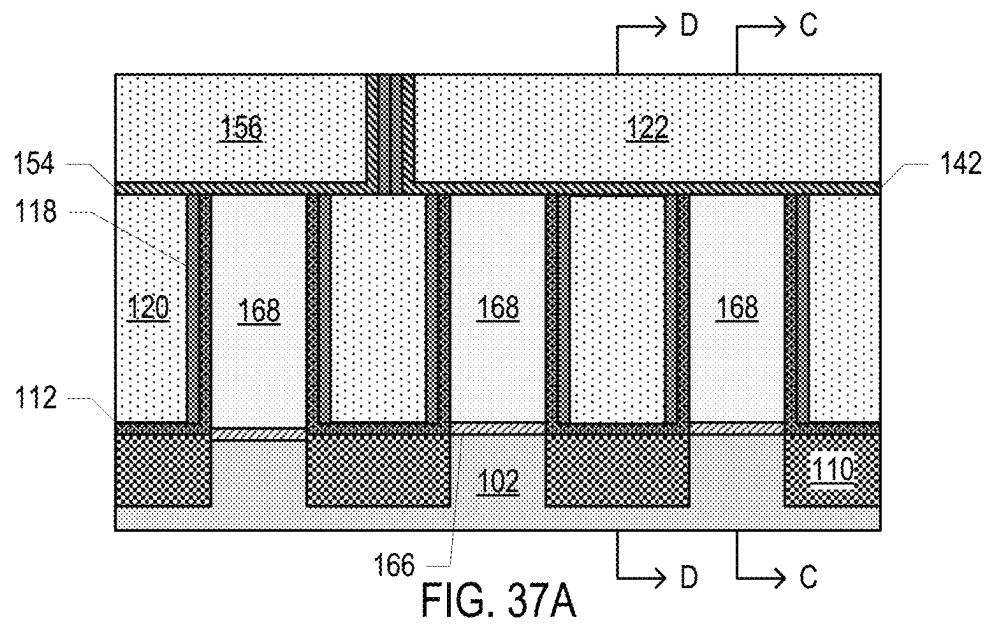


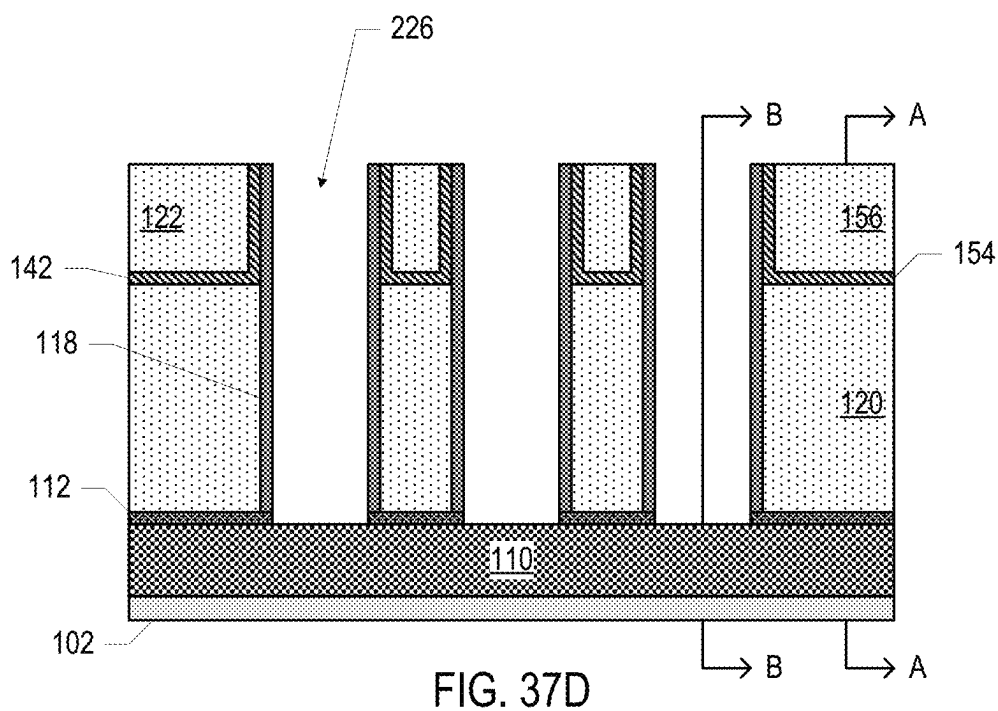
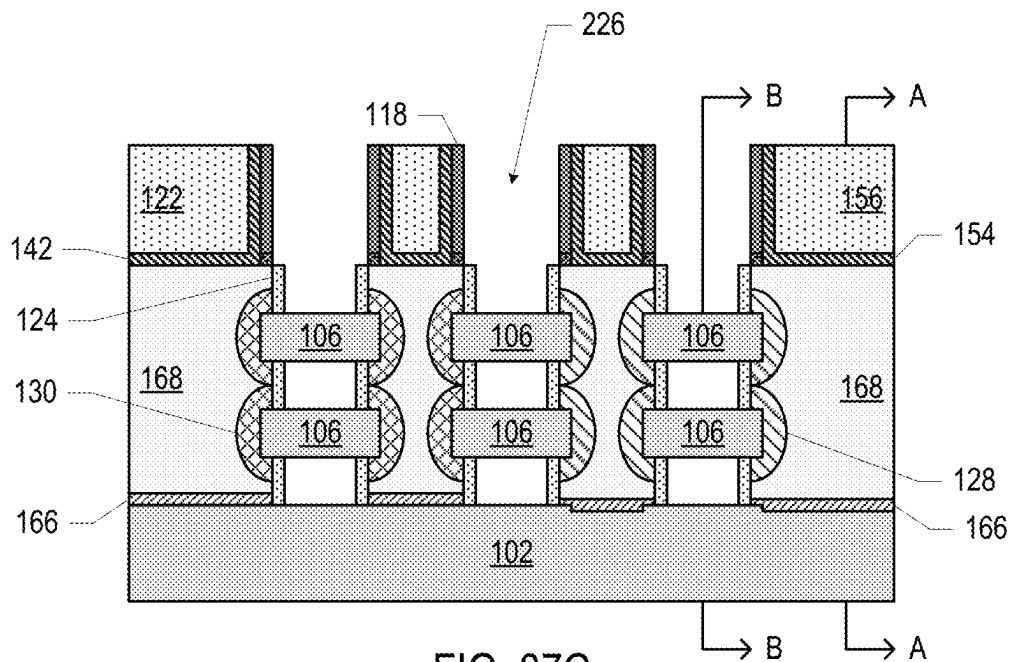
FIG. 35B

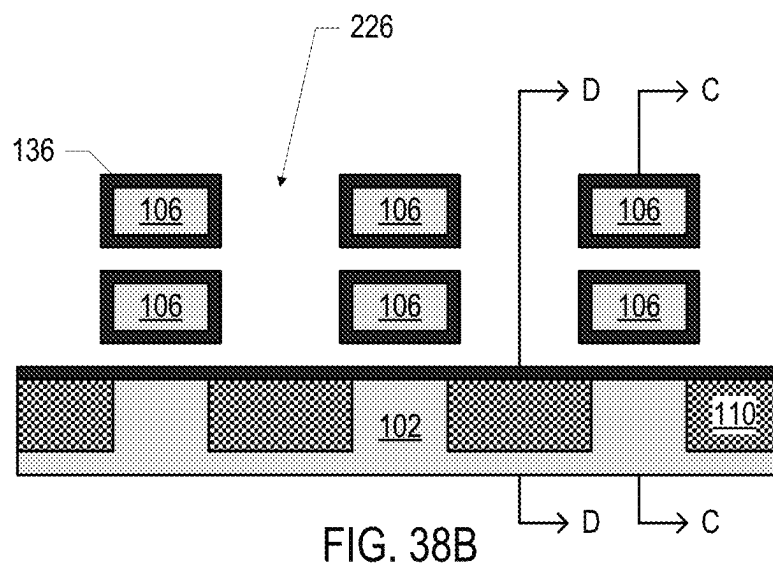
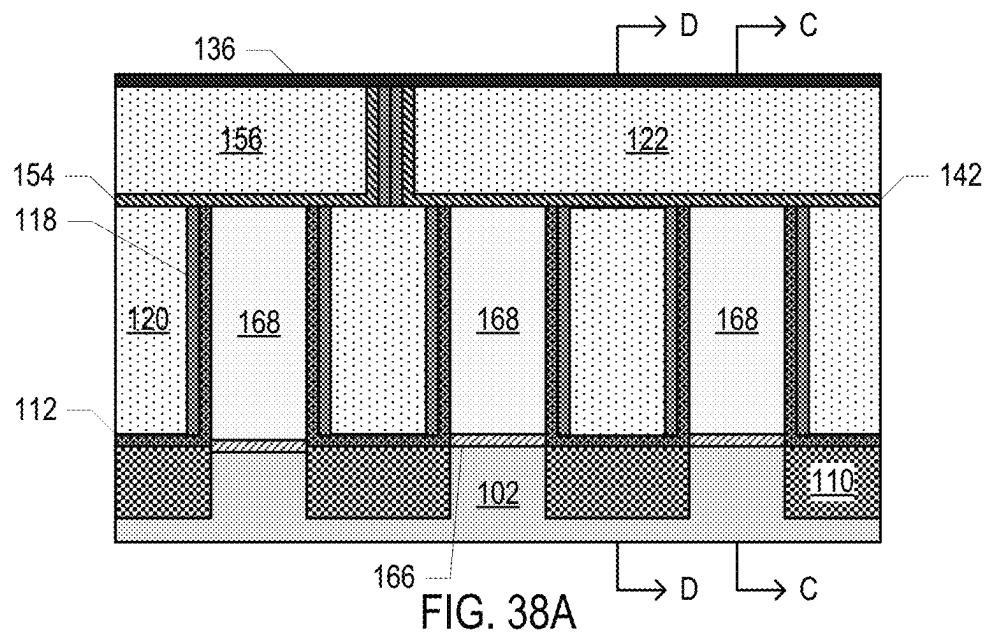


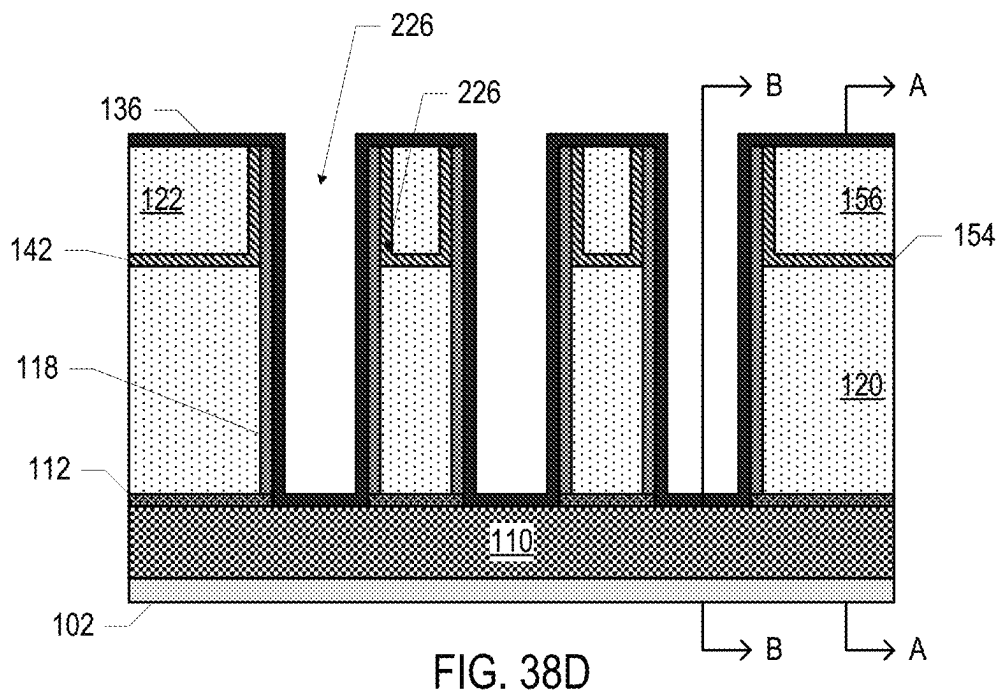
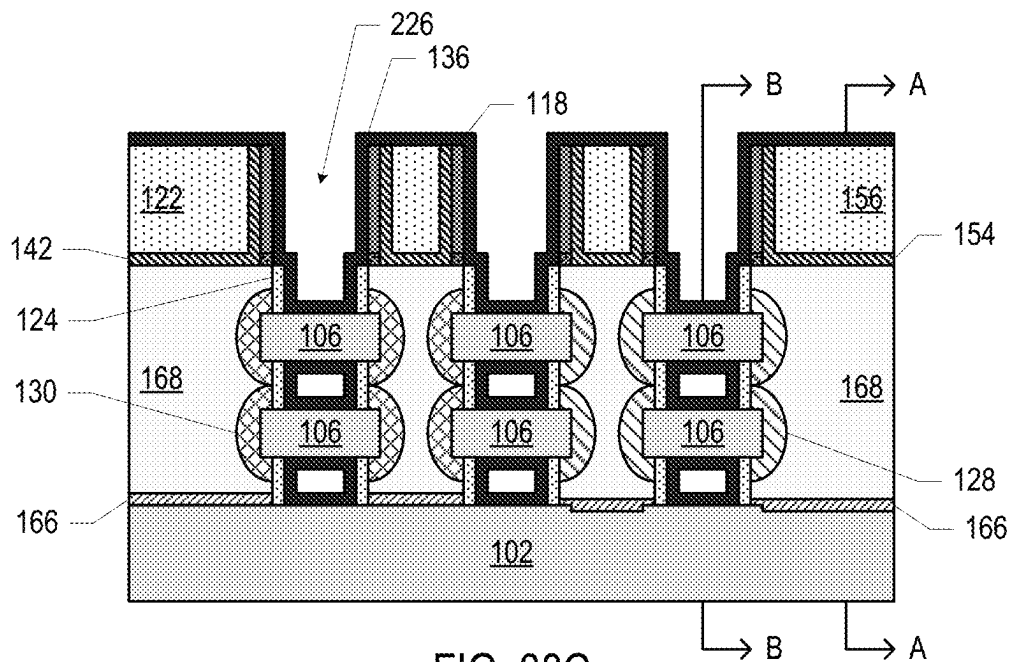


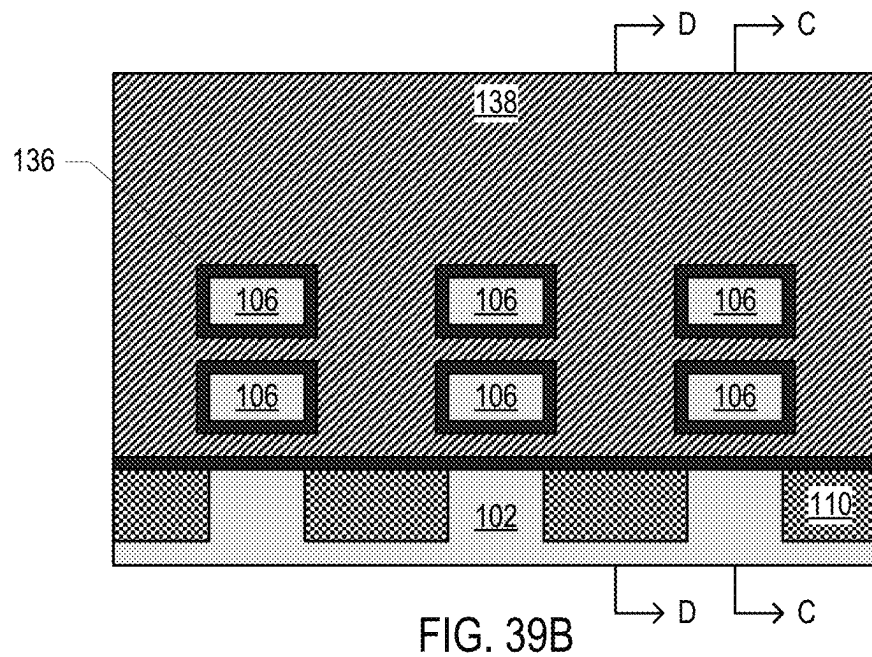
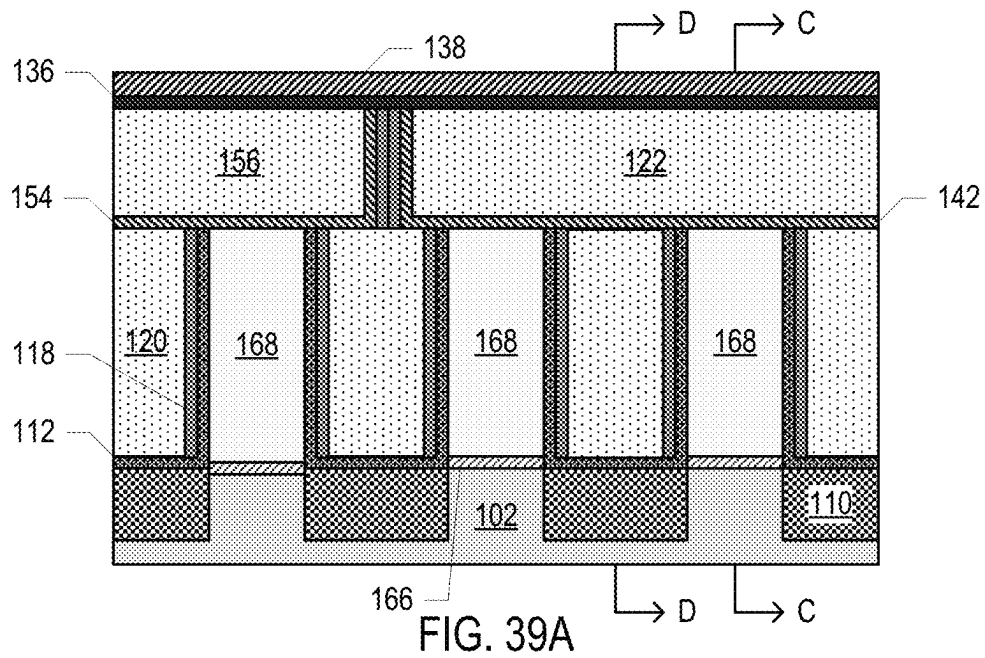


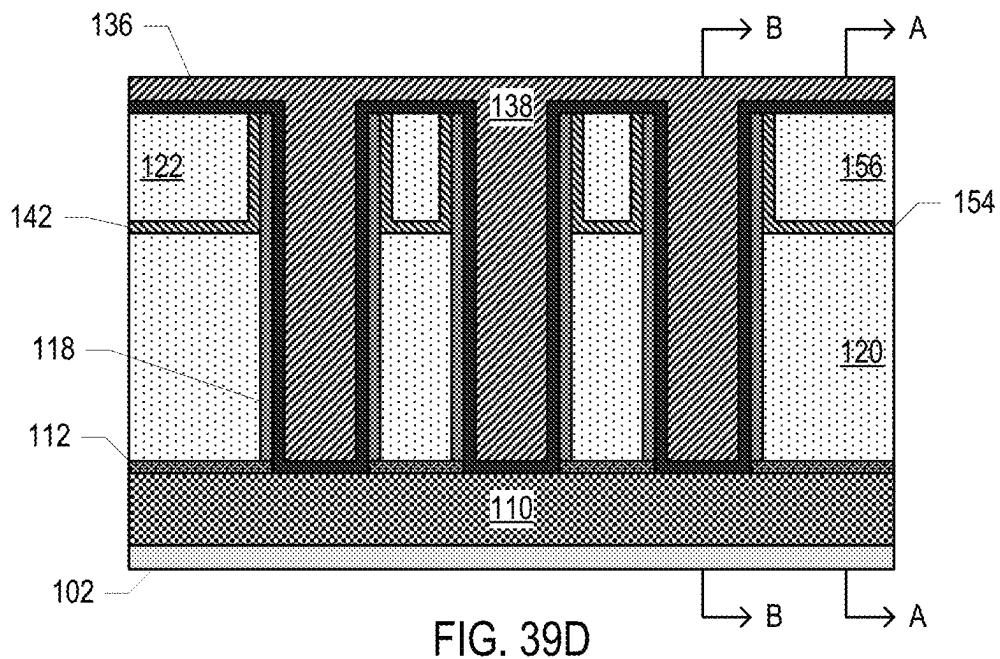
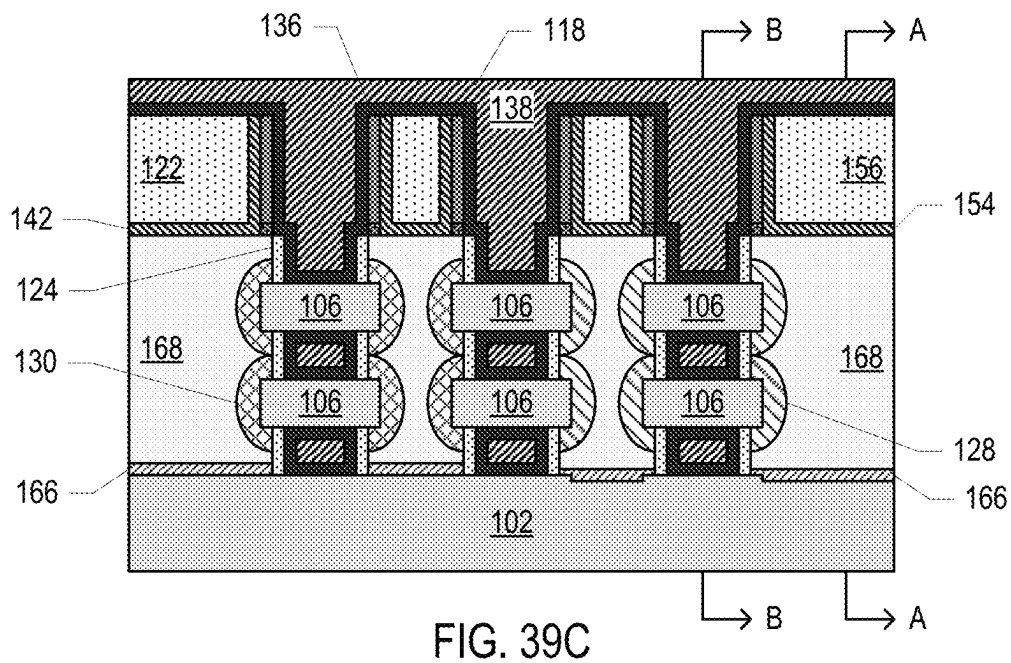


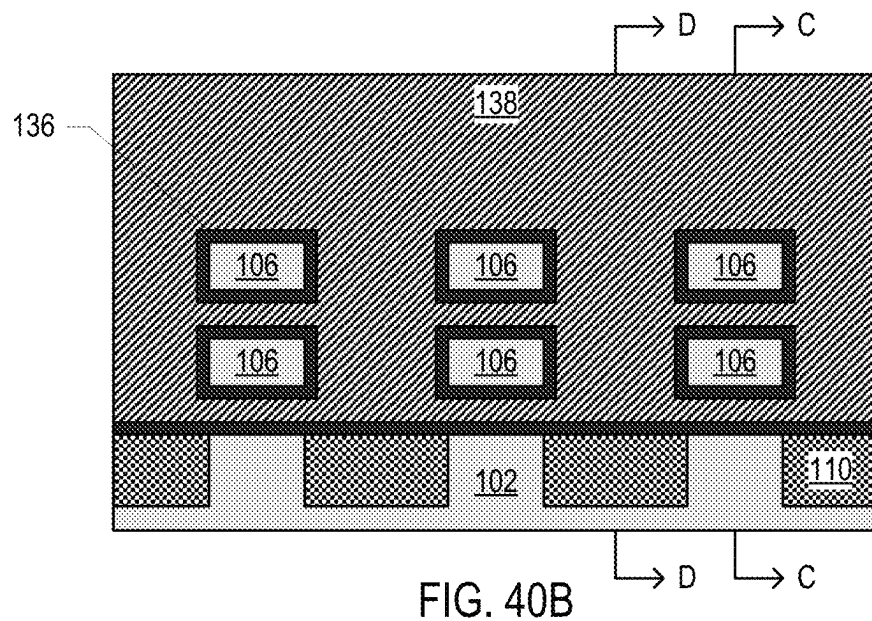
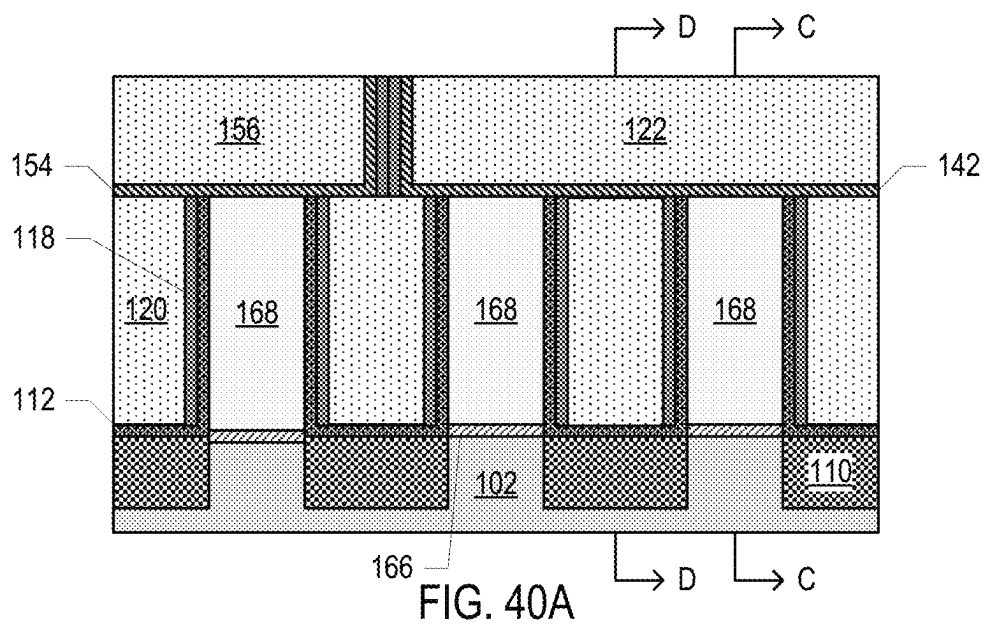


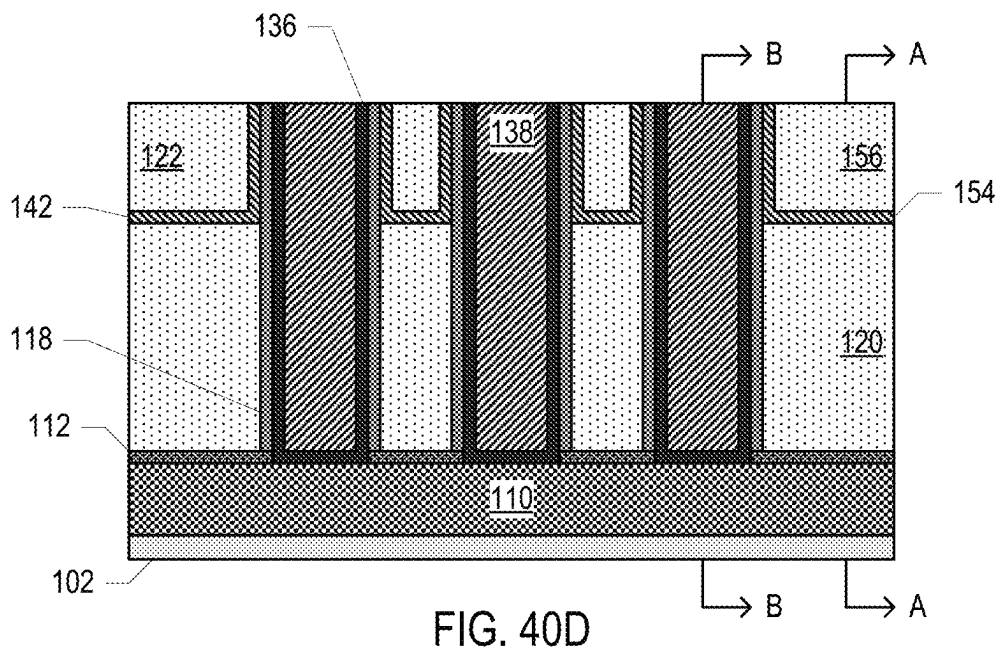
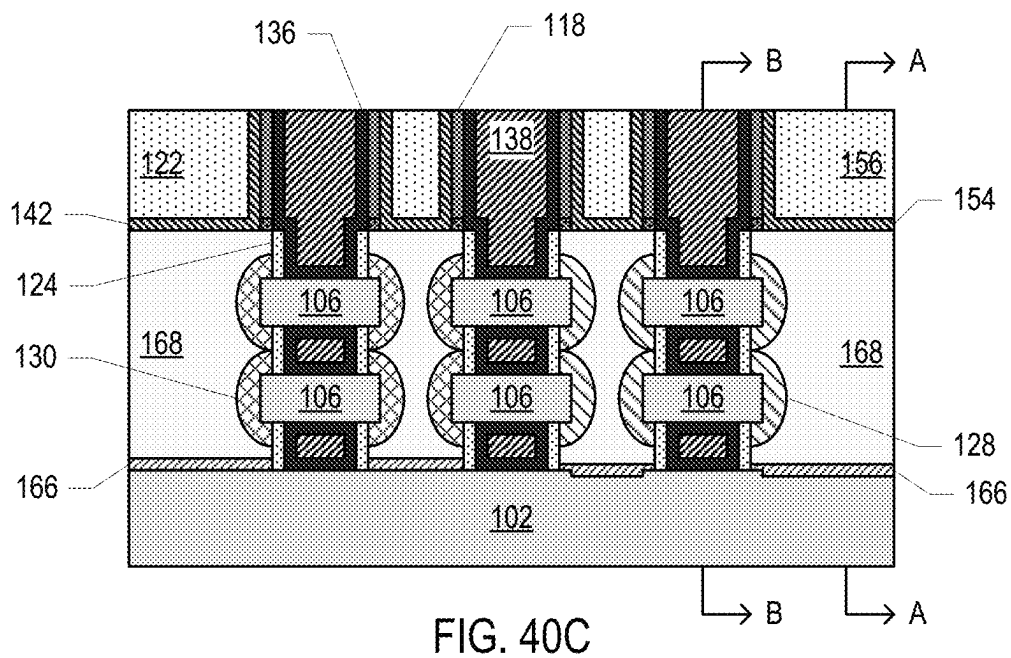


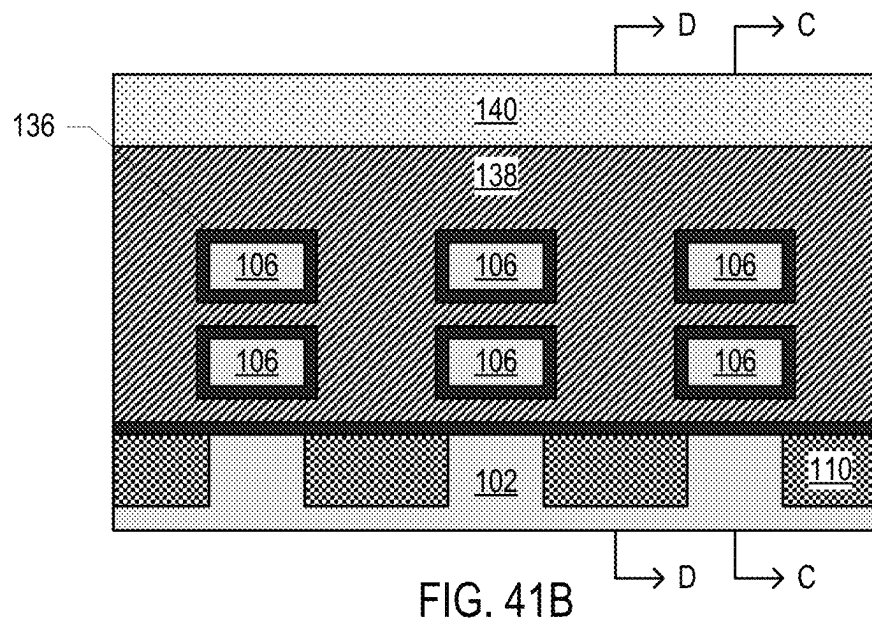
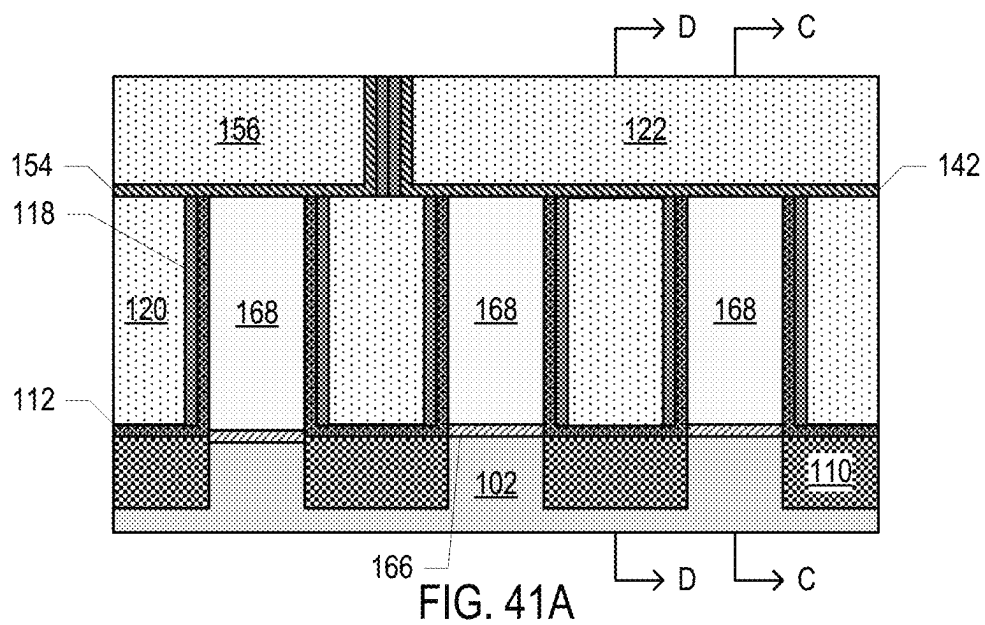


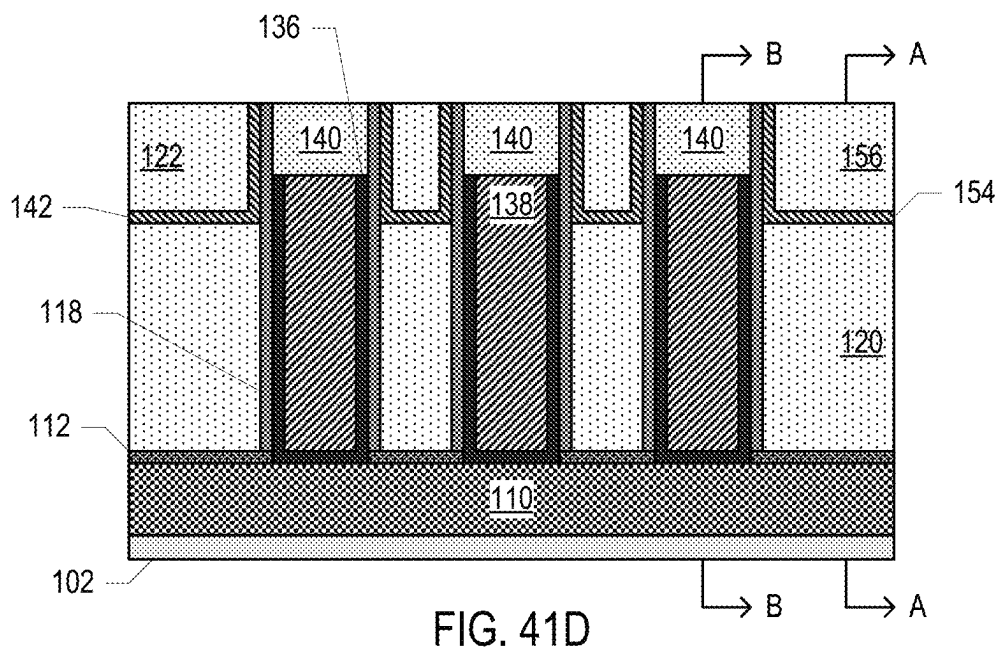
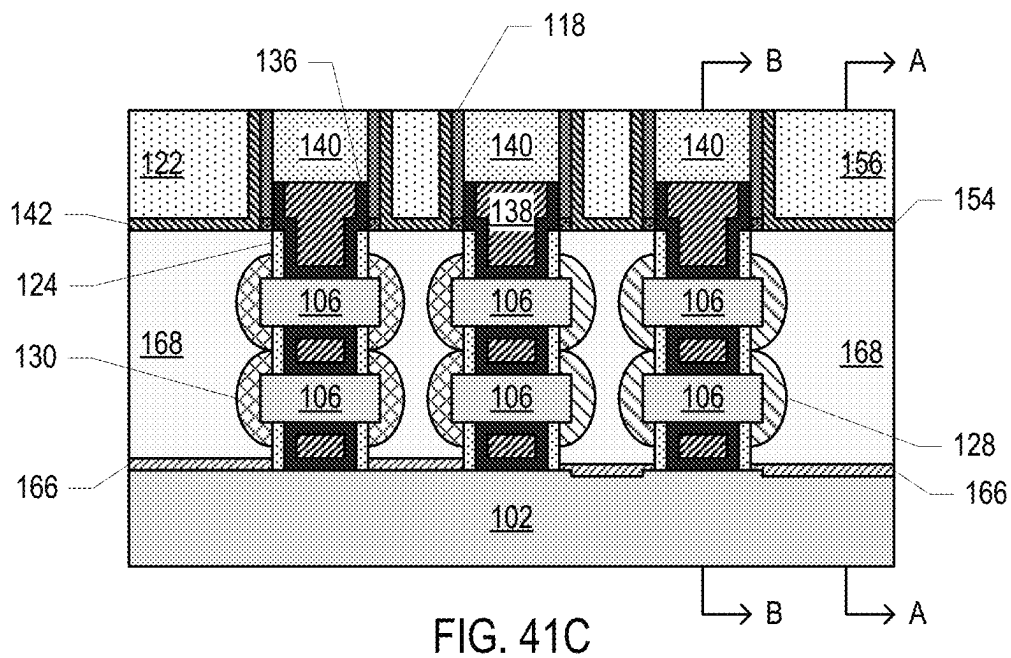


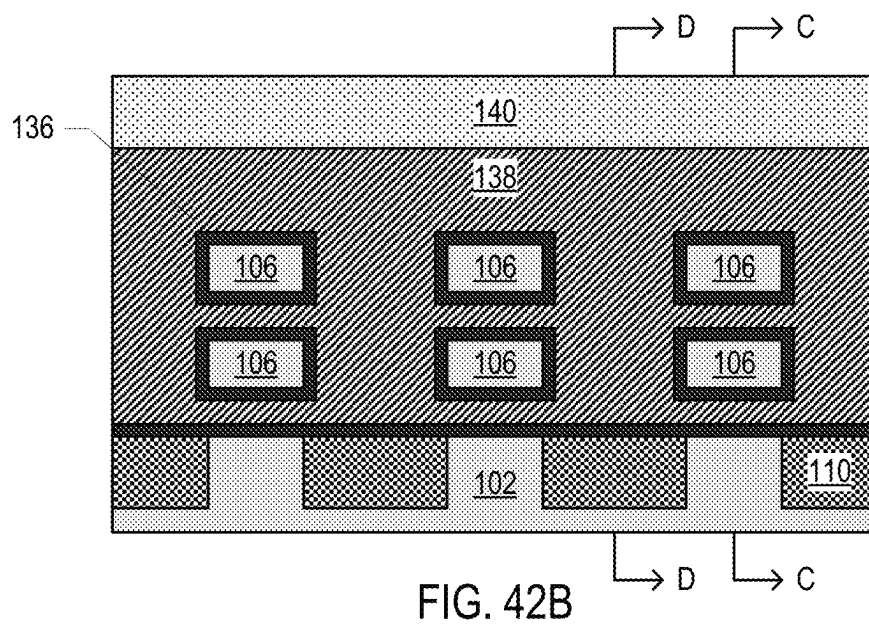
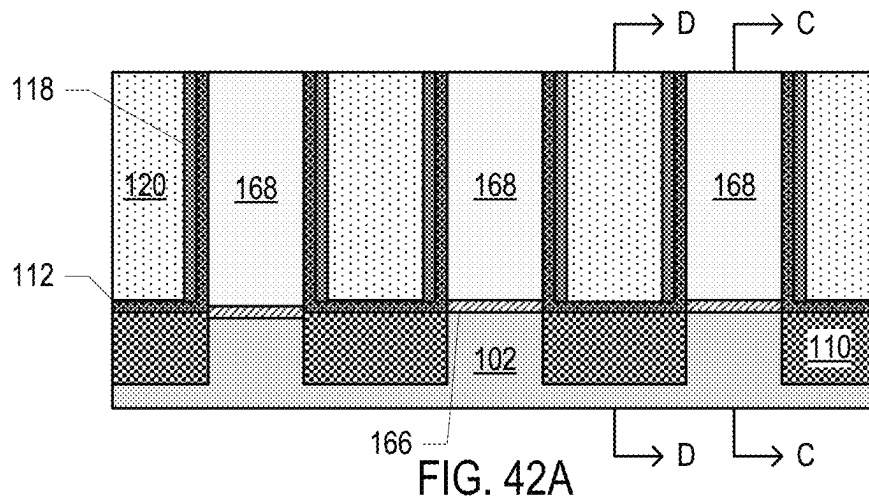


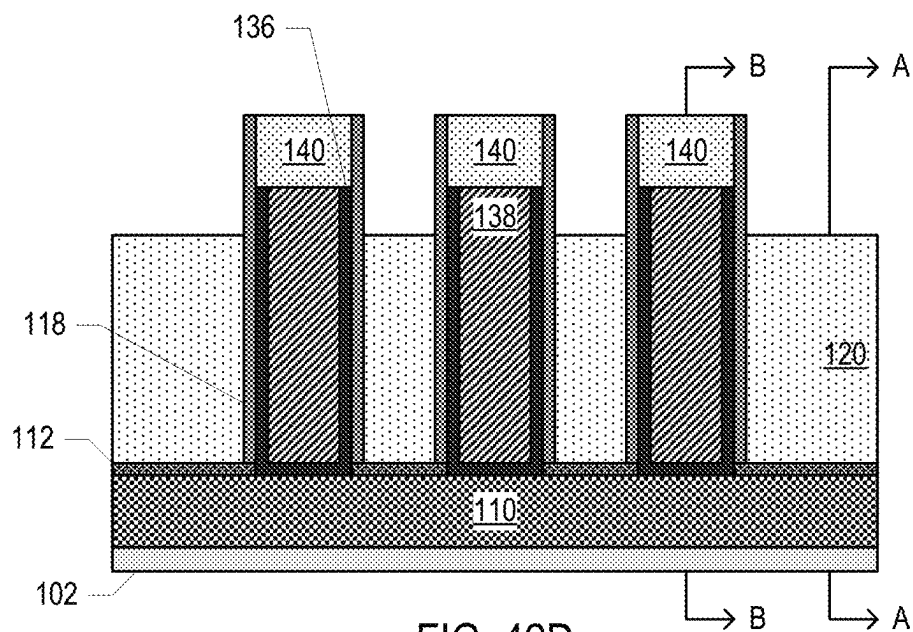
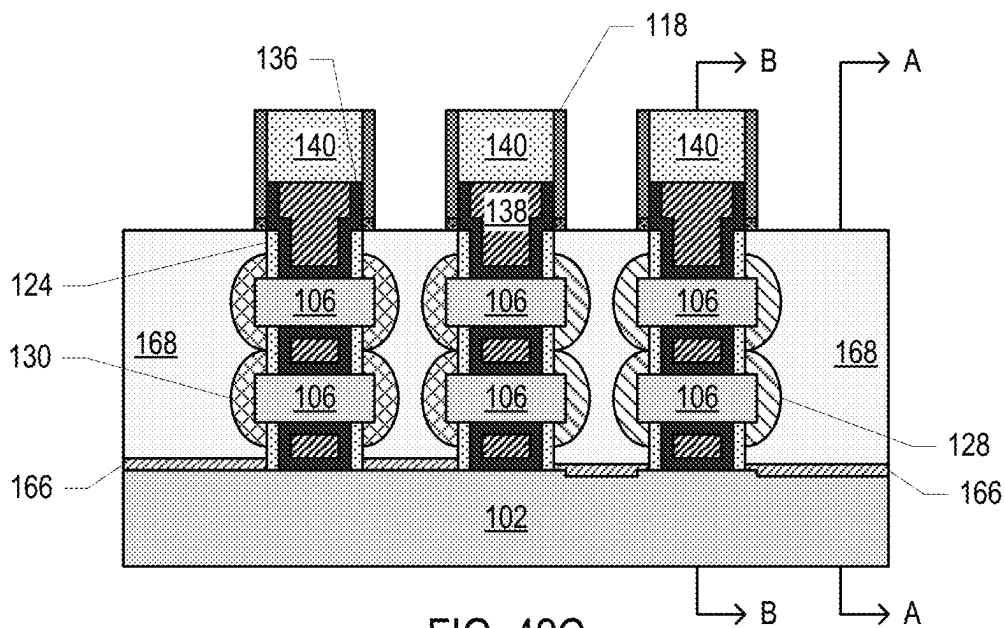


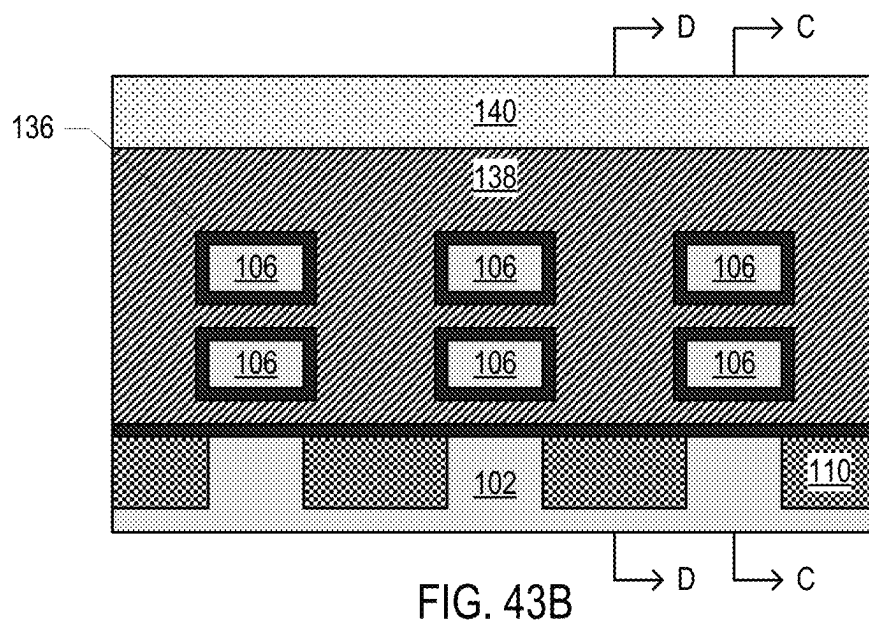
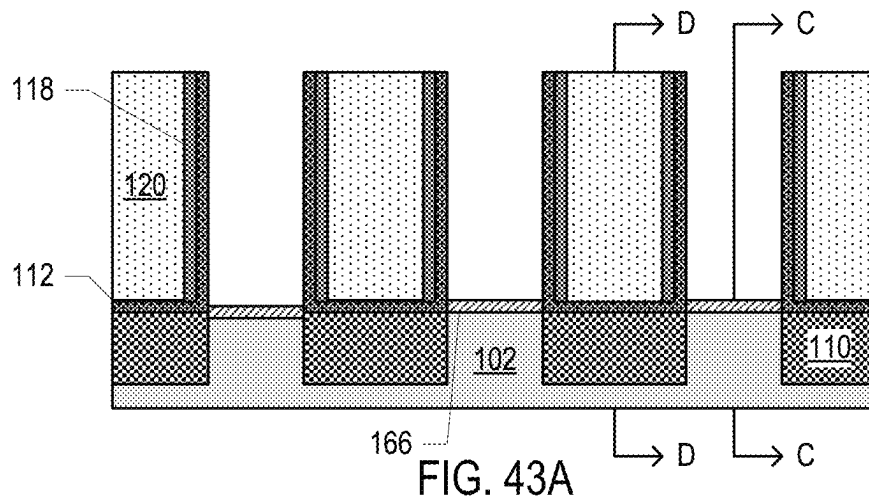


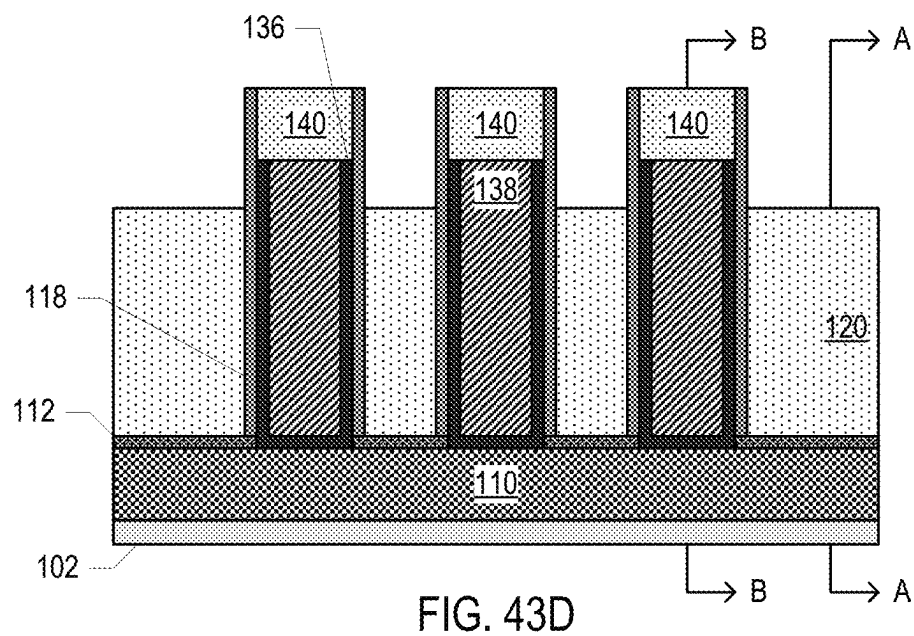
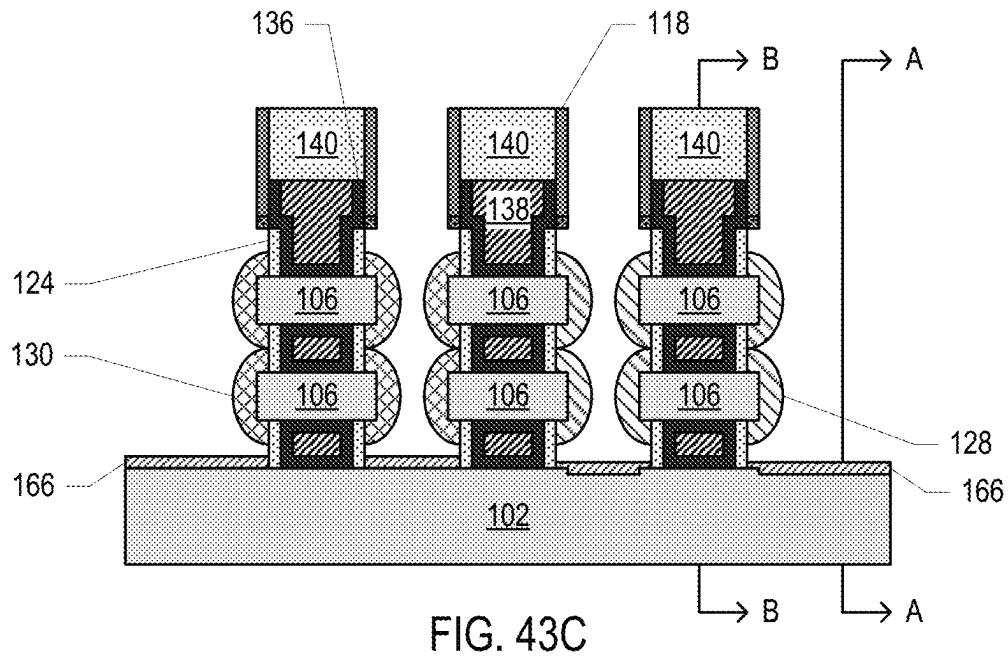


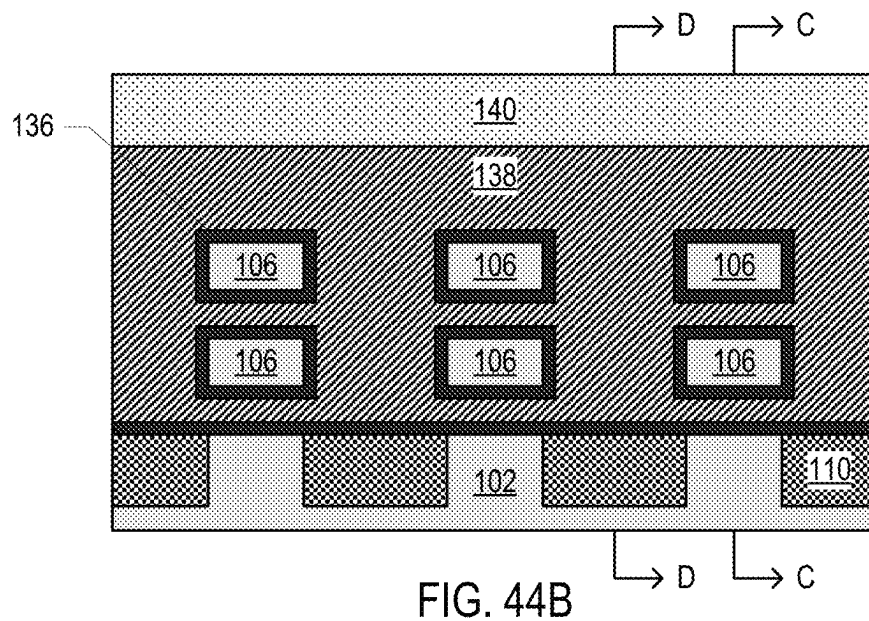
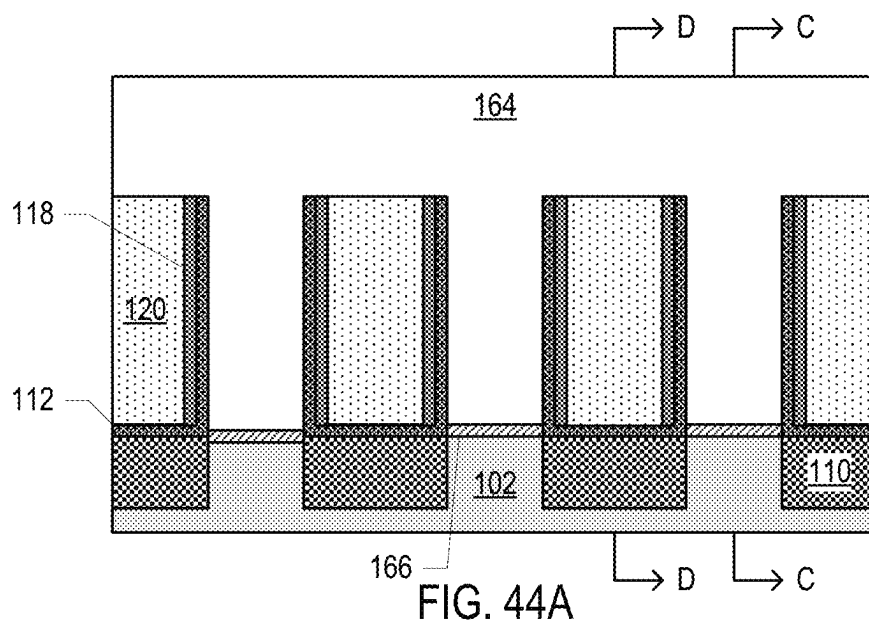












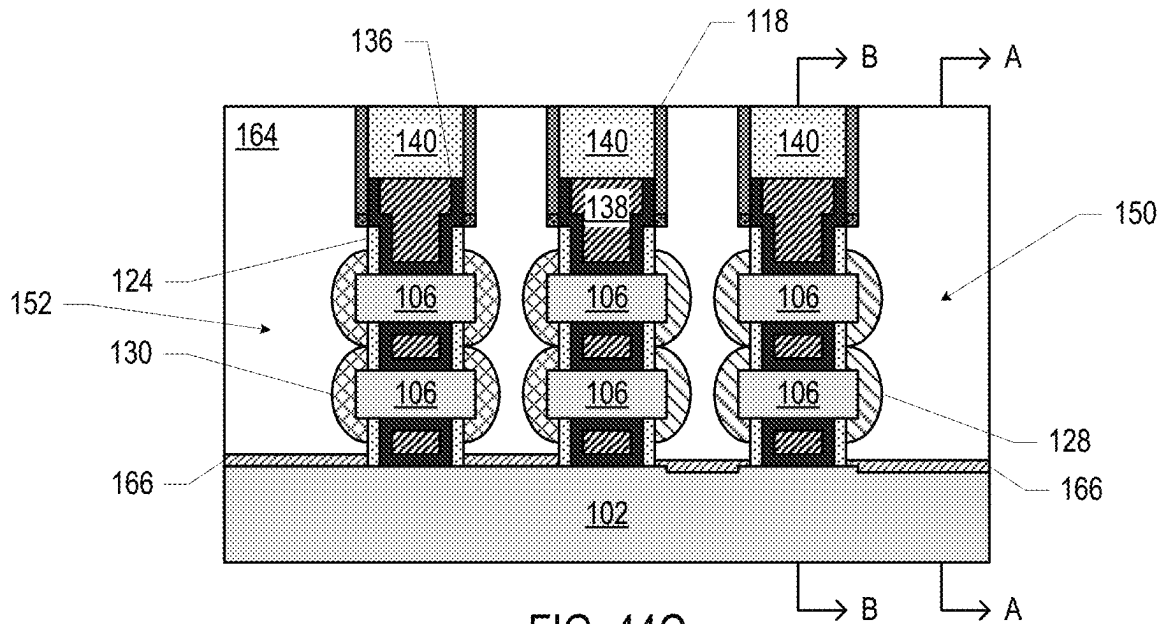


FIG. 44C

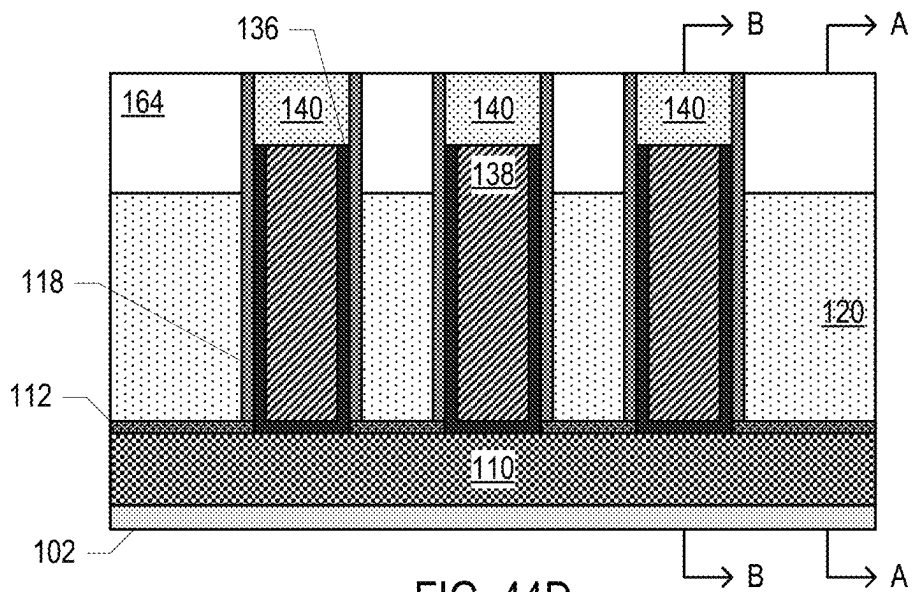
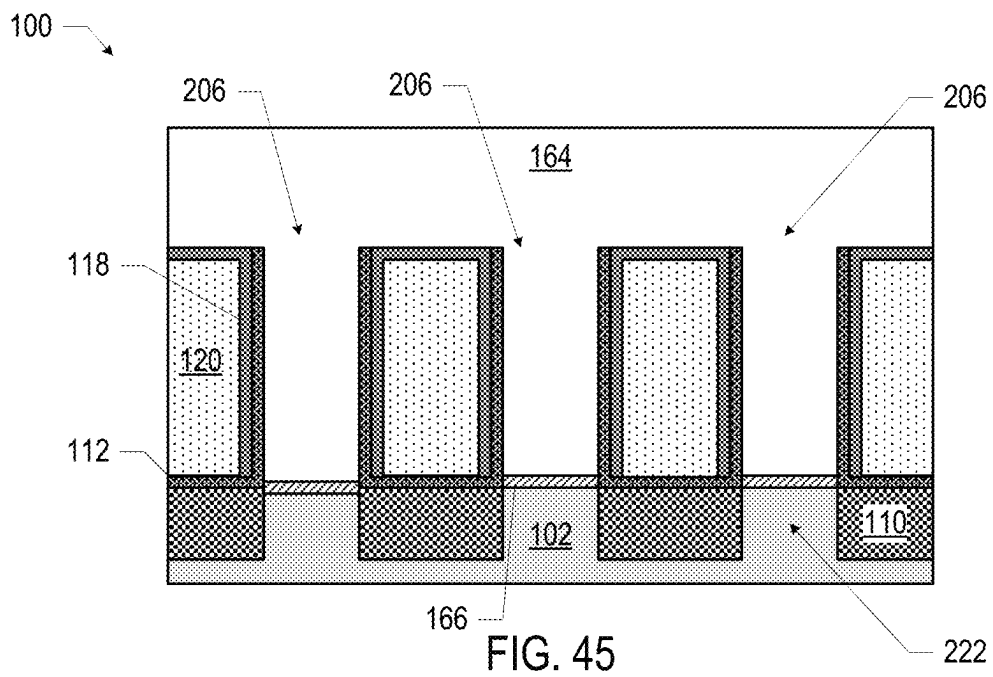
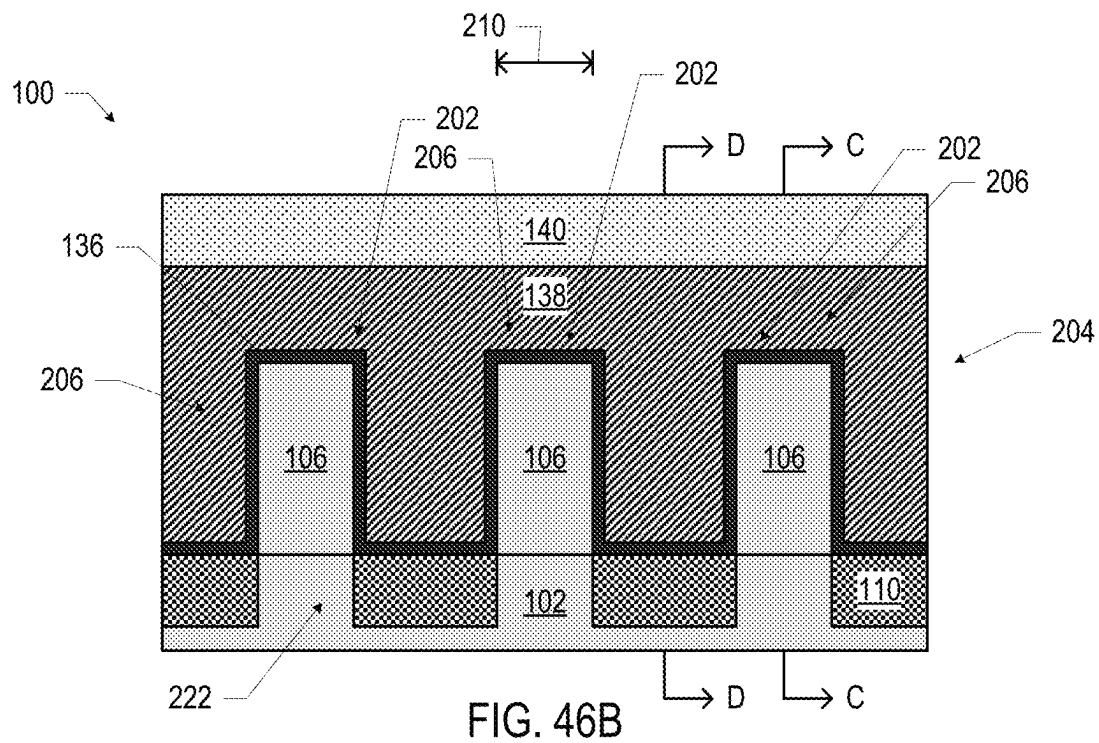
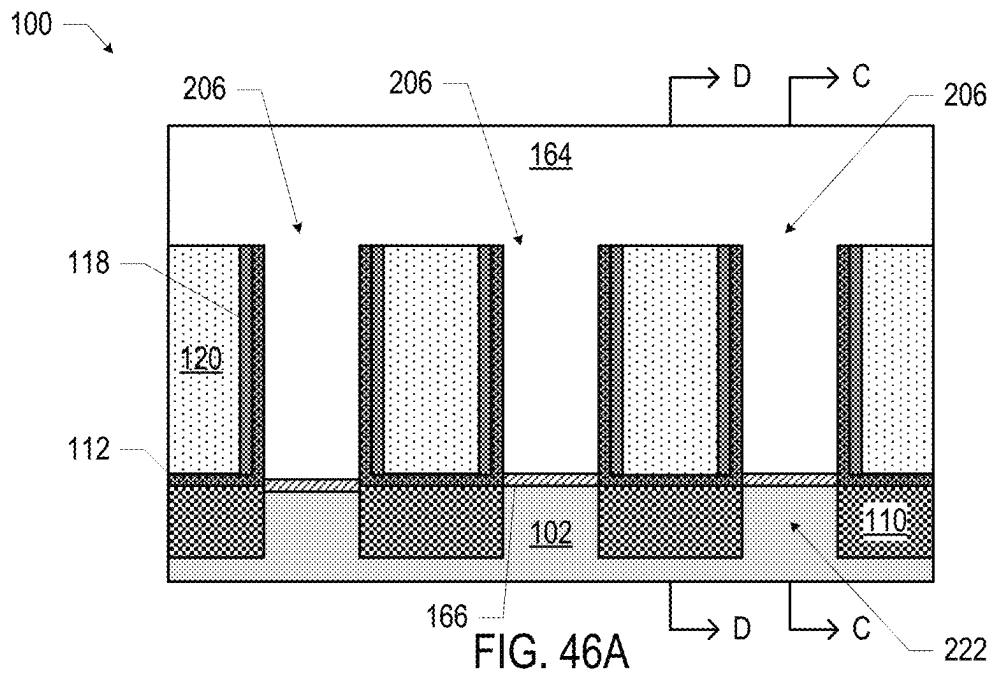
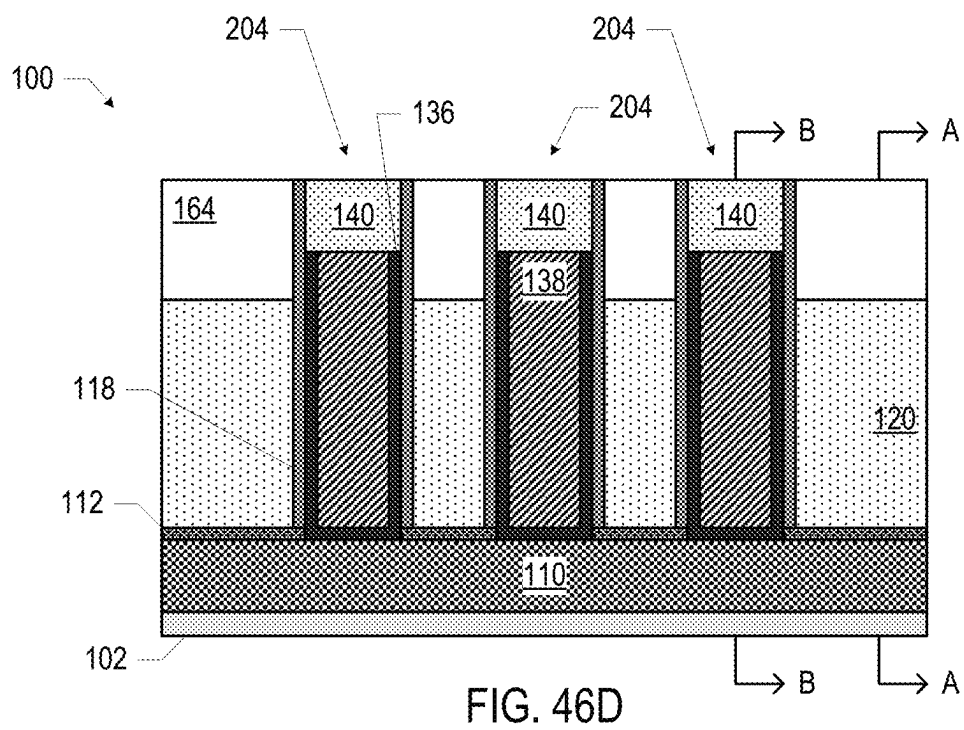
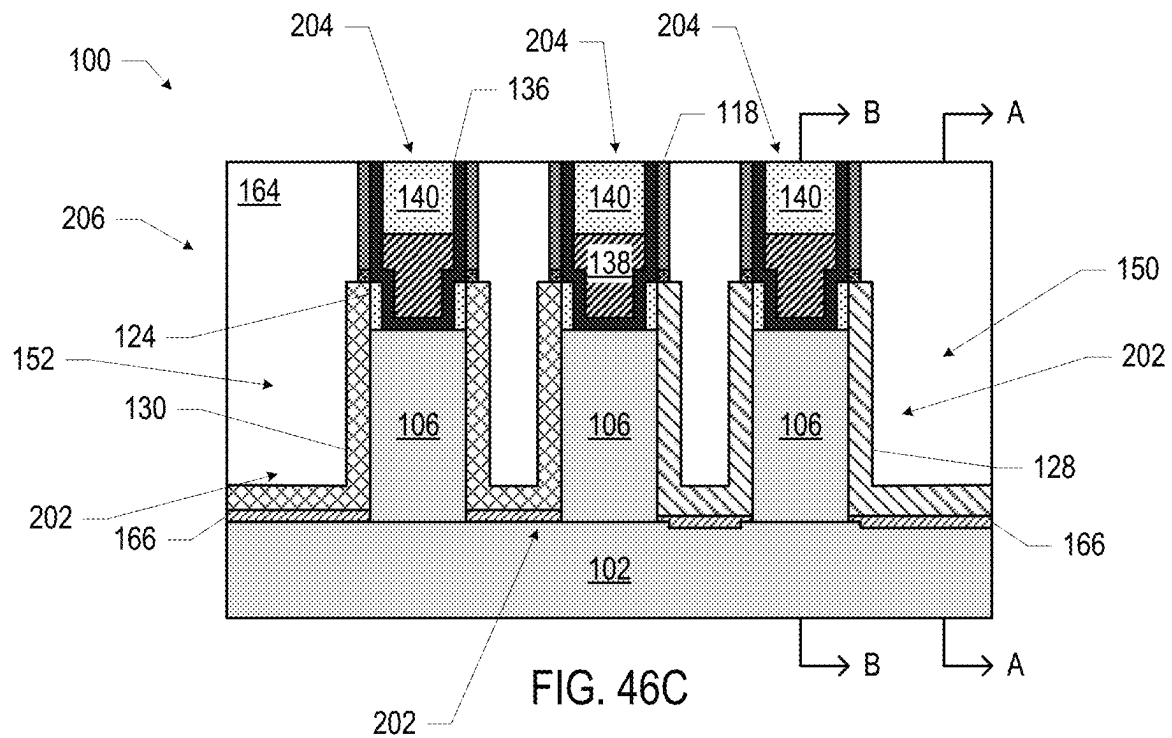


FIG. 44D







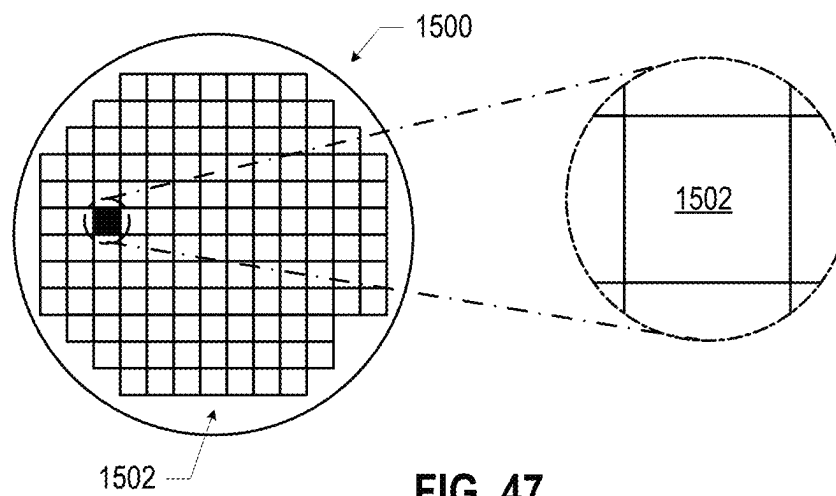


FIG. 47

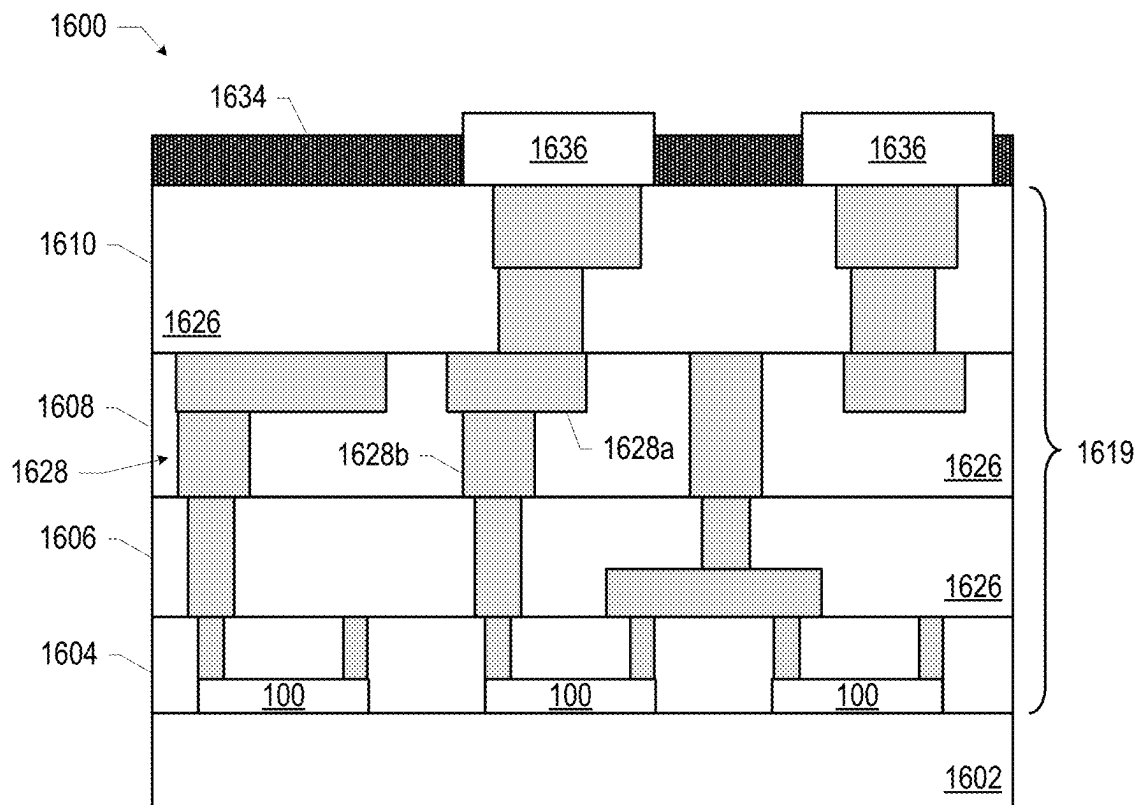


FIG. 48

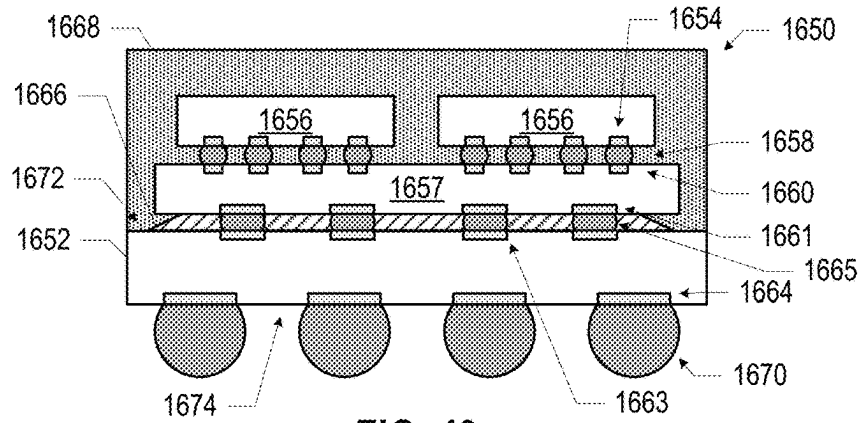


FIG. 49

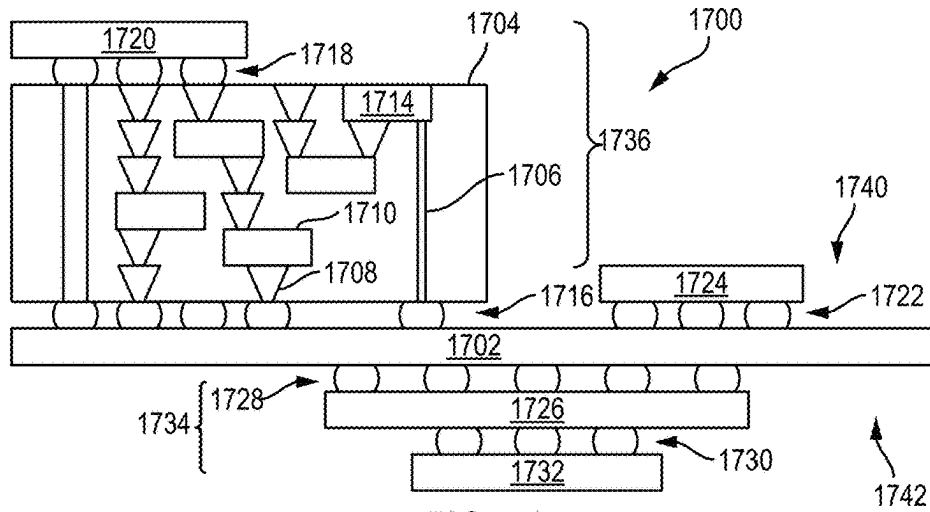


FIG. 50

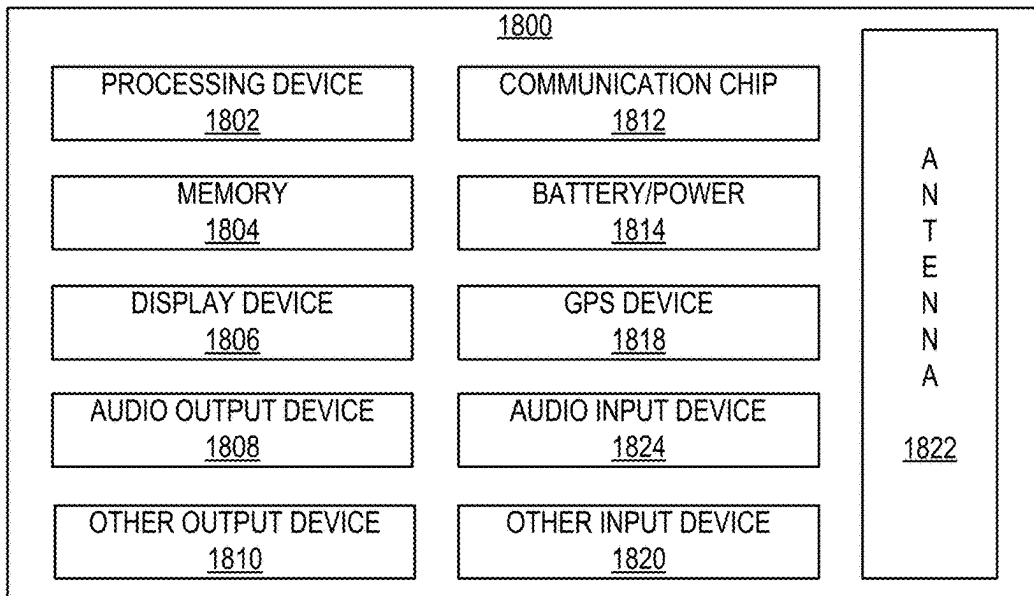


FIG. 51

SOURCE/DRAIN REGIONS IN INTEGRATED CIRCUIT STRUCTURES

BACKGROUND

Electronic components may include active electrical elements, such as transistors. The design of these elements may impact the size, performance, and reliability of the electronic component.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example, not by way of limitation, in the figures of the accompanying drawings.

FIGS. 1A-1G are cross-sectional views of an integrated circuit (IC) structure, in accordance with various embodiments.

FIGS. 2A-2D, 3A-3D, 4A-4D, 5A-5D, 6A-6D, 7A-7D, 8A-8D, 9A-9D, 10A-10D, 11A-11D, 12A-12D, 13A-13D, 14A-14D, 15A-15D, 16A-16D, 17A-17D, 18A-18D, 19A-19D, 20A-20D, 21A-21D, 22A-22D, 23A-23D, 24A-24D, 25A-25D, 26A-26D, 27A-27D, 28A-28D, 29A-29D, 30A-30D, 31A-31D, 32A-32D, 33A-33D, 34A-34D, 35A-35D, 36A-36D, 37A-37D, 38A-38D, 39A-39D, 40A-40D, 41A-41D, 42A-42D, 43A-43D, and 44A-44D are cross-sectional views of stages in an example process of manufacturing the IC structure of FIGS. 1A-1F, in accordance with various embodiments.

FIG. 45 is a cross-sectional view of another example of an IC structure, in accordance with various embodiments.

FIGS. 46A-46D are cross-sectional views of another example of an IC structure, in accordance with various embodiments.

FIG. 47 is a top view of a wafer and dies that may include an IC structure in accordance with any of the embodiments disclosed herein.

FIG. 48 is a side, cross-sectional view of an IC component that may include an IC structure in accordance with any of the embodiments disclosed herein.

FIG. 49 is a side, cross-sectional view of an IC package that may include an IC structure in accordance with any of the embodiments disclosed herein.

FIG. 50 is a side, cross-sectional view of an IC component assembly that may include an IC structure in accordance with any of the embodiments disclosed herein.

FIG. 51 is a block diagram of an example electrical device that may include an IC structure in accordance with any of the embodiments disclosed herein.

DETAILED DESCRIPTION

Disclosed herein are source/drain regions in integrated circuit (IC) structures, as well as related methods and components. For example, in some embodiments, an IC structure may include: a channel region including a semiconductor material; and a source/drain region at a side face of the channel region, wherein the source/drain region includes a semiconductor portion and a contact metal, and the semiconductor portion is between the contact metal and the semiconductor material.

Gate-all-around (GAA) transistors and fork-sheet (FS) transistors may include a vertically oriented stack of lateral semiconductor channels (e.g., semiconductor wires, such as

semiconductor ribbons) wrapped or partially wrapped by gate material. During operation, current may flow through these semiconductor channels, modulated by electrical signals applied to the gate and proximate source/drain (S/D) regions. However, as the number of semiconductor wires increases (e.g., to boost the drive current), the depth of the vertical stack increases and the resistance of the electrical connection to the semiconductor wires may become a performance and manufacturing bottleneck. For example, some fabrication techniques may require a deep etch into the epitaxially formed S/D regions to form S/D contacts; such techniques may need to be extremely anisotropic so that the epitaxial S/D regions are etched only vertically, without undesirable lateral etch, and may also need to avoid damaging any local gate spacers. Under real world manufacturing conditions, the failure of an etch to meet criteria like these may compromise device performance or may render such devices impractical to manufacture.

Disclosed herein are novel IC structures that may provide improved S/D contacts relative to previous approaches. The S/D contact structures disclosed herein may exhibit reduced external resistance without adding significant manufacturing complexity (thus speeding adoption and reducing cost), and may allow for improved process control relative to conventional techniques.

In the following detailed description, reference is made to the accompanying drawings that form a part hereof wherein like numerals designate like parts throughout, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made, without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense.

Various operations may be described as multiple discrete actions or operations in turn, in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed, and/or described operations may be omitted in additional embodiments.

For the purposes of the present disclosure, the phrase “A and/or B” means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The phrase “A or B” means (A), (B), or (A and B). The drawings are not necessarily to scale. Although many of the drawings illustrate rectilinear structures with flat walls and right-angle corners, this is simply for ease of illustration, and actual devices made using these techniques will exhibit rounded corners, surface roughness, and other features.

The description uses the phrases “in an embodiment” or “in embodiments,” which may each refer to one or more of the same or different embodiments. Furthermore, the terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. When used to describe a range of dimensions, the phrase “between X and Y” represents a range that includes X and Y. As used herein, the term “insulating” means “electrically insulating” unless otherwise specified. For convenience, the phrase “FIG. 1” may be used to refer to the collection of drawings of FIGS. 1A-1G, the phrase “FIG. 2” may be used to refer to the collection of drawings of FIGS. 2A-2D, etc.

FIG. 1 provides cross-sectional views of an IC structure 100, in accordance with various embodiments. In particular, FIG. 1A is a cross-sectional view taken through the section A-A of FIGS. 10 and 1D (perpendicular to the longitudinal axis of a channel region 202, and across the source/drain (S/D) regions 150/152 of different channel regions 202), FIG. 1B is a cross-sectional view taken through the section B-B of FIGS. 10 and 1D (perpendicular to the longitudinal axis of a channel region 202, and across a gate 204 spanning multiple channel regions 202), FIG. 1C is a cross-sectional view taken through the section C-C of FIGS. 1A and 1B (along the longitudinal axis of a channel region 202), and FIG. 1D is a cross-sectional view taken through the section D-D of FIGS. 1A and 1B (between adjacent channel regions 202, parallel to the longitudinal axis of the channel regions 202). FIGS. 1E and 1F share the perspective of FIG. 10, and illustrate alternate arrangements of the semiconductor regions 128/130. The “A,” “B,” “C,” and “D” sub-figures of FIGS. 2-44 share the same perspectives as those of the sub-figures “A,” “B,” “C,” and “D” of FIG. 1, respectively. Although various ones of the accompanying drawings depict a particular number of device regions 206 (e.g., three), channel regions 202 (e.g., three) in a device region 206, and a particular arrangement of channel materials 106 (e.g., two wires) in a channel region 202, this is simply for ease of illustration, and an IC structure 100 may include more or fewer device regions 206 and/or channel regions 202, and/or other arrangements of channel materials 106.

A device region 206 may be oriented vertically relative to an underlying base 102, with multiple device regions 206 arrayed along the base 102. The base 102 may be a semiconductor substrate composed of semiconductor material systems including, for example, n-type or p-type materials systems (or a combination of both). The base 102 may include, for example, a crystalline substrate formed using a bulk silicon. The base 102 may include a layer of silicon dioxide on a bulk silicon or gallium arsenide substrate. The base 102 may include a converted layer (e.g., a silicon layer that has been converted to silicon dioxide during an oxygen-based annealing process). In some embodiments, the base 102 may be formed using alternative materials, which may or may not be combined with silicon, that include but are not limited to germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide. Further materials classified as group II-VI, III-V, or IV may also be used to form the base 102. Although a few examples of materials from which the base 102 may be formed are described here, any material or structure that may serve as a foundation for an IC structure 100 may be used. The base 102 may be part of a singulated die (e.g., the dies 1502 of FIG. 47) or a wafer (e.g., the wafer 1500 of FIG. 47). In some embodiments, the base 102 may itself include an interconnect layer, an insulation layer, a passivation layer, an etch stop layer, additional device layers, etc. As shown in FIG. 1, the base 102 may include pedestals 222, around which a dielectric material 110 may be disposed; the dielectric material 110 may include any suitable material, such as a shallow trench isolation (STI) material (e.g., an oxide material, such as silicon oxide).

The IC structure 100 may include one or more device regions 206 having channel material 106 with a longitudinal axis (into the page from the perspective of FIGS. 1A and 1B, and left-right from the perspective of FIGS. 1C and 1D). The channel material 106 of a device region 206 may be arranged in any of a number of ways. For example, FIG. 1 illustrates the channel material 106 of the device regions 206 as including multiple semiconductor wires (e.g., nanowires or

nanoribbons in GAA, forksheet, double-gate, or pseudo double-gate transistors). Although various ones of the accompanying drawings depict a particular number of wires in the channel material 106 of a device region 206, this is simply for ease of illustration, and a device region 206 may include more or fewer wires as the channel material 106. In other embodiments, the channel material 106 of one or more of the device regions 206 may include a semiconductor fin instead of or in addition to one or more semiconductor wires; examples of such embodiments are discussed below with reference to FIG. 46. More generally, any of the IC structures 100 disclosed herein or substructures thereof (e.g., the S/D regions 150/152, discussed below) may be utilized in a transistor having any desired architecture, such as forksheet transistors, double-gate transistors, or pseudo double-gate transistors. In some embodiments, the channel material 106 may include silicon and/or germanium. In some embodiments, the channel material 106 may include indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide, or further materials classified as group II-VI, III-V, or IV. In some embodiments, the channel material 106 may include a semiconducting oxide (e.g., indium gallium zinc oxide). In some embodiments, the material composition of the channel material 106 used in different ones of the wires in a particular device region 206 may be different, or may be the same.

The channel material 106 may be in contact with a gate dielectric 136. In some embodiments, the gate dielectric 136 may surround the channel material 106 (e.g., when the channel material 106 includes wires, as shown in FIG. 1), while in other embodiments, the gate dielectric 136 may not surround the channel material 106 (e.g., when the channel material 106 includes a fin, as discussed below with reference to FIG. 46, or in forksheet, double-gate, or pseudo double-gate transistors). The gate dielectric 136 may include one layer or a stack of layers. The one or more layers may include silicon oxide, silicon dioxide, silicon carbide, and/or a high-k dielectric material. The high-k dielectric material may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Examples of high-k materials that may be used in the gate dielectric 136 include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and lead zinc niobate. In some embodiments, an annealing process may be carried out on the gate dielectric 136 to improve its quality when a high-k material is used.

The gate dielectric 136 may be disposed between the channel material 106 and a gate metal 138. In some embodiments, the gate metal 138 may surround the channel material 106 (e.g., when the channel material 106 includes wires, as shown in FIG. 1), while in other embodiments, the gate metal 138 may not surround the channel material 106 (e.g., when the channel material 106 includes a fin, as discussed below with reference to FIG. 46, or in forksheet, double-gate, or pseudo double-gate transistors). Together, the gate metal 138 and the gate dielectric 136 may provide a gate 204 for the associated channel material 106 in an associated channel region 202, with the electrical impedance of the channel material 106 modulated by the electrical potential applied to the associated gate 204 (through gate contacts 140). The gate metal 138 may include at least one p-type work function metal or n-type work function metal (or both),

depending on whether the transistor of which it is a part is to be a p-type metal oxide semiconductor (PMOS) or an n-type metal oxide semiconductor (NMOS) transistor. In some implementations, the gate metal **138** may include a stack of two or more metal layers, where one or more metal layers are work function metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included for other purposes, such as a barrier layer (e.g., tantalum, tantalum nitride, an aluminum-containing alloy, etc.). In some embodiments, a gate metal **138** may include a resistance-reducing cap layer (e.g., copper, gold, cobalt, or tungsten). For a PMOS transistor, metals that may be used for the gate metal **138** include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides (e.g., ruthenium oxide), and any of the metals discussed herein with reference to an NMOS transistor (e.g., for work function tuning). For an NMOS transistor, metals that may be used for the gate metal **138** include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide), and any of the metals discussed above with reference to a PMOS transistor (e.g., for work function tuning). In some embodiments, the gate metal **138** may include grading (increasing or decreasing) of the concentration of one or more materials therein. Dielectric material **118** and/or dielectric material **124** may separate the gate dielectric **136**, the gate metal **138**, the gate contact **140** from the proximate S/D regions **150/152**. The dielectric materials **118** and **124** may include silicon nitride, silicon oxide, silicon carbide, silicon nitride doped with carbon, silicon oxide doped with carbon, silicon oxynitride, or silicon oxynitride doped with carbon, for example.

Source/drain (S/D) regions **150/152** may be in electrical contact with the longitudinal ends of the channel material **106**, allowing current to flow from one S/D region **150/152** to another S/D region **150/152** through the channel material **106** (upon application of appropriate electrical potentials to the S/D regions **150/152**) during operation. The S/D regions **150/152** may include semiconductor regions **128/130**, respectively; as discussed further below with reference to FIGS. 2-44, the semiconductor regions **128** may have a particular dopant type (i.e., n-type or p-type) while the semiconductor regions **130** may have the opposite dopant type (i.e., p-type or n-type, respectively); reference may be made herein to “n-type” or “p-type” S/D regions **150/152** according to the dopant type of the semiconductor regions **128/130**, respectively. The particular arrangement of S/D regions **150/152** in the accompanying drawings is simply illustrative, and any desired arrangement may be used (e.g., by appropriate selective masking).

In some embodiments, the semiconductor regions **128/130** may include a silicon alloy such as silicon germanium or silicon carbide. In some embodiments, semiconductor regions **128/130** may include dopants such as boron, arsenic, or phosphorous. In some embodiments, the semiconductor regions **128/130** may include one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. For PMOS transistors, semiconductor regions **128/130** may include, for example, group IV semiconductor materials such as silicon, germanium, silicon germanium, germanium tin, or silicon germanium alloyed with carbon. Example p-type dopants in silicon, silicon germanium, and germanium include boron, gallium, indium, and aluminum. For NMOS transistors, semiconductor regions **128/130** may include, for example, group III-V semiconductor materials such as indium, aluminum, arsenic,

phosphorous, gallium, and antimony, with some example compounds including indium aluminum arsenide, indium arsenide phosphide, indium gallium arsenide, indium gallium arsenide phosphide, gallium antimonide, gallium aluminum antimonide, indium gallium antimonide, or indium gallium phosphide antimonide. The semiconductor regions **128/130** may be in contact with the adjacent channel material **106**, as shown.

The S/D regions **150/152** may also include a contact metal **164**. The contact metal **164** may be in contact with the semiconductor regions **128/130** of the S/D regions **150/152**, and may extend above the semiconductor regions **128/130** to facilitate electrical contact with the S/D regions **150/152** (e.g., through a conductive via in contact with the contact metal **164**, not shown). The surface of the semiconductor regions **128/130** in contact with the contact metal **164** may be undulating, as shown. In some embodiments, the surface of the semiconductor regions **128/130** may have protrusions toward the contact metal **164**, with individual protrusions corresponding to an individual associated channel material **106**. In some embodiments, the individual protrusions in a semiconductor region **128/130** associated with adjacent channel materials **106** may contact each other, while in other embodiments, the individual protrusions in a semiconductor region **128/130** may not contact each other. FIG. 10 and others of the accompanying drawings illustrate an embodiment in which the individual protrusions in a semiconductor region **128/130** associated with adjacent channel materials **106** may just contact each other, while in other embodiments, the individual protrusions may not contact each other (e.g., as shown in FIG. 1E) or may contact each other more fully (e.g., as shown in FIG. 1F). The protrusions in the semiconductor regions **128/130** may allow at least some contact metal **164** to be located between adjacent ones of the protrusions (e.g., as shown in FIGS. 10, 1E, and 1F). In some embodiments, the semiconductor regions **128/130** may be provided by a more uniform layer of material (e.g., as shown in FIG. 1G), instead of distinct protrusions. Although FIG. 1A (and others of the accompanying drawings) depicts a single portion of contact metal **164** spanning (“shorting”) multiple S/D regions **150/152**, this is simply illustrative, and the contact metal **164** may be arranged so as to isolate and connect various ones of the S/D regions **150/152** as desired.

The contact metal **164** may itself be a stressed material, and thus may impart stress to the surrounding elements, including the channel material **106**. The stress in the contact metal **164** may be selected so as to impart compressive stress to the channel material **106** in a PMOS transistor or to impart tensile stress to the channel material **106** in an NMOS transistor. In some embodiments, the contact metal **164** may include binary titanium alloys having alloyed or implanted elements (e.g., zirconium, tantalum, tin, molybdenum, manganese, niobium, hafnium, iron, indium, gold, silver, palladium, platinum, chromium, copper, or cobalt). For example, these alloyed or implanted elements may comprise 10 atomic-percent to 20 atomic-percent of the contact metal **164**. In some embodiments, additional stress on the channel material **106** may be imparted by a sacrificial dielectric material utilized during manufacturing (as discussed below with reference to the sacrificial material **168**). Various ones of the embodiments disclosed herein may enable a channel stress between -500 megapascals and -1500 megapascals in a PMOS transistor, and between 500 megapascals and 1500 megapascals in an NMOS transistor. Relative to conventional approaches, the additional channel stress achieved by the embodiments disclosed herein, and/or the increased contact area between the contact metal **164** and the semi-

conductor regions **128/130**, may contribute to significant improvements in device drive and mobility.

The S/D regions **150/152** may be partially laterally confined by insulating material regions including dielectric material **112**, dielectric material **118**, and dielectric material **120** between adjacent device regions **206**. As shown in FIG. 1A, in some embodiments, the dielectric material **112** may have a U-shaped cross-section, with “spacers” formed of the dielectric material **118** thereon, and the dielectric material **120** therebetween.

The S/D regions **150/152** disclosed herein may increase the contact area between the contact metal **164** and the adjacent semiconductor regions **128/130** relative to conventional structures, substantially reducing the contact resistance and thereby improving device performance (e.g., improving power efficiency). As discussed below with reference to FIGS. 2-44, the S/D regions **150/152** of the IC structure **100** may be manufactured by growing “epitaxial nubs” as the semiconductor regions **128/130**, and then filling the volume between opposing sets of semiconductor regions **128/130** with the contact metal **164**. Such manufacturing techniques do not add significant complexity to a manufacturing flow, and thus may be quickly and efficiently adopted.

In the IC structure **100** of FIG. 1, a dielectric material **166** may be present between the S/D regions **150/152** and the base **102**; the presence of such a dielectric material **166** may help isolate the S/D regions **150/152** from the underlying material, and thus mitigate or eliminate the formation of an undesirable parasitic channel in the underlying material, as discussed above. The dielectric material **166** may include any suitable dielectric material, such as an oxide (e.g., silicon oxide). Together, a channel material **106**, gate dielectric **136**, gate metal **138**, and associated S/D regions **150/152** may form a transistor.

The dimensions of the elements of the IC structure of FIG. 1 (and others of the embodiments disclosed herein) may take any suitable form. For example, in some embodiments, a gate length **208** of a gate **204** may be between 3 nanometers and 100 nanometers; different ones of the gates **204** in a device region **206** may have the same gate length **208**, or different gate lengths **208**, as desired. In some embodiments, the width **210** of the channel material **106** may be between 3 nanometers and 30 nanometers. In some embodiments, the thickness **212** of the channel material **106** may be between 1 nanometer and 500 nanometers (e.g., between 40 nanometers and 400 nanometers when the channel material **106** is a fin, and between 5 nanometers and 40 nanometers when the channel material **106** is a wire). In some embodiments in which a channel region **202** includes semiconductor wires, the spacing **214** between adjacent ones of the wires in a channel region **202** may be between 5 nanometers and 40 nanometers.

In some embodiments, the IC structure **100** may be part of a memory device, and transistors of the IC structure **100** may store information in the IC structure **100** or facilitate access to (e.g., read and/or write) storage elements of the memory device. In some embodiments, the IC structure **100** may be part of a processing device. In some embodiments, the IC structure **100** may be part of a device that includes memory and logic devices (e.g., in a single die **1502**, as discussed below), such as a processor and cache. More generally, the IC structures **100** disclosed herein may be part of memory devices, logic devices, or both.

FIGS. 2-44 illustrate stages in an example process for manufacturing the IC structure **100** of FIG. 1. Although the operations of the process may be illustrated with reference to particular embodiments of the IC structures **100** disclosed

herein, the process of FIGS. 2-44 and variants thereof may be used to form any suitable IC structure. Operations are illustrated a particular number of times and in a particular order in FIGS. 2-44, but the operations may be reordered and/or repeated as desired (e.g., with different operations performed in parallel when manufacturing multiple IC structures **100** simultaneously).

FIG. 2 illustrates an assembly including a base **102** and a stack of material layers on the base **102**. The stack of material layers may include one or more layers of the channel material **106** spaced apart from each other (and from the base **102**) by intervening layers of sacrificial material **104**. The size and arrangement of the material layers in the stack of the assembly of FIG. 2 corresponds to the desired size and arrangement of the channel material **106** in the IC structure **100**, as will be discussed further below, and thus the material layers in the assembly of FIG. 2 may vary from the particular embodiment illustrated in FIG. 2. For example, the thickness of a layer of channel material **106** may correspond to the channel thickness **212** discussed above (though the thickness of the layer of channel material **106** may differ from the final channel thickness **212** due to material lost during processing, etc.), and the thickness of a layer of sacrificial material **104** may correspond to the wire spacing **214** discussed above (though the thickness of the layer of sacrificial material **104** may differ from the final wire spacing **214** due to material lost during processing, etc.). The sacrificial material **104** may be any material that may be appropriately selectively removed in later processing operations (as discussed below with reference to FIG. 30). For example, the sacrificial material **104** may be silicon germanium, and the channel material **106** may be silicon. In another example, the sacrificial material **104** may be silicon dioxide and the channel material **106** may be silicon or germanium. In another example, the sacrificial material **104** may be gallium arsenide and the channel material **106** may be indium gallium arsenide, germanium, or silicon germanium. The assembly of FIG. 2 may be formed using any suitable deposition techniques, such as chemical vapor deposition (CVD), metalorganic vapor phase epitaxy (MOVPE), molecular-beam epitaxy (MBE), physical vapor deposition (PVD), atomic layer deposition (ALD), or a layer transfer process.

FIG. 3 illustrates an assembly subsequent to forming a patterned hardmask **108** on the assembly of FIG. 2. Forming the patterned hardmask **108** may include depositing the hardmask (using any suitable method) and then selectively removing portions of the hardmask **108** (e.g., using lithographic techniques) to form the patterned hardmask **108**. In some embodiments, the pattern of the patterned hardmask **108** may first be formed in another material on the initially deposited hardmask, and then the pattern may be transferred from the other material into the hardmask **108**. The locations of the hardmask **108** may correspond to the device regions **206** in the IC structure **100**, as discussed further below. In the embodiment of FIG. 3, the hardmask **108** may be patterned into multiple parallel rectangular portions (corresponding to the fins **220** discussed below).

FIG. 4 illustrates an assembly subsequent to forming fins **220** in the material stack of the assembly of FIG. 2, in accordance with the pattern of the patterned hardmask **108**. Etch techniques may be used to form the fins **220**, including wet and/or dry etch schemes, as well as isotropic and/or anisotropic etch schemes. The fins **220** may include the sacrificial material **104** and the channel material **106**, as well as a portion of the base **102**; the portion of the base **102** included in the fins **220** provides a pedestal **222**. The width

of the fins **220** may be equal to the width **210** of the channel material **106**, as discussed above. Any suitable number of fins **220** may be included in the assembly of FIG. **4** (e.g., more or fewer than 3). Although the fins **220** depicted in FIG. **4** (and others of the accompanying drawings) are perfectly rectangular, this is simply for ease of illustration, and in practical manufacturing settings, the shape of the fins **220** may not be perfectly rectangular. For example, the fins **220** may be tapered, widening toward the base **102**. The top surface of the fins **220** may not be flat, but may be curved, rounding into the side surfaces of the fins **220**, and these non-idealities may carry over into subsequent processing operations. In some embodiments, the pitch **101** of the fins **220** may be between 20 nanometers and 50 nanometers (e.g., between 20 nanometers and 40 nanometers).

FIG. **5** illustrates an assembly subsequent to forming a dielectric material **110** on the base **102** of the assembly of FIG. **4**, between the fins **220**. The dielectric material **110** may include any suitable material, such as an STI material (e.g., an oxide material, such as silicon oxide). The dielectric material **110** may be formed by blanket-depositing the dielectric material **110** and then recessing the dielectric material **110** back to a desired thickness. In some embodiments, the thickness of the dielectric material **110** may be selected so that the top surface of the dielectric material **110** is approximately coplanar with the top surface of the pedestals **222**. In some embodiments, the height **103** of a fin **220** above the top surface of the dielectric material **110** may be between 40 nanometers and 100 nanometers (e.g., between 50 nanometers and 70 nanometers).

FIG. **6** illustrates an assembly subsequent to forming a conformal layer of a dielectric material **112** over the assembly of FIG. **5**. The dielectric material **112** may be formed using any suitable technique (e.g., ALD). The dielectric material **112** may include any suitable material (e.g., silicon oxide).

FIG. **7** illustrates an assembly subsequent to forming a dielectric material **114** over the assembly of FIG. **6**. The dielectric material **114** may extend over the top surfaces of the fins **220**, as shown, and may serve as a “dummy gate.” The dielectric material **114** may include any suitable material (e.g., polysilicon).

FIG. **8** illustrates an assembly subsequent to forming a patterned hardmask **116** on the assembly of FIG. **7**. The hardmask **116** may include any suitable materials (e.g., silicon nitride, carbon-doped silicon oxide, or carbon-doped silicon oxynitride). The hardmask **116** may be patterned into strips that are oriented perpendicular to the longitudinal axis of the fins **220** (into and out of the page in accordance with the perspective of FIGS. **8C** and **8D**), corresponding to the locations of the gates **204** in the IC structure **100**, as discussed further below.

FIG. **9** illustrates an assembly subsequent to etching the dielectric material **114** (the “dummy gate”) of the assembly of FIG. **8** using the patterned hardmask **116** as a mask. The locations of the remaining dielectric material **114** may correspond to the locations of the gates **204** in the IC structure **100**, as discussed further below.

FIG. **10** illustrates an assembly subsequent to depositing a conformal layer of dielectric material **118** on the assembly of FIG. **9**, and then performing a directional “downward” etch to remove the dielectric material **118** on horizontal surfaces, leaving the dielectric material **118** as “spacers” on side faces of exposed surfaces, as shown. The dielectric material **118** may be deposited to any desired thickness using any suitable technique (e.g., ALD). The dielectric material **118** may include any suitable dielectric material

(e.g., silicon oxycarbonitride). The dielectric material **118** may border the fins **220** in the volumes that will be replaced by the S/D regions **150/152**, as discussed below.

FIG. **11** illustrates an assembly subsequent to depositing a dielectric material **120** on the assembly of FIG. **10**. The dielectric material **120** may be blanket deposited over the assembly of FIG. **10** and then the dielectric material **120** may be polished (e.g., by chemical mechanical polishing (CMP)) or otherwise recessed back so that the top surface of the dielectric material **120** is coplanar with the top surface of the patterned hardmask **116**, as shown in FIGS. **11C** and **11D**. The dielectric material **120** may include any suitable material (e.g., an oxide, such as silicon oxide).

FIG. **12** illustrates an assembly subsequent to depositing a hardmask **126** on the assembly of FIG. **11**. The hardmask **126** may have any suitable material composition; for example, in some embodiments, the hardmask **126** may include titanium nitride.

FIG. **13** illustrates an assembly subsequent to patterning the hardmask **126** of the assembly of FIG. **12** so as to selectively remove the hardmask **126** in areas that will correspond to the S/D regions **152**, while otherwise leaving the hardmask **126** in place. Any suitable patterning technique (e.g., a lithographic technique) may be used to pattern the hardmask **126**. The particular arrangement of the S/D regions **152** in an IC structure **100** (and thus the particular layout of the patterned hardmask **126**) depicted in various ones of the accompanying figures is simply illustrative, and any desired arrangement may be used.

FIG. **14** illustrates an assembly subsequent to recessing the exposed dielectric material **120** of the assembly of FIG. **13** (i.e., the dielectric material **120** not protected by the hardmask **126**). Any suitable selective etch technique may be used to recess the exposed dielectric material **120**, such as an isotropic etch. In the areas not protected by the hardmask **126**, the dielectric material **120** may remain.

FIG. **15** illustrates an assembly subsequent to removing some of the dielectric material **118** exposed in the assembly of FIG. **14**. This operation may enlarge the “canyons” between adjacent portions of hardmask **116**/dielectric material **114**, facilitating subsequent operations. In some embodiments, the removal of some of the dielectric material **118** may be achieved by a partial isotropic etch (e.g., a nitride partial isotropic etch when the dielectric material **118** includes a nitride).

FIG. **16** illustrates an assembly subsequent to further recessing the exposed dielectric material **120** of the assembly of FIG. **15** (i.e., the dielectric material **120** not protected by the hardmask **126**). Any suitable selective etch technique may be used to recess the exposed dielectric material **120**, such as an isotropic etch. In the areas not protected by the hardmask **126**, the dielectric material **120** may remain.

FIG. **17** illustrates an assembly subsequent to conformally depositing additional dielectric material **118** on the assembly of FIG. **16**, and then performing another directional “downward” etch to remove the dielectric material **118** on horizontal surfaces, “repairing” the dielectric material **118** as “spacers” on side faces of exposed surfaces, as shown. The etch of FIG. **17** (e.g., a reactive ion etch (RIE)) may also remove the dielectric material **112** from the top faces of the sacrificial material **104**, as shown.

FIG. **18** illustrates an assembly subsequent to removing most of the portions of the sacrificial material **104** and the channel material **106** in the assembly of FIG. **17** that are not covered by the hardmask **126** to form open volumes **224** (e.g., using any suitable etch techniques). Some of the bottommost portion of the sacrificial material **104** may

remain, as shown. These open volumes **224** may correspond to the locations of the S/D regions **152** in the IC structure **100**, as discussed further below, and are self-aligned to the dielectric material **112**, as shown.

FIG. **19** illustrates an assembly subsequent to recessing the exposed sacrificial material **104** of the assembly of FIG. **18**, without simultaneously recessing the exposed channel material **106** (as shown in FIG. **19C**). Any suitable selective etch technique may be used. Since this partial lateral recess of the exposed sacrificial material **104** is self-aligned to the exposed channel material **106**, the recess of the exposed sacrificial material **104** may be uniform across the width of the channel material **106** (i.e., left-right from the perspective of FIG. **19A**).

FIG. **20** illustrates an assembly subsequent to conformally depositing a dielectric material **124** over the assembly of FIG. **19**. The dielectric material **124** may include any suitable material (e.g., a low-k dielectric material) and may be deposited so as to fill the recesses formed by recessing the exposed sacrificial material **104** (as discussed above with reference to FIG. **19**). In some embodiments, conformally depositing the dielectric material **124** may include multiple rounds of deposition (e.g., three rounds) of one or more dielectric materials.

FIG. **21** illustrates an assembly subsequent to recessing the dielectric material **124** of the assembly of FIG. **20**. Any suitable selective etch technique may be used to recess the exposed dielectric material **124**, such as an isotropic etch. The dielectric material **124** may remain on side surfaces of the sacrificial material **104** proximate to the open volumes **224**, as shown in FIG. **21C**. The amount of recess may be such that the recessed surface of the dielectric material **124** is flush with (not shown) or slightly beyond the side surface of the channel material **106**, as shown in FIG. **21C**. Excessive recess of the exposed dielectric material **124** beyond the side surface of the channel material **106** may result in device performance degradation (e.g., due to elevated parasitic contact-to-gate coupling capacitance) and/or device defect (e.g., due to contact-to-gate short).

FIG. **22** illustrates an assembly subsequent to forming a dielectric material **166** on the base **102** in the open volumes **224** of the assembly of FIG. **21**. The dielectric material **166** may include any suitable material, and in some embodiments, may be formed by initially blanket-depositing the dielectric material **166** and then recessing the dielectric material **166** back to a desired thickness on the base **102**.

FIG. **23** illustrates an assembly subsequent to forming the semiconductor regions **130** in the open volumes **224** of the assembly of FIG. **22**. The semiconductor regions **130** may be formed by epitaxial growth (e.g., an initial nucleation operation to provide a seed layer, followed by a primary epitaxy operation in which the remainder of the semiconductor regions **130** are formed on the seed layer), but unlike conventional approaches, the semiconductor regions **130** do not fill the volumes **224**. Instead, the semiconductor regions **130** form around the exposed ends of the channel material **106**, with the growth stopped before the volumes **224** are filled, forming semiconductor regions **130** having the undulating shape shown. In some embodiments, the different protrusions (or “epi nubs”) of the semiconductor regions **130** may correspond to the different channel materials **106**, and as discussed above with reference to FIGS. **10**, **1E**, and **1F**, the portions of the semiconductor regions **130** corresponding to different ones of the channel materials **106** may or may not contact each other. In some embodiments, the different semiconductor regions **130** in a single open volume **224** (corresponding to the channel materials **106** on either side of

the open volume **224**) may not contact each other (i.e., a “gap” may remain between one semiconductor region **130** and another semiconductor region **130** in an open volume **224**; as discussed below, this gap may be filled with contact metal **164** in subsequent operations to complete the S/D regions **152**. In some embodiments, the semiconductor region **130** in an open volume **224** may be provided by a thin epitaxial layer having a U-shaped cross-section (e.g., as illustrated in FIG. **1G**). In some embodiments, the semiconductor regions **130** may include an n-type epitaxial material (e.g., heavily in-situ phosphorous-doped material for use in an NMOS transistor).

FIG. **24** illustrates an assembly subsequent to filling the remainder of the open volumes **224** with a sacrificial material **168**. The sacrificial material **168** may be a dielectric material, such as carbon-doped oxide, fluorine-doped silicon dioxide, porous silicon dioxide, spin-on silicon-based polymeric dielectric, or spin-on organic polymeric dielectrics (e.g., polyimide, polynorbornenes, benzocyclobutene, polytetrafluoroethylene, etc.), and may be removed in subsequent operations, as discussed below. In some embodiments, the sacrificial material **168** may be cured by ultraviolet (UV) radiation and/or by annealing (e.g., at a temperature up to 400 degrees Celsius). The sacrificial material **168** may itself be a stressed material, and thus may impart stress to the surrounding elements, including the channel material **106**. The stress in the sacrificial material **168** may be selected so as to impart compressive stress to the channel material **106** in a PMOS transistor or to impart tensile stress to the channel material **106** in an NMOS transistor.

FIG. **25** illustrates an assembly subsequent to depositing a conformal layer of a dielectric material **142** on the assembly of FIG. **24**. The dielectric material **142** may be a contact etch stop layer (CESL), and may be formed of any suitable material (e.g., silicon nitride).

FIG. **26** illustrates an assembly subsequent to depositing a dielectric material **122** on the assembly of FIG. **25**, and then polishing the dielectric material **122** and the dielectric material **142** to expose the hardmask **126**. In some embodiments, the dielectric material **122** may be a pre-metal dielectric (PMD), such as an oxide material (e.g., silicon oxide).

FIG. **27** illustrates an assembly subsequent to removing the hardmask **126** from the assembly of FIG. **26**, then depositing and patterning a hardmask **127**. The hardmask **127** may have any suitable material composition; for example, in some embodiments, the hardmask **127** may include titanium nitride. The hardmask **127** may be patterned so as to selectively remove the hardmask **127** in areas that will correspond to the S/D regions **150**, while otherwise leaving the hardmask **127** in place. Any suitable patterning technique (e.g., a lithographic technique) may be used to pattern the hardmask **127**. As noted above, the particular arrangement of the S/D regions **150** in an IC structure **100** (and thus the particular layout of the patterned hardmask **127**) depicted in various ones of the accompanying figures is simply illustrative, and any desired arrangement may be used.

FIG. **28** illustrates an assembly subsequent to recessing the exposed dielectric material **120** (i.e., the dielectric material **120** not protected by the hardmask **127**) of the assembly of FIG. **27**. Any suitable selective etch technique may be used to recess the exposed dielectric material **120**, such as an isotropic etch.

FIG. **29** illustrates an assembly subsequent to removing some of the dielectric material **118** exposed in the assembly of FIG. **28**. This operation may enlarge the “canyons”

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between adjacent portions of hardmask **116**/dielectric material **114**, facilitating subsequent operations. In some embodiments, the removal of some of the dielectric material **118** may be achieved by a partial isotropic etch (e.g., a nitride partial isotropic etch when the dielectric material **118** includes a nitride).

FIG. **30** illustrates an assembly subsequent to further recessing the exposed dielectric material **120** of the assembly of FIG. **29** (i.e., the dielectric material **120** not protected by the hardmask **127**). Any suitable selective etch technique may be used to recess the exposed dielectric material **120**, such as an isotropic etch.

FIG. **31** illustrates an assembly subsequent to conformally depositing additional dielectric material **118** on the assembly of FIG. **30**, and then performing another directional “downward” etch to remove the dielectric material **118** on horizontal surfaces, “repairing” the dielectric material **118** as “spacers” on side faces of exposed surfaces, as shown. The etch of FIG. **31** (e.g., an RIE) may also remove the dielectric material **112** from the top faces of the sacrificial material **104**, as shown.

FIG. **32** illustrates an assembly subsequent to removing the portions of the sacrificial material **104** and the channel material **106** in the assembly of FIG. **31** that are not covered by the hardmask **127** to form open volumes **225** (e.g., using any suitable etch techniques). These open volumes **225** may correspond to the locations of the S/D regions **150** in the IC structure **100**, as discussed further below, and are self-aligned to the dielectric material **112**, as shown.

FIG. **33** illustrates an assembly subsequent to recessing the exposed sacrificial material **104** of the assembly of FIG. **32**, without simultaneously recessing the exposed channel material **106**, conformally depositing a dielectric material **124**, and recessing the dielectric material **124**. These operations may take any of the forms discussed above with reference to FIGS. **19-21**. The dielectric material **124** may remain on side surfaces of the sacrificial material **104** proximate to the open volumes **225**, as shown in FIG. **33C**.

FIG. **34** illustrates an assembly subsequent to forming a dielectric material **166** on the base **102** in the open volumes **225**, forming the semiconductor regions **128** in the open volumes **225** of the assembly of FIG. **33**, forming a sacrificial material **168** to fill the remainder of the open volumes **225**, depositing a conformal layer of a dielectric material **154**, and depositing a dielectric material **156**. The semiconductor regions **128** may be formed by epitaxial growth (e.g., an initial nucleation operation to provide a seed layer, followed by a primary epitaxy operation in which the remainder of the semiconductor regions **128** are formed on the seed layer), but as discussed above with reference to FIG. **23** and unlike conventional approaches, the semiconductor regions **128** do not fill the volumes **225**. Instead, the semiconductor regions **128** form around the exposed ends of the channel material **106**, with the growth stopped before the volumes **225** are filled, forming semiconductor regions **128** having the undulating shape shown. In some embodiments, the different protrusions (or “epi nubs”) of the semiconductor regions **128** may correspond to the different channel materials **106**, and as discussed above with reference to FIGS. **10, 1E, and 1F**, the portions of the semiconductor regions **128** corresponding to different ones of the channel materials **106** may or may not contact each other. In some embodiments, the different semiconductor regions **128** in a single open volume **226** (corresponding to the channel materials **106** on either side of the open volume **226**) may not contact each other (i.e., a “gap” may remain between one semiconductor region **128** and another semiconductor

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region **128** in an open volume **225**; as discussed below, this gap may be filled with contact metal **164** in subsequent operations to complete the S/D regions **150**. In some embodiments, the semiconductor regions **128** may include a p-type epitaxial material (e.g., heavily in-situ boron-doped material for use in a PMOS transistor). In some embodiments, the semiconductor region **128** in an open volume **225** may be provided by a thin epitaxial layer having a U-shaped cross-section (e.g., as illustrated in FIG. **1G**). In some implementations, the semiconductor regions **128** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some embodiments, the epitaxially deposited silicon alloy may be doped in-situ with dopants such as boron, arsenic, or phosphorous. In some embodiments, the semiconductor regions **128** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. The sacrificial material **168** may include any of the dielectric materials discussed above with reference to FIG. **24**, and may be cured by UV radiation and/or by annealing (e.g., at a temperature up to 400 degrees Celsius). As discussed above, the sacrificial material **168** may itself be a stressed material, and thus may impart stress to the surrounding elements, including the channel material **106**. The stress in the sacrificial material **168** may be selected so as to impart compressive stress to the channel material **106** in a PMOS transistor or to impart tensile stress to the channel material **106** in an NMOS transistor. The dielectric material **154** may be a CESL, and may be formed of any suitable material (e.g., silicon nitride). In some embodiments, the dielectric material **156** may be a PMD, such as an oxide material (e.g., silicon oxide).

FIG. **35** illustrates an assembly subsequent to polishing the hardmask **127**, the dielectric material **122**, the dielectric material **142**, the dielectric material **154**, and the dielectric material **156** of the assembly of FIG. **34** (e.g., using a CMP technique) to expose the hardmask **116** above the channel regions **202**.

FIG. **36** illustrates an assembly subsequent to removing the hardmask **116**, the dielectric material **114** (the “dummy gate”), and the dielectric material **112** from the assembly of FIG. **35** to form open volumes **226**. Any suitable etch techniques may be used.

FIG. **37** illustrates an assembly subsequent to “release” of the channel material **106** in the assembly of FIG. **36** by removal of the sacrificial material **104**. Any suitable etch technique may be used.

FIG. **38** illustrates an assembly subsequent to forming a conformal gate dielectric **136** over the assembly of FIG. **37**. The gate dielectric **136** may be formed using any suitable technique (e.g., ALD), and may include any of the materials discussed herein with reference to the gate dielectric **136**.

FIG. **39** illustrates an assembly subsequent to forming a gate metal **138** over the assembly of FIG. **38**. The gate metal **138** may include any one or more material layers, such as any of the materials discussed herein with reference to the gate metal **138**.

FIG. **40** illustrates an assembly subsequent to polishing the gate metal **138** and the gate dielectric **136** of the assembly of FIG. **39** to remove the gate metal **138** and the gate dielectric **136** over the dielectric material **122** and the dielectric material **156**. Any suitable polishing technique, such as a CMP technique, may be used.

FIG. **41** illustrates an assembly subsequent to recessing the gate metal **138** and the gate dielectric **136** (e.g., using one or more etch techniques) to form recesses in the assembly of FIG. **40**, and then forming gate contacts **140** in the recesses.

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The gate contacts **140** may include any one or more materials (e.g., an adhesion liner, a barrier liner, one or more fill metals, etc.).

FIG. **42** illustrates an assembly subsequent to patterning the dielectric material **122**, the dielectric material **142**, the dielectric material **154**, and the dielectric material **156** of the assembly of FIG. **41** to form recesses. Any suitable etch technique may be used.

FIG. **43** illustrates an assembly subsequent to removing the recesses with contact metal **164**. As shown, the contact metal **164** may contact the semiconductor regions **128/130**, and may complete the S/D regions **150/152**. The contact metal **164** may include any one or more materials (e.g., an adhesion liner, a barrier liner, one or more fill metals, etc.). As noted above, the contact metal **164** may itself be a stressed material, and thus may impart stress to the surrounding elements, including the channel material **106**. The stress in the contact metal **164** may be selected so as to impart further compressive stress to the channel material **106** in a PMOS transistor or to impart further tensile stress to the channel material **106** in an NMOS transistor. The contact metal **164** may include any suitable material (e.g., any of the materials discussed above with reference to FIG. **1**). In some embodiments, the contact metal **164** may be cured or annealed after deposition (e.g., at a temperature up to 450 degrees Celsius). The assembly of FIG. **44** may take the form of the IC structure **100** of FIG. **1**.

FIG. **44** illustrates an assembly subsequent to filling the recesses with contact metal **164**. As shown, the contact metal **164** may contact the semiconductor regions **128/130**, and may complete the S/D regions **150/152**. The contact metal **164** may include any one or more materials (e.g., an adhesion liner, a barrier liner, one or more fill metals, etc.). As noted above, the contact metal **164** may itself be a stressed material, and thus may impart stress to the surrounding elements, including the channel material **106**. The stress in the contact metal **164** may be selected so as to impart further compressive stress to the channel material **106** in a PMOS transistor or to impart further tensile stress to the channel material **106** in an NMOS transistor. The contact metal **164** may include any suitable material (e.g., any of the materials discussed above with reference to FIG. **1**). In some embodiments, the contact metal **164** may be cured or annealed after deposition (e.g., at a temperature up to 450 degrees Celsius). The assembly of FIG. **44** may take the form of the IC structure **100** of FIG. **1**.

In the process discussed above with reference to FIGS. **2-44**, the semiconductor regions **128/130** are formed before the “replacement gate” operations illustrated in FIGS. **35-39**; in other embodiments, the semiconductor regions **128/130** may be formed after the operations illustrated in FIGS. **35-39**. Instead of forming the semiconductor regions **128/130** as discussed above with reference to FIGS. **23** and **34**, only the sacrificial material **168** may fill the open volumes **224** and **225**; after the “replacement gate” operations illustrated in FIGS. **35-39**, the sacrificial material **168** may be removed and the semiconductor regions **128/130** formed, followed by formation of the contact metal **164**. In such embodiments, the stress imparted to the channel material **106** by the sacrificial material **168** may be “locked” into place by the “replacement gate” operations illustrated in FIGS. **35-39**, and then additional stress may be imparted to the channel material **106** by the formation of the contact metal **164**.

In some embodiments, the repeated deposition and etching operations around the dielectric material **118** may be performed such that a “cap” of the dielectric material **118** extends over the insulating material **120**. FIG. **45** is a side, cross-sectional view of such an IC structure **100**, sharing the perspective of the “A” sub-figures herein. The resulting dielectric material **118** may have the same of an upside down “U” and may be nested in the U-shaped dielectric material **112**. Any of the embodiments disclosed herein may include a dielectric material **118** having the structure of FIG. **45**.

As noted above, in some embodiments, the channel material **106** may have any desired arrangement. For example, FIG. **46** illustrates an IC structure **100** in which the channel material **106** is arranged as a fin, instead of as one or more nanowires; in other embodiments, the channel material **106** may include a fin and nanowires, or other

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arrangements. An IC structure **100** like that of FIG. **46** may be manufactured using the manufacturing processes disclosed herein as appropriate (e.g., omitting the “release” operations, etc.).

The IC structures **100** disclosed herein may be included in any suitable electronic component. FIGS. **47-51** illustrate various examples of apparatuses that may include any of the IC structures **100** disclosed herein.

FIG. **47** is a top view of a wafer **1500** and dies **1502** that may include one or more IC structures **100** in accordance with any of the embodiments disclosed herein. The wafer **1500** may be composed of semiconductor material and may include one or more dies **1502** having IC structures (e.g., the IC structures **100** disclosed herein) formed on a surface of the wafer **1500**. Each of the dies **1502** may be a repeating unit of a semiconductor product that includes any suitable IC. After the fabrication of the semiconductor product is complete, the wafer **1500** may undergo a singulation process in which the dies **1502** are separated from one another to provide discrete “chips” of the semiconductor product. The die **1502** may include one or more IC structures **100** (e.g., as discussed below with reference to FIG. **48**), one or more transistors (e.g., some of the transistors discussed below with reference to FIG. **48**) and/or supporting circuitry to route electrical signals to the transistors, as well as any other IC components. In some embodiments, the wafer **1500** or the die **1502** may include a memory device (e.g., a random access memory (RAM) device, such as a static RAM (SRAM) device, a magnetic RAM (MRAM) device, a resistive RAM (RRAM) device, a conductive-bridging RAM (CBRAM) device, etc.), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuit element. Multiple ones of these devices may be combined on a single die **1502**. For example, a memory array formed by multiple memory devices may be formed on a same die **1502** as a processing device (e.g., the processing device **1802** of FIG. **51**) or other logic that is configured to store information in the memory devices or execute instructions stored in the memory array.

FIG. **48** is a side, cross-sectional view of an IC component **1600** that may include one or more IC structures **100** in accordance with any of the embodiments disclosed herein. One or more of the IC components **1600** may be included in one or more dies **1502** (FIG. **47**). The IC component **1600** may be formed on a substrate **1602** (e.g., the wafer **1500** of FIG. **47**) and may be included in a die (e.g., the die **1502** of FIG. **47**). The substrate **1602** may take the form of any of the embodiments of the base **102** disclosed herein.

The IC component **1600** may include one or more device layers **1604** disposed on the substrate **1602**. The device layer **1604** may include features of one or more IC structures **100**, other transistors, diodes, or other devices formed on the substrate **1602**. The device layer **1604** may include, for example, source and/or drain (S/D) regions, gates to control current flow between the S/D regions, S/D contacts to route electrical signals to/from the S/D regions, and gate contacts to route electrical signals to/from the S/D regions (e.g., in accordance with any of the embodiments discussed above with reference to the IC structures **100**). The transistors that may be included in a device layer **1604** are not limited to any particular type or configuration, and may include any one or more of, for example, planar transistors, non-planar transistors, or a combination of both. Planar transistors may include bipolar junction transistors (BJT), heterojunction bipolar transistors (HBT), or high-electron-mobility transistors (HEMT). Non-planar transistors may include FinFET transistors, such as double-gate transistors or tri-gate tran-

sistors, and wrap-around or all-around gate transistors, such as nanoribbon and nanowire transistors (e.g., as discussed above with reference to the IC structures 100).

Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the devices (e.g., the IC structures 100) of the device layer 1604 through one or more interconnect layers disposed on the device layer 1604 (illustrated in FIG. 48 as interconnect layers 1606-1610). For example, electrically conductive features of the device layer 1604 (e.g., the gate contacts and the S/D contacts) may be electrically coupled with the interconnect structures 1628 of the interconnect layers 1606-1610. The one or more interconnect layers 1606-1610 may form a metallization stack (also referred to as an "ILD stack") 1619 of the IC component 1600. Although FIG. 48 depicts an ILD stack 1619 at only one face of the device layer 1604, in other embodiments, an IC component 1600 may include two ILD stacks 1619 such that the device layer 1604 is between the two ILD stacks 1619.

The interconnect structures 1628 may be arranged within the interconnect layers 1606-1610 to route electrical signals according to a wide variety of designs (in particular, the arrangement is not limited to the particular configuration of interconnect structures 1628 depicted in FIG. 48). Although a particular number of interconnect layers 1606-1610 is depicted in FIG. 48, embodiments of the present disclosure include IC components having more or fewer interconnect layers than depicted.

In some embodiments, the interconnect structures 1628 may include lines 1628a and/or vias 1628b filled with an electrically conductive material such as a metal. The lines 1628a may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the substrate 1602 upon which the device layer 1604 is formed. For example, the lines 1628a may route electrical signals in a direction in and out of the page from the perspective of FIG. 48. The vias 1628b may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the substrate 1602 upon which the device layer 1604 is formed. In some embodiments, the vias 1628b may electrically couple lines 1628a of different interconnect layers 1606-1610 together.

The interconnect layers 1606-1610 may include a dielectric material 1626 disposed between the interconnect structures 1628, as shown in FIG. 48. In some embodiments, the dielectric material 1626 disposed between the interconnect structures 1628 in different ones of the interconnect layers 1606-1610 may have different compositions; in other embodiments, the composition of the dielectric material 1626 between different interconnect layers 1606-1610 may be the same.

A first interconnect layer 1606 may be formed above the device layer 1604. In some embodiments, the first interconnect layer 1606 may include lines 1628a and/or vias 1628b, as shown. The lines 1628a of the first interconnect layer 1606 may be coupled with contacts (e.g., the S/D contacts or gate contacts) of the device layer 1604.

A second interconnect layer 1608 may be formed above the first interconnect layer 1606. In some embodiments, the second interconnect layer 1608 may include vias 1628b to couple the lines 1628a of the second interconnect layer 1608 with the lines 1628a of the first interconnect layer 1606. Although the lines 1628a and the vias 1628b are structurally delineated with a line within each interconnect layer (e.g., within the second interconnect layer 1608) for the sake of clarity, the lines 1628a and the vias 1628b may be structur-

ally and/or materially contiguous (e.g., simultaneously filled during a dual-damascene process) in some embodiments.

A third interconnect layer 1610 (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer 1608 according to similar techniques and configurations described in connection with the second interconnect layer 1608 or the first interconnect layer 1606. In some embodiments, the interconnect layers that are "higher up" in the metallization stack 1619 in the IC component 1600 (i.e., farther away from the device layer 1604) may be thicker.

The IC component 1600 may include a solder resist material 1634 (e.g., polyimide or similar material) and one or more conductive contacts 1636 formed on the interconnect layers 1606-1610. In FIG. 48, the conductive contacts 1636 are illustrated as taking the form of bond pads. The conductive contacts 1636 may be electrically coupled with the interconnect structures 1628 and configured to route the electrical signals of device layer 1604 to other external devices. For example, solder bonds may be formed on the one or more conductive contacts 1636 to mechanically and/or electrically couple a chip including the IC component 1600 with another component (e.g., a circuit board). The IC component 1600 may include additional or alternate structures to route the electrical signals from the interconnect layers 1606-1610; for example, the conductive contacts 1636 may include other analogous features (e.g., posts) that route the electrical signals to external components. In embodiments in which the IC component 1600 includes an ILD stack 1619 at each opposing face of the device layer 1604, the IC component 1600 may include conductive contacts 1636 on each of the ILD stacks 1619 (allowing interconnections to the IC component 1600 to be made on two opposing faces of the IC component 1600).

FIG. 49 is a side, cross-sectional view of an example IC package 1650 that may include one or more IC structures 100 in accordance with any of the embodiments disclosed herein. In some embodiments, the IC package 1650 may be a system-in-package (SiP).

The package substrate 1652 may be formed of a dielectric material (e.g., a ceramic, a buildup film, an epoxy film having filler particles therein, glass, an organic material, an inorganic material, combinations of organic and inorganic materials, embedded portions formed of different materials, etc.), and may have conductive pathways extending through the dielectric material between the face 1672 and the face 1674, or between different locations on the face 1672, and/or between different locations on the face 1674. These conductive pathways may take the form of any of the interconnects 1628 discussed above with reference to FIG. 48.

The package substrate 1652 may include conductive contacts 1663 that are coupled to conductive pathways (not shown) through the package substrate 1652, allowing circuitry within the dies 1656 and/or the interposer 1657 to electrically couple to various ones of the conductive contacts 1664.

The IC package 1650 may include an interposer 1657 coupled to the package substrate 1652 via conductive contacts 1661 of the interposer 1657, first-level interconnects 1665, and the conductive contacts 1663 of the package substrate 1652. The first-level interconnects 1665 illustrated in FIG. 49 are solder bumps, but any suitable first-level interconnects 1665 may be used. In some embodiments, no interposer 1657 may be included in the IC package 1650; instead, the dies 1656 may be coupled directly to the conductive contacts 1663 at the face 1672 by first-level interconnects 1665. More generally, one or more dies 1656

may be coupled to the package substrate **1652** via any suitable structure (e.g., a silicon bridge, an organic bridge, one or more waveguides, one or more interposers, wirebonds, etc.).

The IC package **1650** may include one or more dies **1656** coupled to the interposer **1657** via conductive contacts **1654** of the dies **1656**, first-level interconnects **1658**, and conductive contacts **1660** of the interposer **1657**. The conductive contacts **1660** may be coupled to conductive pathways (not shown) through the interposer **1657**, allowing circuitry within the dies **1656** to electrically couple to various ones of the conductive contacts **1661** (or to other devices included in the interposer **1657**, not shown). The first-level interconnects **1658** illustrated in FIG. **49** are solder bumps, but any suitable first-level interconnects **1658** may be used. As used herein, a “conductive contact” may refer to a portion of conductive material (e.g., metal) serving as an interface between different components; conductive contacts may be recessed in, flush with, or extending away from a surface of a component, and may take any suitable form (e.g., a conductive pad or socket).

In some embodiments, an underfill material **1666** may be disposed between the package substrate **1652** and the interposer **1657** around the first-level interconnects **1665**, and a mold compound **1668** may be disposed around the dies **1656** and the interposer **1657** and in contact with the package substrate **1652**. In some embodiments, the underfill material **1666** may be the same as the mold compound **1668**. Example materials that may be used for the underfill material **1666** and the mold compound **1668** are epoxy mold materials, as suitable. Second-level interconnects **1670** may be coupled to the conductive contacts **1664**. The second-level interconnects **1670** illustrated in FIG. **49** are solder balls (e.g., for a ball grid array arrangement), but any suitable second-level interconnects **16770** may be used (e.g., pins in a pin grid array arrangement or lands in a land grid array arrangement). The second-level interconnects **1670** may be used to couple the IC package **1650** to another component, such as a circuit board (e.g., a motherboard), an interposer, or another IC package, as known in the art and as discussed below with reference to FIG. **50**.

The dies **1656** may take the form of any of the embodiments of the die **1502** discussed herein (e.g., may include any of the embodiments of the IC component **1600**). In embodiments in which the IC package **1650** includes multiple dies **1656**, the IC package **1650** may be referred to as a multi-chip package (MCP). The dies **1656** may include circuitry to perform any desired functionality. For example, or more of the dies **1656** may be logic dies (e.g., silicon-based dies), and one or more of the dies **1656** may be memory dies (e.g., high bandwidth memory). In some embodiments, the die **1656** may include one or more IC structures **100** (e.g., as discussed above with reference to FIG. **47** and FIG. **48**).

Although the IC package **1650** illustrated in FIG. **49** is a flip chip package, other package architectures may be used. For example, the IC package **1650** may be a ball grid array (BGA) package, such as an embedded wafer-level ball grid array (eWLB) package. In another example, the IC package **1650** may be a wafer-level chip scale package (WLCS) or a panel fanout (FO) package. Although two dies **1656** are illustrated in the IC package **1650** of FIG. **49**, an IC package **1650** may include any desired number of dies **1656**. An IC package **1650** may include additional passive components, such as surface-mount resistors, capacitors, and inductors disposed on the first face **1672** or the second face **1674** of the package substrate **1652**, or on either face of the interposer

1657. More generally, an IC package **1650** may include any other active or passive components known in the art.

FIG. **50** is a side, cross-sectional view of an IC component assembly **1700** that may include one or more IC packages or other electronic components (e.g., a die) including one or more IC structures **100** in accordance with any of the embodiments disclosed herein. The IC component assembly **1700** includes a number of components disposed on a circuit board **1702** (which may be, e.g., a motherboard). The IC component assembly **1700** includes components disposed on a first face **1740** of the circuit board **1702** and an opposing second face **1742** of the circuit board **1702**; generally, components may be disposed on one or both faces **1740** and **1742**. Any of the IC packages discussed below with reference to the IC component assembly **1700** may take the form of any of the embodiments of the IC package **1650** discussed above with reference to FIG. **49** (e.g., may include one or more IC structures **100** in a die).

In some embodiments, the circuit board **1702** may be a printed circuit board (PCB) including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1702**. In other embodiments, the circuit board **1702** may be a non-PCB substrate.

The IC component assembly **1700** illustrated in FIG. **50** includes a package-on-interposer structure **1736** coupled to the first face **1740** of the circuit board **1702** by coupling components **1716**. The coupling components **1716** may electrically and mechanically couple the package-on-interposer structure **1736** to the circuit board **1702**, and may include solder balls (as shown in FIG. **50**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

The package-on-interposer structure **1736** may include an IC package **1720** coupled to a package interposer **1704** by coupling components **1718**. The coupling components **1718** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1716**. Although a single IC package **1720** is shown in FIG. **50**, multiple IC packages may be coupled to the package interposer **1704**; indeed, additional interposers may be coupled to the package interposer **1704**. The package interposer **1704** may provide an intervening substrate used to bridge the circuit board **1702** and the IC package **1720**. The IC package **1720** may be or include, for example, a die (the die **1502** of FIG. **47**), an IC component (e.g., the IC component **1600** of FIG. **48**), or any other suitable component. Generally, the package interposer **1704** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the package interposer **1704** may couple the IC package **1720** (e.g., a die) to a set of BGA conductive contacts of the coupling components **1716** for coupling to the circuit board **1702**. In the embodiment illustrated in FIG. **50**, the IC package **1720** and the circuit board **1702** are attached to opposing sides of the package interposer **1704**; in other embodiments, the IC package **1720** and the circuit board **1702** may be attached to a same side of the package interposer **1704**. In some embodiments, three or more components may be interconnected by way of the package interposer **1704**.

In some embodiments, the package interposer **1704** may be formed as a PCB, including multiple metal layers separated from one another by layers of dielectric material and

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interconnected by electrically conductive vias. In some embodiments, the package interposer **1704** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, an epoxy resin with inorganic fillers, a ceramic material, or a polymer material such as polyimide. In some embodiments, the package interposer **1704** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The package interposer **1704** may include metal lines **1710** and vias **1708**, including but not limited to through-silicon vias (TSVs) **1706**. The package interposer **1704** may further include embedded devices **1714**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) devices, and memory devices. More complex devices such as radio frequency devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the package interposer **1704**. The package-on-interposer structure **1736** may take the form of any of the package-on-interposer structures known in the art.

The IC component assembly **1700** may include an IC package **1724** coupled to the first face **1740** of the circuit board **1702** by coupling components **1722**. The coupling components **1722** may take the form of any of the embodiments discussed above with reference to the coupling components **1716**, and the IC package **1724** may take the form of any of the embodiments discussed above with reference to the IC package **1720**.

The IC component assembly **1700** illustrated in FIG. **50** includes a package-on-package structure **1734** coupled to the second face **1742** of the circuit board **1702** by coupling components **1728**. The package-on-package structure **1734** may include an IC package **1726** and an IC package **1732** coupled together by coupling components **1730** such that the IC package **1726** is disposed between the circuit board **1702** and the IC package **1732**. The coupling components **1728** and **1730** may take the form of any of the embodiments of the coupling components **1716** discussed above, and the IC packages **1726** and **1732** may take the form of any of the embodiments of the IC package **1720** discussed above. The package-on-package structure **1734** may be configured in accordance with any of the package-on-package structures known in the art.

FIG. **51** is a block diagram of an example electrical device **1800** that may include one or more IC structures **100** in accordance with any of the embodiments disclosed herein. For example, any suitable ones of the components of the electrical device **1800** may include one or more of the IC component assemblies **1700**, IC packages **1650**, IC components **1600**, or dies **1502** disclosed herein. A number of components are illustrated in FIG. **51** as included in the electrical device **1800**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the electrical device **1800** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single system-on-a-chip (SoC) die.

Additionally, in various embodiments, the electrical device **1800** may not include one or more of the components illustrated in FIG. **51**, but the electrical device **1800** may include interface circuitry for coupling to the one or more components. For example, the electrical device **1800** may

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not include a display device **1806**, but may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **1806** may be coupled. In another set of examples, the electrical device **1800** may not include an audio input device **1824** or an audio output device **1808**, but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1824** or audio output device **1808** may be coupled.

The electrical device **1800** may include a processing device **1802** (e.g., one or more processing devices). As used herein, the term “processing device” or “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processing device **1802** may include one or more digital signal processors (DSPs), application-specific integrated circuits (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The electrical device **1800** may include a memory **1804**, which may itself include one or more memory devices such as volatile memory (e.g., dynamic random access memory (DRAM)), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some embodiments, the memory **1804** may include memory that shares a die with the processing device **1802**. This memory may be used as cache memory and may include embedded dynamic random access memory (eDRAM) or spin transfer torque magnetic random access memory (STT-MRAM).

In some embodiments, the electrical device **1800** may include a communication chip **1812** (e.g., one or more communication chips). For example, the communication chip **1812** may be configured for managing wireless communications for the transfer of data to and from the electrical device **1800**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

The communication chip **1812** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultra mobile broadband (UMB) project (also referred to as “3GPP2”), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **1812** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **1812** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial

Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **1812** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **1812** may operate in accordance with other wireless protocols in other embodiments. The electrical device **1800** may include an antenna **1822** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

In some embodiments, the communication chip **1812** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **1812** may include multiple communication chips. For instance, a first communication chip **1812** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **1812** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication chip **1812** may be dedicated to wireless communications, and a second communication chip **1812** may be dedicated to wired communications.

The electrical device **1800** may include battery/power circuitry **1814**. The battery/power circuitry **1814** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the electrical device **1800** to an energy source separate from the electrical device **1800** (e.g., AC line power).

The electrical device **1800** may include a display device **1806** (or corresponding interface circuitry, as discussed above). The display device **1806** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display.

The electrical device **1800** may include an audio output device **1808** (or corresponding interface circuitry, as discussed above). The audio output device **1808** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds.

The electrical device **1800** may include an audio input device **1824** (or corresponding interface circuitry, as discussed above). The audio input device **1824** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

The electrical device **1800** may include a GPS device **1818** (or corresponding interface circuitry, as discussed above). The GPS device **1818** may be in communication with a satellite-based system and may receive a location of the electrical device **1800**, as known in the art.

The electrical device **1800** may include an other output device **1810** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1810** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

The electrical device **1800** may include an other input device **1820** (or corresponding interface circuitry, as discussed above). Examples of the other input device **1820** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as

a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

The electrical device **1800** may have any desired form factor, such as a handheld or mobile electrical device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultra mobile personal computer, etc.), a desktop electrical device, a server device or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable electrical device. In some embodiments, the electrical device **1800** may be any other electronic device that processes data.

The following paragraphs provide various examples of the embodiments disclosed herein.

Example 1 is an integrated circuit (IC) structure, including: a channel region including a semiconductor material; and a source/drain region at a side face of the channel region, wherein the source/drain region includes a semiconductor portion and a contact metal, and the semiconductor portion is between the contact metal and the semiconductor material.

Example 2 includes the subject matter of Example 1, and further specifies that the contact metal includes titanium.

Example 3 includes the subject matter of any of Examples 1-2, and further specifies that the contact metal includes an element, and the element includes zirconium, tantalum, tin, molybdenum, manganese, niobium, hafnium, iron, indium, gold, silver, palladium, platinum, chromium, copper, or cobalt.

Example 4 includes the subject matter of Example 3, and further specifies that the element is present in the contact metal at an amount between 10 atomic-percent and 20 atomic-percent.

Example 5 includes the subject matter of any of Examples 1-4, and further specifies that the contact metal is stressed.

Example 6 includes the subject matter of any of Examples 1-5, and further specifies that the semiconductor material is stressed.

Example 7 includes the subject matter of any of Examples 1-6, and further specifies that the semiconductor material has a stress between -500 megapascals and -1500 megapascals, or between 500 megapascals and 1500 megapascals.

Example 8 includes the subject matter of any of Examples 1-7, and further specifies that the semiconductor material includes a semiconductor fin.

Example 9 includes the subject matter of any of Examples 1-8, and further specifies that the semiconductor material includes multiple semiconductor wires.

Example 10 includes the subject matter of any of Examples 1-9, and further specifies that the semiconductor portion is part of a semiconductor layer having a U-shaped cross-section in the source/drain region.

Example 11 includes the subject matter of Example 10, and further specifies that the semiconductor layer includes an epitaxial layer.

Example 12 includes the subject matter of any of Examples 1-9, and further specifies that the semiconductor portion includes a first semiconductor portion and a second semiconductor portion, and the contact metal is at least partially between the first semiconductor portion and the second semiconductor portion.

Example 13 includes the subject matter of Example 12, and further specifies that the first semiconductor portion does not contact the second semiconductor portion.

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Example 14 includes the subject matter of Example 12, and further specifies that the first semiconductor portion contacts the second semiconductor portion.

Example 15 includes the subject matter of any of Examples 12-14, and further specifies that the semiconductor material includes a first semiconductor wire and a second semiconductor wire.

Example 16 includes the subject matter of Example 15, and further specifies that the first semiconductor portion contacts the first semiconductor wire, and the second semiconductor portion contacts the second semiconductor wire.

Example 17 includes the subject matter of any of Examples 15-16, and further specifies that the first semiconductor portion includes a first sub-portion and a second sub-portion, the first sub-portion is between a pair of dielectric spacers, and the first sub-portion is between the second sub-portion and the first semiconductor wire.

Example 18 includes the subject matter of any of Examples 15-17, and further specifies that the second semiconductor portion includes a first sub-portion and a second sub-portion, the first sub-portion is between a pair of dielectric spacers, and the first sub-portion is between the second sub-portion and the second semiconductor wire.

Example 19 includes the subject matter of any of Examples 15-18, and further specifies that the first semiconductor wire and the second semiconductor wire are arranged in a vertical array.

Example 20 includes the subject matter of any of Examples 1-17, and further specifies that the channel region is a first channel region, the semiconductor material is a first semiconductor material, the semiconductor portion is a first semiconductor portion, and the IC structure further includes: a second channel region including a second semiconductor material; wherein the source/drain region is between the first channel region and the second channel region, the source/drain region includes a second semiconductor portion, and the second semiconductor portion is between the contact metal and the second semiconductor material.

Example 21 includes the subject matter of Example 20, and further specifies that the second semiconductor portion includes a third semiconductor portion and a fourth semiconductor portion, and the contact metal is at least partially between the third semiconductor portion and the fourth semiconductor portion.

Example 22 includes the subject matter of Example 21, and further specifies that the third semiconductor portion does not contact the fourth semiconductor portion.

Example 23 includes the subject matter of Example 21, and further specifies that the third semiconductor portion contacts the fourth semiconductor portion.

Example 24 includes the subject matter of any of Examples 20-23, and further specifies that the second semiconductor material includes a third semiconductor wire and a fourth semiconductor wire.

Example 25 includes the subject matter of Example 24, and further specifies that the third semiconductor portion contacts the third semiconductor wire, and the fourth semiconductor portion contacts the fourth semiconductor wire.

Example 26 includes the subject matter of any of Examples 24-25, and further specifies that the third semiconductor portion includes a first sub-portion and a second sub-portion, wherein the first sub-portion is between a pair of dielectric spacers, and the first sub-portion is between the second sub-portion and the third semiconductor wire.

Example 27 includes the subject matter of any of Examples 24-26, and further specifies that the fourth semiconductor portion includes a first sub-portion and a second

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sub-portion, the first sub-portion is between a pair of dielectric spacers, and the first sub-portion is between the second sub-portion and the fourth semiconductor wire.

Example 28 includes the subject matter of any of Examples 21-27, and further specifies that the contact metal is between the first semiconductor portion and the third semiconductor portion.

Example 29 includes the subject matter of any of Examples 21-28, and further specifies that the contact metal is between the second semiconductor portion and the fourth semiconductor portion.

Example 30 includes the subject matter of any of Examples 1-29, and further specifies that the source/drain region is a first source/drain region, and the IC structure further includes: a second source/drain region, adjacent to the first source/drain region; and an insulating material region at least partially between the first source/drain region and the second source/drain region.

Example 31 includes the subject matter of Example 30, and further specifies that the insulating material region includes a first insulating material and a second insulating material, wherein the first insulating material has a U-shaped cross-section, and the first insulating material is between the second insulating material and first source/drain region.

Example 32 includes the subject matter of any of Examples 1-31, and further includes: a base region; and an insulating material region between the base region and the source/drain region.

Example 33 includes the subject matter of Example 32, and further specifies that the base region includes a semiconductor material.

Example 34 is an integrated circuit (IC) structure, including: a first channel region including a first semiconductor material; a second channel region including a second semiconductor material; and a source/drain region between the first channel region and the second channel region, wherein the source/drain region includes a first semiconductor region proximate to the first channel region, the source/drain region includes a second semiconductor region proximate to the second channel region, the source/drain region includes a contact metal between the first semiconductor region and the second semiconductor region, and the contact metal is stressed.

Example 35 includes the subject matter of Example 34, and further specifies that the contact metal includes titanium.

Example 36 includes the subject matter of any of Examples 34-35, and further specifies that the contact metal includes an element, and the element includes zirconium, tantalum, tin, molybdenum, manganese, niobium, hafnium, iron, indium, gold, silver, palladium, platinum, chromium, copper, or cobalt.

Example 37 includes the subject matter of Example 36, and further specifies that the element is present in the contact metal at an amount between 10 atomic-percent and 20 atomic-percent.

Example 38 includes the subject matter of any of Examples 34-37, and further specifies that the first semiconductor material is stressed.

Example 39 includes the subject matter of any of Examples 34-38, and further specifies that the first semiconductor material has a stress between -500 megapascals and -1500 megapascals, or between 500 megapascals and 1500 megapascals.

Example 40 includes the subject matter of any of Examples 38-39, and further specifies that the second semiconductor material is stressed.

Example 41 includes the subject matter of any of Examples 38-40, and further specifies that the second semiconductor material has a stress between -500 megapascals and -1500 megapascals, or between 500 megapascals and 1500 megapascals.

Example 42 includes the subject matter of any of Examples 34-41, and further specifies that the first semiconductor material includes a semiconductor fin.

Example 43 includes the subject matter of any of Examples 34-42, and further specifies that the first semiconductor material includes multiple semiconductor wires.

Example 44 includes the subject matter of any of Examples 42-43, and further specifies that the second semiconductor material includes a semiconductor fin.

Example 45 includes the subject matter of any of Examples 42-44, and further specifies that the second semiconductor material includes multiple semiconductor wires.

Example 46 includes the subject matter of any of Examples 34-41, and further specifies that the first semiconductor region and the second semiconductor region are part of a semiconductor layer having a U-shaped cross-section in the source/drain region.

Example 47 includes the subject matter of Example 46, and further specifies that the semiconductor layer includes an epitaxial layer.

Example 48 includes the subject matter of any of Examples 34-47, and further specifies that the first semiconductor region includes portions between pairs of dielectric spacers.

Example 49 includes the subject matter of any of Examples 34-48, and further specifies that the first semiconductor region contacts the first semiconductor material.

Example 50 includes the subject matter of any of Examples 34-49, and further specifies that the source/drain region is a first source/drain region, and the IC structure further includes: a second source/drain region, adjacent to the first source/drain region; and an insulating material region at least partially between the first source/drain region and the second source/drain region.

Example 51 includes the subject matter of Example 50, and further specifies that the insulating material region includes a first insulating material and a second insulating material, wherein the first insulating material has a U-shaped cross-section, and the first insulating material is between the second insulating material and first source/drain region.

Example 52 includes the subject matter of any of Examples 34-51, and further includes: a base region; and an insulating material region between the base region and the source/drain region.

Example 53 includes the subject matter of Example 52, and further specifies that the base region includes a semiconductor material.

Example 54 is an integrated circuit (IC) structure, including: a first channel region including a first semiconductor material; a second channel region including a second semiconductor material; and a source/drain region between the first channel region and the second channel region, wherein the source/drain region includes a first semiconductor region proximate to the first channel region, the source/drain region includes a second semiconductor region proximate to the second channel region, the source/drain region includes a contact metal between the first semiconductor region and the second semiconductor region, and the contact metal extends above the first semiconductor region and the second semiconductor region.

Example 55 includes the subject matter of Example 54, and further specifies that the contact metal includes titanium.

Example 56 includes the subject matter of any of Examples 54-55, and further specifies that the contact metal includes an element, and the element includes zirconium, tantalum, tin, molybdenum, manganese, niobium, hafnium, iron, indium, gold, silver, palladium, platinum, chromium, copper, or cobalt.

Example 57 includes the subject matter of Example 56, and further specifies that the element is present in the contact metal at an amount between 10 atomic-percent and 20 atomic-percent.

Example 58 includes the subject matter of any of Examples 54-57, and further specifies that the contact metal is stressed.

Example 59 includes the subject matter of any of Examples 54-58, and further specifies that the first semiconductor material is stressed.

Example 60 includes the subject matter of any of Examples 54-59, and further specifies that the first semiconductor material has a stress between -500 megapascals and -1500 megapascals, or between 500 megapascals and 1500 megapascals.

Example 61 includes the subject matter of any of Examples 59-60, and further specifies that the second semiconductor material is stressed.

Example 62 includes the subject matter of any of Examples 59-61, and further specifies that the second semiconductor material has a stress between -500 megapascals and -1500 megapascals, or between 500 megapascals and 1500 megapascals.

Example 63 includes the subject matter of any of Examples 54-62, and further specifies that the first semiconductor material includes a semiconductor fin.

Example 64 includes the subject matter of any of Examples 54-63, and further specifies that the first semiconductor material includes multiple semiconductor wires.

Example 65 includes the subject matter of any of Examples 63-64, and further specifies that the second semiconductor material includes a semiconductor fin.

Example 66 includes the subject matter of any of Examples 63-65, and further specifies that the second semiconductor material includes multiple semiconductor wires.

Example 67 includes the subject matter of any of Examples 54-66, and further specifies that the first semiconductor region and the second semiconductor region are part of a semiconductor layer having a U-shaped cross-section in the source/drain region.

Example 68 includes the subject matter of Example 67, and further specifies that the first semiconductor region includes an epitaxial layer.

Example 69 includes the subject matter of any of Examples 54-68, and further specifies that the first semiconductor region includes portions between pairs of dielectric spacers.

Example 70 includes the subject matter of any of Examples 54-69, and further specifies that the source/drain region is a first source/drain region, and the IC structure further includes: a second source/drain region, adjacent to the first source/drain region; and an insulating material region at least partially between the first source/drain region and the second source/drain region.

Example 71 includes the subject matter of Example 70, and further specifies that the insulating material region includes a first insulating material and a second insulating material, wherein the first insulating material has a U-shaped cross-section, and the first insulating material is between the second insulating material and first source/drain region.

Example 72 includes the subject matter of any of Examples 54-71, and further includes: a base region; and an insulating material region between the base region and the source/drain region.

Example 73 includes the subject matter of Example 72, and further specifies that the base region includes a semiconductor material.

Example 74 is an electronic assembly, including: a die including the IC structure of any of Examples 1-73; and a support electrically coupled to the die.

Example 75 includes the subject matter of Example 74, and further specifies that the support includes a package substrate.

Example 76 includes the subject matter of any of Examples 74-75, and further specifies that the support includes an interposer.

Example 77 includes the subject matter of any of Examples 74-75, and further specifies that the support includes a printed circuit board.

Example 78 includes the subject matter of any of Examples 74-77, and further includes: a housing around the die and the support.

Example 79 includes the subject matter of Example 78, and further specifies that the housing is a handheld computing device housing.

Example 80 includes the subject matter of Example 78, and further specifies that the housing is a server housing.

Example 81 includes the subject matter of any of Examples 78-80, and further includes: a display coupled to the housing.

Example 82 includes the subject matter of Example 81, and further specifies that the display is a touchscreen display. The invention claimed is:

1. An integrated circuit (IC) structure, comprising:

a first channel region including a first semiconductor material;

a second channel region including a second semiconductor material; a

a source/drain region between the first channel region and the second channel region, wherein the source/drain region includes a first semiconductor region proximate to the first channel region, a second semiconductor region proximate to the second channel region, a third semiconductor region, and a contact metal at least partially between the first semiconductor region and the second semiconductor region;

a dielectric layer that is physically in contact with the third semiconductor region; and

a substrate,

wherein the first semiconductor region is between a portion of the contact metal and the first channel region, the second semiconductor region is between the portion of the contact metal and the second channel region, the first semiconductor region abuts a first surface of the contact metal, the second semiconductor region abuts a second surface of the contact metal, the first surface and the second surface of the contact metal are opposing, the third semiconductor region is between the first semiconductor region and the second semiconductor region, and the dielectric layer is between the contact metal and the substrate.

2. The IC structure of claim 1, wherein the first semiconductor material is stressed.

3. The IC structure of claim 1, wherein the first semiconductor material includes a semiconductor fin.

4. The IC structure of claim 1, wherein the first channel region includes multiple semiconductor wires.

5. The IC structure of claim 1, wherein the contact metal includes titanium.

6. The IC structure of claim 1, wherein the contact metal includes an element, and the element includes zirconium, tantalum, tin, molybdenum, manganese, niobium, hafnium, iron, indium, gold, silver, palladium, platinum, chromium, copper, or cobalt.

7. The IC structure of claim 6, wherein the element is present in the contact metal at an amount between 10 atomic-percent and 20 atomic-percent.

8. The IC structure of claim 1, wherein the contact metal is stressed.

9. The IC structure of claim 2, wherein the first semiconductor material has a stress between -500 megapascals and -1500 megapascals, or between 500 megapascals and 1500 megapascals.

10. The IC structure of claim 1, wherein a cross section of a combination of the first semiconductor region, the second semiconductor region, and the third semiconductor region has a U shape.

11. An integrated circuit (IC) structure, comprising:

a first channel region including a first semiconductor material;

a second channel region including a second semiconductor material; and

a source/drain region between the first channel region and the second channel region, wherein the source/drain region includes a first semiconductor region proximate to the first channel region, a second semiconductor region proximate to the second channel region, a third semiconductor region, and a contact metal at least partially between the first semiconductor region and the second semiconductor region; and

a dielectric layer that is physically in contact with the third semiconductor region and is below the third semiconductor region,

wherein the first semiconductor region is between a portion of the contact metal and the first channel region, the second semiconductor region is between the portion of the contact metal and the second channel region, the first semiconductor region abuts a first surface of the contact metal, the second semiconductor region abuts a second surface of the contact metal, the first surface and the second surface of the contact metal are opposing, the third semiconductor region is between the first semiconductor region and the second semiconductor region, and the contact metal extends above the first semiconductor region and the second semiconductor region.

12. The IC structure of claim 11, wherein the contact metal is stressed.

13. The IC structure of claim 11, wherein the first semiconductor material and the second semiconductor material are stressed.

14. The IC structure of claim 11, wherein the first semiconductor region includes an epitaxial layer.

15. The IC structure of claim 11, wherein the contact metal includes titanium.

16. The IC structure of claim 11, wherein the contact metal includes an element, and the element includes zirconium, tantalum, tin, molybdenum, manganese, niobium, hafnium, iron, indium, gold, silver, palladium, platinum, chromium, copper, or cobalt.

17. The IC structure of claim 16, wherein the element is present in the contact metal at an amount between 10 atomic-percent and 20 atomic-percent.

18. The IC structure of claim **13**, wherein the first semiconductor material or the semiconductor material has a stress between -500 megapascals and -1500 megapascals, or between 500 megapascals and 1500 megapascals.

19. The IC structure of claim **11**, wherein a cross section 5 of a combination of the first semiconductor region, the second semiconductor region, and the third semiconductor region has a U shape.

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